

Grade 7 (Year 8) | Phase 3 / Cycle 2 | AI Solution Builder

Tooling Guide - Practical Kit for Term 1

Teachable Machine · Scratch · Pre-generated output packs · Card decks

TOOLING GUIDE

This guide equips the Grade 7 practical labs. The kit is deliberately small so it works on one teacher device and a

projector, keeps every student account-free, uses no personal data, and always has an unplugged fallback.

Confirm each tool against your school / ADEK approved-tools list and test it on the classroom network before

the lesson.

How the kit maps to the unit

The unit teaches AI as a workflow: input to process to output. Teachable Machine makes that pipeline visible; Scratch

lets students build one; pre-generated output packs let them judge outputs safely; card decks make workflows and

risk concrete with no device at all.

Golden rules (every tool, every lesson)

- The teacher operates any live tool; students direct it but never log in.
- No personal data - never faces, voices or names. Use objects, gestures, drawings or neutral text.
- Verify before you trust: check every output against the task and a source.
- Every activity has an unplugged fallback, so a blocked tool never blocks the lesson.
- Clear any trained model or project at the end of the lesson.

Google Teachable Machine

What it is: A free, browser-based tool to train a simple image / sound / pose classifier with no account. Training

happens on the device and the samples do not leave it.

What it teaches: Makes the input -> process -> output pipeline visible and shows how the examples you give shape

what a model learns.

Access & accounts: teachablemachine.withgoogle.com - open in a browser, teacher-operated on the board. No

login. Works on the teacher device; students direct it, they do not sign in.

Safety: Teachable Machine trains in the browser and the samples stay on the teacher's device. Use objects, hand

gestures or drawings only - never faces, voices or names. Clear the project at the end of the lesson.

Used in: Main L2, L5, L6, L8, L10; Bridging W1, W3, W5.

Alternative digital tool: If Teachable Machine is blocked, a school-approved image-classification demo or a

simple spreadsheet 'rule table' (a lookup that returns a label for typed features) shows the same input-process-

output pipeline on the teacher device.

Unplugged fallback: A card-deck 'human classifier': the class agrees a rule from example cards, then tests it on new cards.

Scratch (with optional Machine Learning for Kids)

What it is: A free block-based programming environment for building multi-step projects. An offline app is available

so no account is needed. Machine Learning for Kids can plug a trained model into a Scratch project (teacher-managed key).

What it teaches: Lets students build a working input -> process -> output workflow themselves, and (with ML

for Kids) have a project actually use a model they trained.

Prerequisite: Assumes basic Scratch familiarity (adding blocks, ask/say, simple if-blocks) from earlier grades.

If students are new to Scratch, use L3 for a short guided orientation first, or run the paper 'block cards' fallback.

Access & accounts: scratch.mit.edu or the offline Scratch app. Account-free for the base tool.

ML for Kids

(machinelearningforkids.co.uk) needs a teacher-managed setup - check your approved-tools list first.

Safety: No personal data in projects. If connecting a model, reuse the non-personal Teachable Machine model

from the same unit.

Saving across weeks (no accounts): projects built in L3 are reopened in L7-L10. Because students do not log in,

save each project as a local .sb3 file (File > Save to your computer in the offline Scratch app) on the teacher

device or a shared class drive, named by group. Reopen with File > Load from your computer.

Keep only non-personal projects.

Connect or simulate - which one? By default students SIMULATE the model: the Scratch project checks the

input against their own rule or a small lookup list (if-blocks). 'Connecting' a trained Teachable Machine model

requires Machine Learning for Kids with a teacher-managed key; use it only if it is on your approved-tools list.

Tell students which part of the build is real AI (the trained model) and which is a stand-in (the simulated rule).

Used in: Main L3, L6, L7, L8, L9, L10; Bridging W2, W5.

Alternative digital tool: If Scratch is unavailable, a school-approved block editor (e.g.

Blockly-based) or a flow-

chart app can build the same input-process-output sequence.

Unplugged fallback: Paper 'block cards': students arrange printed instruction blocks into a working sequence

on the desk.

Pre-generated output packs

What it is: Sets of AI outputs the teacher prepares BEFORE the lesson - strong and weak versions of the same task,

or outputs from data that differ in a known way - printed or shown on the board.

What it teaches: Lets students evaluate, compare and triage real outputs against criteria with zero live-tool risk and no waiting.

Access & accounts: Prepared in advance by the teacher (using the school-approved assistant on the teacher

device) and printed. Students never touch a tool.

Safety: Teacher curates every output in advance, so nothing unexpected or personal reaches students.

Used in: Main L4, L6, L9; Bridging W3, W4, W6.

Unplugged fallback: This IS the offline route - it needs no live tool.

Card decks

What it is: Printed card sets: workflow-example cards, system cards, risk-scenario cards and 'block' cards. Used for

sorting, matching, rule-building and risk-spotting on the desk.

What it teaches: Makes workflow structure, classification and risk concrete and fully unplugged; the default fallback for every AI activity.

Access & accounts: School-printed. No devices, no accounts. Always available.

Safety: Fully offline; no data of any kind involved.

Used in: Main L1, L5, L7; Bridging W4, W6 (and the fallback everywhere).

Unplugged fallback: This IS the unplugged tool.

Lab-by-lab tool map

Lab Primary tool Students produce

Bridging W1 - Lab Google Teachable Machine (clean vs messy data)

A comparison log showing how data quality changed the output

Bridging W2 - Lab Scratch (structured solution from guided inputs)

A structured Scratch solution and a short explanation.

Bridging W3 - Lab Teachable Machine retrain + pre-generated packs (inputs change outputs)

A before/after record explaining how input changes improved

Bridging W4 - Lab Card decks (bias-scenario cards) + pre-generated packs

A fairness record: the bias, its cause, who is affected and

Bridging W5 - Integrated Teachable Machine + Scratch (integrated: design & build)

A designed and built integrated solution with a strand-tagge

Bridging W6 - Integrated Pre-generated packs + card decks (integrated: evaluate, justify, risk)

An evaluated, justified solution with a risk check and readi

Main L1 - Lab Card decks (workflow-example cards) A sorted card layout plus a difference log.

Main L2 - Lab Google Teachable Machine (teacher-operated) A trained 3-class model and a completed

input-process-output

Main L3 - Lab Scratch (input -> process -> output build) A working three-stage Scratch project and a

short run log.

Main L4 - Lab Pre-generated output packs (evaluation clinic) A scored, ranked output pack with reasons and

one improvemen

Main L5 - Lab Card decks (risk-scenario cards) + pre-generated output packs + Teachable

Machine (live bias demo: skew -> cause -> fix)

A risk-spotting record with risks, who-is-affected and a pro

Main L6 - Lab Teachable Machine + Scratch + pre-generated packs (integrated run)

A run-and-check record across all four strands.

Main L7 - Lab Card decks (plan cards) + Scratch storyboard A planned card layout plus a Scratch

storyboard with a fallb

Main L8 - Lab Scratch (build) with optional Teachable Machine model

A working solution plus a build-and-test log with one logged

Main L9 - Lab Scratch + pre-generated comparison packs (refine)

A test-set accuracy count, a refinement note and a completed

Main L10 - Lab Showcase the Scratch / Teachable Machine solution + peer-review cards

A demonstrated solution, completed peer-review cards and a T

Setup checklist (before the term)

- Bookmark Teachable Machine and test the webcam on the teacher device; prepare non-personal example sets.
- Install the offline Scratch app (or confirm scratch.mit.edu loads); save the starter projects.
- Pre-generate and print the output packs for L4, L6, L9 and Bridging W3, W4, W6.
- Print the card decks: workflow-example, system, risk-scenario, bias-scenario, plan and block cards.
- Confirm all four tools against the approved-tools list; note any that need a teacher-managed key (e.g. ML for Kids).

Connectivity & browser notes (for IT)

Teachable Machine bandwidth: the site and its model library load once per device (about 5-10 MB). After that, training

and testing run locally in the browser - images are NOT uploaded - so one teacher device on a basic school

connection is enough. Allow headroom if several classes load the site at the same moment.

Browsers: use an up-to-date Chrome (recommended) or Edge for Teachable Machine and webcam capture. Scratch

runs in Chrome, Edge or Firefox, or fully offline via the installed Scratch app - prefer the offline app where

connectivity is limited.

If the network blocks a tool, every activity still runs on its unplugged card-deck fallback (see each lesson's 'Unplugged

fallback'), so a connectivity problem never blocks the lesson.

Grade 7 (Year 8) | Phase 3 / Cycle 2 | AI Solution Builder

Bridging Course - AI as Workflows - Input, Process,

Output

Transition-Year readiness | Weeks 1-6 (Core + Lab)

TEACHER PACK

1. About this Teacher Pack

This pack contains everything needed to teach the Grade 7 bridging program: a start-of-year diagnostic and six

weeks that rebuild each of the four strands to Grade 6 Advanced mastery before the Grade 7 unit begins. Weeks

1-4 rebuild one strand each; Weeks 5-6 integrate them in a mastery task.

How to use this pack

- Teach each week as Core (period 1) then Lab (period 2) on the same outcome.
- Keep all AI use teacher-mediated and account-free; enter no personal data.
- Build the evaluate-verify-risk habit into every AI activity; provide the unplugged fallback where tools are unavailable.
- Use the per-lesson formative check to confirm the outcome before moving on.
- Pair this Teacher Pack with the Student Lesson Content and Student Worksheets (same order).

- Privacy is technical, not only a rule: teach de-identification, data minimisation and retention/deletion at an Emerging level; note the UAE PDPL and ADEK policy.
- Simulated vs trained: the paper 'sorter' is a simulated lookup table - it cannot generalise to unseen items, so it is not a real trained model. Name which part of a build is a stand-in.
- Scratch prerequisite: assumes basic Scratch familiarity; save projects as local .sb3 files (offline Scratch app, account-free) to reopen across weeks.

Safeguarding & responsible use

Grade 7 designs, builds and evaluates simple AI workflows. Keep all AI use teacher-mediated and account-free for

students; prefer a board demonstration, curated or pre-generated output sets, or a card-based 'simulated AI' routine.

Never enter personal data (names, faces, voices) into a prompt or a model. Build the evaluate-verify-risk habit into

every AI activity: check each output against the task and a trusted source, because a confident output can still be

wrong. Every AI activity has a non-AI (unplugged) fallback so no lesson depends on tool access.

Materials & tools (UAE-accessible)

Tool / material What it is & why Access route Unplugged fallback

Workflow demo set +

example cards

A board demo of an input-process-output

workflow plus printed example cards.

School-printed; teacher-

run; no accounts.

Sort and map familiar

workflows on paper.

School-approved child-

safe assistant

An approved generative tool, teacher-

operated only, for curated demos on non-

personal data.

School-approved; teacher

device only.

Use pre-generated output

sets from the teacher.

Pre-generated output

sets

Curated good/weak outputs from the same

task for evaluation and triage.

Prepared by the teacher in

advance.

Same sets, printed.

Simulated-AI cards Card decks that mimic an AI step so students practise the workflow unplugged.

School-printed. This is itself the unplugged route.

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Program at a glance

Key terms - consistent definitions (use these throughout the term)

classifier / sorter - the process step that assigns each input to one class. Our paper 'sorter' is a simulated

classifier (a lookup table), not a trained model.

training data vs testing/input data - training data is the labelled examples a model learned from; testing (input)

data is the new, unseen items you feed the finished model. Keep them separate.

accuracy - the count-correct measure (right out of total, as a fraction or percentage). Do not call this 'reliability'.

reliable / reliability - consistency across more than one input.

bias - a repeating, systematic error that makes a system perform worse for one group of cases than another.

fairness - treating comparable cases or groups consistently; can be measured by comparing accuracy across groups.

model - learns patterns from training data and can generalise to unseen inputs; a fixed lookup table is not a model.

Week Strand / session Target (prior-grade Adv.)

Focus

Wk 1 AI Conceptual Understanding AIF·CU·G6·A Explains how training data affects output quality and performance

Wk 2 AI Solution Design & Development AIF·SD·G6·A Builds structured AI-assisted solutions using guided inputs and data

Wk 3 Critical Evaluation & Informed Interaction

AIF·CE·G6·A Explains how changes to inputs or data improve outputs

Wk 4 Ethics & Governance AIF·GE·G6·A Identifies biased outputs and suggests simple ways to improve fairness and safety

Wk 5-6 Integrated Prerequisite Mastery Task AIF·CU/SD/CE/GE·G6·A Design, build, evaluate & justify an AI solution

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Start-of-Year Diagnostic - Grade 6 Advanced readiness (all four strands)

Administered at the very start of the year, outside timetabled teaching hours (about 30 minutes, in small groups or stations), before bridging teaching begins. Each strand is recorded as mastery / partial / not-yet; no grade is reported to students.

Strand-specific routing (when strand scores differ)

The diagnostic reports a score per strand, so route per strand - not by one overall number. A student ready on some strands

but weak on others joins only the Bridging weeks for their weak strands: W1 Conceptual Understanding, W2 Solution

Design, W3 Critical Evaluation, W4 Ethics & Governance.

Per-strand rule of thumb: 40% or above - may skip that strand's Bridging week; 15-39% - join that strand's focus-group work;

below 15% - needs the Primer for that strand first. Re-check any borderline strand (about 38-42%) after Week 3 and adjust.

Strand Prior-grade Advanced skill probed Baseline probe Record

AI Conceptual

Understanding

Explains how training data affects output quality and performance.

Can the student demonstrate:

Explains how training data affects output quality and performance.

mastery /

partial / not-yet

AI Solution Design &

Development

Builds structured AI-assisted solutions using guided inputs and data.

Can the student demonstrate: Builds structured AI-assisted solutions using guided inputs and data.

mastery /

partial / not-yet

Critical Evaluation &

Informed Interaction

Explains how changes to inputs or data improve outputs.

Can the student demonstrate:

Explains how changes to inputs or data improve outputs.

mastery /

partial / not-yet

Ethics & Governance Identifies biased outputs and suggests simple ways to improve fairness and

safety.

Can the student demonstrate:

Identifies biased outputs and
suggests simple ways to improve
fairness and safety.

mastery /

partial / not-yet

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Week 1 - Core | AI Conceptual Understanding — Explains how training data
affects output quality and performance

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Prerequisite - How Models Learn (condensed, 20 min, teach before W1 Lab)

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Before the first sort/train activity, deliver this quick foundation (full lesson:
Grade7_New_Concept_Lesson_How_Models_Learn; interactive: 'How Models
Learn' HTML). Cover, with one example each: (1) rule-based vs learned - a person
writes rules, a model learns patterns from labelled examples; (2) training vs
inference - taught once, then used without changing; (3) a model stores patterns,
not copies of the examples; (4) same input gives the same answer, and a
confidence score (e.g. 0.90) is per-input sureness, not accuracy. Check: students
name one rule-based and one learned system and explain how training differs
from using.

Strand: AI Conceptual Understanding | Framework ref(s): AIF·CU·G6·A | 1 x 45 min

Objectives & success criteria

- Explain how the training data behind a model affects the quality and performance of its output.
- Analyse a worked input-process-output example to reason about how the training data shaped the result.
- Justify to a partner, using evidence from the example, why the training data made the output better or worse.

Student-facing success criteria:

- I can explain how the data a model learned from changes the output it gives.
- I can trace an example from input to output and show where the training data made a difference.
- I can convince my partner, using the example, why the training data helped or hurt the output.

Prior knowledge & Prerequisite Primer (10 min)

Assumes: This week rebuilds a skill from earlier grades before the new unit begins. In Grade 4
you explained how patterns

in data influence outcomes; in Grade 5 you explained how data quality affects AI outputs; in

Grade 6 you explained how

training data affects output quality and performance. Today you make that Grade 6 idea secure
and apply it to workflows.

Activate: Show two curated outputs from the same task on the board - one from a model trained
on a rich, balanced set of

examples and one from a model trained on a thin, narrow set - and ask which is more useful and why.

Teach: Name the chain: training data feeds the model, the model runs the process, and the process shapes the output. Model the reasoning on the two board examples - point to what the thin data set was missing and how that shows up in its output.

Apply: In pairs, students look at one more curated pair and say, in one sentence, how the training data explains the difference in the outputs.

Check: Cold-call two pairs to explain the link from data to output. Listen for the full chain - data shapes the model, the model shapes the output - not just 'one is better'. If a pair stops at the output, re-model tracing it back to the data before moving on.

Critical & analytical thinking focus

Analyse - reason about how the system/workflow produces outcomes (Understand/Analyse | DOK 3 | SOLO: Relational)

The technical concept (teacher background)

A model learns patterns from its training data, so the quality, range and balance of that data set a ceiling on how well the model can perform. Fed a rich, representative set, a model tends to produce accurate, useful outputs; fed a thin, narrow or skewed set, it produces gaps, errors or bias - even when the input and the process are identical. In a workflow this matters because the training data acts on the process step: it is the hidden influence behind the output long before a user sees it.

The reasoning move here is analysis - tracing an output back through the process to the data that shaped it. Keep all examples pre-generated and teacher-operated, account-free, with no personal data; any model discussed is trained only on non-personal examples. Cognitive demand for this lesson: Understand/Analyse | DOK 3 | SOLO: Relational.

Lesson plan

Phase | time Teacher (I do / we do) Students Key questions (targeted thinking)

Hook / 6 min Show two curated outputs for one task, Study both outputs and jot • Both models did the same task -

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made by models trained on different data sets (rich vs thin). Ask students which output they would rely on. one difference they notice between them.

so why did the outputs come out so different?

Teach the data-

to-output chain /

9 min

Model the chain training data -> model -> process -> output. Think aloud: name what the training data set was missing and trace how that gap reaches the output.

Copy the chain into their notes and label where the training data acts on the workflow.

- At which step does the training data do its work - and where does its effect show up?

Investigate

worked

examples / 14

min

Hand out curated example cards (input, the training-data note, and the output).

Circulate and prompt students to reason from data to output, not from opinion. Keep any live tool teacher-operated only.

In pairs, map each example from input to output and write how the training data explains the output quality.

- What in the training data would you change to improve this output, and why?

Explain to a

partner / 11 min

Set up a structured pair-explain: each partner justifies one example aloud using evidence from the card, then swaps. Prompt for the full relational chain.

Explain one example to a partner, pointing to the training data as the cause of the output quality; listen and challenge in return.

- Convince your partner: how do you know the training data - not the

input - caused this?

Check and close

/ 5 min

Run the exit check; collect the mapped example cards and the exit tickets as evidence. Flag verify-before-trust: a confident output can still rest on weak data.

Complete the exit ticket: one sentence linking a training-data feature to an output quality, with a reason.

- Explain how the training data affected this output's quality - in one clear sentence.

Anticipated misconceptions

Misconception How to address it

If the output is wrong, the person just typed a bad input. Show a clear, identical input feeding two models with different

training data and different outputs; the input was fine, so the cause must lie in the data and the process. Trace it back together.

More training data always means a better output. Compare a large but skewed data set with a smaller balanced

one; agree that balance and relevance matter, not size alone.

Name 'representative' as the quality that counts.

The training data and the input are the same thing. Separate the two on the board: the input is what a user gives

today; the training data is what the model learned from before.

Have pairs relabel any example that confused them.

Hands-on activity guide

Overview: Pairs investigate curated worked examples, mapping each from input to output and reasoning about how the

training data shaped the output's quality, then justify one example aloud to a partner using evidence from the card.

Grouping / timing: Pairs for the investigation and the explain, whole class for the debrief.

Tool: Pre-generated example cards on paper plus an optional teacher-operated board demo (a 'simulated AI' routine).

Account-free, no logins, no personal data - students never enter names or use a live model themselves.

Facilitation steps:

1. Read the example card: the input, the training-data note, and the output.
2. Map the example from input, through the process, to the output.
3. Write how the training data explains the quality of the output - what it had, or what it was missing.

4. Compare with a second card and note how a different data set changed the output.
5. Explain one example to your partner, pointing to the training data as the cause.
6. Check each other's reasoning: could the input alone explain it, or does it have to be the data?

Unplugged fallback: The cards are already paper-based, so no device is needed. If the board demo is unavailable, read one worked example aloud - input, training-data note and output - and have pairs reason from the printed cards only.

Formative assessment (aligned to the objectives)

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Task / item Model response Marking note

Explain how the training data affected the output in one worked example.

The model was trained mostly on photos of red apples, so its output labelled a green apple wrong - the training data was missing that example, and the gap shows up in the output.

Look for a clear link from a training-data feature to the output quality; the reason must name the data, not just the result.

Given two models with different training data on the same task, predict which output is stronger and justify it.

The model trained on a balanced set of many voices will handle new cases better, because it learned a wider range of patterns than the narrow one.

One mark for the correct prediction, one for a justification that references the range or balance of the training data.

Explain why the same input can give a poor output.

The input was fine, but the model was trained on thin, one-sided data, so its process could not handle the case and the output came out weak.

Accept answers that locate the cause in the training data or the process, not in the input alone.

- Targets: the Grade 6 skill being rebuilt - explaining how training data affects output quality and performance - applied to a workflow and reasoned through by analysis.

- Check: an exit ticket plus circulating listen-ins during the pair-explain, questioned against the training data -> model -> process -> output chain.
- Success criteria: the student links a specific training-data feature to an output's quality and justifies it with evidence from the example.
- Evidence: collected mapped example cards and the exit tickets.
- Feedback / adjustment: if students stop at 'one output is better', re-model tracing an output back to the training data and have them add the missing data-to-output link before the next lesson.

Differentiation

Criterion Emerging (expected) Approaching Not yet
 Linking data to output Explains how one training-data feature affected the output quality, with support.

Names a difference in the outputs but not the data cause.

Cannot connect the training data to the output.

Analysing the workflow Traces an example from input to output and locates where the data acts, with guidance.

Maps the example but misplaces where the data has its effect.

Cannot map the example through the workflow.

Justifying to a partner Justifies one example aloud using evidence from the card.

States an opinion without evidence from the example.

Does not justify a choice.

Representation: pair each example card with the visible training data -> model -> process -> output diagram; pre-teach 'training data', 'balanced' and 'representative' with simple icons.

Action & expression: let students show the data-to-output link by annotating the card, speaking it aloud to a scribe, or writing it - accept any mode for the exit ticket.

People of Determination: provide a reduced set of cards with picture support, sentence starters ('The output is weak because the

training data ...'), extra processing time, and a peer partner for the explain step.

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Week 1 - Lab | AI Conceptual Understanding — Explains how training data affects output quality and performance (Lab)

Strand: AI Conceptual Understanding | Framework ref(s): AIF·CU·G6·A | 1 x 45 min

Objectives & success criteria

- Investigate a set of worked examples to test how changing the training data changes the output quality.
- Map how the training data flows through the input-process-output workflow to shape the result.
- Justify a conclusion to a partner about which training data produced the better output, using evidence.

Student-facing success criteria:

- I can investigate examples and show how different training data changed the output.
- I can map how the training data acts inside the input-process-output workflow.
- I can justify to a partner which data gave the better output and back it with evidence.

Prior knowledge & Prerequisite Primer (10 min)

Assumes: In the Core period you rebuilt the Grade 6 skill - explaining how training data affects output quality and

performance - by tracing worked examples. That skill grew from Grade 4 (patterns in data influence outcomes) and Grade 5

(data quality affects AI outputs). Today you apply it more deeply and hands-on.

Activate: Show one input feeding two curated 'model kitchens' whose only difference is the training data used, then reveal their

two different outputs and ask what could possibly explain the gap.

Teach: Model an investigate-and-map routine: for one worked pair, map the shared input, the different training data, the process

and the two outputs onto a single sheet, then state which data won and why.

Apply: In pairs, students map one worked pair onto the sheet and write a one-sentence claim about which training data produced the better output.

Check: Check that each pair has pinned the cause on the training data, not the input; redirect any pair that credits the input, since the input was held identical.

Critical & analytical thinking focus

Analyse - reason about how the system/workflow produces outcomes (Understand/Analyse | DOK 3 | SOLO: Relational)

The technical concept (teacher background)

This Lab does not introduce a new concept - it deepens the same idea by controlled comparison. When the input and the

process are held identical and only the training data changes, any difference in the output must come from the data. That is

a fair test, and it is the clearest way to see how training data sets the quality and performance of a model. Students

investigate curated example pairs, map how the data flows through the input-process-output workflow, and justify an evidence-based conclusion to a partner. The habit - hold things constant, change one thing, explain the result - is the foundation of later evaluation work. All examples are pre-generated and teacher-operated, account-free, with no personal data; any model discussed is trained only on non-personal examples. Cognitive demand for this lesson:

Understand/Analyse | DOK 3 | SOLO: Relational.

Lesson plan

Phase | time Teacher (I do / we do) Students Key questions (targeted thinking)

Recap and set up

/ 5 min

Recap the training data -> model -> process -> output chain and the safe-use charter. Hand out the mapping sheets and the curated example-pair packs.

Read the first pair and predict which training data will give the better output.

- The input is the same for both - so what is the only thing that can explain a different output?

Model

investigate-and-

map / 8 min

Think aloud through one pair: map the shared input, the two training data sets, the process and the two outputs, then state which data won and the evidence for it.

Watch, then copy the mapping headings onto their own sheet.

- Because we changed only the training data, what does the different output prove?

Investigation

station 1 / 14 min

Circulate as pairs work the first two example pairs; prompt for evidence

Map each pair and write how the training data explains the

- What is your evidence that the training data - not chance - caused 10

from the data, not opinion. Keep all tools teacher-operated only. difference in output quality. this difference? Rotate and conclude / 13 min

Rotate pairs to a new pack; prompt them to compare across packs and build a justified conclusion.

Map the new pair, then write a conclusion naming which training data produced the better output and why.

- Across every pair you mapped, what quality of the training data mattered most?

Justify and close / 5 min

Collect the mapping sheets; take two conclusions and test them against the class evidence; flag verify-before-trust.

Complete the exit ticket: one conclusion about training data and output quality, with its evidence.

- Justify which training data gave the better output, and name your evidence.

Anticipated misconceptions

Misconception How to address it

The better output just got lucky. Point to the fair test: the input and process were identical, so the result is not luck - it traces to the training data. Have pairs name the controlled variable on their sheet.

You can tell the training data is good just by reading the output.

Show a fluent, confident output that came from biased data; agree the output alone is not proof, so we check what the model learned from. Reinforce verify-before-trust.

If two outputs differ, the model changed. Clarify that the model recipe was the same - only its training data

differed - so the data is the cause; re-map one pair to make this

visible.

Hands-on activity guide

Overview: Pairs run a controlled comparison across curated example-pair packs, mapping how identical inputs but different training data lead to different outputs, then justify an evidence-based conclusion about which data produced the better result.

Grouping / timing: Pairs at rotating stations, whole class for the debrief.

Tool: Pre-generated example-pair packs on paper or on a teacher-operated board (a 'simulated AI' comparison routine).

Account-free, no logins, no personal data.

Facilitation steps:

1. Read the first example pair and predict which training data gives the better output.
2. Map the shared input, the two training data sets, the process and the two outputs onto your sheet.
3. Write how the training data explains the difference in output quality.
4. Rotate to a new pack and map it the same way.
5. Write a conclusion naming which training data produced the better output, with your evidence.
6. Check the fair test: was the input really identical, so the data is the true cause?

Unplugged fallback: The packs are already paper-based, so no device is required. If the board demo is unavailable, use

the printed example pairs only and map the workflow by hand.

Lab tool integration (practical, hands-on)

Tool: Google Teachable Machine (clean vs messy data)

Teacher setup: Prepare two small example sets for the same class: one clean, one deliberately messy (blurry/mixed).

What students do:

1. Train the model on the clean set; test and read the output confidence.
2. Retrain the same class on the messy set.
3. Test the same unseen example and compare the two outputs.
4. Explain how the training data changed the output quality.
5. Write the rule: better examples, better output.

Students produce: A comparison log showing how data quality changed the output.

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Data-safety: Teachable Machine trains in the browser and the samples stay on the teacher's device. Use objects, hand gestures or drawings only - never faces, voices or names. Clear the project at the end of the lesson.

Unplugged fallback: Card-deck classifier trained on clear vs contradictory example cards.

Same model, clean vs messy training data, produces different output quality.

Formative assessment (aligned to the objectives)

Task / item Model response Marking note

Map one example pair and state which training data gave the better output.

Both used the same input, but the balanced data set produced the accurate output while the narrow one missed the case; the balanced data won.

One mark for a correct map, one for naming the training data as the cause with evidence.

Explain why a different output proves something about the training data here.

The input and process were kept the same, so the only thing left to explain the different output is the training data.

Look for reference to the controlled fair test - identical input and process, so the data is the cause.

Write a conclusion about what quality of training data mattered most, using your maps as evidence.

A balanced, representative data set mattered most, because in every pair it handled the new cases the narrow set got wrong.

Accept a conclusion that cites the mapped evidence and names balance or representativeness, not just size.

- Targets: the rebuilt Grade 6 skill - explaining how training data affects output quality and performance - applied through a controlled comparison and justified with evidence.
- Check: circulating evidence-prompts during the stations plus an exit ticket stating one conclusion about training data and output quality.
- Success criteria: the student maps the workflow, pins the output difference on the training data, and justifies a conclusion with mapped evidence.
- Evidence: collected mapping sheets and the exit tickets.
- Feedback / adjustment: if pairs credit the input or chance, re-show the fair test - identical input, different data - and have them relocate the cause before the next lesson.

Differentiation

Criterion Emerging (expected) Approaching Not yet
Investigating the fair test Recognises that only the training data changed and uses that to explain the output, with support.

Notices the outputs differ
but not that the data was the
only change.

Cannot identify what
caused the difference.

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Mapping the workflow Maps input, training data, process
and output onto the sheet, with
guidance.

Maps some stages but
misplaces where the data
acts.

Cannot complete a
usable map.

Justifying a conclusion States a conclusion about the
training data backed by mapped
evidence.

States a conclusion but
without evidence from the
maps.

No conclusion offered.

Representation: side-by-side example cards, a pre-formatted mapping sheet with the four
stages labelled, and the training data ->
model -> process -> output diagram on display.

Action & expression: pairs may record their map in writing, by annotating the cards, or by
explaining it to a scribing teacher.

People of Determination: fewer example pairs, a partly completed mapping sheet, sentence
starters ('The better output came from
... because the training data ...'), and extra time to rotate.

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Week 2 - Core | AI Solution Design & Development — Builds structured AI-
assisted solutions using guided inputs and data

Strand: AI Solution Design & Development | Framework ref(s): AIF·SD·G6·A | 1 x 45 min

Objectives & success criteria

- Design a structured, multi-step AI-assisted workflow that uses guided inputs and a small
curated dataset to complete a
set task.

- Build and run the workflow on a teacher-mediated or simulated routine, producing an output for
a given input.

- Explain how the guided input fields and the data shape the output, and verify the output
against the task.

Student-facing success criteria:

- I can plan a step-by-step workflow that takes a guided input and gives a useful output.
- I can build and run my workflow using our simulated routine to get an output.

- I can explain how my input fields and the data changed the output, and check the output is right.

Prior knowledge & Prerequisite Primer (10 min)

Assumes: In Grade 5 you learned to refine prompts to improve the quality of an output. In Grade 6 you built on that by

making structured AI-assisted solutions - using guided inputs (set fields, not just free text) and a small set of data to

complete a task. This week rebuilds that G6 skill before the new unit so it is secure.

Activate: Show two ways of asking for the same thing on the board: a messy free-text request and a tidy guided form with set fields

(material, size, clean or dirty). Ask which one a system could act on more reliably, and why.

Teach: Model the shape of a structured solution: a guided input feeds a process, the process uses a small data table, and it

returns an output. Walk one example end to end on the board, naming each part aloud.

Apply: In pairs, students fill one guided input card for a sample item and predict the output the data table would give.

Check: Cold-call two pairs to read their input fields and predicted output; check they can name the input, the data used and the

output as separate parts. If a pair treats the whole thing as one step, re-model the three parts before moving on.

Critical & analytical thinking focus

Apply & create - design and build a structured solution/workflow (Apply/Create | DOK 3 | SOLO: Relational)

The technical concept (teacher background)

A structured AI-assisted solution is a workflow with clear parts: a guided input (fixed fields rather than open free text), a

process that draws on data, and an output. Guiding the input - asking for exactly the fields the task needs - makes the

process more reliable, because the system is not left to guess what the user meant. The data is the small set of labelled

examples the process matches against; cleaner, fuller data usually gives a better output.

Building this solution is an

Apply/Create task: students structure the steps themselves rather than following one given to them. Keep every routine

teacher-operated and account-free, use only non-personal data, and verify each output against the task - a neat output can

still be wrong. Cognitive demand for this lesson: Apply/Create | DOK 3 | SOLO: Relational.

Lesson plan

Phase | time Teacher (I do / we do) Students Key questions (targeted thinking)

Hook / 6 min Present a real task - sorting classroom

waste into the right recycling bin - and

show one messy free-text request

beside one guided input card. Ask

which the class could act on reliably.

Vote and give one reason; jot which input they would trust and why.

- Which input gives the process exactly what it needs, and which leaves it guessing?

Model the structure / 10 min

Draw the workflow: guided input -> process -> output, with a small curated data table feeding the process. Walk one item end to end, thinking aloud at each stage.

Copy the three-part workflow shape and label where the data is used.

- At which stage does the data get used, and what would happen without it?

Design the guided input / 10 min

Hand out the design sheet. Guide pairs to choose the few input fields the task really needs (material, size, clean or

In pairs, design a guided input card - list the fields and why each one matters to the

- Which fields change the answer, and which are just noise you can drop?

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dirty) and to justify each field. output.

Build and run /

14 min

Run the simulated routine (a teacher-operated card-and-table 'classifier', or a board demo of a school-approved tool trained on non-personal item photos). Circulate and prompt verification.

Feed three items through the workflow, record each

output with the bin and reason, then check one output against the task.

- Your workflow says this bin - what is your evidence it is correct?

Check and close

/ 5 min

Collect design sheets and outputs. Run the exit check and flag verify-before-trust for the Lab.

Complete the exit ticket:

name your input fields, the data used and one output, and say how you checked it.

- Explain how your guided input and the data produced this output.

Anticipated misconceptions

Misconception How to address it

If I just type more words, the AI will do a better job. Compare a long free-text request with a short guided input that

has the right fields; show the guided input gives a more reliable output because the process receives exactly what it needs, not more words to sift.

The data does not matter - the AI already knows everything.

Swap in a data table with a gap or a wrong row and run the same input; show the output changes. Establish that the output can only be as good as the data behind it.

Building a solution means one big step. Have students point to the separate stages - input, process using

data, output; redraw any one-step plan as a labelled multi-step workflow before they build.

Hands-on activity guide

Overview: Pairs design a guided input card for a real sorting task, then build and run a small structured workflow on a

simulated routine, recording each output with a reason and verifying one against the task.

Grouping / timing: Pairs for the design and build, whole class for the model and the debrief.

Tool: A teacher-operated 'simulated AI' routine: a printed data table plus a matching card set that stands in for a classifier

trained on non-personal item examples. Optionally a board demo of a school-approved, teacher-run tool. Account-free, no

logins, no personal data - students never enter names, faces or private details.

Facilitation steps:

1. Read the task and agree what a good output looks like (the right bin, with a reason).

2. Design your guided input card - choose only the fields the task needs and say why each matters.

3. Build your workflow shape: guided input -> process using the data table -> output.

4. Run three items through the workflow and record each output with its reason.

5. Verify one output against the task and a trusted reference - is it actually correct?

6. Note one change to your input fields or data that would make the output more reliable.

Unplugged fallback: The routine already runs on paper cards and a printed data table, so no device is needed. If the board

demo is unavailable, pairs match their input cards against the data table by hand and record the output - the structured-

solution skill is unchanged.

Formative assessment (aligned to the objectives)

Task / item Model response Marking note

Design a guided input for the sorting task
and name the fields you chose.

Fields: material (plastic, paper, metal, glass),
size (small or large), clean or dirty - because
these are what decide the bin.

Award marks for fields that
genuinely affect the output; each
field should have a reason tied to
the task, not just be listed.

Run one item through your workflow and
record the output with a reason.

Input: plastic, small, clean -> the data table
maps clean plastic to the recycling bin ->

Output: recycling bin, because clean plastic
can be recycled.

One mark for a correct output that
traces back through the process
and the data, with a reason - not a
bare answer.

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Explain how the data changed the output,
using one example.

When the table had no row for dirty plastic,
the workflow could not sort it; adding that
row let it send dirty plastic to general waste,
so the data decided the output.

Look for a clear link between a
change in the data and a change
in the output; reject 'the AI just
knows'.

- Targets: designing and building a structured AI-assisted workflow that uses guided inputs and data, and explaining how those parts shape the output.
- Check: circulating build-and-verify prompts during the run, plus an exit ticket naming the input fields, the data used and one checked output.
- Success criteria: the student designs a guided input, runs it through a data-driven process to a correct output, and explains how the input and data produced it.
- Evidence: collected design sheets, the recorded outputs with reasons, and the exit tickets.
- Feedback / adjustment: if a pair builds a one-step plan or ignores the data, re-model the three labelled stages and have them rerun one item using the data table before the Lab.

Differentiation

Criterion Emerging (expected) Approaching Not yet

Designing guided inputs Chooses sensible input fields for the task and gives a reason for each, with support.

Chooses fields but reasons are missing, or includes fields that do not affect the output.

Cannot choose task-relevant fields.

Building and running the workflow

Builds a multi-step workflow and runs an item through it to a correct output, with guidance.

Runs the workflow but skips a stage or misuses the data.

Cannot produce an output from an input.

Explaining input and data Explains how the guided input and the data shaped the output for one example.

Describes the output but not how the input or data produced it.

No link between input, data and output.

Representation: display the guided input -> process -> output diagram, provide the data table in large print with icons, and pre-teach the words field, data and output with simple examples.

Action & expression: pairs may build the workflow by arranging cards, annotating the design sheet, or explaining aloud to the teacher who scribes; accept any mode for the exit ticket.

People of Determination: provide a part-completed design sheet with two fields already given, a reduced data table, sentence

starters ('This item goes to ... because the data says ...'), and extra processing time.

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Week 2 - Lab | AI Solution Design & Development — Builds structured AI-assisted solutions using guided inputs and data (Lab)

Strand: AI Solution Design & Development | Framework ref(s): AIF·SD·G6·A | 1 x 45 min

Objectives & success criteria

- Extend a structured AI-assisted workflow to handle new and harder inputs, adjusting the guided input fields or the data as needed.
- Test the workflow with fresh cases, identify where the output fails, and refine one part to improve it.
- Justify the design choices, explaining how the guided inputs and the data produce a reliable output.

Student-facing success criteria:

- I can improve my workflow so it handles new, trickier inputs.
- I can test my workflow, find where it goes wrong, and fix one part.
- I can explain why my input fields and data make the output reliable.

Prior knowledge & Prerequisite Primer (10 min)

Assumes: In Grade 5 you refined prompts to improve output quality, and in Grade 6 you built structured AI-assisted

solutions using guided inputs and data. In this week's Core period you designed and ran a small sorting workflow. Today you

make that same solution more robust.

Activate: Show a tricky input that breaks the Core workflow - an item that fits no row in the data table, such as a coffee cup that is

part paper, part plastic - and ask what the workflow should do.

Teach: Model a test-and-refine loop: run a hard case, find the stage that failed (a missing input field or a data gap), change one

thing, then run it again and compare.

Apply: In pairs, students run one tricky case through their Core workflow and name the stage that fails.

Check: Check each pair can point to the failing stage and name one change that would fix it; redirect any pair who only says 'the AI

is wrong' back to the input or the data.

Critical & analytical thinking focus

Apply & create - design and build a structured solution/workflow (Apply/Create | DOK 3 | SOLO: Relational)

The technical concept (teacher background)

Refining a structured solution means improving one part at a time and testing the effect. When an output is wrong, a missing input field or a data gap is ONE common cause you can often trace - but not the only one. Outputs can also be wrong because the labels were noisy, the data was unrepresentative or biased, the item is genuinely ambiguous, or the model is uncertain (a low confidence score). Treat a missing field or data gap as a first hypothesis to test, not the whole explanation. Adding a field, correcting a data row, or adding an example can let the workflow handle more cases reliably. This is still Apply/Create at DOK 3 - students are not meeting a new idea but building and justifying a more robust version of the same structured solution. Good design is judged against the task: does the output stay correct for new, less predictable inputs? Keep the routine teacher-operated and account-free, use only non-personal data, and verify every output against the task and a trusted reference before trusting it. Cognitive demand for this lesson: Apply/Create | DOK 3 | SOLO: Relational.

Lesson plan

Phase | time Teacher (I do / we do) Students Key questions (targeted thinking)

Recap and set

up / 5 min

Recap the guided input -> process ->

output shape and the safe-use rules.

Hand out the test log and a set of tricky

test cases designed to stress the Core workflow.

Get out their Core design

sheet and data table, and

read the first tricky case.

- Before you run it, where do you predict this hard case will break - the input or the data?

Model test-and-

refine / 8 min

Think aloud through one hard case: run

it, find the failing stage, change one

thing (add a field or a data row), rerun,

and log the improvement.

Copy the test-and-refine

loop headings into their log:

case, what failed, one

change, result.

- Which single change fixed it, and

how do you know it did not break
the other cases?

Test station / 14 Circulate as pairs run the tricky cases Run at least three tricky • What is your
evidence that the

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min through their workflow; prompt them to
locate the failing stage precisely and to
change only one thing at a time.

cases, logging what failed
and the stage it failed at for
each.

fault is in the data and not the
input?

Refine and re-run

/ 13 min

Support pairs to make one targeted
refinement and rerun the failed cases;
prompt them to verify each new output
against the task.

Make one refinement, rerun
the cases it should now
handle, and record whether
the output is correct.

- Did your change fix the hard case
without spoiling an easy one?

Show me.

Debrief and

close / 5 min

Collect the logs. Take two refinements
and test the reasoning against the class
evidence; reinforce verify-before-trust.

Complete the exit ticket: one
refinement you made and
why it improved the
workflow.

- Justify one change you made -
how did it make the output more
reliable?

Anticipated misconceptions

Misconception How to address it

If the output is wrong, the whole workflow is broken and I
should start again.

Trace the fault to a single stage - usually a missing input field or a
data gap - and show that changing one part fixes it; rebuilding

from scratch is rarely needed.

I should change lots of things at once to fix it fast. Model changing one thing at a time and rerunning, so you can tell

which change actually helped; a fair test isolates one variable.

Once it works on one hard case, it works on everything. Have pairs rerun the easy cases after refining; show a change can

fix one case and break another, so the whole set must be re-tested against the task.

Hands-on activity guide

Overview: Pairs stress-test their Core workflow with tricky cases, locate the stage that fails, make one targeted refinement

to the guided input or the data, rerun, and justify why the workflow is now more reliable.

Grouping / timing: Pairs at the test station, whole class for the model and the debrief.

Tool: The same teacher-operated 'simulated AI' routine from the Core period - printed data table plus card set - now run

against a curated set of tricky test cases. Optionally a teacher-run board demo of a school-approved tool. Account-free, no logins, no personal data.

Facilitation steps:

1. Set up your Core workflow: guided input card, data table and output slot.
2. Run each tricky test case and watch for the output that is wrong or missing.
3. For each failure, log what went wrong and the stage it failed at - input or data.
4. Choose one targeted change: add or fix an input field, or correct or add a data row.
5. Rerun the failed cases and the easy cases, and record whether each output is now correct.
6. Write one sentence justifying why your refinement made the workflow more reliable.

Unplugged fallback: The test cases and data table are already on paper, so no device is needed. If any board demo is

unavailable, pairs match the tricky cards against their data table by hand, refine a row, and rerun - the test-and-refine skill is unchanged.

Lab tool integration (practical, hands-on)

Tool: Scratch (structured solution from guided inputs)

Teacher setup: Provide a Scratch starter with labelled input slots and a guided step list.

What students do:

1. Read the guided inputs and decide the order of steps.
2. Build the structured workflow block by block.
3. Run it and check the output matches the goal.
4. Improve one step so the output is clearer.
5. Explain why your structure works.

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Students produce: A structured Scratch solution and a short explanation.

Data-safety: No personal data; neutral inputs only.

Unplugged fallback: Sequence printed block-cards into the structured solution.

Guided inputs assembled into a structured multi-step Scratch workflow.

Formative assessment (aligned to the objectives)

Task / item Model response Marking note

Run a tricky case and log what failed and the stage it failed at.

Case: a paper-and-plastic coffee cup. It failed because the data table had no row for a mixed material (stage: data), so the output was blank.

One mark for a correctly located failure - the stage named must match the actual cause, not just 'output wrong'.

Make one refinement and show it fixed the case without breaking an easy one.

I added a 'mixed material -> general waste' row. The coffee cup now outputs general waste, and the plain plastic case still outputs recycling, so both are correct.

Award the mark only if the refinement is a single targeted change and the easy case is re-checked and still correct.

Justify why your workflow is now more reliable, using the data or the input.

It is more reliable because the data now covers mixed items, so the guided input for a coffee cup maps to a correct output instead of a gap.

Look for a justification that links a specific change in the input or data to a more reliable output; reject vague 'it just works now'.

- Targets: extending and refining a structured AI-assisted solution to handle harder inputs, and justifying how the guided input and data make the output reliable.
- Check: circulating test-and-refine prompts during the station, plus an exit ticket naming one refinement and its justification.
- Success criteria: the student locates a failing stage, makes one targeted refinement that fixes the hard case without breaking an easy one, and justifies why.
- Evidence: collected test logs showing failures, refinements and re-run results, and the exit tickets.
- Feedback / adjustment: if a pair changes several things at once or cannot locate the fault, re-model the one-change-at-a-time

loop and have them isolate a single stage before the next unit lesson.

Differentiation

Criterion Emerging (expected) Approaching Not yet

Locating the failure Runs a tricky case and names the stage that failed - input or data - with support.

Notices the output is wrong but mislocates or cannot name the stage.

Cannot tell that or where the workflow failed.

Refining the workflow Makes one targeted change and Makes a change but alters Cannot make a change

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reruns to check the case is fixed, with guidance.

several parts at once or does not rerun.

that improves the output.

Justifying the design Explains how the change to the input or data made the output more reliable.

Describes the change but not why it improved reliability.

No justification linking the change to the output.

Representation: display the test-and-refine loop as a simple cycle diagram, provide the data table in large print, and pre-teach the words refine, test case and reliable with examples.

Action & expression: pairs may refine by moving or rewriting cards, annotating the log, or explaining the change aloud to the teacher who scribes.

People of Determination: provide fewer, clearly labelled test cases, a part-completed log with one worked example, sentence

starters ('It failed at the ... I changed ... so now ...'), and extra time to rerun.

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Week 3 - Core | Critical Evaluation & Informed Interaction — Explains how changes to inputs or data improve outputs

Strand: Critical Evaluation & Informed Interaction | Framework ref(s): AIF·CE·G6·A | 1 x 45 min

Objectives & success criteria

- Explain how a change to the input or data can improve the output of a simple AI workflow.

- Judge two outputs from the same workflow against agreed criteria and justify which is better with evidence.

- Use input-process-output language to say what was changed and why the output improved.

Student-facing success criteria:

- I can explain how changing the input improved the output of a workflow.

- I can compare two outputs against agreed criteria and say which is better, with evidence.

- I can point to the exact change I made and link it to the improvement in the output.

Prior knowledge & Prerequisite Primer (10 min)

Assumes: This rebuilds a skill from before. In Grade 4 you explained how data influences the outcome of a solution; in

Grade 6 you explained how changes to inputs or data improve outputs. Today you use that same skill inside an input-process-output workflow.

Activate: Show two outputs from one workflow on the board - a vague first try and a stronger second try - and ask: which is better, and what do you think was changed to get it?

Teach: Name the move out loud - a workflow has an input, a process and an output; if we change the input while the process stays the same, the output can get better, and we can prove it against a criterion.

Apply: In pairs, look at the two board outputs and write one criterion that tells a better output from a weaker one (for example, clearer, more accurate, more complete).

Check: Cold-call two pairs to read their criterion; the rung is secure when a pair can name a change to the input and link it to a better output against their criterion.

Critical & analytical thinking focus

Evaluate - judge outputs and solutions against criteria, with evidence (Evaluate | DOK 3 | SOLO: Relational)

The technical concept (teacher background)

In a workflow the process stage is fixed, but the input is not. A machine-learning process maps whatever it is given to a likely

output, so the quality of the input strongly shapes the quality of the output - clearer, fuller or more relevant data usually

yields a better result, while vague or biased data degrades it. Improving an output therefore often means improving the

input rather than rebuilding the process. To claim an improvement fairly, learners need criteria: agreed, observable

measures such as accuracy, clarity, relevance or completeness. Judging two outputs against the same criteria, with

evidence for each judgement, is genuine evaluation - not a matter of taste. This is the foundation of iteration in the Grade 7

unit: change the input, re-run, compare against criteria, and keep the change only if the evidence shows the output actually

improved. Every output is still checked before it is trusted.

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An evaluation ladder: judgement, criterion, evidence.

Lesson plan

Phase | time Teacher (I do / we do) Students Key questions (targeted thinking)

Hook / 6 min Show two board outputs from the same

curated workflow - a weak first output

and a stronger second - and think aloud

that only the input was changed.

Study both outputs and call

out which one looks better

and one reason why.

The process did not change - so

why is the second output better?

Direct teach / 12

min

Define criteria (accuracy, clarity, relevance, completeness); we-do

judging the two outputs against one

shared criterion and record the

evidence.

Copy the criteria and help

score the two outputs

against the shared criterion.

What evidence lets us say this

output is more accurate, not just

nicer?

Guided evaluate

/ 12 min

Hand out a curated set of before-and-

after outputs; model justifying the first

judgement, then release pairs to judge

the rest against the agreed criteria.

In pairs, judge each output

pair against the criteria and

write the evidence for your

choice.

What was changed in the input,

and how did that change improve

the output?

Explain the link /

10 min

Prompt pairs to state the change and

the improvement as one cause-and-

effect sentence; circulate and listen for

the link and the evidence.

Say aloud: I changed the input by ..., so the output improved because

Can you point to the exact change and the exact improvement?

Check and close

/ 5 min

Collect the exit ticket; recap that a better input can improve the output and that we prove it against criteria - preview tomorrow's Lab evaluation panel.

Complete the exit ticket naming one input change and the improvement it caused.

Name your criterion and the evidence for a better output - go.

Anticipated misconceptions

Misconception How to address it

A better output means the AI got smarter or the process changed.

Show that only the input was changed while the process card stayed identical; the output improved because a better input flows through the same process.

One output is better just because I like it more. Replace taste with criteria: agree an observable measure such as

accuracy or completeness, and require evidence from the output for every judgement.

Any change to the input will always improve the output. Show a change that made an output worse; explain that we keep a

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change only if the evidence against the criteria shows a real improvement, and we still check the output.

Hands-on activity guide

Overview: Students evaluate a curated set of before-and-after outputs from one workflow, judging each pair against agreed criteria and justifying which output is better with evidence.

Grouping / timing: Pairs for the evaluate; whole class for the debrief.

Tool: A teacher-operated board demo with a curated, pre-generated set of before-and-after outputs (no accounts, no live typing by students) - so the class evaluates safe, prepared results and no personal data is entered.

Facilitation steps:

1. Read the two outputs your teacher shows for the same workflow.
2. Use the agreed criteria - accuracy, clarity, relevance, completeness.
3. Judge which output is better on each criterion and mark it.
4. Write the evidence from the output that supports your judgement.
5. Say what was changed in the input to make the better output.
6. Check the better output against the task - does it actually answer it?

Unplugged fallback: Simulated-AI cards: a fixed 'process' rule sits in an envelope. A pair posts a weak input card, then a clearer input card, and compares the two output cards the envelope returns - the same evaluate-against-criteria routine with no device at all.

Formative assessment (aligned to the objectives)

Task / item Model response Marking note

Judge two outputs from the same workflow against one agreed criterion and mark the better one.

On accuracy, output B is better: it names the correct capital while output A leaves it blank - so B is more accurate.

Award full marks when the judgement names the criterion and cites evidence from the output; a choice with no evidence is partial.

Explain how the input was changed to improve the output.

The first input was one vague word; the second input added a clear, complete question, so the process returned a fuller and more accurate output.

Look for a named input change linked by cause and effect to a named improvement, not a general 'it got better'.

Write one criterion you would use to tell a better output from a weaker one, and why.

Completeness - a better output answers the whole task, so I can check whether every part of the question was covered.

Accept any observable, checkable criterion with a reason; gently redirect vague criteria such as

'nicer' to something measurable.

- Targets: the prior-grade Advanced outcome - explains how changes to inputs or data improve outputs - rebuilt in the workflow context.

- Check: pair evaluation of curated before-and-after outputs against agreed criteria, sampled by cold-call.

- Success criteria: the student judges outputs against criteria with evidence and links a named input change to the improvement.

- Evidence: the completed evaluation table and the one cause-and-effect sentence on the worksheet.

- Feedback / adjustment: if a judgement is taste-based, re-show both outputs and ask the student to quote the evidence for one criterion before deciding again.

Differentiation

Criterion Emerging (expected) Approaching Not yet

Judges against criteria Judges outputs against agreed criteria and picks the better one

Picks a better output but

does not use the criteria

Picks by preference with

no criterion

Justifies with evidence Cites evidence from the output for each judgement

Gives a reason but no

evidence from the output

Gives no reason for the

judgement

Links change to Names the input change and links it Names a change but the link Does not connect the

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improvement to the better output is vague change to the output

Representation: pair a before-and-after output panel with the written criteria list so the comparison is visible and the measures are always in view.

Action and expression: let students show judgement by marking a table, speaking the comparison to a partner, or sorting output cards - not only by writing.

People of Determination: provide a criteria checklist with tick boxes and sentence starters ('Output B is better on ... because ...');

allow extra time and a calm partner for the explain step.

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Week 3 - Lab | Critical Evaluation & Informed Interaction — Explains how changes to inputs or data improve outputs (Lab)

Strand: Critical Evaluation & Informed Interaction | Framework ref(s): AIF·CE·G6·A | 1 x 45 min
Objectives & success criteria

- Evaluate a curated set of outputs from a workflow against agreed criteria and rank them.
- Justify each judgement with specific evidence from the output and name the input change behind the best result.
- Iterate once: change an input, re-run the simulated workflow, and evaluate whether the output actually improved.

Student-facing success criteria:

- I can rank a set of outputs against agreed criteria and explain my order.
- I can justify every judgement with evidence I can point to in the output.
- I can change an input, compare the new output against the criteria, and say whether it really improved.

Prior knowledge & Prerequisite Primer (10 min)

Assumes: This rebuilds a prior-grade skill. Grade 4: you explained how data influences the outcome of a solution. Grade 6:

you explained how changes to inputs or data improve outputs. In the Core period you judged two outputs against criteria with evidence; now you evaluate a whole set and iterate.

Activate: Replay the Core before-and-after panel and ask one pair to state, in ten seconds, the change and the improvement it caused.

Teach: Show a set of four curated outputs from one workflow and model ranking them against the agreed criteria, quoting evidence for the top and bottom.

Apply: In pairs, agree the order of two of the four outputs and write the evidence for your top choice.

Check: Thumbs up or down - does the rest of the class agree with your top choice? Ask one pair to defend their order with evidence.

Critical & analytical thinking focus

Evaluate - judge outputs and solutions against criteria, with evidence (Evaluate | DOK 3 | SOLO: Relational)

The technical concept (teacher background)

Evaluating a set means applying the same criteria across several outputs and ranking them, then justifying the order with

evidence - the DOK 3 move of judging solutions against criteria. Iteration adds the causal step: change one input, hold the

process constant, re-run, and evaluate whether the new output beats the old one on the criteria.

This is how real teams

improve AI solutions - not by guessing, but by making a deliberate change and testing it against agreed measures. It also

guards against a trap: a change can look better yet fail a criterion, or improve one criterion while hurting another, so

evidence and criteria keep the judgement honest. In this Lab students need no accounts and no live tools; a simulated-AI card deck or a teacher-run prepared output set reproduces the same behaviour safely, so every learner can rank, justify and iterate, with verify-before-trust built into each judgement and no personal data ever entered.

Lesson plan

Phase | time Teacher (I do / we do) Students Key questions (targeted thinking)

Recap and set-

up / 5 min

Replay the Core panel; hand each pair a curated set of four output cards from one workflow and the agreed criteria strip.

Recall the criteria and lay the four output cards out ready to rank.

What criteria are we using, and what counts as evidence?

Model the

ranking / 10 min

Think aloud ranking the four outputs against the criteria, quoting the evidence for the strongest and weakest.

Watch and copy the ranking method for the demonstrated criterion.

Which output wins on this criterion, and what evidence proves it?

Rank and justify /

12 min

Release pairs to rank all four outputs and write the evidence for each judgement; circulate and press for evidence, not taste.

In pairs, rank the four outputs and write the evidence that supports your order.

Could someone else check your ranking using only your evidence?

Iterate once and

re-evaluate / 12

Prompt pairs to change one input, re-

run the simulated workflow for a new

Change one input, run it

again, and decide against the

Did your change improve the

output on the criteria, or only look

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min output, and judge it against the set;

model verify-before-trust on the new

output.

criteria whether the new

output is really better.

different?

Share and check

/ 6 min

Take two rankings to the board; collect

the exit ticket naming the change that

most improved an output and the

evidence for it.

Share your ranking and

complete the exit ticket.

What single input change most

improved an output, and how do

you know?

Anticipated misconceptions

Misconception How to address it

The output at the top is best because it is the longest or

most detailed.

Rank two outputs where the longer one is less accurate; show

that length is not a criterion and evidence must match the agreed

measures.

If my new output looks different, I have improved it. Judge the new output against the criteria

beside the old one; a

change counts as an improvement only when the evidence shows

it beats the old output on a criterion.

One change can improve every criterion at once. Show a change that improves clarity but drops

completeness;

explain that improvements can trade off, so we weigh the

evidence across criteria and still verify the output.

Hands-on activity guide

Overview: Students evaluate a curated set of outputs from one workflow, rank them against

agreed criteria with evidence,

then iterate once - changing an input and re-judging whether the output truly improved.

Grouping / timing: Pairs for the rank and iterate; whole class for the board share.

Tool: A simulated-AI card deck (a fixed 'process' rule in an envelope) or a teacher-run prepared output set - fully account-

free and device-free, so students evaluate and iterate without entering any personal data.

Facilitation steps:

1. Lay out the set of outputs from the same workflow.
2. Rank them against the agreed criteria - accuracy, clarity, relevance, completeness.
3. Write the evidence from the output for each judgement.
4. Change one input and run it through the process again.
5. Judge the new output against the criteria - is it really better?
6. Check the best output against a trusted answer before you trust it.

Unplugged fallback: The card deck is already the unplugged version: no screens are required. If a live approved demo is

used instead, the same card routine is the fallback so every pair can still rank, justify and iterate.

Lab tool integration (practical, hands-on)

Tool: Teachable Machine retrain + pre-generated packs (inputs change outputs)

Teacher setup: Have a trained model and a pack showing outputs before and after an input change.

What students do:

1. Test the current model and note the output.
2. Change the inputs (add clearer examples to one class) and retrain.
3. Test again and compare with the before output from the pack.
4. Explain which input change improved the output and why.
5. Record the improvement with evidence.

Students produce: A before/after record explaining how input changes improved outputs.

Data-safety: Teachable Machine trains in the browser and the samples stay on the teacher's device. Use objects, hand

gestures or drawings only - never faces, voices or names. Clear the project at the end of the lesson.

Unplugged fallback: Compare before/after outputs from the printed pack only.

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Change the inputs, retrain, and the output improves - shown before and after.

Formative assessment (aligned to the objectives)

Task / item Model response Marking note

Rank a set of outputs against the agreed criteria and justify the top choice with evidence.

My order is C, A, D, B. C is top because it is accurate and complete - it names the correct fact and answers every part of the task.

Full marks when the order is defended with evidence tied to the criteria, not to length or preference.

Change one input, re-run, and decide

against the criteria whether the output improved.

I added the missing date to the input; the new output is more complete because it now covers the part it left out before, so it improved on completeness.

Look for a named input change, a re-judgement against a criterion, and a clear yes or no with evidence.

Explain a case where a changed input did not improve the output.

When I shortened the input it became unclear, so the output missed the main point and scored worse on accuracy - that change was not worth keeping.

Accept any honest example where the evidence shows the change failed a criterion; this shows evaluation, not just optimism.

- Targets: the same prior-grade Advanced outcome - explains how changes to inputs or data improve outputs - applied by ranking a set and iterating.
- Check: pair ranking of a curated output set plus one input-change iteration, observed during the build.
- Success criteria: the student ranks outputs against criteria with evidence and judges whether a changed input actually improved the output.
- Evidence: the completed ranking table and the before-and-after iteration row on the worksheet.
- Feedback / adjustment: if a ranking rests on length or taste, ask the student to quote the criterion and the evidence for two neighbouring outputs before re-ordering.

Differentiation

Criterion Emerging (expected) Approaching Not yet

Ranks against criteria Ranks the set against agreed criteria and defends the order

Ranks the set but the criteria are applied loosely

Orders by preference with no criteria

Justifies with evidence Cites specific evidence from the output for each judgement

Gives evidence for some judgements only

Gives judgements with
no evidence
Iterates and re-judges Changes an input and judges the
new output against the criteria
Changes an input but does
not re-judge fairly
Does not connect the
change to the criteria

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Representation: give a criteria matrix (outputs down the side, criteria across the top) so the whole set can be compared in one view.

Action and expression: accept a spoken defence of the ranking, a card arrangement, or a written matrix as evidence of the same reasoning.

People of Determination: pre-fill one criterion column so the student focuses on the rest; provide a ranking frame ('I put ... above ... because the evidence shows ...').

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Week 4 - Core | Ethics & Governance - Identifies biased outputs and suggests simple ways to improve fairness and safety

Strand: Ethics & Governance | Framework ref(s): AIF·GE·G6·A | 1 x 45 min

Objectives & success criteria

- Identify biased or unfair outputs in a curated AI workflow and explain how the bias shows up in the output.
- Trace a spotted bias back to the input or the process stage of the workflow and justify where it began.
- Suggest at least two simple, reasoned measures that would make the workflow fairer or safer.

Student-facing success criteria:

- I can spot when an AI output looks biased or unfair and say how I can tell.
- I can point to where in the workflow - the input or the process - the bias probably started.
- I can suggest two simple ways to make the workflow fairer or safer and give a reason for each.

Prior knowledge & Prerequisite Primer (10 min)

Assumes: This week rebuilds a skill from earlier grades. In Grade 4 you judged fairness by looking at the input data. In

Grade 5 you explained how misinformation can spread and proposed ways to verify a claim. In Grade 6 you began to identify

biased outputs and to suggest simple ways to improve fairness and safety. Today we make that Grade 6 skill secure again before the new unit.

Activate: Show two curated AI outputs made for the same task - one fair, one clearly skewed (for example, a job-advert generator that only pictures men as engineers). Ask: which output would be unfair to someone, and how can you tell?

Teach: Model the spot-trace-fix routine on the board: name the unfairness in plain words, trace it back to the input or the process

of the workflow, then propose one simple measure that would reduce it, with a reason.

Apply: In pairs, students look at one short curated output and write one sentence naming a possible bias and one sentence suggesting a fix.

Check: Cold-call two pairs to name the bias and where it started. Listen for a bias that is described in the output and located at a workflow stage. If a pair only says 'it is wrong', prompt them to say who it is unfair to and why.

Critical & analytical thinking focus

Evaluate & justify - reason about risk, ethics and governance (Evaluate | DOK 3 | SOLO: Relational)

The technical concept (teacher background)

Bias in an AI workflow is a pattern that makes the output systematically unfair to a person or group. It usually enters at the input stage - the training data or the prompt over-represents some groups and under-represents others - or at the process stage, where the model amplifies patterns in that data. Because an AI output can look confident and neat, bias is easy to

miss, which is why we evaluate outputs against the task and a fairness lens rather than trusting the surface. Simple

mitigations map to the workflow: broaden or balance the input data, adjust the prompt, add a human check on the output,

or restrict the tool's use. Use consistent definitions: bias is a repeating, systematic error that makes a system perform

worse for one group of cases than another; fairness means treating comparable cases or groups consistently, and it can be

MEASURED - split the test set by a characteristic, count correct in each group, and compare the rates (see

Grade7_Fairness_and_Bias_Technical_Note). This bridging lesson revises the Grade 6 skill of identifying biased outputs and

proposing fairness and safety measures, applied to the unit's input-process-output frame. All examples are teacher-

mediated, account-free, pre-generated, and contain no personal data. Cognitive demand:

Evaluate | DOK 3 | SOLO:

Relational.

Quantify the bias (applied numeracy).

Make the bias a number students compute. On each bias-scenario card, have pairs count correct ÷ total for

each group and write it as a percentage – for example Class B 1 of 4 = 25% versus Class A 4 of 4 = 100%; the gap is the bias.

Then name the cause (too few Class B examples in the data) and the fix (add balanced examples), showing the

target count rising to, say, 3 of 4 = 75%. Students state the before/after percentage and the one change that improved fairness.

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Enrichment (optional, teacher-led, 5–8 min): show the same skew live on a Teachable Machine image classifier

trained on a deliberately skewed set – test it, record the rare-class score, then retrain with balanced examples

and re-test so the score rises. No personal data – objects, drawings or gestures only.

Lesson plan

Phase | time Teacher (I do / we do) Students Key questions (targeted thinking)

Hook / 6 min Display the two curated outputs from

the same task, one fair and one

skewed. Think aloud about who the

skewed output would be unfair to,

without naming the fix yet.

Watch, then jot on a sticky

note who they think the

skewed output treats

unfairly.

- Who would this output be unfair to, and what in it tells you that?

Model spot-

trace-fix / 9 min

Model the routine on one output: name

the bias, trace it to the input or process

of the workflow, then propose one

measure with a reason. Keep any tool

teacher-operated.

Copy the three headings -

Spot, Trace, Fix - and follow

the model on a guided

example.

- At which stage - input or process - did this bias most likely begin?

Analyse the

scenario set / 15

min

Hand out the curated scenario cards

(each a short workflow plus its output).

Circulate and prompt students to justify

who is affected and why. Keep all

examples pre-generated and personal-

data-free.

In pairs, analyse two scenarios: spot the bias or safety risk, trace where it began, and note who it affects.

- What is your evidence this is bias and not just one odd answer?

Propose fairness

measures / 10

min

Bring pairs together to share fixes. Sort proposed measures against the workflow stages - fix the input, fix the process, add an output check, restrict use.

For each scenario, write at least two simple measures that would make it fairer or safer, each with a reason.

- Does your fix change the input, the process, or add a check on the output?

Check and close

/ 5 min

Run the exit check; collect the scenario logs as evidence. Reaffirm verify-before-trust: a confident output can still be biased.

Complete the exit ticket: one biased output, where it began, and one reasoned fix.

- Name one bias you found and one simple way to make that workflow fairer.

Anticipated misconceptions

Misconception How to address it

If the AI is unfair, it must be doing it on purpose. Explain that bias usually comes from unbalanced input data or

from the process amplifying patterns, not from intent. Trace one scenario back to its input to show a cause without blame.

A confident, well-written output cannot be biased. Show a fluent output that still favours one group; agree that

surface polish is not fairness, and that outputs are checked against a fairness lens (verify-before-trust).

The only fix is to switch the AI off. Broaden the options: balance the input data, adjust the prompt,

add a human check on the output, or restrict use. Map each fix to a workflow stage so students see a range of measures.

Hands-on activity guide

Overview: Students analyse a set of curated AI-workflow scenarios, identify biased or unsafe outputs, trace each back to

its workflow stage, and propose reasoned fairness and safety measures.

Grouping / timing: Pairs for the scenario analysis, whole class for the fixes debrief.

Tool: A pre-printed scenario card set plus a teacher-operated board demo (a 'simulated AI' paper routine). Account-free

and personal-data-free - students never log in or enter names, and every output is pre-generated.

Facilitation steps:

1. Read your first scenario card - the short workflow and the output it produced.

2. Spot the bias or safety risk in the output and write who it is unfair to.

3. Trace it back: mark whether it most likely began at the input or the process.

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4. Propose at least two simple measures that would make it fairer or safer.

5. Write one reason for each measure - why would it help?

6. Move to your second scenario and repeat, then check: would your fix still work for a new case?

Unplugged fallback: The scenario cards are already paper-based, so no device is needed. If the board demo is

unavailable, read one scenario aloud, model the spot-trace-fix routine on the board, and have pairs work the printed cards by hand.

Formative assessment (aligned to the objectives)

Task / item Model response Marking note

Identify the bias in a curated output and say who it is unfair to.

The advert generator only shows men as engineers, which is unfair to women and girls because it suggests the job is not for them.

One mark for a bias named in the output plus a group it

disadvantages; the reason must reference the output, not a guess.

Trace the bias to a workflow stage and justify your choice.

It probably began at the input: the examples the model learned from mostly showed men as engineers, so the process repeated that pattern.

Accept input or process with a justification that links the stage to the observed bias. Reject 'the output' alone as the origin.

Suggest two simple measures to improve fairness or safety, each with a reason.

Balance the input data so it shows engineers of all genders, because the model learns from what it is given; and add a human check before the advert is used, because a person can catch what the model missed.

One mark per measure that is simple, mapped to a workflow stage, and supported by a reason.

- Targets: identifying biased or unsafe outputs, tracing bias to a workflow stage, and proposing reasoned fairness and safety measures - the revised Grade 6 skill in the workflow context.
- Check: circulating justification prompts during the scenario analysis plus an exit ticket naming one bias, its stage, and one fix.
- Success criteria: the student identifies a real bias, locates its stage with a reason, and proposes at least two simple measures each backed by a reason.
- Evidence: collected scenario logs and the exit tickets.
- Feedback / adjustment: if students name a bias but cannot locate or fix it, re-model tracing one scenario to its input and co-write one measure before the paired Lab period.

Differentiation

Criterion Emerging (expected) Approaching Not yet

Identifying biased outputs Names a bias in the output and who it disadvantages, with support.

Senses something is unfair but cannot say who is affected.

Does not identify a bias in the output.

Tracing to a workflow stage Locates the bias at the input or process and gives a simple reason, with guidance.

Names a stage but without a reason, or mislocates it.

Cannot link the bias to a workflow stage.

Proposing fairness measures

Suggests two simple, reasoned measures that would improve fairness or safety.

Suggests one measure, or a measure without a reason.

Offers no workable measure.

Representation: pair each scenario card with the input-process-output diagram and a short fairness-lens checklist (who is

included, who is left out, who could be harmed); pre-teach bias and fairness with simple icons.

Action & expression: students may show their analysis by writing, annotating the scenario card, or explaining aloud to the teacher

who scribes; accept any mode for the exit ticket.

People of Determination: provide a reduced scenario set with picture support, sentence starters ('This is unfair to ... because ...', 'A

fix is ... because ...'), extra processing time, and a peer partner where helpful.

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Week 4 - Lab | Ethics & Governance - Identifies biased outputs and suggests simple ways to improve fairness and safety (Lab)

Strand: Ethics & Governance | Framework ref(s): AIF·GE·G6·A | 1 x 45 min

Objectives & success criteria

- Evaluate a more complex AI-workflow scenario for biased outputs and safety or governance concerns.
- Justify which concern matters most and where in the workflow it should be addressed.
- Design a short responsible-use plan of reasoned measures and test it against the scenario.

Student-facing success criteria:

- I can evaluate a harder scenario and find its bias and safety concerns.
- I can say which concern matters most and where in the workflow to fix it, with reasons.
- I can write a short responsible-use plan and check it against the scenario.

Prior knowledge & Prerequisite Primer (10 min)

Assumes: In the Core period you practised the spot-trace-fix routine: naming a bias, tracing it to the input or process of a

workflow, and proposing simple fairness measures. This Lab rebuilds the same Grade 6 skill more deeply, so you can apply

it to a harder case and justify your approach. It rests on the earlier chain - judging fairness from input data in Grade 4 and

explaining and verifying misinformation risks in Grade 5.

Activate: Show one richer curated scenario - an AI workflow that helps shortlist school-club applicants - whose output quietly

favours one group and also raises a safety concern. Ask which problem you would tackle first and why.

Teach: Model an evaluate-and-plan routine: list every concern (bias, safety, governance), weigh which matters most, decide the

workflow stage to address it, and draft a short responsible-use plan with reasons.

Apply: In pairs, students list two concerns in the scenario and rank which they would fix first, with one reason.

Check: Check that each pair has separated bias from a safety or governance concern and can justify their ranking. Redirect any

pair that lists concerns without a reason for their order.

Critical & analytical thinking focus

Evaluate & justify - reason about risk, ethics and governance (Evaluate | DOK 3 | SOLO: Relational)

The technical concept (teacher background)

Real AI workflows rarely carry a single, tidy problem. A scenario can hold a fairness bias, a safety risk, and a governance gap

at once - for example, unbalanced training data, a confident but unverified output, and no rule about who may act on it.

Evaluating such a case means separating the concerns, weighing their impact on people, and deciding where in the input-

process-output workflow each is best addressed, then justifying the order. A responsible-use plan is a short set of reasoned

measures - broaden the input data, add a verification or human check, and set a governance rule such as teacher-mediated

use - tested against the scenario to confirm it would help. This Lab does not introduce a new concept; it deepens the

revised Grade 6 skill toward a more complex case, matching the Evaluate | DOK 3 | SOLO: Relational demand. All scenarios

are pre-generated, teacher-mediated, account-free, and contain no personal data.

Lesson plan

Phase | time Teacher (I do / we do) Students Key questions (targeted thinking)

Recap and set up

/ 5 min

Recap spot-trace-fix and hand out the

richer scenario pack and the

responsible-use plan sheet. Restate the

safe-use charter.

Read the scenario and

predict which concern will

matter most before

analysing.

- Before you dig in, which concern

do you expect to be the biggest,

and why?

Model evaluate-

and-plan / 9 min

Think aloud through the scenario: list

bias, safety and governance concerns,

weigh them, choose the workflow stage

for each, and start a responsible-use plan with reasons.

Copy the plan headings -
Concern, Stage, Measure,
Reason - and follow the
model.

- Which concern would cause the most harm to a person if it went unfixed?

Deep evaluation

/ 15 min

Circulate as pairs evaluate the full scenario; press for justification and evidence, not opinion. Keep all

List every bias, safety and governance concern you find and mark the workflow stage

- What is your evidence, and would this concern still stand for a different user?

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examples pre-generated and personal-data-free.

for each.

Design the plan /

11 min

Prompt pairs to turn their evaluation into a short responsible-use plan and to justify the order of measures. Sort plans against the workflow stages.

Write a plan of at least three reasoned measures and rank which to do first, with a justification.

- Why does your first measure come before the others?

Debrief and

close / 5 min

Collect the plans; take two and test them against the scenario as a class.

Reaffirm verify-before-trust and teacher-mediated use.

Complete the exit ticket:

your top-priority concern, its

stage, and the measure you would take first.

- Name your highest-priority concern and the first measure in your plan.

Anticipated misconceptions

Misconception How to address it

You should just fix every problem at once, in any order. Model weighing concerns by harm to people; agree that ordering measures by impact and by workflow stage is part of a justified plan, not an afterthought.

Bias and safety are the same thing. Separate them with the scenario: bias is unfair treatment of a

group; a safety concern is possible harm from a wrong or misused output. Have pairs label each concern by type.

A governance rule is just extra paperwork. Show a case where a simple rule - teacher-mediated use, no

personal data, a required human check - would have prevented harm; treat governance as a real mitigation, not red tape.

Hands-on activity guide

Overview: Pairs evaluate one richer AI-workflow scenario for bias, safety and governance concerns, weigh which matters most, and design a short responsible-use plan of reasoned measures tested against the scenario.

Grouping / timing: Pairs for the evaluation and plan, whole class for the debrief.

Tool: A pre-generated richer scenario pack plus a teacher-operated board demo (a 'simulated AI' paper routine). Account-

free, no logins, no personal data - every output is prepared in advance by the teacher.

Facilitation steps:

1. Read the richer scenario and predict which concern will matter most.
2. List every bias, safety and governance concern you can find in the output and workflow.
3. Mark the workflow stage - input, process or output - where each concern is best addressed.
4. Weigh the concerns by how much they could harm a person, and rank them.
5. Write a responsible-use plan of at least three reasoned measures.
6. Test the plan against the scenario: would it still protect a different user? Adjust if not.

Unplugged fallback: The scenario pack is already paper-based, so no device is required. If the board demo is unavailable,

read the scenario aloud, model the evaluate-and-plan routine on the board, and have pairs build their plan on the printed sheet.

Lab tool integration (practical, hands-on)

Tool: Card decks (bias-scenario cards) + pre-generated packs

Teacher setup: Print bias-scenario cards and include a pair of outputs where one is unfair due to skewed data.

What students do:

1. Read a bias-scenario card and the paired outputs.
2. Identify which output is biased and what in the data likely caused it.
3. Say who could be treated unfairly.
4. Suggest one simple change to improve fairness or safety.
5. Record your fix and why it helps.

Students produce: A fairness record: the bias, its cause, who is affected and a fix.

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Data-safety: Unplugged; outputs pre-curated.

Unplugged fallback: This is the unplugged route.

Skewed data leads to a biased output; a simple fix improves fairness.

Formative assessment (aligned to the objectives)

Task / item Model response Marking note

Evaluate the scenario and list one bias concern and one safety or governance concern.

Bias: the shortlist favours applicants from one school, which is unfair to the others.

Safety or governance: the output is acted on with no human check, so a wrong shortlist could go unnoticed.

One mark for a correctly typed bias concern and one for a distinct safety or governance concern, each grounded in the scenario.

Rank which concern to address first and justify your order.

Fix the biased input data first, because it affects every applicant and shapes the whole output; the missing human check comes next as a safety net.

Accept any defensible order that is justified by harm to people or by workflow stage; the justification is what earns the mark.

Write a responsible-use plan of three reasoned measures for this workflow.

Balance the input data so all schools are represented, because the model learns from what it is given; add a required human check on the shortlist, because a person can catch errors; and set a rule that the tool is teacher-mediated, because that keeps use

accountable.

One mark per measure that is simple, mapped to a stage, and supported by a reason; the three should cover fairness, safety and governance.

- Targets: evaluating a complex workflow for bias, safety and governance concerns and designing a justified responsible-use plan - the revised Grade 6 skill applied to a harder case.
- Check: circulating justification and evidence prompts during the evaluation plus an exit ticket naming the top concern, its stage, and the first measure.
- Success criteria: the student separates concern types, ranks them with a reason, and produces a plan of at least three reasoned measures that fits the scenario.
- Evidence: collected evaluation sheets and responsible-use plans, and the exit tickets.
- Feedback / adjustment: if plans list measures without justification, model weighing two concerns by harm and re-order one plan together before moving on to the unit.

Differentiation

Criterion Emerging (expected) Approaching Not yet

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Evaluating for concerns Identifies a bias concern and a distinct safety or governance concern in the scenario, with support.

Identifies concerns but blurs bias with safety, or finds only one.

Cannot identify a concern in the scenario.

Justifying priority and stage Ranks concerns and links each to a workflow stage with a reason, with guidance.

Ranks or locates concerns but without a clear reason.

Cannot rank concerns or link them to a stage.

Designing a responsible-use plan

Writes at least three reasoned measures covering fairness, safety and governance.

Writes one or two measures,

or measures without reasons.

Offers no workable plan.

Representation: pair the scenario with the input-process-output diagram, a concern-type key (bias, safety, governance), and a partly framed plan template on display.

Action & expression: pairs may build the plan in writing, by annotating the scenario, or by dictating to the teacher; ranking may be shown by ordering cards.

People of Determination: provide a shorter scenario, a plan template with the first measure modelled, sentence starters ('The biggest concern is ... because ...', 'I would do ... first because ...'), and extra time.

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Week 5 - Integrated Task (Design & Build) | Integrated Prerequisite Mastery Task -

Design, Build, Evaluate & Justify an AI Solution

Strand: Integration (all four prior-grade strands) | Framework ref(s): AIF·CU·G6·A AIF·SD·G6·A AIF·CE·G6·A AIF·GE·G6·A | 1 x 45 min

Objectives & success criteria

- Analyse how the training data and design choices in a simulated AI sorter shape the outcomes it produces.
- Design and build, in pairs, a structured input-process-output workflow that completes a real sorting task using guided example data.
- Document the training set, the sorting rule and one test case as evidence the solution works.

Student-facing success criteria:

- I can explain how the example data and the choices we make change what our sorter decides.
- I can design and build a simple input-process-output workflow that sorts items correctly.
- I can record our training examples, our rule and one test so someone else can see it works.

Prior knowledge & Prerequisite Primer (10 min)

Assumes: In Weeks 1-4 you rebuilt four Grade 6 Advanced skills: explaining how training data affects output quality (CU),

building structured AI-assisted solutions from guided inputs and data (SD), explaining how changes to inputs or data

improve outputs (CE), and identifying biased outputs and suggesting fairness and safety fixes (GE). This two-week task

brings all four together in one solution.

Activate: Show a two-item board demo of a simulated sorter: feed one item card in, reveal the small table of example items it was

'trained' on, and reveal the bin it chooses. Ask what would happen if the example data were different.

Teach: Model the whole workflow shape once - input item card, then a simulated model (a lookup table of example items and their

bins that the pair curates), then the output bin. Show live how adding or removing one example changes a decision.

Apply: In pairs, students add one example to a starter training table and predict how it changes one sort.

Check: Ask two pairs to name the input, the process and the output of the demo and to state one way the training data changed a

decision; re-model if a pair confuses the training data with the item being sorted.

Critical & analytical thinking focus

Analyse, Evaluate and Create - sustained higher-order thinking across all four strands

(Analyse/Evaluate/Create | DOK 3-4 |

SOLO: Relational to Extended abstract)

The technical concept (teacher background)

A classifier is a workflow that takes an input, compares it against patterns learned from example data, and returns a label

as its output. Here the model is a simulated one - a lookup table of example items and their correct bins that each pair

curates by hand - so students can see exactly how the training data drives every decision. That makes the core Grade 6 idea

visible: the quantity, balance and quality of the examples shape output quality and performance.

Building the workflow

(input card, simulated model, output bin) rehearses structured, data-guided solution design.

Everything is account-free and

uses only non-personal item descriptions, never names or faces, and every output is checked against a trusted sorting

guide before it is trusted. Cognitive demand for this lesson: Analyse/Create | DOK 2-3 | SOLO: Multistructural to Relational.

Lesson plan

Phase | time Teacher (I do / we do) Students Key questions (targeted thinking)

Launch the task /

6 min

Introduce the integrated mastery task and the paired brief. Run the two-item simulated-sorter board demo and reveal the example data behind one decision.

Watch, then name the input, the process and the output on a mini whiteboard.

- Where did the sorter's decision actually come from - the item or the examples?

Analyse the data

/ 12 min

Hand each pair a starter training table and a set of item cards. Model how removing one example changes a sort;

prompt pairs to test a change (CU step).

In pairs, change one example and record how it changes one decision and why.

- Which change to your examples would most improve the sorter, and how do you know?

Design the Model drawing the input-process- Draw their own input- • What is your input, what does

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workflow / 10

min

output map and writing the one rule the simulated model follows. Circulate to check the maps (SD step).

process-output map and write the rule their sorter will follow.

your process do, and what is your output?

Build and test /

12 min

Support pairs to finish a small, balanced training table and run three item cards through the workflow, logging each result. Keep any live tool teacher-operated.

Build the sorter, run three test cards, and log the output bin for each.

- Did every test land in the correct bin - and if not, where did it go wrong?

Check and close

/ 5 min

Collect the build sheets. Run the exit check and preview Week 6 (evaluate, justify, risks, reflection and showcase).

Complete the exit ticket:

name one design choice and how it shaped an output.

- Name one thing about your data or design that changed what your

sorter decided.

Anticipated misconceptions

Misconception How to address it

The AI figures items out by itself - our examples don't really matter.

Remove all matching examples for one item and run it; the sorter fails or guesses. Show that the curated example data is what drives every decision.

The training examples and the item we sort are the same thing.

Separate the two piles physically: the training table stays fixed while the test item is fed in. Have pairs point to each before every run.

More examples always means a better sorter. Add many examples of one bin only and show the sorter now over-

picks it; agree that balance and quality matter, not just quantity.

Hands-on activity guide

Overview: Across the week's two periods, pairs analyse how example data shapes a simulated sorter, then design and

build their own input-process-output sorting workflow and test it against sample item cards.

Grouping / timing: Pairs for the analyse-design-build; whole class for the launch demo and the close.

Tool: A 'simulated AI' sorter - a paper training table (example items and their bins) plus item cards, run by hand or on a teacher-operated board. Account-free, no logins, and only non-personal item descriptions - never names, faces or private details.

Facilitation steps:

1. Watch the launch demo and name the input, the process and the output of the simulated sorter.
2. Analyse the starter training table: change one example and record how it changes one decision (CU step).
3. Draw your input-process-output map and write the one rule your sorter will follow (SD step).
4. Build your sorter by curating a small, balanced training table of example items and bins.
5. Run three test item cards through your workflow and log the output bin for each.
6. Check each output against the trusted sorting guide and fix any example that caused a wrong sort.

Unplugged fallback: The whole build already runs on paper cards and a hand-run lookup table, so no device is needed. If

the board demo is unavailable, sort the cards by hand against the training table and log the results on the build sheet.

Lab tool integration (practical, hands-on)

Tool: Teachable Machine + Scratch (integrated: design & build)

Teacher setup: Provide the full kit: Teachable Machine open, Scratch starter, plan cards.

What students do:

1. In pairs, plan the small solution with the plan cards (goal, inputs, steps, checks, risk).
2. Train a simple non-personal model in Teachable Machine.
3. Build the workflow in Scratch that uses or simulates the model.
4. Run it once and confirm an input produces an output.
5. Save your build and note what each strand contributed.

Students produce: A designed and built integrated solution with a strand-tagged note.

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Data-safety: Teachable Machine trains in the browser and the samples stay on the teacher's device. Use objects, hand gestures or drawings only - never faces, voices or names. Clear the project at the end of the lesson. Non-personal model only.

Unplugged fallback: Build the whole thing with card-decks and block-cards.

Integrated build: plan cards to a trained model to a Scratch workflow.

Formative assessment (aligned to the objectives)

Task / item Model response Marking note

Change one training example and explain how it changes one of the sorter's decisions.

We added 'battery -> recycling point' to the table, so now the battery card is sorted there instead of general waste, which matches our guide.

One mark for a correct linked change (example, then changed decision) with a reason that references the data.

Draw or describe the input, process and output of your sorting workflow.

Input: an item card. Process: the simulated model compares it to our example table.

Output: the bin it belongs in.

One mark each for a correct input, process and output that fit a workflow shape.

Run one test card and say whether the output is correct, using the sorting guide.

The 'newspaper' card was sorted to recycling, which matches the guide, so the output is correct.

Accept a logged result checked against the trusted guide; it must reference verify-before-trust, not

a guess.

- Targets: the CU-analyse and SD-build steps - analysing how data and design choices shape outcomes, and designing and building a working input-process-output sorting workflow.
- Check: circulating design-and-build prompts during the pair work plus an exit ticket naming one design choice and its effect.
- Success criteria: the pair curates a balanced training table, maps a correct input-process-output workflow, and links one data or design choice to a changed output.
- Evidence: collected build sheets - the training table, the workflow map, and the three logged test results.
- Feedback / adjustment: if a pair cannot link data to outcome, re-run one card with and without a key example before Week 6 and have them log the difference.

Differentiation

Criterion Emerging (expected) Approaching Not yet

Analysing data and design Links one data or design choice to a changed output, with support.

Notices a change but cannot say why it happened.

Cannot connect the examples to a decision.

Building the workflow Maps and builds a working input-process-output sorter using guided

Builds part of the workflow, or the rule is unclear.

No usable workflow, or the parts are mixed up.

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example data.

Testing and recording Runs three test cards and logs each output against the guide, with guidance.

Runs a test but does not check it against the guide.

Does not produce a usable test record.

Representation: display the input-process-output map, a worked example table, and the trusted sorting guide; pre-teach 'training data', 'model' and 'bin' with simple icons.

Action & expression: pairs may build with physical cards, on a printed table, or on the board, and may explain a decision by speaking, drawing or writing.

People of Determination: provide a partly completed training table, a reduced card set, sentence starters ('When we added ..., the sorter ...'), and extra time to run tests, with a peer scribe where helpful.

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Week 6 - Integrated Task (Evaluate, Justify & Showcase) | Integrated Prerequisite

Mastery Task - Design, Build, Evaluate & Justify an AI Solution

Strand: Integration (all four prior-grade strands) | Framework ref(s): AIF·CU·G6·A AIF·SD·G6·A AIF·CE·G6·A AIF·GE·G6·A | 1 x 45 min

Objectives & success criteria

- Evaluate the pair's sorter against a clear accuracy criterion, then improve it and justify the change with evidence.
- Identify one risk in the solution and propose a responsible-use mitigation, giving reasons.
- Complete a structured reflection across the four skills and present the solution at the readiness showcase.

Student-facing success criteria:

- I can test our sorter against a criterion, make one improvement, and prove with evidence that it helped.
- I can name one risk in our sorter and suggest a fair, safe fix, and say why it helps.
- I can reflect on all four skills and show our solution is ready for the Grade 7 unit.

Prior knowledge & Prerequisite Primer (10 min)

Assumes: In Week 5 you designed and built your simulated sorter and logged three test results.

You also rebuilt the Grade

6 Advanced skills for explaining how input or data changes improve outputs (CE) and for spotting biased outputs and

improving fairness and safety (GE). This week you use them to evaluate, fix and justify your solution.

Activate: Show a sorter that gets four of six test cards right and ask: is that good enough, and how would you make it better?

Teach: Model the evaluate-improve-justify loop - score the sorter against an accuracy criterion, change one example or input, re-score, and state the evidence that the change helped.

Apply: In pairs, students score their Week 5 sorter on a six-card test set and mark which cards it gets wrong.

Check: Ask two pairs for their score and one wrong card; check they can name a data or input change that would fix it, not just say 'try again'.

Critical & analytical thinking focus

Analyse, Evaluate and Create - sustained higher-order thinking across all four strands

(Analyse/Evaluate/Create | DOK 3-4 |

SOLO: Relational to Extended abstract)

The technical concept (teacher background)

Evaluation turns a build into an improvement cycle: you set a criterion (here, accuracy on a fixed test set), measure the

solution against it, change one thing, and measure again. Because the model is a simulated lookup table, the cause of any error is visible - usually a missing, wrong or unbalanced example - so students can justify each change with evidence rather than guesswork. Risk work sits alongside: an unbalanced training set produces biased outputs, and over-trusting a confident sort can be unsafe for items like hazardous waste, so responsible-use measures include balancing the examples, adding a 'not sure - ask a human' route, and verify-before-trust. All work stays account-free and free of personal data.

Cognitive demand for this lesson: Evaluate | DOK 3 | SOLO: Relational.

Lesson plan

Phase | time Teacher (I do / we do) Students Key questions (targeted thinking)

Recap and set

the criterion / 6

min

Recap the Week 5 builds. Agree the accuracy criterion and hand out the six-card test set and the risk-and-reflection sheet.

Score their sorter on the six test cards and circle any it gets wrong.

- Does your sorter meet the criterion yet - and which cards let it down?

Evaluate and

improve / 12 min

Model changing one example or input to fix a wrong card, then re-scoring.

Circulate to prompt evidence-based changes (CE step).

Change one example or input, re-run the failed cards, and log the new score.

- What is your evidence that your change improved the output, not just changed it?

Find the risk / 10

min

Model spotting a biased or unsafe output and proposing a mitigation with a reason. Prompt pairs to inspect their own training set (GE step).

Identify one risk in their sorter and write a responsible-use fix with a reason.

- Who could your sorter treat unfairly, or which item could it handle unsafely, and how would you guard against it?

Reflect and gate

/ 12 min

Guide the four-skill reflection and run the readiness gate check against the Complete the structured reflection and self-check

- Across the four skills, which was hardest, and how did you work

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criteria. Set up the showcase. their solution against the readiness gate.

through it?

Showcase and

close / 5 min

Run a short pair-to-pair showcase; collect all evidence sheets as the readiness record.

Present the solution: input-process-output, one improvement with evidence, and one risk and its fix.

- In one line, why is your solution ready for the Grade 7 unit?

Anticipated misconceptions

Misconception How to address it

A change that moves the output means the sorter improved.

Compare scores before and after against the criterion; only a change that raises the score on the same test set counts as an improvement. Keep the evidence.

If the sorter is usually right, we don't need to worry about fairness or safety.

Show one biased case - a whole item type it never sorts right - and one unsafe case, like a battery going to general waste; agree these need mitigations even at high accuracy.

Reflection just means saying it went well. Model a real reflection naming the hardest skill, how they worked

through it, one improvement and one risk; require evidence, not praise.

Hands-on activity guide

Overview: Across the week's two periods, pairs evaluate their Week 5 sorter against an accuracy criterion, make one evidence-based improvement, identify a risk and propose a responsible-use fix, then reflect across the four skills and showcase the solution at the readiness gate.

Grouping / timing: Pairs for the evaluate-improve-risk-reflect work; pair-to-pair for the showcase; whole class for the close.

Tool: The same 'simulated AI' sorter from Week 5 - a paper training table and item cards run by hand or on a teacher-operated board - plus a fixed six-card test set. Account-free, no logins, and only non-personal item descriptions.

Facilitation steps:

1. Score your sorter on the six-card test set and circle any cards it gets wrong.
2. Change one example or input to fix a wrong card, then re-run and log the new score (CE step).
3. Write the evidence that your change improved the output, checked against the sorting guide.
4. Identify one risk - a biased or unsafe output - and propose a responsible-use fix with a reason (GE step).
5. Complete the four-skill reflection: the hardest skill and how you worked through it, one improvement, one risk to guard against.
6. Showcase your solution to another pair and self-check it against the readiness gate.

Unplugged fallback: The evaluation, risk work and reflection all run on paper with the card sets, so no device is needed. If the board is unavailable, score and re-run the cards by hand and present from the evidence sheets.

Lab tool integration (practical, hands-on)

Tool: Pre-generated packs + card decks (integrated: evaluate, justify, risk)

Teacher setup: Have each pair's Week-5 build, an evaluation pack and a risk-scenario deck.

What students do:

1. Run your build and evaluate the outputs against the criteria.
2. Justify one improvement you make, with evidence.
3. Draw a risk-scenario card and check your solution against it.
4. Confirm no personal data is used and name one mitigation.
5. Complete the four-strand readiness reflection.

Students produce: An evaluated, justified solution with a risk check and readiness reflection.

Data-safety: Risk check confirms no personal data; outputs curated.

Unplugged fallback: Evaluate the paper version using printed outputs and risk cards.

Evaluate against criteria, justify a change, check the risk, confirm readiness.

Formative assessment (aligned to the objectives)

Task / item Model response Marking note

Score your sorter, make one change, and show with evidence that the output improved.

We scored 4 out of 6; the 'battery' card went to general. We added 'battery -> recycling point' to the table and re-scored 5 out of 6, so the change improved accuracy against the test set.

One mark for a before-and-after score plus a change justified by the test evidence, not opinion.

Identify one risk in your sorter and propose a responsible-use fix with a reason.

Our table had no compost examples, so compost items were always mis-sorted - a biased output. We added balanced compost examples and a 'not sure - ask a human' route, because a confident sort can still be wrong.

Accept a real risk (bias or safety) with a mitigation and a reason tied to fairness, safety or verify-before-trust.

Reflect across the four skills: name the hardest and how you worked through it.

Evaluating was hardest because our first change did not help; we compared scores and found the real wrong example, then fixed that. One improvement: balance the table.

One risk: over-trusting the sorter for hazardous items.

Look for the hardest skill named with a concrete strategy, plus one improvement and one risk.

- Targets: the CE-evaluate and GE-risk steps plus the four-skill reflection and readiness gate - evaluating against a criterion and improving with justified evidence, identifying a risk with a reasoned mitigation, and confirming readiness for the Grade 7 unit.

- Check: circulating evidence-and-reason prompts during the pair work, the four-skill reflection, and a pair-to-pair showcase

scored against the readiness gate.

- Success criteria: the pair raises its accuracy score with a justified change, names one risk with a reasoned responsible-use fix, and completes an honest four-skill reflection.
- Evidence: collected test-score logs (before and after), the risk-and-mitigation note, the structured reflection, and the showcase readiness record.
- Feedback / adjustment: if a pair changes the output without improving the score, re-model comparing scores against the test set and have them redo one evidence-based change before the readiness sign-off.

Differentiation

Criterion Emerging (expected) Approaching Not yet

Evaluating and justifying Scores against the criterion, makes Makes a change but the No evaluation, or

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one change, and cites evidence it improved the output, with support. evidence is missing or it does not raise the score. changes are guesses with no evidence.

Risk and mitigation Identifies one real risk and proposes a responsible-use fix with a reason.

Names a risk but the fix or the reason is unclear.

Cannot identify a risk in the solution.

Reflection and readiness Completes the four-skill reflection and shows the solution meets the readiness gate.

Reflects partly, or the readiness check is incomplete.

No usable reflection or readiness evidence.

Representation: display the accuracy criterion, the six-card test set, a risk-matrix prompt (bias and safety), and reflection sentence stems.

Action & expression: pairs may present the showcase by speaking, by walking through their sheets, or with a simple poster; reflections may be written or spoken to a scribe.

People of Determination: provide a partly completed test log, a reduced test set, a risk card bank to choose from, reflection stems

('The hardest skill was ... We worked through it by ...'), and extra time for the showcase.

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Readiness Gate

Level What it looks like Action

Mastery Meets the prior-grade Advanced bar independently within the integrated task.

Proceed to framework content (Week 7).

Partial Meets the bar with support or on some steps of the task. Targeted Prerequisite Primer on that strand in the relevant Week 7+ lesson.

Not yet Does not yet meet the bar on that strand. Monitored small-group Primer support into Term 1; re-checked in context.

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Standards & references

This pack is designed against the following international standards for curriculum and assessment quality.

Reference Used for

ADEK (2026). K-12 AI Fluency Framework (Draft V4). Source of all learning outcomes, strands, tiers and codes.

AERA, APA & NCME (2014). Standards for Educational and Psychological Testing.

Validity, reliability, and fairness of assessment.

Assessment Reform Group (2002). Assessment for Learning: 10 Principles.

Formative assessment that supports, not just measures, learning.

OECD (2005). Formative Assessment: Improving Learning in Secondary Classrooms.

Evidence base for embedding formative assessment.

Anderson & Krathwohl (2001). Revised Bloom's Taxonomy. Cognitive-demand classification of objectives.

Webb, N. (1997/2002). Depth of Knowledge. Depth of cognitive processing and grading of demand.

Biggs & Collis (1982). SOLO Taxonomy. Structure of observed learning; anchors success criteria.

CAST (2018). Universal Design for Learning Guidelines v2.2. Access, representation, and multiple means of engagement.

Grade 7 (Year 8) | Phase 3 / Cycle 2 | AI Solution Builder

Bridging Course - AI as Workflows - Input, Process, Output

Transition-Year readiness | Weeks 1-6 (Core + Lab) |

Foundation - How Models Learn (quick version, read before Week 1)

- Rules or learning? A rule-based system follows fixed rules a person wrote (if metal -> Recycle). A

learned model is shown many labelled examples and adjusts inside until it predicts well.

- Training vs using: a model is trained once from labelled examples, then used (inference) on new

inputs without changing. It stores patterns, not copies of your examples.

- Same input, same answer: a classifier repeats the same output for the same input. A confidence

score like 0.90 says how sure it is about THIS answer - not how often it is right, and a confident answer

can still be wrong.

My reflection: one thing I now understand about how models learn is ...

Week 1 - Core | AI Conceptual Understanding — Explains how training data affects output quality and performance

I can...

- I can explain how the data a model learned from changes the output it gives.

- I can trace an example from input to output and show where the training data made a difference.

- I can convince my partner, using the example, why the training data helped or hurt the output.

Words to know

- training data - the many examples a model learns from before you ever use it

- output quality - how accurate, useful and fair the model's result is

- balanced - a data set that includes a fair range of examples, not just one kind

- representative - training data that matches the real cases the model will meet

- analyse - to break something down and reason about why it turned out the way it did

Before the new unit starts, we are making one big idea rock-solid: the data a model learned from decides how

good its answers can be. Today you become the detective who traces an output back to that data.

The data behind the answer

Every AI model learns from training data before you type a single thing. That data is the hidden ingredient. If a

model learned from a rich, balanced set of examples, it usually gives accurate, useful outputs. If it learned from

a thin or one-sided set, it has gaps - and those gaps show up as mistakes, even when your input is perfect. So

when an output looks wrong, a good detective does not just blame the input. They ask: what did this model learn

from, and what was missing?

Same shape, hidden cause

In a workflow the shape is always input, then process, then output. The training data works inside the process

step, quietly steering the result. That is why two models can take the exact same input and give very different

outputs - their training data was different. Your job is to trace the output back through the process to the data

that shaped it, and to explain the link out loud. Remember the charter: the examples are made by your teacher,

use no personal data, and a confident-looking output still has to be checked before you trust it. The training data sits behind the process step and quietly shapes the output.

Why it matters

Why it matters - across the UAE, teams train models on data for things like Arabic-English translation, spotting

faults in solar panels, and sorting recycling; if that training data is thin or one-sided, the tool lets people down,

so understanding this helps you question the AI around you.

Let's try it - hands-on

1. Read the example card: the input, the training-data note, and the output.
2. Map the example from input, through the process, to the output.
3. Write how the training data explains the output's quality.
4. Compare a second card and note how different data changed the output.
5. Explain one example to your partner, pointing to the training data as the cause.

Think about it

Think about it - if two models take the same input but give different outputs, what does that tell you about the

data each one learned from?

My reflection

My reflection: One way training data affected an output I looked at today was ...

Week 1 - Lab | AI Conceptual Understanding — Explains how training data affects output quality and performance (Lab)

Human in the loop: the AI tool does the sorting or drafting, but a person decides what to give it, checks each output, and makes the final call - especially on low-confidence results.

I can...

- I can investigate examples and show how different training data changed the output.
- I can map how the training data acts inside the input-process-output workflow.
- I can justify to a partner which data gave the better output and back it with evidence.

Words to know

- fair test - a test where you change one thing only, so you know what caused the result
- controlled - kept the same on purpose, so it cannot be the reason for a difference
- map - to lay out each stage of a workflow so you can see how it connects
- evidence - what you actually observed, used to back up your conclusion
- conclusion - the idea you decide on after weighing the evidence

In the Core period you traced outputs back to their training data. Today you run the fair test yourself - keep the

input the same, change only the data, and prove which data makes the better output.

Change one thing, prove the cause

A good investigator uses a fair test. Here, the input and the process are kept exactly the same for both examples

- the only thing that changes is the training data. So if the two outputs come out different, there is only one thing left to blame: the data the model learned from. That is powerful. It turns 'I think this data is better' into 'I can prove this data is better', because you controlled everything else. Mapping the workflow makes the proof visible to anyone who reads it.

Map it, then justify it

On your mapping sheet you lay out the shared input, the two training data sets, the process and the two

outputs. When it is all in front of you, you can point to exactly where the data made the difference and say which

data won. Then you justify it to a partner using the map as evidence - not a feeling. Remember the charter: the

examples are made by your teacher, use no personal data, and even a fluent, confident output has to be

checked before you trust the data behind it.

A fair test holds everything constant except the one thing you are testing - here, the training data.

Why it matters

Why it matters - engineers building UAE tools, from smart-traffic systems in Abu Dhabi to Arabic-English

translators, test different training data before they choose one; running a fair test on the data is exactly how they

decide which model to trust.

Let's try it - hands-on

1. Read the first example pair and predict which training data gives the better output.
2. Map the shared input, the two data sets, the process and the two outputs.
3. Write how the training data explains the difference in output quality.
4. Rotate to a new pack and map it the same way.
5. Write a conclusion naming which training data won, and justify it to your partner.

Today's tool - hands-on

Tool: Google Teachable Machine (clean vs messy data)

1. Train the model on the clean set; test and read the output confidence.
2. Retrain the same class on the messy set.
3. Test the same unseen example and compare the two outputs.
4. Explain how the training data changed the output quality.
5. Write the rule: better examples, better output.

You will make: A comparison log showing how data quality changed the output.

Stay safe: Teachable Machine trains in the browser and the samples stay on the teacher's device. Use objects, hand

gestures or drawings only - never faces, voices or names. Clear the project at the end of the lesson.

Same model, clean vs messy training data, produces different output quality.

Think about it

Think about it - if you wanted to prove that a data set was fairer, what one thing would you keep the same and

what one thing would you change?

My reflection

My reflection: The training data that produced the better output today was ..., and my evidence was ...

Week 2 - Core | AI Solution Design & Development — Builds structured AI-assisted solutions using guided inputs and data

I can...

- I can plan a step-by-step workflow that takes a guided input and gives a useful output.
- I can build and run my workflow using our simulated routine to get an output.
- I can explain how my input fields and the data changed the output, and check the output is right.

Words to know

- structured solution - a workflow built from clear steps that work together to finish a task
- guided input - an input with set fields (like a form), not just free typing
- data - the small set of labelled examples the process matches your input against
- process - the middle step that turns your input into an output using the data
- verify - to check an output against the task and a trusted source before you trust it

Today you become a solution builder. Instead of just asking an AI a question, you design a small workflow that

takes a tidy, guided input and uses data to produce a reliable answer to a real task.

Guide the input, and the output gets better

Imagine asking a helper to sort waste. If you just say 'sort this', the helper has to guess. If you hand over a card

that already asks the right things - what material, what size, clean or dirty - the helper knows exactly what to do.

That is a guided input: set fields instead of loose words. When your input is clear and your workflow uses good

data, the output is far more reliable. Building this well is a real design skill, not just typing a question.

The output is only as good as the data

Your workflow uses a small data table to decide the answer. If the table is missing a case or has a wrong row,

the output will be wrong too - even if it looks confident. So you check two things: is the data right for this task,

and does the output actually match what the task needs? Remember the class rules: the routine is run by your

teacher, there is no personal data anywhere, and a neat-looking answer still has to be verified before you rely on

it.

A structured solution: a guided input feeds a process that uses data to give an output.

Why it matters

Why it matters - across the UAE, structured AI workflows sort recycling, route customer requests at government service desks and match people to services; the person who can design the guided input and check the data is

the one who makes the tool actually work.

Let's try it - hands-on

1. Agree what a good output looks like for the sorting task.
2. Design a guided input card - choose only the fields the task needs and say why each matters.
3. Build your workflow: guided input, then process using the data table, then output.
4. Run three items through and record each output with a reason.
5. Verify one output against the task, then note one change that would make it more reliable.

Think about it

Think about it - if two workflows used the same input but different data tables, could they give different outputs?

How would you decide which one to trust?

My reflection

My reflection: One field I put in my guided input, and why it mattered to the output, was ...

Week 2 - Lab | AI Solution Design & Development — Builds structured AI-assisted solutions using guided inputs and data (Lab)

Human in the loop: the AI tool does the sorting or drafting, but a person decides what to give it, checks each output, and makes the final call - especially on low-confidence results.

I can...

- I can improve my workflow so it handles new, trickier inputs.
- I can test my workflow, find where it goes wrong, and fix one part.
- I can explain why my input fields and data make the output reliable.

Words to know

- test case - an input you try on purpose to see if the workflow copes
- refine - to improve one part of a solution and check the effect
- failing stage - the exact step where the output went wrong
- reliable - gives a correct output again and again, even on new inputs
- justify - to give clear reasons for a design choice, backed by evidence

In the Core period you built a workflow that worked on easy items. Today you are the tester and the engineer -

you throw hard cases at it, find where it breaks, and refine one part so it handles more.

Find the failing stage, do not start again

When an output is wrong, it is tempting to scrap everything. Do not. A structured solution has clear stages, so

you can trace the fault to one of them. Usually it is a missing input field or a gap in the data.

Once you know the

failing stage, you change just that one thing and run the case again. This is how real engineers work: they do not

rebuild the whole system, they find the one part that needs fixing and prove that the fix worked.

Change one thing, then re-test everything

If you change five things at once, you will not know which one helped. So change one thing, rerun the hard case, and then rerun the easy cases too - because a fix can quietly break something that used to work. Only when the whole set passes can you say your workflow is more reliable. Keep to the class rules: the routine is teacher-run, there is no personal data, and every output is verified against the task before you trust it. The test-and-refine loop: run a case, find the failing stage, change one thing, rerun.

Why it matters

Why it matters - teams building AI services in the UAE, from smart-city sorting to service-desk routing, spend most of their time testing and refining, not building from scratch; the skill of finding the failing stage and fixing one part is exactly what keeps those systems reliable.

Let's try it - hands-on

1. Set up your Core workflow: guided input, data table and output.
2. Run each tricky test case and spot the output that is wrong or missing.
3. Log what failed and the stage it failed at.
4. Make one targeted change to the input fields or the data.
5. Rerun the hard and easy cases, then justify why your workflow is now more reliable.

Today's tool - hands-on

Tool: Scratch (structured solution from guided inputs)

1. Read the guided inputs and decide the order of steps.
2. Build the structured workflow block by block.
3. Run it and check the output matches the goal.
4. Improve one step so the output is clearer.
5. Explain why your structure works.

You will make: A structured Scratch solution and a short explanation.

Stay safe: No personal data; neutral inputs only.

Guided inputs assembled into a structured multi-step Scratch workflow.

Think about it

Think about it - how many easy cases should you re-test after a fix before you are confident the workflow is reliable? What would convince you?

My reflection

My reflection: One change I made to my workflow today, and why it made the output more reliable, was ...

Week 3 - Core | Critical Evaluation & Informed Interaction — Explains how changes to inputs or data improve outputs

I can...

- I can explain how changing the input improved the output of a workflow.
- I can compare two outputs against agreed criteria and say which is better, with evidence.
- I can point to the exact change I made and link it to the improvement in the output.

Words to know

- input - the data that goes into a workflow, such as a question, a word or a photo
- output - what the workflow gives back: an answer, a label or new content
- criteria - agreed, checkable measures of a good output, such as accuracy, clarity, relevance or completeness
- evidence - the part of the output you point to that proves your judgement
- improve - to make the output better against the criteria by changing the input

You already know how to spot a better output. Today you prove it - you change the input, judge the two outputs

against clear criteria, and show your evidence.

A better input, a better output

The process in a workflow does not change on its own - but the input can. When you give the process a clearer,

fuller or more relevant input, it usually gives back a better output. That is why skilled users work on the input

first: fix the question, add the missing detail, remove what is unclear. The process stays the same; only what you

feed it changes. If the output improves, the change was worth keeping - and if it does not, you try a different

change.

Judge with criteria, not taste

Saying one output is better because you like it more is not evaluation. Real evaluation uses criteria - measures

you agree on first, like accuracy, clarity, relevance and completeness. For each criterion you point to the

evidence in the output: this one names the correct fact, that one answers the whole question.

Then your

judgement holds up because anyone can check it. Always check the better output against the task before you

trust it - a confident output can still be wrong.

Same process, a better input - and a better output to judge against criteria.

Why it matters

Teams building AI services in the UAE improve results by fixing the input and testing against clear criteria - from

Arabic-English translation to a smart-city help service. Judging outputs fairly is how they decide a change is a

real improvement.

Let's try it - hands-on

1. Read the two outputs your teacher shows for the same workflow.
2. Use the agreed criteria - accuracy, clarity, relevance, completeness.
3. Judge which output is better on each criterion and mark it.
4. Write the evidence from the output that supports your judgement.
5. Say what was changed in the input to make the better output.

Think about it

If a change to the input makes the output worse, what should you do with that change?

My reflection

My reflection: the criterion that helped me most when comparing outputs was ... because ...

Week 3 - Lab | Critical Evaluation & Informed Interaction — Explains how changes to inputs or data improve outputs (Lab)

Human in the loop: the AI tool does the sorting or drafting, but a person decides what to give it, checks each output, and makes the final call - especially on low-confidence results.

I can...

- I can rank a set of outputs against agreed criteria and explain my order.
- I can justify every judgement with evidence I can point to in the output.
- I can change an input, compare the new output against the criteria, and say whether it really improved.

Words to know

- rank - to put outputs in order from best to weakest against the criteria
- criteria - the agreed, checkable measures of a good output
- evidence - the part of the output you point to that proves your judgement
- iterate - to change an input, run it again, and re-judge the new output
- trade-off - when a change improves one criterion but weakens another

You judged two outputs in the Core. Today you become the evaluator - rank a whole set against clear criteria,

then change an input and test whether you can make the best one better.

Rank the set, show the evidence

Evaluating a set means using the same criteria on every output and putting them in order. The top output is not

the longest or the one you like - it is the one the evidence supports on accuracy, clarity, relevance and

completeness. For each place in your ranking you point to the proof in the output. That way anyone can check

your order using only your evidence. A ranking you can defend is far stronger than a favourite you cannot explain.

Iterate: change, re-run, re-judge

Once you know which output is best, try to beat it. Change one input, keep the process the same, and run it

again. Then judge the new output against the criteria beside the old one. Sometimes it improves; sometimes a

change helps one criterion but hurts another - that is a trade-off. A change only counts as an improvement when

the evidence shows it. And before you trust the winner, check it against something you know is correct.

Why it matters

This is exactly how UAE teams improve an AI service before launch: they change an input, test the new output

against agreed criteria, and keep only the changes the evidence supports. Fair evaluation protects the people who will use the tool.

Let's try it - hands-on

1. Lay out the set of outputs from the same workflow.
2. Rank them against the agreed criteria - accuracy, clarity, relevance, completeness.
3. Write the evidence from the output for each judgement.
4. Change one input and run it through the process again.
5. Judge the new output against the criteria - is it really better?

Today's tool - hands-on

Tool: Teachable Machine retrain + pre-generated packs (inputs change outputs)

1. Test the current model and note the output.
2. Change the inputs (add clearer examples to one class) and retrain.
3. Test again and compare with the before output from the pack.
4. Explain which input change improved the output and why.
5. Record the improvement with evidence.

You will make: A before/after record explaining how input changes improved outputs.

Stay safe: Teachable Machine trains in the browser and the samples stay on the teacher's device. Use objects, hand gestures or drawings only - never faces, voices or names. Clear the project at the end of the lesson.

Change the inputs, retrain, and the output improves - shown before and after.

Think about it

If a change improves clarity but drops completeness, is the output better or worse overall - and how would you decide?

My reflection

My reflection: iterating on the input taught me that a real improvement is ...

Week 4 - Core | Ethics & Governance - Identifies biased outputs and suggests simple ways to improve fairness and safety

I can...

- I can spot when an AI output looks biased or unfair and say how I can tell.
- I can point to where in the workflow - the input or the process - the bias probably started.
- I can suggest two simple ways to make the workflow fairer or safer and give a reason for each.

Words to know

- bias - a repeating, systematic error that makes a system work less well for one group of cases than another, usually

because of the examples it learned from

- fairness - treating comparable cases (or groups) consistently, so no group of inputs is unfairly favoured or left out;

fairness can also be measured by comparing how well the system works for one group against another

- safety - keeping people from harm, including from a misleading output
- trace - follow a problem back to where it started in the workflow

- measure - a simple action you take to fix or reduce a problem

Sometimes an AI output looks neat and sure of itself but is quietly unfair to a group of people.

Today you

become a fairness checker - spotting bias in a workflow, working out where it started, and suggesting simple

ways to make it fairer and safer.

Where bias hides in a workflow

Every AI tool is a workflow: input, process, output. Bias usually sneaks in at the input, when the examples the

model learned from show some people far more than others. The process then repeats that pattern, and the

output comes out unfair. Because the output can look polished, the bias is easy to miss. Your job is to look past

the polish, ask who might be treated unfairly, and trace the problem back to the stage where it began.

Spot, trace, fix

A fairness checker uses three steps. Spot: name what is unfair in the output and who it affects.

Trace: decide

whether the bias began at the input or the process, and say why. Fix: suggest a simple measure

- balance the

input data, change the prompt, add a human check, or limit how the tool is used - and give a reason. Remember

the class rules: outputs are made by the teacher, no personal data goes in, and a confident output still gets

checked before anyone trusts it.

Bias usually enters at the input and flows through to an unfair output - trace it back to fix it.

Put a number on the bias

You can measure bias, not just describe it. On each bias-scenario card, count how many items each group gets

right and turn it into a percentage: $\text{correct} \div \text{total}$.

Example: a sorter gets Class B right on only 1 of 4 cards = 25%, while Class A gets 4 of 4 = 100%. That gap is the

bias. The cause is in the data – too few Class B examples. The fix: add balanced Class B examples so the count

rises – for example 3 of 4 = 75%.

Enrichment: your teacher may show the same skew live on Teachable Machine – see the low score, name the

cause, retrain, and watch the score climb. No personal data – objects or drawings only.

Why it matters

Why it matters - across the UAE, AI helps sort job applications, translate between Arabic and English, and

answer questions at government service desks. If a workflow is biased, real people can be treated unfairly, so

learning to spot and fix bias helps keep these tools fair for everyone.

Let's try it - hands-on

1. Read your scenario card - the short workflow and the output it produced.
2. Spot the bias or safety risk and write who it is unfair to.
3. Trace it back: did it begin at the input or the process?
4. Suggest two simple measures to make it fairer or safer.
5. Write one reason for each measure, then test it on your second scenario.

Think about it

Think about it - if an AI output looks confident and well written, does that make it fair? How would you check?

My reflection

My reflection: One bias I spotted today, and one simple way I would fix it, was ...

Week 4 - Lab | Ethics & Governance - Identifies biased outputs and suggests simple ways to improve fairness and safety (Lab)

Human in the loop: the AI tool does the sorting or drafting, but a person decides what to give it, checks each output, and makes the final call - especially on low-confidence results.

I can...

- I can evaluate a harder scenario and find its bias and safety concerns.
- I can say which concern matters most and where in the workflow to fix it, with reasons.
- I can write a short responsible-use plan and check it against the scenario.

Words to know

- evaluate - judge something carefully against clear criteria before deciding
- governance - the rules about who may use a tool and how, to keep it accountable
- priority - the concern you decide to deal with first because it matters most
- responsible-use plan - a short set of measures that make a workflow fairer and safer
- justify - give clear reasons for the choice or order you decided on

Real AI tools often carry more than one problem at once. Today you take on a harder case - evaluating a

workflow for bias, safety and governance concerns, deciding which matters most, and designing a short plan to make it responsible.

One workflow, several concerns

A single AI workflow can hold more than one problem. It might be biased because its input data over-represents

one group, unsafe because a wrong output could be acted on without checking, and poorly governed because

there is no rule about who may use it. A good evaluator separates these concerns instead of lumping them

together, then works out where in the input-process-output workflow each one is best addressed. Naming the

type of each concern is the first step to fixing it well.

Weigh it, then plan it

Not every concern is equally urgent. You weigh them by how much they could harm a person, then decide the

order to act. Your responsible-use plan lists simple measures - balance the input data, add a human check, set

a rule such as teacher-mediated use - and gives a reason for each. Finally you test the plan against the scenario:

would it still protect a different user? Remember the charter: outputs are teacher-made, no personal data goes

in, and confident outputs are always verified before anyone trusts them.

Privacy is also a technical measure, not only a classroom rule. De-identify data by removing anything that

names a person, minimise data by collecting only what the task needs, and delete data once it is no longer

needed (retention and deletion). In the UAE this is backed by the Personal Data Protection Law (PDPL) and ADEK

policy.

Why it matters

Why it matters - when the UAE builds AI into services like school admissions, healthcare queues or translation,

teams must weigh fairness, safety and governance together; learning to evaluate a workflow and plan

responsible measures is exactly what keeps those services trustworthy.

Let's try it - hands-on

1. Read the richer scenario and predict which concern matters most.
2. List every bias, safety and governance concern you can find.
3. Mark the workflow stage where each concern is best addressed.
4. Rank the concerns by how much they could harm a person.
5. Write a responsible-use plan of at least three reasoned measures and test it against the scenario.

Today's tool - hands-on

Tool: Card decks (bias-scenario cards) + pre-generated packs

1. Read a bias-scenario card and the paired outputs.
2. Identify which output is biased and what in the data likely caused it.
3. Say who could be treated unfairly.
4. Suggest one simple change to improve fairness or safety.
5. Record your fix and why it helps.

You will make: A fairness record: the bias, its cause, who is affected and a fix.

Stay safe: Unplugged; outputs pre-curated.

Skewed data leads to a biased output; a simple fix improves fairness.

Think about it

Think about it - if you could only fix one concern in a workflow today, how would you decide which one, and how

would you justify leaving the others for later?

My reflection

My reflection: The concern I judged most important, and the first measure in my plan, was ...

Week 5 - Integrated Task (Design & Build) | Integrated Prerequisite Mastery Task -

Design, Build, Evaluate & Justify an AI Solution

I can...

- I can explain how the example data and the choices we make change what our sorter decides.
- I can design and build a simple input-process-output workflow that sorts items correctly.
- I can record our training examples, our rule and one test so someone else can see it works.

Words to know

- training data - the example items and their correct bins that teach our sorter how to decide
- model - the part that compares a new item to the examples to choose a bin
- workflow - the input-process-output shape our solution follows
- test card - an item we feed in to check whether the sorter gets it right
- sorting guide - the trusted answer key we check every output against

This week you and your partner become AI solution builders. You will design and build a small sorting helper

that takes an item in, decides where it belongs, and gives you the answer - just like a real AI workflow.

Your examples do the deciding

Our sorter is a simulated AI: it has no magic, just a table of example items and the bin each one belongs in.

When a new item card comes in, the model compares it to those examples and picks a bin. That means the

examples you choose are doing the deciding. If your examples are unbalanced - say ten recycling items and only

one compost item - your sorter will lean towards recycling. Choosing good, balanced examples is how you make

the output good. The 'better examples, better results' idea is shared with real AI - but our paper sorter is only a

simulated lookup table, not a real trained model.

Our simulated sorter vs a real trained model

Our sorter is a simulated lookup table: it can only decide items that are already written in its table, and it gives

the same bin every time for the same item (it is deterministic). A real trained model is different in one key way - it

learns patterns and can generalise, giving a sensible bin for an item it has never seen before.

That generalisation

to unseen inputs is exactly what a lookup table cannot do, so a lookup table is NOT machine learning. Use the

sorter to understand the workflow shape, and remember which part is a stand-in for real AI.

Build it as a workflow

A strong solution has a clear shape: input, process, output. Your input is an item card. Your process is the

model checking it against your training table. Your output is the bin. When you draw this map and write the one

rule your sorter follows, anyone can see how your solution works and test it. Build small, run a few test cards,

and always check each answer against the trusted sorting guide before you trust it - a confident sort can still be wrong.

Your sorter is a workflow: an item goes in, the model checks it against your examples, and a bin comes out.

Why it matters

Why it matters - across the UAE, smart bins and sorting systems help cities like Abu Dhabi recycle more and waste less; the same input-process-output thinking you use this week sits behind those real solutions.

Let's try it - hands-on

1. Watch the demo and name the input, the process and the output.
2. Change one example in the starter table and see how it changes a decision.
3. Draw your input-process-output map and write your sorting rule.
4. Build your training table with balanced example items and bins.
5. Run three test cards and check each output against the sorting guide.

Today's tool - hands-on

Tool: Teachable Machine + Scratch (integrated: design & build)

1. In pairs, plan the small solution with the plan cards (goal, inputs, steps, checks, risk).
2. Train a simple non-personal model in Teachable Machine.
3. Build the workflow in Scratch that uses or simulates the model.

Note: 'simulate the model' means the Scratch project itself checks the item against your rule/table (no account or model connection needed). Only if your school has set up Machine Learning for Kids would you 'connect' the trained model. Either way, the sorting rule is the simulated part - name it as a stand-in for real AI.

4. Run it once and confirm an input produces an output.
5. Save your build and note what each strand contributed.

You will make: A designed and built integrated solution with a strand-tagged note.

Stay safe: Teachable Machine trains in the browser and the samples stay on the teacher's device. Use objects, hand gestures or drawings only - never faces, voices or names. Clear the project at the end of the lesson. Non-personal model only.

Integrated build: plan cards to a trained model to a Scratch workflow.

Think about it

Think about it - if you could add only three more examples to your table, which three would make your sorter fairest, and why?

My reflection

My reflection: One design choice that changed what our sorter decided was ...

Week 6 - Integrated Task (Evaluate, Justify & Showcase) | Integrated Prerequisite
Mastery Task - Design, Build, Evaluate & Justify an AI Solution

I can...

- I can test our sorter against a criterion, make one improvement, and prove with evidence that it helped.
- I can name one risk in our sorter and suggest a fair, safe fix, and say why it helps.
- I can reflect on all four skills and show our solution is ready for the Grade 7 unit.

Words to know

- criterion - the clear target we judge our sorter against, such as accuracy
- evidence - the before-and-after scores that prove a change really helped
- bias - when a sorter is unfair to a whole group of items because of the examples it learned from
- mitigation - a fix that reduces a risk, like balancing the examples or asking a human
- readiness gate - the check that shows our solution and our skills are ready for the Grade 7 unit

This week you prove your solution works. You will test your sorter, make it better, guard it against a risk, and

show it is ready - the same evaluate-and-improve habit real AI builders use every day.

Improve it, then prove it

Making a change is easy; proving it helped is the real skill. First you set a criterion - here, how many test cards

your sorter gets right. You score it, change one example or input, and score it again. If the score goes up on the

same test set, you have evidence the change improved the output. If it does not, you learned something too,

and you try a different change. This is exactly how AI builders turn a first attempt into a solution they can trust -

never by guessing, always by measuring.

Guard against the risk

Even an accurate sorter can be unfair or unsafe. If your examples missed a whole item type, that type gets

sorted wrongly every time - that is a biased output. And a confident wrong sort can be dangerous, like sending a

battery to general waste. So you name the risk and propose a responsible-use fix with a reason: balance your

examples, add a 'not sure - ask a human' route, and verify every important output against a trusted guide.

Naming risks and fixing them is part of building AI responsibly.

Score, change one thing, score again - if the number goes up, you have evidence it improved.

Why it matters

Why it matters - the UAE expects AI to be used responsibly, from smart-city services to sustainability projects;

learning to evaluate, improve and guard your own solution is exactly what a responsible AI builder does.

Let's try it - hands-on

1. Score your sorter on the six-card test set and circle the ones it gets wrong.
2. Change one example or input, re-run, and check if your score went up.

3. Write the evidence that your change improved the output.
4. Name one risk in your sorter and a fair, safe fix, with a reason.
5. Reflect on all four skills and show your solution is ready.

Today's tool - hands-on

Tool: Pre-generated packs + card decks (integrated: evaluate, justify, risk)

1. Run your build and evaluate the outputs against the criteria.
2. Justify one improvement you make, with evidence.
3. Draw a risk-scenario card and check your solution against it.
4. Confirm no personal data is used and name one mitigation.
5. Complete the four-strand readiness reflection.

You will make: An evaluated, justified solution with a risk check and readiness reflection.

Stay safe: Risk check confirms no personal data; outputs curated.

Evaluate against criteria, justify a change, check the risk, confirm readiness.

Think about it

Think about it - if two changes both raised your score by the same amount, how would you decide which one is

the better fix?

My reflection

My reflection: The skill I found hardest was ..., and the way I worked through it was ...

Grade 7 (Year 8) | Phase 3 / Cycle 2 | AI Solution Builder

Bridging Course - AI as Workflows - Input, Process,

Output

Transition-Year readiness | Weeks 1-6 (Core + Lab)

2

Week 1 - Core - Worksheet | AI Conceptual Understanding — Explains how training data affects output quality and performance

Name: _____ Class: _____ Date: _____

Follow the arrow from the output back through the process to the training data that shaped it.

Trace it back

For each example, write how the training data affected the output's quality.

Example What the training data had or was

missing

Effect on the output

Example A

Example B

Example C

My explanation

Pick one example. In two or three sentences, explain to a reader how its training data shaped the output.

Input or data?

An output came out weak but the input was clear. Explain why the training data is the more likely cause.

Exit ticket

Exit ticket - Explain, in one clear sentence, how the training data affected the quality of one output you studied today, and give your reason.

3

Challenge (stretch)

Challenge - Take a harder case: a model trained on a large data set that is still one-sided (for example, mostly one accent or one type of image). Predict how its output will fail on a new case, and justify your approach using the idea of representative data. This stretches toward the Proficient outcome of explaining how data selection shapes a model's performance.

4

Week 1 - Lab - Worksheet | AI Conceptual Understanding — Explains how training data affects output quality and performance (Lab)

Name: _____ Class: _____ Date: _____

Same input, same process, different training data - map it to prove where the difference comes from.

Fair-test map

For each pair, fill in what changed and what stayed the same, then record the effect on the output.

Example pair Training data (what changed) Effect on output quality

Pair 1

Pair 2

Pair 3

My conclusion

Use your maps as evidence. Which training data produced the better output, and why?

Was it fair?

Explain how you know the difference came from the training data and not from the input.

Tool log - Google Teachable Machine (clean vs messy data)

Record as you work. Goal: A comparison log showing how data quality changed the output.

Step What I did What I saw / result

1 Train the model on the clean set; test and read the output

5

c

2 Retrain the same class on the messy set.

3 Test the same unseen example and compare the two outputs.

4 Explain how the training data changed the output quality.

5 Write the rule: better examples, better output.

Safety check (tick): I used no personal data (no faces, voices or names). ☐

Responsible-use & human-in-the-loop reflection

- Where did a person decide or check in today's workflow (not the AI)?

- Could this workflow treat someone unfairly or be misused, and what would you check?

- Creator's reflection: what did you build or design today, and how is it different from what an AI would produce alone?

Exit ticket

Exit ticket - Name which training data gave the better output in one pair, justify it with your mapped evidence, and say why the input cannot be the cause.

Challenge (stretch)

Challenge - Design your own fair test for a harder case: choose one quality of the training data to change (for example, adding a wider range of examples), predict how the output should improve, and describe how you would map and compare the before and after to prove it. This stretches toward the Proficient outcome of evaluating how data choices shape a model's performance.

6

Week 2 - Core - Worksheet | AI Solution Design & Development — Builds structured AI-assisted solutions using guided inputs and data

Name: _____ Class: _____ Date: _____

Feed each item through the same three steps and read the output from the data.

Design my guided input

List the fields your workflow needs the user to fill in, and give one reason each matters to the output.

Field Why it matters to the output

Run the workflow

Feed three items through your workflow. Record the output and a reason for each, using the data table.

Item (guided input) Output (bin) Reason from the data

Verify it

Pick one output. How did you check it was correct for the task?

Exit ticket

Exit ticket - Name your guided input fields, the data your workflow used and one output it produced, then explain how you checked the output was right.

7

Challenge (stretch)

Challenge - Apply the skill to a more complex case and justify the approach: add a fourth field or a new material to your data, run an item that would have failed before, and explain why your redesigned workflow now handles it. This stretches toward the Proficient outcome of building and justifying structured AI solutions for less predictable inputs.

8

Week 2 - Lab - Worksheet | AI Solution Design & Development — Builds structured AI-assisted solutions using guided inputs and data (Lab)

Name: _____ Class: _____ Date: _____

A hard case reveals a data gap; adding one row fixes the output.

Test log

Run each tricky case. Record what failed and tick the stage where it failed.

Test case What failed Stage (input / data)

Case 1

Case 2

Case 3

My refinement

Describe the one change you made, then record what happened when you reran the hard case and one easy case.

Justify it

Why is your workflow now more reliable? Point to the input or the data.

Tool log - Scratch (structured solution from guided inputs)

Record as you work. Goal: A structured Scratch solution and a short explanation.

Step What I did What I saw / result

1 Read the guided inputs and decide the order of steps.

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2 Build the structured workflow block by block.

3 Run it and check the output matches the goal.

4 Improve one step so the output is clearer.

5 Explain why your structure works.

Safety check (tick): I used no personal data (no faces, voices or names). ☐

Responsible-use & human-in-the-loop reflection

- Where did a person decide or check in today's workflow (not the AI)?

- Could this workflow treat someone unfairly or be misused, and what would you check?

• Creator's reflection: what did you build or design today, and how is it different from what an AI would produce alone?

Exit ticket

Exit ticket - Name one refinement you made, the stage it fixed, and explain how it made the output more reliable without breaking an easy case.

Challenge (stretch)

Challenge - Apply the skill to a more complex case and justify the approach: design one input that would defeat a classmate's refined workflow, then propose the change that would fix it and explain why it works. This stretches toward the Proficient outcome of building and justifying structured AI solutions that stay reliable on unpredictable inputs.

10

Week 3 - Core - Worksheet | Critical Evaluation & Informed Interaction — Explains how changes to inputs or data improve outputs

Name: _____ Class: _____ Date: _____

Judge the two outputs

Mark which output is better on each criterion, then write the evidence from the output.

Criterion Better output (A or B) Evidence from the output

Accuracy

Clarity

Completeness

Explain the improvement

Write one sentence linking the input change to the better output. Use because or so.

Exit ticket

Name one change you could make to an input and the improvement it would make to the output, and say which criterion proves it.

Challenge (stretch)

Challenge (stretch) - Two outputs improve on different criteria: one is more accurate, the other more complete. Explain how you would decide which is the better output overall, and justify your choice with evidence.

11

Week 3 - Lab - Worksheet | Critical Evaluation & Informed Interaction — Explains how changes to inputs or data improve outputs (Lab)

Name: _____ Class: _____ Date: _____

One workflow, four outputs, ranked against the same criteria.

Rank the outputs

Put the outputs in order against the criteria and write the evidence for each place.

Rank Output Evidence against the criteria

1 (best)

2

3

4 (weakest)

Iterate once

Change one input, run it again, and judge the new output against the criteria.

Before the change After the change

Input

Output

Better? Which criterion and evidence

When a change does not help

Write one change that did not improve the output, and the evidence that shows it.

Tool log - Teachable Machine retrain + pre-generated packs (inputs change outputs)

Record as you work. Goal: A before/after record explaining how input changes improved outputs.

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Step What I did What I saw / result

1 Test the current model and note the output.

2 Change the inputs (add clearer examples to one class) and re

3 Test again and compare with the before output from the pack.

4 Explain which input change improved the output and why.

5 Record the improvement with evidence.

Safety check (tick): I used no personal data (no faces, voices or names). ☐

Responsible-use & human-in-the-loop reflection

• Where did a person decide or check in today's workflow (not the AI)?

• Could this workflow treat someone unfairly or be misused, and what would you check?

• Creator's reflection: what did you build or design today, and how is it different from what an AI would produce alone?

Exit ticket

You changed one input and re-ran the workflow. Explain in one sentence whether the output improved and name the criterion and evidence that proves it.

Challenge (stretch)

Challenge (stretch) - Design a fair test to decide which of two input changes improves an output more. Say what you would keep the same, what you would measure, and what evidence would settle it.

13

Week 4 - Core - Worksheet | Ethics & Governance - Identifies biased outputs and suggests simple ways to improve fairness and safety

Name: _____ Class: _____ Date: _____

Spot, trace, fix

For each scenario, name the bias, mark where it began, and note who it affects.

Scenario Bias I spotted (and who it affects) Stage (input / process)

Scenario 1

Scenario 2

My fairness measures

Write at least two simple measures to make a workflow fairer or safer. Give a reason for each.

Verify it

Pick one fix. How would you check it actually made the output fairer for a new case?

Exit ticket

Exit ticket - Name one biased output, say whether the bias began at the input or the process, and give one simple, reasoned way to make that workflow fairer or safer.

Challenge (stretch)

Challenge - Take a harder case where two biases appear in the same workflow. Decide which to fix first and justify your

order, then explain what a fair UAE service built on this workflow would need to guarantee. This stretches toward the

Proficient idea of evaluating and improving a workflow against explicit fairness criteria.

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Week 4 - Lab - Worksheet | Ethics & Governance - Identifies biased outputs and suggests simple ways to improve fairness and safety (Lab)

Name: _____ Class: _____ Date: _____

Sort each concern by type and stage, then order your measures into a responsible-use plan.
Evaluate the scenario

List each concern, mark its type and the workflow stage where you would address it.

Concern I found Type (bias / safety / governance) Stage (input / process / output)

My responsible-use plan

Write at least three measures. Rank them and give a reason for each and for the order.

Test it

Would your plan still protect a different user? Explain how you would check.

Tool log - Card decks (bias-scenario cards) + pre-generated packs

Record as you work. Goal: A fairness record: the bias, its cause, who is affected and a fix.

Step What I did What I saw / result

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1 Read a bias-scenario card and the paired outputs.

2 Identify which output is biased and what in the data likely

3 Say who could be treated unfairly.

4 Suggest one simple change to improve fairness or safety.

5 Record your fix and why it helps.

Safety check (tick): I used no personal data (no faces, voices or names). ☐

Responsible-use & human-in-the-loop reflection

- Where did a person decide or check in today's workflow (not the AI)?

• Could this workflow treat someone unfairly or be misused, and what would you check?

• Creator's reflection: what did you build or design today, and how is it different from what an AI would produce alone?

Exit ticket

Exit ticket - Name your highest-priority concern, say which workflow stage you would address it at, and state the first measure in your plan with a reason.

Challenge (stretch)

Challenge - Take the scenario one step further: a new group of users is added whose needs conflict with your current plan.

Redesign one measure to keep the workflow fair for everyone and justify the trade-off you made. This stretches toward the

Proficient idea of evaluating and governing an AI workflow against competing fairness and safety criteria.

16

Week 5 - Integrated Task (Design & Build) - Worksheet | Integrated Prerequisite

Mastery Task - Design, Build, Evaluate & Justify an AI Solution

Name: _____ Class: _____ Date: _____

Our training table

List the example items your sorter learns from and the correct bin for each. Keep the bins balanced.

Example item Bin (recycle / compost / general)

Our workflow map

Write your input, your process and your output, then the one rule your sorter follows.

Test log

Run three item cards and record the output bin and whether it matches the guide.

Test card Output bin Matches guide? (yes / no)

Tool log - Teachable Machine + Scratch (integrated: design & build)

Record as you work. Goal: A designed and built integrated solution with a strand-tagged note.

Step What I did What I saw / result

1 In pairs, plan the small solution with the plan cards (goal,

2 Train a simple non-personal model in Teachable Machine.

3 Build the workflow in Scratch that uses or simulates the mod

4 Run it once and confirm an input produces an output.

5 Save your build and note what each strand contributed.

Safety check (tick): I used no personal data (no faces, voices or names). ☐

Responsible-use & human-in-the-loop reflection

• Where did a person decide or check in today's workflow (not the AI)?

• Could this workflow treat someone unfairly or be misused, and what would you check?

• Creator's reflection: what did you build or design today, and how is it different from what an AI would produce alone?

Exit ticket

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Exit ticket - Name one design choice you made (an example you added, removed or changed) and explain how it changed one of your sorter's outputs.

Challenge (stretch)

Challenge - Add a new item type your table has never seen (for example a battery or a coffee cup) and predict how your sorter will handle it. Explain what examples you would need to sort it reliably. This stretches toward the Proficient idea of designing workflows that handle new, unseen inputs.

18

Week 6 - Integrated Task (Evaluate, Justify & Showcase) - Worksheet | Integrated Prerequisite Mastery Task - Design, Build, Evaluate & Justify an AI Solution

Name: _____ Class: _____ Date: _____

Score and improve

Score your sorter, make one change, and re-score. Show the evidence it improved.

Stage Score (out of 6) Change I made

Before

After

Risk and fix

Name one risk in your sorter and a responsible-use fix, and give your reason.

Four-skill reflection

Answer all three: the hardest skill and how you worked through it; one improvement you would make; one risk you would guard against.

Tool log - Pre-generated packs + card decks (integrated: evaluate, justify, risk)

Record as you work. Goal: An evaluated, justified solution with a risk check and readiness reflection.

Step What I did What I saw / result

1 Run your build and evaluate the outputs against the criteria

2 Justify one improvement you make, with evidence.

3 Draw a risk-scenario card and check your solution against it

4 Confirm no personal data is used and name one mitigation.

5 Complete the four-strand readiness reflection.

Safety check (tick): I used no personal data (no faces, voices or names). ☐

Responsible-use & human-in-the-loop reflection

- Where did a person decide or check in today's workflow (not the AI)?

- Could this workflow treat someone unfairly or be misused, and what would you check?

- Creator's reflection: what did you build or design today, and how is it different from what an AI would produce alone?

Exit ticket

Exit ticket - In one or two sentences, say why your solution is ready for the Grade 7 unit, naming one improvement you made and one risk you guarded against.

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Challenge (stretch)

Challenge - Optimise your sorter for a second criterion - efficiency, usability or fairness - and describe the trade-off: what

got better, what got worse, and why you would still choose it. This stretches toward the Proficient idea of evaluating solutions against more than one criterion.

Grade 7 (Year 8) | Phase 3 / Cycle 2 | AI Solution Builder

Main Course, Term 1 - AI as Workflows - Input, Process, Output

Emerging tier | 10 weeks (Core + Lab) |

TEACHER PACK

1. About this Teacher Pack

Lesson timing - read first (resolves the 45-minute-period fit)

Every Core and Lab lesson fits a single 45-minute period. The 10-minute 'Prior knowledge & Prerequisite Primer' runs as pre-lesson time (a starter/Do-Now, or the last 10 minutes of the previous session) and is NOT counted inside the 45 minutes. Choose ONE option below for the whole term; Option 1 is the default.

- Option 1 (default) - primer separate. The 45-minute period runs five phases exactly as written: Hook 6 · Frame/Teach 8 · Explore/Build (ONE main activity) 14 · Apply/Consolidate 12 · Check & close 5 = 45.
- Option 2 - primer inside 45 minutes (use when no pre-lesson time exists). Compress to: Recall/primer 5 · Hook 5 · Frame/Teach 7 · Explore/Build 13 · Apply/Consolidate 10 · Check & close 5 = 45.
- One activity per Lab. Each Lab lists a 'Hands-on activity guide' (unplugged card routine) and a 'Lab tool integration' (Teachable Machine / Scratch). These are the SAME task at two levels - run ONE within the period, not both.
- Label change. Wherever a lesson shows 'Prior knowledge & Prerequisite Primer (10 min)', read it as '(10 min, pre-lesson - not counted in the 45-minute period)'. Full detail and worked examples are in [Grade7_Revised_Lesson_Timings](#).

Unit map - how the ten weeks build the workflow theme

The term runs one idea - AI as a workflow of input -> process -> output - and deepens it each week. Foundation: How Models Learn (rule-based vs learned, training vs inference, confidence). W1 Launch: name input/process/output. W2 Understand: trace the three-stage shape. W3 Design & build a workflow. W4 Judge & improve outputs. W5 Risks, ethics & responsible use. W6 Bring the four strands together on one task. W7-W9 Signature Solution: plan -> build -> refine. W10 Showcase & reflect.

Strand emphasis by week: CU (Conceptual Understanding) leads Foundation-W2; SD (Solution Design) leads W3, W7-W9; CE (Critical Evaluation) leads W4, W9; GE (Ethics & Governance) leads W5; W1, W6 and W10 integrate all four. Every Lab also carries a responsible-use reflection so ethics runs throughout, not only in W5.

Technology note - a technology-agnostic design

The tools named here (Teachable Machine, Scratch, the pre-generated output packs)

are examples that make the workflow visible; they are not the curriculum. If a tool is unavailable or blocked, substitute a school-approved equivalent (any no-account image-classifier or block editor) or run the unplugged card-deck fallback listed in every lesson. The learning outcome - designing, building and evaluating a workflow - stays the same whichever tool is used.

This pack contains everything needed to teach the Grade 7 main course, Term 1: ten weeks of Emerging-tier

lessons on the unit 'AI as Workflows - Input, Process, Output', each delivered as a Core period and a paired Lab

period on the same outcome. Weeks 7-10 run the signature solution end-to-end.

Where this sits - Grade 6 to Grade 7. This Emerging-tier term builds directly on Grade 6

Advanced. In Grade 6,

students learned that AI systems are trained on data and began describing simple input-output behaviour;

Grade 7 now treats AI as a full input-process-output workflow that students design, build, evaluate and use

responsibly. Weeks 1-2 deliberately re-activate and consolidate that Grade 6 foundation - the workflow shape

and the trained-model idea - before Weeks 3-10 extend it into building, judging, risk, and a signature solution.

Each week's outcome is one clear, named step up in independence and rigour from the week before.

How to use this pack

- Teach each week as Core (period 1) then Lab (period 2) on the same outcome.
- Keep all AI use teacher-mediated and account-free; enter no personal data.
- Build the evaluate-verify-risk habit into every AI activity; provide the unplugged fallback where tools are unavailable.
- Use the per-lesson formative check to confirm the outcome before moving on.
- Pair this Teacher Pack with the Student Lesson Content and Student Worksheets (same order).

Running the Lab (which hands-on activity?): each Lab lists a 'Hands-on activity guide' (unplugged card routine)

AND a 'Lab tool integration' (Teachable Machine / Scratch). These are the SAME task on two levels, not two

separate tasks - run ONE. Use the tool integration when the tool is available and time allows; otherwise run

the unplugged activity guide, which is the fallback. Never run both in one 45-minute period.

Scratch prerequisite: students should have basic Scratch familiarity (adding blocks, using ask/say and simple if-

blocks) from earlier grades. If not, spend the first Scratch Lab (L3) on a short guided orientation before the

build, or use the paper 'block cards' fallback.

Saving projects across weeks (accounts-free): because students do not log in, save each Scratch project as a local .sb3 file on the teacher device or a shared class drive, named by group, and reopen it in L7-L10. The

offline Scratch app saves and loads .sb3 files without any account. Keep only non-personal projects.

Timing: the 10-minute Prerequisite Primer runs as separate pre-lesson time; the five in-period phases already

total 45 minutes. See Grade7_Revised_Lesson_Timings for the corrected phase template.

Safeguarding & responsible use

Grade 7 designs, builds and evaluates simple AI workflows. Keep all AI use teacher-mediated and account-free for

students; prefer a board demonstration, curated or pre-generated output sets, or a card-based 'simulated AI' routine.

Never enter personal data (names, faces, voices) into a prompt or a model. Teach privacy as a technical concept too,

not only a rule: de-identification (remove anything that names a person), data minimisation (collect only what the task

needs), and retention/deletion (keep data only as long as needed, then delete it). These are backed in the UAE by the

Personal Data Protection Law (PDPL) and ADEK data-protection policy. Build the evaluate-verify-risk habit into every

AI activity: check each output against the task and a trusted source, because a confident output can still be wrong.

Every AI activity has a non-AI (unplugged) fallback so no lesson depends on tool access.

Materials & tools (UAE-accessible)

Tool / material What it is & why Access route Unplugged fallback

Workflow demo set +

example cards

A board demo of an input-process-output

workflow plus printed example cards.

School-printed; teacher-

run; no accounts.

Sort and map familiar

workflows on paper.

School-approved child-

safe assistant

An approved generative tool, teacher-

operated only, for curated demos on non-personal data.

School-approved; teacher

device only.

Use pre-generated output

sets from the teacher.

Pre-generated output
sets

Curated good/weak outputs from the same
task for evaluation and triage.

Prepared by the teacher in
advance.

Same sets, printed.

Simulated-AI cards Card decks that mimic an AI step so
students practise the workflow unplugged.

School-printed. This is itself the
unplugged route.

Program at a glance

Key terms - consistent definitions (use these throughout the term)

classifier - the process step that assigns each input to one of a fixed set of classes (this is the
standard term; the

paper 'sorter' cards are a simulated classifier). Classification is one AI task among others such
as generation,
ranking and translation.

class / label - one of the named groups a classifier chooses between (e.g. Recycle, Compost,
Landfill); the label

is the answer it outputs.

training - the one-off process of showing a model many labelled examples so it learns patterns;
inference is

using the finished model on a new input. Training happens before the lesson; students run
inference.

inference - using the finished, already-trained model on a new input to produce an output; unlike
training, the

model's parameters do not change during inference.

training data vs testing/input data - training data is the labelled examples the model learned
from; testing (or

input) data is the new, unseen items you feed the finished model. Always keep them separate (a
train/test split).

accuracy - the count-correct measure: how many test items the model got right out of the total,
written as a

fraction or percentage. Use 'accuracy' for this, never 'reliability'.

reliable / reliability - consistency: the workflow completes the task across more than one input.
Keep 'reliable'

for consistency only.

confidence score - a number the model reports with each answer showing how sure it is (e.g.
0.90). It is NOT the

same as accuracy and a high score does not mean the answer is correct.

model - a system that has learned patterns from training data and can generalise to unseen
inputs. A fixed

lookup table is NOT a model, because it cannot generalise.

black box - when you can see the input and output but not the reason inside for the decision.
bias - a repeating, systematic error that makes a system perform worse for one group of cases than another,
usually from skewed training data.

fairness - treating comparable cases or groups consistently; it can be measured by comparing accuracy across

groups (see the Fairness & Bias Technical Note).

Lesson Strand Framework ref(s)

F Foundation - How Models Learn AI Conceptual Understanding

AIF·CU·G7·E

L1 Unit Launch - AI as Workflows - Input, Process, Output

Integration AIF·CU·G7·E AIF·SD·G7·E

AIF·CE·G7·E AIF·GE·G7·E

L2 Understanding the System - Input, Process, Output AI Conceptual Understanding

AIF·CU·G7·E

L3 Designing & Building a Workflow AI Solution Design & Development

AIF·SD·G7·E

L4 Judging & Improving Outputs Critical Evaluation & Informed Interaction

AIF·CE·G7·E

L5 Risks, Ethics & Responsible Use Ethics & Governance AIF·GE·G7·E

L6 Bringing the Strands Together Integration AIF·CU·G7·E AIF·SD·G7·E

AIF·CE·G7·E AIF·GE·G7·E

L7 Signature Solution - Planning Integration AIF·CU·G7·E AIF·SD·G7·E

AIF·CE·G7·E AIF·GE·G7·E

L8 Signature Solution - Building Integration AIF·CU·G7·E AIF·SD·G7·E

AIF·CE·G7·E AIF·GE·G7·E

L9 Signature Solution - Refining & Finishing Integration AIF·CU·G7·E AIF·SD·G7·E

AIF·CE·G7·E AIF·GE·G7·E

L10 Showcase, Assessment & Reflection Integration AIF·CU·G7·E AIF·SD·G7·E

AIF·CE·G7·E AIF·GE·G7·E

Foundation Core - How Models Learn (teach before Week 1)

Strand: AI Conceptual Understanding | Framework ref(s): AIF·CU·G7·E | 1 x 45 min (+10-min primer, pre-

lesson) | Interactive lesson: 'How Models Learn'

Why first. Students meet the Rule-vs-Learned card sort (Deck A) in Week 1 and train a classifier from Week 2.

This foundation gives them the concept of how a model learns from data BEFORE those activities, so the sort

and the training make sense. Teach it as a Foundation session in the first timetabled slot (or fold into Week 1 as a double period).

Objectives & success criteria

- Distinguish a rule-based system from a learned (trained) model, with an example of each.
- Explain the difference between training (teaching the model) and inference (using it).
- State what a trained model stores - learned patterns/parameters - and that it does NOT keep copies of the training examples.
- Explain that a classifier gives the same output for the same input (deterministic); variation belongs to generative models that sample.
- Describe a train/test split and why a model is tested on unseen data; read a confidence score as distinct from accuracy.

Student-facing success criteria:

- I can say whether a system follows fixed rules or has learned from examples, and why.
- I can explain the difference between training a model and using (inference) a model.
- I can explain that a model stores learned patterns, not saved examples, and why a classifier repeats the same answer.
- I can say what a train/test split is and read a confidence score.

The technical concept (teacher background)

Rule-based vs learned. A rule-based system does exactly what a person wrote (if metal -> Recycle). A learned

model is shown many labelled examples and adjusts internal numbers (parameters) until it predicts well -

nobody writes the rule by hand.

Training vs inference. Training is the one-off teaching phase where the model updates its parameters from

labelled examples. Inference is every later use: the finished model labels a new input without changing.

What a model stores. A trained model does not save the training photos or sentences; it stores learned patterns

as parameters. That is why it answers about new inputs but cannot show you the originals.

Determinism & confidence. A classifier is deterministic: the same input gives the same output and the same

confidence score every time. A confidence score (e.g. 0.90) shows how sure the model is for THIS input; it is not

accuracy (a count of correct answers over a whole test set).

Train/test split. To check a model really learned rather than memorised, hold back some labelled examples as a

test set and measure accuracy only on those unseen items.

Lesson plan (45 min, Option 1). Hook 6: show one rule-based and one learned example, ask 'who wrote the rule?'. Frame/Teach 8: model training vs inference on the board. Explore/Build 14: pairs sort Deck A cards into 'fixed rules' vs 'learns from examples' and justify. Apply/Consolidate 12: run a tiny train/test split with cards (hold back 3 of 12) and read a confidence figure. Check & close 5: exit ticket.

Anticipated misconceptions

- 'The model remembers the examples.' - No: it stores patterns/parameters, not the originals.
 - 'A high confidence score means it is correct.' - No: confidence is per-input sureness, not accuracy; a model can be confidently wrong.
 - 'Rules and learning are the same.' - A person writes rules; a model learns patterns from data.
- Many real

systems mix both (see Deck A borderline cards).

Formative check (exit ticket)

- Name one rule-based system and one learned system, and say how you can tell them apart.
- In one sentence, what is the difference between training and inference?
- A classifier gives 'cat, 0.95' for a photo. If you run the same photo again, what will it give - and why?
- We had 12 labelled cards. Why hold back 3 as a test set instead of testing on all 12?

Cross-curricular links: Maths - accuracy as a fraction/percentage; Science - train/test split as a fair test;

Language - classifiers behind Arabic-English translation. Full detail:

Grade7_New_Concept_Lesson_How_Models_Learn.

Week 1 - Core | Unit Launch - AI as Workflows - Input, Process, Output |

Interactive lesson: 'AI as a Workflow'

Real-World Connection (UAE): in 2017 the UAE appointed a Minister of State for Artificial Intelligence and set a National AI Strategy looking to 2031. Many everyday UAE services already run as input -> process -> output AI workflows.

Prerequisite: teach the Foundation lesson 'How Models Learn' before this week's Deck A Rule-vs-Learned sort, so students already know how a model learns from data. (Interactive version: 'How Models Learn' HTML.)

Strand: Integration | Framework ref(s): AIF·CU·G7·E AIF·SD·G7·E AIF·CE·G7·E AIF·GE·G7·E | 1 x 45 min

Objectives & success criteria

- Generate at least two focused questions about how AI works as a workflow of input, process and output.
- Explore and classify everyday examples as input, process or output, justifying each choice.
- Co-construct the unit success criteria and the class responsible-use norms for AI tools.

Student-facing success criteria:

- I can ask at least two good questions about how AI turns an input into an output.
- I can sort examples into input, process and output and say why.
- I can help write our unit success criteria and our safe, responsible-use rules for AI.

Prior knowledge & Prerequisite Primer (10 min)

Assumes: In Grade 6 you learned that AI systems are trained on data and that the data you give them affects the output you

get back. You have used a simple AI tool with your teacher and talked about giving clear instructions.

Activate: Show a familiar everyday machine on the board - a vending machine or the school printer - and ask: what goes in, what happens inside, and what comes out?

Teach: Name the three parts of any workflow - input (what goes in), process (the steps that change it) and output (what comes out)

- and model them on the vending-machine example. Say explicitly where the analogy breaks: a vending machine follows fixed rules

a person wrote, while an AI model has learned its patterns from examples - so use the machine only for the input-process-output

shape, not to explain how AI 'decides'.

Apply: In pairs, students pick one everyday machine and label its input, process and output on a mini whiteboard.

Check: Cold-call two pairs to name each part; listen for correct use of input, process and output.

If a pair mixes up process and

output, re-model with the vending machine before moving on.

Critical & analytical thinking focus

Question & analyse - surface prior knowledge and frame inquiry (Understand/Analyse | DOK 1-2 | SOLO: Uni to

Multistructural)

The technical concept (teacher background)

An AI system is a workflow: it takes an input, runs a process, and returns an output. What makes the process 'AI' is that it

usually relies on a model trained on many examples to recognise patterns, rather than on rules a person wrote by hand. A

classifier gives the SAME output for the same input - it is deterministic; only generative models that sample can vary. Either

way a confident output can still be wrong - which is why every result is checked against the task and a trusted source (verify-

before-trust). This launch lesson is about surfacing what students already know and framing good questions, not building

yet. Keep all tool use teacher-operated, account-free and free of personal data. Cognitive demand for this lesson:

Understand/Analyse | DOK 1-2 | SOLO: Uni to Multistructural.

This builds on the training-data work from Bridging Week 1 (AI Conceptual Understanding - explains how training data

affects output quality and performance): Main Term 1 assumes students already know that a model's training data shapes

its output, and this launch lesson gives that idea a workflow vocabulary - input, process and output.

Lesson plan

Phase | time Teacher (I do / we do) Students Key questions (targeted thinking)

Hook / 6 min Show a short, curated stimulus on AI as

a workflow (a board demo of one input

becoming one output). Ask students

what went in and what came out.

Watch, then jot one thing

they noticed and one thing

they wonder on a sticky note.

- What did the system take in, and what did it give back?

Frame the

inquiry / 8 min

Introduce input - process - output as

the shape of every AI workflow. Model

Turn their 'I wonder' note into

at least two questions about

- Which part of the workflow does your question ask about - input, turning a wondering into a clear question (What... How... Why...).

AI as a workflow. process or output?

Explore and sort

/ 14 min

Hand out a curated card set of everyday

examples. Circulate and prompt

justifications; keep any live tool

teacher-operated only.

In pairs, sort each card as

input, process or output and

write one reason for each

sort.

- Why is this an input and not an output? Convince your partner.

Co-build wall

and charter / 12

min

Scribe a shared vocabulary wall (input,

process, output, model, data) and co-

write the class safe-use charter.

Preview the signature activity.

Contribute definitions in

their own words and propose

one responsible-use rule (no

personal data, verify before trust, teacher-operated tools).

- What is one rule that keeps us safe when we use an AI tool in class?

Check and close

/ 5 min

Run the exit check; collect the sticky-note questions and the sort cards as evidence.

Complete the exit ticket: one classified example with a reason, plus one question they still have.

- Give one example and say whether it is an input, a process or an output - and why.

Anticipated misconceptions

Misconception How to address it

AI just knows the answer - it doesn't do steps. Show the same input twice and reveal the hidden middle: the

model processes the input before any output appears. Name the process step on the board.

The output is always right because the computer said it. Display a curated output that is confidently wrong and check it

against the task and a trusted source. Establish verify-before-trust as a class norm.

Input and output are basically the same thing. Use a concrete before-and-after pair (a question in, an answer

out) and have students point to which is which; re-sort any cards that were misplaced.

A lookup table (or a set of matching cards) is machine learning.

Contrast the two: a lookup table only returns answers already written in it and cannot handle a new item; a trained model learns patterns and generalises to unseen inputs. Remove all matching examples for one item and run it - the lookup 'sorter' fails or guesses. That gap is the difference between a stand-in and real machine learning.

Hands-on activity guide

Overview: The class opens the unit with a curated stimulus, then explores and sorts everyday examples into input, process

and output before co-building the vocabulary wall and safe-use charter and previewing the signature activity.

Grouping / timing: Pairs for the sort, whole class for the wall, charter and debrief.

Tool: A teacher-operated board demo plus a pre-printed, curated card set (a 'simulated AI' paper routine). Account-free

and no personal data - students never log in or enter names.

Facilitation steps:

1. Watch the stimulus and note one thing you noticed and one thing you wonder.
2. Turn your wondering into at least two clear questions about AI as a workflow.
3. With your partner, sort each example card as input, process or output.
4. Write one reason for each card - why does it belong there?
5. Add one word and its meaning to the class vocabulary wall.
6. Propose one responsible-use rule for our safe-use charter.

Unplugged fallback: The whole lesson already runs unplugged on paper cards and a board demo. If the demo tool is unavailable, describe one worked input-output example aloud and have pairs sort the printed cards - no device needed.

Formative assessment (aligned to the objectives)

Task / item Model response Marking note

Write two questions about how AI works as a workflow.

How does the process step change my input into an output? What data was the model trained on?

Accept any two clear questions that name input, process and/or output. Look for question form, not perfect wording.

Classify three examples as input, process or output and give a reason.

A typed prompt = input (what I give the system). Sorting photos into groups = process (the steps that change the input). A finished summary = output (what comes back).

One mark per correct sort with a matching reason; the reason must reference giving-in, changing, or coming-out.

Suggest one responsible-use rule and say why it keeps us safe.

Never type a real name or photo into the tool, because that is personal data and it should stay private.

Accept any rule aligned to no-personal-data, verify-before-trust, or teacher-operated tools, with a reason.

- Targets: the three launch objectives - questioning, classifying with justification, and co-constructing the success criteria and responsible-use norms.
- Check: exit ticket plus circulating listen-ins during the pair sort, questioned against the input-process-output frame.
- Success criteria: the student generates two on-topic questions, sorts examples correctly with reasons, and contributes to the charter.
- Evidence: collected sticky-note questions, completed sort cards, and the co-built vocabulary wall and charter.
- Feedback / adjustment: if pairs mix up process and output, re-model with one clear before-and-after pair and re-sort two cards before the next lesson.

Differentiation

Criterion Emerging (expected) Approaching Not yet

Questioning the workflow Generates two on-topic questions about input, process or output, with guidance.

Generates one on-topic question, or needs prompting.

No question, or questions unrelated to the workflow.

Classifying with justification Sorts examples as input, process or output and gives a simple reason for each, with support.

Sorts correctly but reasons are missing or vague.

Sorts incorrectly and cannot give a reason.

Responsible-use contribution

Suggests one sensible rule for the charter using the safe-use ideas modelled.

Repeats a rule when prompted.

Does not contribute a rule.

Representation: pair the spoken stimulus with the printed cards, the vocabulary wall and a visible input-process-output diagram;
pre-teach the three words with simple icons.

Action & expression: let students show sorting by placing cards, speaking their reason aloud, or writing - accept any mode for the exit ticket.

People of Determination: provide a reduced card set with picture support, sentence starters ('This is an input because...'), extra processing time, and a peer scribe where helpful.

Week 1 - Lab | Unit Launch - AI as Workflows - Input, Process, Output (Lab) |

Interactive lesson: 'Compare and Log the Difference'

Strand: Integration | Framework ref(s): AIF·CU·G7·E AIF·SD·G7·E AIF·CE·G7·E AIF·GE·G7·E |
1 x 45 min

Objectives & success criteria

- Compare paired examples of AI workflows and record the differences you observe.
- Trace each example through input, process and output, noting where the differences appear.
- Draw one evidence-based conclusion about what makes a workflow's output better or worse.

Student-facing success criteria:

- I can compare two AI examples and write down how they differ.
- I can trace each example through input, process and output.
- I can use what I observed to say what made one output better.

Prior knowledge & Prerequisite Primer (10 min)

Assumes: In the Core period you named the three parts of a workflow - input, process and output - and sorted examples into them. You also agreed the class safe-use charter.

Activate: Show two curated outputs made from almost the same input (one clear prompt, one vague prompt) and ask which is more useful and why.

Teach: Model a compare-and-log routine: put two examples side by side, trace each through input-process-output, and record one difference in a log table.

Apply: In pairs, students trace one pair of examples and log a single observed difference.

Check: Check that each pair has located the difference at the correct workflow stage (usually the input or the process); redirect any pair that logged only the output.

Critical & analytical thinking focus

Question & analyse - surface prior knowledge and frame inquiry (Understand/Analyse | DOK 1-2 | SOLO: Uni to

Multistructural)

The technical concept (teacher background)

Comparing workflows is how we start to see that the input and the process shape the output.

Two AI runs from nearly

identical inputs can differ because one input was clearer, or because the model processed it differently. In this Lab

students do not meet a new concept - they investigate the same input-process-output idea more deeply by comparing

curated example pairs and logging differences as evidence. This is the habit behind later evaluation work: notice a

difference, locate where in the workflow it came from, and describe it in plain words. All

examples are pre-generated and

teacher-operated, account-free, with no personal data. Cognitive demand for this lesson:

Understand/Analyse | DOK 1-2 |

SOLO: Uni to Multistructural.

Lesson plan

Phase | time Teacher (I do / we do) Students Key questions (targeted thinking)

Recap and set up

/ 5 min

Recap input-process-output and the

safe-use charter. Hand out the

comparison log and the curated

example-pair packs.

Read the first example pair

and predict which output will

be more useful.

- Before we look closely, which output do you expect to be better, and why?

Model compare-

and-log / 8 min

Think aloud through one pair: trace

each example from input to process to

output and log one clear difference in

the correct row.

Watch, then copy the

routine's headings into their

own log.

- At which stage - input, process or output - did the difference begin?

Exploration

station 1 / 14 min

Circulate as pairs work the first two

example pairs; prompt for evidence, not

opinion. Keep all tools teacher-

operated only.

Trace each example, then

log at least two observed

differences with the stage

where each appears.

- What is your evidence that this difference is real and not just a guess?

Rotate and log /

13 min

Rotate pairs to a new pack; prompt them to compare across packs and Log differences for the new pair and begin a one-

- Across the pairs you have seen, what one change most improved start a conclusion. sentence conclusion about what improved the output. the output?

Debrief and

close / 5 min

Collect the logs; take two conclusions and test them against the class evidence; flag verify-before-trust.

Complete the exit ticket: one logged difference and the workflow stage it came from.

- Name one difference you logged and where in the workflow it started.

Anticipated misconceptions

Misconception How to address it

A different answer means the AI is broken. Show that small input changes cause different-but-valid outputs;

the point is to explain the difference, not to declare a fault.

The difference is always in the output. Trace a pair back to show the difference usually starts at the input

or the process; have pairs relabel the stage on their log.

Better just means longer or fancier. Compare a long vague output with a short accurate one against

the task; agree that 'better' means fit-for-task, checked against a trusted source.

Hands-on activity guide

Overview: Pairs work through curated example-pair packs, tracing each example through input-process-output and

logging observed differences as evidence, then draw one conclusion about what made an output better.

Grouping / timing: Pairs at rotating stations, whole class for the debrief.

Tool: Pre-generated example-pair packs on paper or on a teacher-operated board (a 'simulated AI' comparison routine).

Account-free, no logins, no personal data.

Facilitation steps:

1. Read the first example pair and predict which output is more useful.
2. Trace each example through input, process and output.
3. Log at least two differences you observe, and mark the stage where each begins.
4. Rotate to a new pack and log differences again.
5. Compare across packs and write one conclusion about what improved the output.
6. Check your conclusion against the task - would it still hold for a new example?

Unplugged fallback: The packs are already paper-based, so no device is required. If the board demo is unavailable, use

the printed example pairs only and log differences by hand.

Lab tool integration (practical, hands-on)

Tool: Card decks (workflow-example cards)

Teacher setup: Print one workflow-example deck per pair (8-10 cards, each a familiar system such as a vending machine, a photo-tagging app, a spell-checker).

What students do:

1. Deal the deck face-up between partners.
2. For each card, agree the input, the process and the output and write them on the card's three blanks.
3. Sort the cards into 'follows fixed rules' and 'learns from examples'.
4. Pick the two hardest-to-place cards and say why.
5. Log at least three differences you noticed between the two piles.

Students produce: A sorted card layout plus a difference log.

Data-safety: Fully unplugged; no devices or data.

Unplugged fallback: This is already the unplugged activity; extend by adding two blank cards and inventing new systems.

Card sort: each workflow card split into input, process and output, sorted into two piles.

Formative assessment (aligned to the objectives)

Task / item Model response Marking note

Log two differences between an example pair and mark the workflow stage for each.

Pair A used a clearer input (stage: input); Pair B's process grouped items differently (stage: process).

One mark per difference correctly described and located at a plausible stage.

Write one conclusion about what made an output better, using your log as evidence.

The clearer input gave a more useful output, because the system had exactly what it needed to work on.

Accept a conclusion that cites a logged difference; it must reference evidence, not opinion.

Explain why two outputs from similar inputs can differ.

The inputs were not identical, and the model processes each one, so the outputs come out different but both can be reasonable.

Look for reference to an input or process difference; not 'the AI is broken'.

- Targets: comparing AI workflow examples and logging observed differences, and drawing an evidence-based conclusion, on the same input-process-output outcome.
- Check: circulating evidence-prompts during the stations plus an exit ticket naming one logged difference and its stage.
- Success criteria: the student logs at least two located differences and states one conclusion backed by their log.
- Evidence: collected comparison logs and the exit tickets.
- Feedback / adjustment: if pairs log only outputs, re-model tracing a pair back to its input before the next lesson and have them relocate one difference.

Differentiation

Criterion Emerging (expected) Approaching Not yet
Comparing examples Logs two differences between a pair, with support.

Logs one difference, or needs prompting.

Does not log a usable difference.

Locating the stage Marks the workflow stage where a difference begins, with guidance.

Marks a stage but often mislocates it.

Cannot link a difference to a stage.

Evidence-based conclusion States one conclusion that cites a logged difference.

States a conclusion but without evidence.

No conclusion offered.

Representation: side-by-side example cards, a pre-formatted log table with stage labels, and the input-process-output diagram on display.

Action & expression: pairs may record differences in writing, by annotating the cards, or by speaking to the teacher who scribes.

People of Determination: fewer example pairs, a partly completed log, sentence starters ('One difference is ... It began at the ...'), and extra time to rotate.

Week 2 - Core | Understanding the System - Input, Process, Output | Interactive lesson: 'The Three-Stage Shape'

Strand: AI Conceptual Understanding | Framework ref(s): AIF·CU·G7·E | 1 x 45 min

Objectives & success criteria

- Describe how an AI system follows a sequence of steps: input, process, output.
- Investigate a worked example and explain the reasoning behind the result to a partner.
- Use the input-process-output vocabulary accurately and connect it to building or evaluating a simple AI solution.

Student-facing success criteria:

- I can name the three stages of an AI workflow - input, process and output - in order.
- I can trace a worked example and explain to a partner how the input became the output.
- I can use the words input, process and output correctly when I talk about a solution.

Prior knowledge & Prerequisite Primer (10 min)

Assumes: In Grade 6 you followed step-by-step instructions (algorithms) and sorted information into groups. You know a computer does exactly what its steps tell it to do, in order.

Activate: Show a familiar machine on the board - a toaster or a vending machine - and ask: what goes in, what happens inside, and what comes out?

Teach: Name the three stages out loud - input (what goes in), process (what happens inside), output (what comes out) - and label the machine on the board. Say explicitly where the analogy breaks: these everyday machines follow fixed rules a person wrote, while an AI model has learned its patterns from examples - so use the machine only for the input-process-output shape, not to explain how AI 'decides'.

Apply: In pairs, label a second everyday machine (a kettle or a pencil sharpener) with input, process and output on a mini whiteboard.

Check: Cold-call two pairs to read their three labels aloud; the rung is secure when a pair can point to each stage in the correct order.

Critical & analytical thinking focus

Analyse - reason about how the system/workflow produces outcomes (Understand/Analyse | DOK 2-3 | SOLO: Relational)

The technical concept (teacher background)

An AI system is a pipeline. It takes an input (data such as text, an image or a number), runs a process (a trained model or set of rules that transforms that input), and returns an output (a prediction, a label or generated content). The process stage is where a machine-learning model applies patterns it learned from many past examples; it does not think or understand, it maps inputs to likely outputs. The same three-stage shape describes almost every AI tool students meet, from a photo filter to a translation app. Making the stages visible helps learners reason about where an error can enter a workflow - a poor input, a limited process or a misread output - which is the foundation for evaluating and improving AI solutions later in the unit.

The same shape in an everyday machine.

Lesson plan

Phase | time Teacher (I do / we do) Students Key questions (targeted thinking)

Hook / 6 min Show a toaster and a translation app side by side; think aloud that both take something in, change it, and give something back.

Watch and call out what goes in and what comes out of each.

What is the same about a toaster and a translation app?

Direct teach / 12 min

Name and draw the three stages - input, process, output - as a labelled flow diagram; we-do one worked example (a translation app) together. Copy the flow diagram and help label the worked example.

Which stage is the part we cannot see inside?

Guided investigate / 12 min

Hand out three pre-generated worked examples; model tracing the first, then release pairs to map input and output on the rest.

In pairs, trace each example

and write what the input and output are.

How did this input turn into that output?

Pair explain / 10

min

Prompt partners to explain one example using cause-and-effect words (because, so, then); circulate and listen for the link.

Take turns explaining to a partner how the input became the output, in order.

Can you explain it without the words input, process or output being wrong?

Check and close

/ 5 min

Collect the exit ticket; recap the three-stage shape and preview that tomorrow's Lab will test it by changing inputs.

Complete the exit ticket naming the three stages and one example input.

Name the three stages in order - go.

Anticipated misconceptions

Misconception How to address it

The AI 'thinks' or 'knows' the answer in the process stage. Explain that the process applies patterns learned from past

examples; it maps an input to a likely output without understanding. Re-voice it as 'it matches, it does not know'.

Input and output are the same thing, or the order does not matter.

Trace one item slowly on the board so students see the input change into a different output; stress the arrows show a fixed order.

The output is always correct because the computer produced it.

Show a pre-generated wrong output next to a correct one and model verify-before-trust: a confident output can still be wrong, so check it against the task.

Hands-on activity guide

Overview: Students investigate three teacher-provided worked examples, map how each input becomes an output, and explain the reasoning to a partner.

Grouping / timing: Pairs for the investigate and explain; whole class for the debrief.

Tool: A teacher-operated board demo with a curated, pre-generated set of example outputs (no accounts, no live typing by students) - so the class studies safe, prepared results and no personal data is entered.

Facilitation steps:

1. Look at the worked example your teacher shows on the board.
2. Find what goes in and write it next to Input.
3. Describe what happens inside and write it next to Process.
4. Find what comes out and write it next to Output.
5. Tell your partner how the input became the output, in order.
6. Check one output against the task - does it actually answer it?

Unplugged fallback: Simulated-AI cards: a 'process' envelope holds a fixed rule (for example, match a word to a category).

A student posts an input card in, the envelope returns the matching output card - the same three-stage shape with no device at all.

Formative assessment (aligned to the objectives)

Task / item Model response Marking note

Label a new worked example (a translation app) with input, process and output.

Input: the English sentence typed in. Process: the model converts it using patterns learned from many translations. Output: the Arabic sentence shown.

Award full marks when all three stages are correct and in order; naming only input and output is partial.

Explain to a partner how the input became the output for the board example.

The input went into the process, which used learned patterns to change it, so the output appeared; without the input there would be no output.

Listen for cause-and-effect language linking the stages, not just the three names.

Use input, process and output correctly in one sentence about a solution idea.

In my step-counter idea the input is my movement, the process counts the steps, and the output is the daily total.

Vocabulary used accurately and in order; gently correct any stage that is swapped.

- Targets: the Emerging outcome - describes how an AI system follows a sequence of steps (input, process, output).
- Check: partner explanation plus a labelled exit ticket, sampled by cold-call.
- Success criteria: the student names all three stages in order and explains how the input became the output.
- Evidence: the completed trace table and the one-sentence explanation on the worksheet.
- Feedback / adjustment: if a stage is swapped, re-run the board example and ask the student to point to each stage before writing again.

Differentiation

Criterion Emerging (expected) Approaching Not yet
Names the stages Names input, process and output in the correct order

Names the stages but muddles the order

Names only one or two stages

Explains the sequence Explains clearly how the input became the output

Describes the stages but the link is vague

Lists stages with no link between them

Uses vocabulary Uses input, process and output accurately and unprompted

Uses the words with prompting

Avoids or misuses the words

Representation: pair the flow diagram with the everyday-machine analogy so the concept is available visually, verbally and through a physical object.

Action and expression: let students show understanding by labelling, speaking to a partner, or arranging physical cards - not only by writing.

People of Determination: provide a pre-labelled stage strip and sentence starters ('The input is ...'); allow extra time and a calm partner for the explain step.

Week 2 - Lab | Understanding the System - Input, Process, Output (Lab) |
Interactive lesson: 'Build, Trace and Test a Workflow'

Builds on the Foundation lesson 'How Models Learn': students now train a classifier - recall training vs inference and that the model stores patterns, not the examples.

Strand: AI Conceptual Understanding | Framework ref(s): AIF·CU·G7·E | 1 x 45 min

Objectives & success criteria

- Build and trace a simple three-stage input-process-output workflow on a hands-on task.
- Compare how the same workflow behaves when the input is changed, and explain why the output changed.
- Use input-process-output vocabulary to describe where an error could enter the workflow and how you would check it.

Student-facing success criteria:

- I can build a workflow with an input, a process and an output, and trace one item through it.
- I can change the input, compare the two outputs, and explain why the output changed.
- I can point to where an error could enter the workflow and say how I would check the output.

Prior knowledge & Prerequisite Primer (10 min)

Assumes: In the Core period you named the three stages of an AI workflow - input, process and output - and traced one worked example.

Activate: Replay the Core flow diagram and ask one pair to narrate it in ten seconds.

Teach: Show that changing the input can change the output while the process stays the same, using two prepared examples on the board.

Apply: In pairs, predict what the output will be for a third input before it is revealed.

Check: Thumbs up or down - did your prediction match? Ask one pair whose prediction missed to say why.

Critical & analytical thinking focus

Analyse - reason about how the system/workflow produces outcomes (Understand/Analyse | DOK 2-3 | SOLO: Relational)

The technical concept (teacher background)

Tracing a workflow means following one item of data through each stage and naming what happens to it: what went in, what

the process did, and what came out. Comparing means running different inputs through the same process and watching

how the output changes. This makes the process stage less of a black box - learners see that a model responds to its input,

and that a small change in input can produce a large change in output. It also shows where things can go wrong: an unclear

input, a process trained on narrow data, or an output read without checking. In this Lab students need no accounts and no

live tools; a simulated-AI card routine or a teacher-run prepared demo reproduces the same behaviour safely, so every

learner can trace and compare, with verify-before-trust built into the routine.

Lesson plan

Phase | time Teacher (I do / we do) Students Key questions (targeted thinking)

Recap and set-

up / 5 min

Replay the Core flow diagram; hand each pair a simulated-AI process card/envelope and a small set of input cards.

Recall the three stages and lay out their input, process and output spaces.

What are the three stages, in order?

Model the trace /

10 min

Think aloud tracing one input card through the process to its output; write each stage under the diagram.

Watch and copy the trace for the demonstrated item.

Which single item are we following all the way through?

Build and trace /

12 min

Release pairs to send their own item through and record each stage; circulate and check the trace is consistent.

In pairs, send one item through your workflow and record input, process and output.

Did the same item stay the same all the way through, or did it change?

Change input and compare /

12 min

Prompt pairs to swap in a different input, keep the process the same, and compare the two outputs; model verify-before-trust on one output.

Change just the input, run it again, and compare the two

outputs; check one against a trusted answer.

The process did not change - so why did the output change?

Share and check

/ 6 min

Take two comparisons to the board; collect the exit ticket and name where an error could enter a workflow.

Share your comparison and complete the exit ticket.

Where in your workflow could an error sneak in?

Anticipated misconceptions

Misconception How to address it

If the output changed, the process must have changed too. Show that only the input was swapped while the process card stayed identical; the output changes because the process responds to its input.

A different input should give the same output if the tool is 'good'.

Run two clearly different inputs and compare; explain that responding to the input is exactly what a working process is meant to do.

Once the workflow runs, the output can be trusted without checking.

Model verify-before-trust: hold one output against a trusted answer; a confident output can still be wrong, so every output is checked.

The model keeps learning from students as they use it.

Teachable Machine learns only while you click Train; after that, testing with new inputs does NOT change it.

Show this: test the trained model on several inputs and confirm the model does not update. Inference (using the model) is separate from training (teaching it).

Hands-on activity guide

Overview: Students build a small three-stage workflow, trace one item through it, then change the input and compare how the output behaves - ending with a verify step.

Grouping / timing: Pairs for the build, trace and compare; whole class for the board share.

Tool: A simulated-AI card deck (a fixed 'process' rule in an envelope) or a teacher-run prepared example set - fully account-free and device-free, so students test behaviour without entering any personal data.

Facilitation steps:

1. Set up your three spaces: input, process and output.
2. Send one item through and write what happens at each stage.
3. Change just the input and send it through again.
4. Compare the two outputs - write what changed.
5. Explain why the output changed even though the process stayed the same.
6. Check one output against a trusted answer before you trust it.

Unplugged fallback: The card deck is already the unplugged version: no screens are required. If a live approved demo is

used instead, the same card routine is the fallback so every pair can still trace and compare.

Lab tool integration (practical, hands-on)

Tool: Google Teachable Machine (teacher-operated)

Teacher setup: Open Teachable Machine (Image Project) on the board. Prepare three classes of NON-personal items in

advance, e.g. three hand gestures or three classroom objects.

What students do:

1. As a class, name the three classes (for example: open hand, fist, thumbs-up).
2. Students direct the teacher to add ~15 example images per class using the webcam pointed at objects or hands only. Aim for a similar number of examples per class, and do not reuse the same example twice.
3. Click Train. Narrate: these examples are the input, training is the process.
4. Test with a new, unseen example and read the confidence output aloud.
5. Trace the full input -> process -> output path on the board and label each stage.

Students produce: A trained 3-class model and a completed input-process-output trace.

Data-safety: Teachable Machine trains in the browser and the samples stay on the teacher's device. Use objects, hand

gestures or drawings only - never faces, voices or names. Clear the project at the end of the lesson.

Unplugged fallback: Run the card-deck 'human classifier': agree a rule from example cards, then test it on new cards.

Teachable Machine: three example classes feed training, which produces a confidence output.

Formative assessment (aligned to the objectives)

Task / item Model response Marking note

Trace one item of data through your built workflow and record each stage.

Input: the word happy. Process: the deck matches the word to a category. Output: the label positive.

Full marks when the same item is followed through all three stages consistently.

Change one input and record how the output changes and why.

When I changed happy to terrible, the output

changed from positive to negative because the process responds to the input.

Look for a correct comparison and a reason tied to the input, not a random guess.

Name one place an error could enter the workflow and how you would check it.

An unclear input could give a wrong output, so I would check the output against a trusted answer before trusting it.

Accept any stage named with a sensible verify-before-trust check.

- Targets: the same Emerging outcome, applied by tracing and comparing how the workflow behaves.
- Check: pair prediction plus a change-the-input comparison, observed during the build.
- Success criteria: the student traces one item through all three stages and explains why a changed input changed the output.
- Evidence: the build-and-trace table and the compare-the-output table on the worksheet.
- Feedback / adjustment: if the comparison is a guess, replay both inputs side by side and ask the student to name what the process did with each.

Differentiation

Criterion Emerging (expected) Approaching Not yet

Builds and traces Builds a three-stage workflow and traces one item through it

Builds the workflow but the trace skips a stage

Needs support to assemble the three stages

Compares behaviour Explains why a changed input changed the output

Notices the output changed but the reason is unclear

Does not connect the change to the input

Reasons about risk Names where an error could enter and a check for it

Names a risk but no check Does not identify a risk

Representation: show a compare panel (same process, two inputs, two outputs) so the cause of a changed output is visible, not only described.

Action and expression: accept a spoken trace, a card arrangement, or a written table as evidence of the same reasoning.

People of Determination: pre-build the process stage so the student focuses on changing inputs; provide a comparison frame ('When I changed ... the output ... because ...').

Week 3 - Core | Designing & Building a Workflow | Interactive lesson: 'Building a Workflow'

Real-World Connection (UAE): Abu Dhabi's smart traffic and transport systems take inputs (sensors and cameras), run a process (predicting flow) and produce outputs (signal timings and route advice) - a real input -> process -> output workflow.

Strand: AI Solution Design & Development | Framework ref(s): AIF·SD·G7·E | 1 x 45 min

Objectives & success criteria

- Build a simple AI solution by following guided input-process-output steps.
- Apply the workflow in a hands-on task and explain the reasoning behind the result.
- Use the key vocabulary of AI solution design accurately and connect it to building or evaluating a solution.

Student-facing success criteria:

- I can build a simple AI workflow by following guided steps for input, process and output.
- I can complete a hands-on task and explain why my workflow gave the result it did.
- I can use words like input, process, output and workflow correctly when I talk about my solution.

Prior knowledge & Prerequisite Primer (10 min)

Assumes: In earlier lessons this term you learned that an AI workflow has three parts - an input you give, a process the AI does, and an output you get back.

Activate: Show a familiar routine on the board - a vending machine: coin and choice in (input), the machine works (process), a snack out (output). Ask students to name the three parts. Remind them where the analogy breaks: the vending machine uses fixed

rules, an AI model uses learned patterns - the analogy shows the shape, not the intelligence.

Teach: Re-establish that a workflow is an ordered set of steps and that in an AI workflow one step is done by an AI. Draw a three-

box strip - Input, Process, Output - and label which box is the AI step.

Apply: In pairs, students turn one everyday task, such as sorting the class library, into a three-box input-process-output strip on a mini-whiteboard.

Check: Cold-call two pairs to read their three boxes aloud; the rung is secure if they place each item in the correct box and can point to where an AI step could sit.

Critical & analytical thinking focus

Apply & create - design and build a structured solution/workflow (Apply/Create | DOK 2-3 | SOLO: Relational)

The technical concept (teacher background)

A workflow is an ordered sequence of steps that turns an input into a useful output. In an AI solution at least one step is

carried out by a model - for example classifying, sorting, summarising or generating. Building a simple solution at this level means composing steps in the right order, not training a model from scratch: the student decides what goes in, uses or is given the AI step, and defines what a good output looks like. Guided steps means the design is scaffolded by a template or a worked routine, with any tool use teacher-mediated. The skill being built is decomposition and sequencing - breaking a task into input, process and output, then checking that each step feeds the next so the whole workflow completes the task reliably.

Verify before you trust: check every output against the task and a trusted source.

Lesson plan

Phase | time Teacher (I do / we do) Students Key questions (targeted thinking)

Hook / 6 min Show a two-step board demo: type a short weather note as input, and the teacher-operated approved tool returns a tidy packing list as output. Ask the class to name the hidden process step. Watch and call out which part is input, process and output.

- Which single step here is done by the AI, and how do you know?

Model the build /

10 min

We-do: on the board build a three-box workflow strip for a real class task, such as turning six messy notes into a sorted to-do list. Think aloud while placing input, the AI process step and the output, and mark the check point.

Copy the strip and annotate each box with what enters and what leaves it.

- What must the input look like so the process step can work?

Guided build / 14

min

Hand out the workflow template.

Circulate as pairs build their own three-step solution using the simulated-AI card routine of pre-generated output cards, account-free and with no

personal data.

In pairs, choose a task, fill in
input, process and output,
run it with the cards, and
record the output.

- Where could your workflow break
if the input changed?

Test and verify /

9 min

Prompt pairs to run their workflow twice
and verify each output against the task
and a trusted reference. Model the
verify-before-trust check on the board.

Test their workflow, compare
the two outputs, and tick or
fix each box.

- Your output looks confident -
what evidence tells you it is
actually correct?

Check and close

/ 6 min

Exit check: each pair states their three
steps and one thing they would
improve. Collect the templates as
evidence.

Say their input-process-
output aloud and name one
improvement.

- If a classmate followed your three
steps, would they get the same
output? Why?

Anticipated misconceptions

Misconception How to address it

The AI does the whole thing, so I do not really build
anything.

Show that the student designs the input and defines the output,
and only the middle step is the model; without the design the AI
step has nothing useful to do.

If the output sounds confident it must be right. Run one demo output that is fluent but wrong,
then practise the

verify-before-trust check against the task and a trusted source.

A workflow can be in any order as long as the steps are
there.

Swap two boxes on the board so the output feeds the input;

students see the workflow break and re-establish that order matters because each step feeds the next.

Hands-on activity guide

Overview: Pairs build and test a small three-step input-process-output workflow that completes one real class task, using

a card-based simulated AI so no accounts or personal data are needed.

Grouping / timing: Pairs for the build and test, whole class for the debrief.

Tool: A simulated-AI card deck of pre-generated output cards prepared by the teacher stands in for the model step -

account-free, no personal data, and it works even with no devices. If a device is used it is the school-approved tool,

teacher-operated on the board only.

Facilitation steps:

1. Choose one task from the task menu, for example sort six book titles into fiction and non-fiction.
2. Write your input in the first box - exactly what you will give the workflow.
3. Place the AI process card that matches your task in the middle box; this is the model step.
4. Draw the output the cards give you in the third box.
5. Run it again with a slightly different input and compare the two outputs.
6. Verify each output against the task and the trusted answer key, then tick or fix each box.

Unplugged fallback: The whole activity is already unplugged using paper cards. For a fully human version, one student

acts as the process and silently sorts the inputs by a rule the pair wrote, while the others check the output.

Formative assessment (aligned to the objectives)

Task / item Model response Marking note

Label a given three-box strip with input, process and output, and circle the AI step.

Box 1 input (the notes), Box 2 process (the AI sorts them) circled, Box 3 output (the sorted list).

Award full marks only if all three labels are correct and the circled AI step is the middle process box.

Write the input your workflow needs to sort class library books, and say why.

Input: a list of book titles with their type, because the process step can only sort what it is given, so the input must include the type.

Look for a clear link between input quality and a working output; reasoning must be present, not just a label.

Your workflow gives a confident but wrong output. State one thing you would do before trusting it.

Check the output against the task and a trusted source, such as the answer key, before using it.

Accept any answer that names a verify-before-trust check; reject 'trust it because it sounds right'.

- Targets: Builds a simple AI solution using guided input-process-output steps and explains the result (AIF·SD·G7·E).

- Check: Pairs run their workflow twice, then complete a two-question exit check on order and verification.

- Success criteria: A correctly sequenced three-box workflow that completes the task, with each box feeding the next and the AI step identified.

- Evidence: Completed workflow templates and exit tickets collected from every pair.

- Feedback / adjustment: If order or the AI step is wrong, re-model with a swapped-box demo next period; if secure, move pairs on to the multi-step challenge.

Differentiation

Criterion Emerging (expected) Approaching Not yet

Sequences the workflow Places input, process and output in the correct order so the task

Orders most steps but one box is out of place

Boxes are unordered or the AI step is missing completes

Explains the result Explains why the workflow gave its output, linking input to output

Describes the output but the reasoning is partial

Cannot say why the output appeared

Uses key vocabulary Uses input, process, output and workflow accurately

Uses one or two terms correctly

Terms are missing or muddled

Representation: Provide the three-box template both as a diagram and as sentence stems such as 'My input is ...', and leave a worked example strip on the desk for reference.

Action & expression: Let pairs show their workflow as drawn boxes, spoken steps, or physically arranged cards - whichever lets them build and explain it.

People of Determination: Pre-cut and colour-code the cards, reduce the task menu to two clear options, pair with a peer scribe, and allow extra time to test and verify.

Week 3 - Lab | Designing & Building a Workflow (Lab) | Interactive lesson: 'Test, Fix and Chain'

Strand: AI Solution Design & Development | Framework ref(s): AIF·SD·G7·E | 1 x 45 min

Objectives & success criteria

- Construct and test a working multi-step input-process-output workflow that completes a chosen task.
- Diagnose where a workflow fails and iterate to improve the output, explaining each change.
- Use AI solution-design vocabulary precisely to justify the order and reliability of the build.

Student-facing success criteria:

- I can build a workflow and test it until it completes my task.
- I can find where my workflow goes wrong and change it to make the output better.
- I can use design words to explain why my steps are in that order and why the workflow is reliable.

Prior knowledge & Prerequisite Primer (10 min)

Assumes: In the Core period you built a three-box input-process-output workflow and ran the verify-before-trust check once.

Activate: Show two versions of the same workflow output on the board - one that completes the task and one that fails - and ask which build is more reliable and how they can tell.

Teach: Re-establish the test-and-iterate loop: run, spot the weak box, change one thing, run again. Stress that improvement comes from changing one box at a time so you know what helped.

Apply: In pairs, students look at a broken three-box strip and say which single box they would change first and why.

Check: The rung is secure when pairs can point to the failing box, usually the input, and name one specific change rather than saying 'redo it all'.

Critical & analytical thinking focus

Apply & create - design and build a structured solution/workflow (Apply/Create | DOK 2-3 | SOLO: Relational)

The technical concept (teacher background)

Building a reliable AI solution is iterative. A first workflow rarely works perfectly, so the builder tests it, isolates the step that weakens the output, changes one variable, and re-tests - a controlled loop rather than random tinkering. Extending to a multi-step workflow adds a second process step whose input is the output of the first; this is chaining, and it only works if

each stage produces output the next stage can use. Testing therefore checks two things: does each stage complete, and does the hand-off between stages hold. Reliability is demonstrated, not assumed - the builder shows the workflow gives a correct output across more than one input, and verifies each output against the task and a trusted source before trusting the whole chain.

Lesson plan

Phase | time Teacher (I do / we do) Students Key questions (targeted thinking)

Recap and set

the build / 5 min

Recap the input-process-output strip from Core and set today's goal: make it work reliably and add a second step. Show the target multi-step example on the board.

Recall their three boxes and state today's build goal.

- What does 'reliable' mean for a workflow, in one sentence?

Test to find the

weak box / 10

min

Model running a workflow, reading the output against the task, and pinpointing the one box that weakens it. Think aloud about changing only one thing.

Run their Core workflow and mark the box that needs work.

- Which single box is letting your output down, and how do you know?

Iterate and

improve / 12 min

Circulate as pairs change one box, re-run with the card deck, and log whether the output improved. Enforce one change at a time.

Change one box, test again, and record the before and after output.

- Did your change help - what is your evidence?

Extend to multi-
step / 12 min

Introduce a second process card so the
output of step one becomes the input of

Add a second process step,
run the full chain, and check

- Where could the join between
your two steps break, and how
step two. Model the hand-off and where
it can break.

the hand-off holds. would you spot it?

Test, verify, close

/ 6 min

Pairs run the full multi-step workflow
across two different inputs and verify
each output; collect the logs as
evidence and share one improvement.

Run the chain twice, verify
each output, and name the
change that helped most.

- Has your workflow completed the
task for more than one input - what
proves it?

Anticipated misconceptions

Misconception How to address it

To fix a broken workflow I should change everything at
once.

Show that changing one box at a time is the only way to know what
helped; demo two changes together where you cannot tell which
one worked.

Adding a second step just means doing the task twice. Show chaining - the output of step one is
the input of step two - so

the second step does something new with the first result, not a
repeat of it.

If it worked once, the workflow is reliable. Run the same workflow on a different input that
breaks it;

establish that reliability means it completes the task across more
than one input.

Hands-on activity guide

Overview: Pairs take their Core workflow, test it to find the weak step, iterate one change at a
time, then extend it into a
working multi-step chained workflow and prove it is reliable.

Grouping / timing: Pairs for building and testing, groups of four for the improvement swap, whole
class for the close.

Tool: The same simulated-AI card deck plus a second set of process cards for the chained step - account-free, no personal data, and it works with no devices. Any live tool stays school-approved and teacher-operated on the board only.

Facilitation steps:

1. Run your Core workflow and read the output against the task.
2. Mark the one box that weakens the output - usually the input.
3. Change only that box, run again, and log the output before and after.
4. Add a second process card so your output becomes the input of a new step.
5. Run the whole chain twice, using two different inputs.
6. Verify each output against the answer key and record which change made the biggest difference.

Unplugged fallback: The activity is already card-based. For a fully human version, one student is process A and another is

process B; the pair passes A's written output to B as B's input, showing the hand-off by hand.

Lab tool integration (practical, hands-on)

Tool: Scratch (input -> process -> output build)

Teacher setup: Open Scratch (offline app or site). Provide a starter project with an input block, an empty process area and an output block.

What students do:

1. Decide a simple task, e.g. sort a typed word into 'question' or 'statement'.
2. Drag an input block (ask and wait) to capture the input.
3. Build the process: two or three 'if' blocks that apply your rule.
4. Add an output block (say) that reports the result.
5. Run it three times, including one tricky input, and note what happened.

Students produce: A working three-stage Scratch project and a short run log.

Data-safety: No personal data in the project; type only neutral example words.

Unplugged fallback: Arrange printed 'block cards' on the desk into the same input -> process -> output sequence.

A Scratch block stack arranged as input, then process (if-blocks), then output.

Formative assessment (aligned to the objectives)

Task / item Model response Marking note

Show a before-and-after output after you changed one box, and name the box.

Before: an unsorted list; After: a sorted list;
box changed: the input, by adding the book type.

Full marks only if exactly one box is named and the two outputs clearly differ because of it.

Draw your multi-step workflow and mark where the output of step one becomes the input of step two.

Two process boxes with an arrow between them labelled 'output of A = input of B'.

Look for a correct hand-off arrow; the join must be labelled, not just implied.

Your workflow worked on one input but failed on another. What does this tell you and what will you do?

It is not yet reliable; I will fix the box that failed and test on both inputs again.

Accept answers that treat one success as not enough and name a re-test; reject 'it works, leave it'.

- Targets: Builds and iterates a multi-step AI workflow using guided steps and explains each change (AIF·SD·G7·E; stretch toward Proficient).
- Check: Test-and-iterate logs plus a run of the full chain across two different inputs.
- Success criteria: A working multi-step workflow whose hand-off holds and that completes the task for more than one input, with each change justified.
- Evidence: Iteration logs and the final workflow diagram collected from every pair.
- Feedback / adjustment: If the hand-off fails, re-model chaining with a single labelled arrow; if secure, ask pairs to add a verify step inside the chain.

Differentiation

Criterion Emerging (expected) Approaching Not yet
Tests and iterates Finds the weak box and improves the output by changing one thing at a time

Improves the output but changes more than one box at once

Cannot locate the weak box or does not re-test

Builds a multi-step workflow Chains a second step so the output of one becomes the input of the next and the task completes

Adds a second step but the hand-off is unclear

Only a single step, or the steps do not connect

Justifies reliability Shows the workflow completes the task for more than one input and verifies each output

Tests once and claims it works

Makes no reliability claim or does not verify

Representation: Provide an iteration log with pre-drawn before and after columns, and a chaining diagram with the hand-off arrow already labelled for pairs who need the scaffold.

Action & expression: Accept the iteration story as a labelled diagram, a filled-in log, or a spoken before-and-after account.

People of Determination: Provide the second process cards pre-sorted, cap the chain at two steps, use a colour-coded arrow for the hand-off, and allow extra testing time with a peer scribe for the log.

Week 4 - Core | Judging & Improving Outputs | Interactive lesson: 'Judging Outputs'

Strand: Critical Evaluation & Informed Interaction | Framework ref(s): AIF·CE·G7·E | 1 x 45 min
Objectives & success criteria

- Identify issues in a set of AI outputs or workflow steps, judging each against agreed criteria.
- Apply the criteria in a hands-on evaluation task and justify each judgement with clear evidence.
- Use the key vocabulary of critical evaluation accurately, and connect judging an output to improving a solution.

Student-facing success criteria:

- I can spot where an AI output or a workflow step has gone wrong.
- I can judge an output against agreed criteria and back my decision with evidence.
- I can use words like criteria, evidence and accuracy correctly when I explain my judgement.

Applied numeracy (Grade 7 depth). Make 'accuracy' a number students compute, not just a word. Accuracy =

$\text{number correct} \div \text{total tested}$, written as a percentage. Worked example: a classifier is tested on 12 items and

gets 9 right, so accuracy = $9 \div 12 = 0.75 = 75\%$. Have pairs compute the accuracy of two sample output sets and

rank them by the number. Then read the confidence score the model reports with each answer (0 to 1, e.g. 0.90

= very sure, 0.55 = unsure) and check that a high confidence score is not a guarantee of a correct answer - so

students still verify. Unplugged fallback: use the printed evaluation pack and a tally sheet to count correct $\div$

total by hand.

Prior knowledge & Prerequisite Primer (10 min)

Assumes: In Grade 6 you learned that an AI workflow has an input, a process and an output, and that outputs are not

always right. You have described what a good result looks like for a simple task.

Activate: Show two versions of the same AI output side by side on the board, one clearly weaker. Ask: which one is better, and how

do you know?

Teach: Introduce criteria - the agreed things we check for. Model judging one output aloud: point to the criterion, then point to the evidence in the output.

Apply: In pairs, students look at a short curated output and name one thing that is good and one thing that could be an issue.

Check: Cold-call three pairs for one issue each. A rung is secure when students name an issue and link it to a criterion, not just say I do not like it.

Critical & analytical thinking focus

Evaluate - judge outputs and solutions against criteria, with evidence (Evaluate | DOK 2-3 | SOLO: Relational)

The technical concept (teacher background)

Evaluation means judging an output against explicit criteria rather than by gut feeling. For an AI workflow, useful criteria

include accuracy (is it factually correct?), relevance (does it answer the task?) and clarity. A model produces an output by

predicting what is likely, not by knowing what is true, so a fluent, confident answer can still contain errors - this is why verify-

before-trust matters. Issues can arise at any stage: a poor input or prompt, a flawed process step, or an output that drifts

from the task. Locating where the issue sits in the input-process-output chain is as important as noticing it. Good

evaluators cite evidence - the specific word, fact or step that fails the criterion - so judgements can be checked and

improved by someone else.

Lesson plan

Phase | time Teacher (I do / we do) Students Key questions (targeted thinking)

Hook / 6 min Show two curated outputs for the same task on the board, one with an error.

Think aloud: I will not guess which is better - I will agree what to check first.

Vote on which output is better, then note that a vote is not evidence.

- What would we need to agree on before we can fairly judge these?

Direct teaching /

10 min

Introduce three agreed criteria:

accuracy, relevance and clarity. Model

Copy the three criteria and

follow the modelled

- Which criterion does this error

break, and what is the evidence?
judging one output - name the criterion,
point to the evidence, give a verdict.

Stress verify-before-trust against a
trusted source.

judgement, marking where
the evidence sits.

Guided practice /

8 min

Hand out a curated output set (teacher
pre-generated, account-free). Walk one
judgement with the class using an
evaluation grid.

In pairs, complete one row of
the grid together: criterion,
verdict, evidence.

- Is that a fault in the output, or in
the input or process step before it?

Independent

hands-on / 14

min

Circulate as pairs judge the remaining
curated outputs against the criteria.

Prompt for evidence, not opinion.

Judge each output against
every criterion and write one
evidence note per

judgement.

- Point to the exact word or step
that made you decide that.

Check & close / 7

min

Collect three verdicts on the same
output and compare evidence. Confirm
the success criteria and preview the
Lab ranking task.

Share one judgement with its
evidence and self-check
against the I can statements.

- Two pairs disagree on the same
output - whose evidence is
stronger, and why?

Anticipated misconceptions

Misconception How to address it

If the AI sounds confident and the writing is neat, the answer must be correct.

Show a fluent output containing a clear factual error. Contrast fluency with accuracy and rehearse verify-before-trust against a trusted source.

Judging is just saying whether I like it or not. Insist every verdict names a criterion and points to evidence.

Reject I do not like it and re-ask: which criterion, and what evidence?

If an output is wrong, the AI itself must be broken. Trace the input-process-output chain together so students see an

issue can start with a weak input or process step, not only the model.

Hands-on activity guide

Overview: Pairs act as an evaluation panel, judging a curated set of AI outputs for one task against three agreed criteria and recording evidence for every verdict.

Grouping / timing: Pairs for the judging; whole class for the agree-the-criteria opener and the evidence debrief.

Tool: A teacher pre-generated output set on paper or the board (account-free, no student logins). Outputs are produced in advance by the teacher on a school-approved, child-safe tool, so no personal data is entered and no student uses a live model.

Facilitation steps:

1. As a class, agree the three criteria you will check: accuracy, relevance and clarity.
2. Read the task the outputs were meant to complete, so you know what good looks like.
3. For each curated output, judge it against every criterion and write a verdict: meets, partly or fails.
4. For every verdict, write one piece of evidence - the exact word, fact or step that made you decide.
5. Check one output against a trusted source to test whether a confident answer is actually accurate.
6. Flag whether each issue sits in the input, the process step or the output, then share your strongest judgement with the class.

Unplugged fallback: If no tool or board display is available, use the printed card set: each card is a pre-written output for the same task, and pairs sort the cards into meets, partly and fails piles with an evidence sticky note on each.

Formative assessment (aligned to the objectives)

Task / item Model response Marking note

Given one curated output and the three criteria, write a verdict for each criterion with evidence.

Accuracy: fails - it says the Burj Khalifa has 100 floors, but the trusted source says 163.

Relevance: meets - it answers the task.

Clarity: meets - it is easy to read.

Award the mark when every verdict names the criterion and cites specific evidence, not opinion.

Point to where an issue in a workflow The issue began at the input - the prompt did Look for the student locating the

output began: input, process or output. not say which city, so the output answered about the wrong place.

issue in the chain, not just naming that the output is wrong.

Explain why a confident, fluent output can still be wrong.

A model predicts likely words, so it can sound sure while stating a fact that is false.

That is why we verify before we trust.

Credit any answer that separates fluency or confidence from accuracy and mentions checking against a source.

- Targets: judging AI outputs and workflow steps against agreed criteria, evidenced (AIF·CE·G7·E - Identifies issues in outputs or workflow steps).

- Check: evidence-tagged evaluation grid completed in pairs, plus cold-call verdicts in the close.

- Success criteria: student identifies each issue, names the criterion it breaks, and cites specific evidence.

- Evidence: the completed evaluation grids and one written verdict-with-evidence per pair collected at the door.

- Feedback / adjustment: if verdicts lack evidence, re-model point to the exact word or step next lesson before ranking begins.

Differentiation

Criterion Emerging (expected) Approaching Not yet

Identifying issues Names a clear issue in an output or workflow step

Names an issue but is vague about where it sits

Cannot yet spot an issue without prompting

Using criteria Judges against all three agreed criteria

Uses one or two criteria Judges by opinion, not

criteria

Evidence Cites specific evidence for each

verdict

Gives evidence for some

verdicts

Gives verdicts without

evidence

Representation: provide the criteria as an icon-labelled checklist and read the task aloud.

Pre-teach accuracy, relevance and clarity with one worked example each.

Action & expression: let students record verdicts by ticking a grid, annotating a printed output, or explaining orally to the teacher rather than writing full sentences.

People of Determination: enlarged high-contrast output cards, sentence starters (This fails ... because ...), extra time, and a partner who scribes evidence for a student who reasons aloud.

Week 4 - Lab | Judging & Improving Outputs (Lab) | Interactive lesson: 'The Evaluation Clinic'

Strand: Critical Evaluation & Informed Interaction | Framework ref(s): AIF·CE·G7·E | 1 x 45 min

Objectives & success criteria

- Apply the agreed criteria to a wider set of AI outputs, identifying issues in the outputs and the workflow steps behind them.

- Rank the outputs from best to worst and justify each placement with evidence.

- Weigh accuracy and usefulness together to argue which output best serves a task, and propose one improvement to the weakest.

Student-facing success criteria:

- I can apply agreed criteria to a larger, messier set of outputs.

- I can rank outputs from best to worst and justify each placement with evidence.

- I can weigh accuracy against usefulness and suggest one way to improve a weak output.

Prior knowledge & Prerequisite Primer (10 min)

Assumes: In the Core period you judged a small curated set of outputs against agreed criteria and backed each verdict with evidence.

Activate: Show the Core criteria and one ranked pair from last period. Ask a pair to explain why one output beat the other.

Teach: Model turning separate verdicts into a ranking: when two outputs both partly meet the criteria, use the evidence to decide which is stronger, then add usefulness as a tie-breaker.

Apply: In pairs, students re-rank two outputs they judged last period and say what evidence moved one above the other.

Check: Listen for a ranking that rests on evidence and a named criterion, not preference. A rung is secure when the tie-breaker is

justified.

Critical & analytical thinking focus

Evaluate - judge outputs and solutions against criteria, with evidence (Evaluate | DOK 2-3 | SOLO: Relational)

The technical concept (teacher background)

Ranking is relative evaluation: instead of judging one output in isolation, you compare several and order them. This is harder

because outputs can fail on different criteria - one may be accurate but incomplete, another fluent but wrong - so you must

decide which criteria carry more weight for the task. Accuracy asks whether the content is correct; usefulness asks whether

it actually helps the user complete the task. The two can conflict: a shorter, less detailed answer may be more useful than a

longer one that buries a key fact. An evaluation clinic also traces issues back through the workflow, distinguishing an output

fault from an input or process fault. Justifying each placement with evidence keeps a ranking defensible and repeatable

rather than a matter of taste.

Lesson plan

Phase | time Teacher (I do / we do) Students Key questions (targeted thinking)

Recap / 5 min Post the agreed criteria and one Core

verdict. Frame today: from single

verdicts to ranking a wider set.

Recall the three criteria and

one issue they found last

period.

- When two outputs both partly meet the criteria, what breaks the tie?

Model the

ranking move / 8

min

Demonstrate ordering three outputs:

line up their verdicts, then use evidence

and usefulness to decide the order.

Follow the model and note

the moment usefulness was

used as a tie-breaker.

- Which matters more here - accuracy or usefulness - and why?

Evaluation clinic

/ 16 min

Issue the wider curated output set

(teacher pre-generated, account-free).

Circulate and push for evidence at every placement.

In pairs, judge the wider set, then rank all outputs from best to worst on a ranking strip.

- Point to the evidence that put this one above that one.

Improve one output / 10 min

Ask each pair to take their lowest-ranked output and propose one concrete fix, naming where the issue sat in the workflow.

Write one improvement and say whether it fixes the input, the process step or the output.

- Would your fix change its rank - and can you prove it?

Clinic debrief & close / 6 min

Compare two pairs' rankings of the same set and resolve differences using evidence. Confirm the success criteria.

Defend one placement with evidence and self-check against the I can statements.

- Your rankings differ - whose evidence should the class trust, and why?

Anticipated misconceptions

Misconception How to address it

Ranking is just putting them in the order I prefer. Require each placement to rest on evidence and a criterion. Ask

which piece of evidence moved one output above the next.

The most detailed or longest output is always the best. Contrast a long output that buries a wrong fact with a short

accurate one. Use the task to decide which is more useful.

If an output ranks low, the fix must be to the output itself. Trace the issue back through input and process. Show that

improving a prompt or a step can raise the rank.

Hands-on activity guide

Overview: Pairs run an evaluation clinic: they judge a wider set of curated outputs against the agreed criteria, rank them

from best to worst with evidence, then propose one improvement to the weakest.

Grouping / timing: Pairs for the clinic and ranking; whole class for the tie-breaker debrief.

Tool: A larger teacher pre-generated output set on cards or a handout (account-free, no student logins). All outputs are made in advance by the teacher on a school-approved, child-safe tool, with no personal data entered and no live model used by students.

Facilitation steps:

1. Lay out the wider set of curated outputs and re-read the task they were meant to complete.
2. Judge every output against the agreed criteria - accuracy, relevance and clarity - and note the verdict.
3. Rank the outputs from best to worst on your ranking strip.
4. Beside each placement, write the evidence that put it there; use usefulness to break any ties.
5. Verify the top-ranked output against a trusted source before you confirm its place.
6. Take your lowest-ranked output, name where the issue sits in the workflow, and propose one improvement.

Unplugged fallback: No display needed - use the printed output cards: pairs physically slide the cards into a top-to-bottom order on the desk and clip an evidence note to each, then move one card up after writing its fix.

Lab tool integration (practical, hands-on)

Tool: Pre-generated output packs (evaluation clinic)

Teacher setup: Prepare a pack of 6-8 outputs for the same task at mixed quality (some accurate, some flawed) and print one pack per group.

What students do:

1. Agree the evaluation criteria as a class (accuracy, usefulness, clarity).
2. In groups, read each output and mark it correct / unclear / incorrect with a reason.
3. Rank the pack from strongest to weakest.
4. Choose one weak output and write the one change that would most improve it.
5. Defend your ranking to another group.

Students produce: A scored, ranked output pack with reasons and one improvement.

Data-safety: All outputs are teacher-curated in advance; students touch no live tool.

Unplugged fallback: This is already tool-free.

Two sample outputs side by side with a criteria checklist for judging each.

Formative assessment (aligned to the objectives)

Task / item Model response Marking note

Rank three given outputs best to worst and justify the top and bottom placements with evidence.

Best: Output C - accurate and answers the task directly (evidence: correct 163 floors, on

topic). Worst: Output A - states a wrong figure (evidence: 100 floors, contradicted by the source).

Award when the ranking is ordered and both named placements cite specific evidence.

Two outputs are close. Explain which is more useful for the task and why, even if it is less detailed.

Output B is more useful: it gives the one fact the task needs clearly, while the longer output hides it among errors.

Credit reasoning that weighs usefulness against accuracy for the specific task, not length alone.

For a low-ranked output, name where the issue began and propose one improvement.

The issue began at the input - the prompt did not name the city. Fix: add the city to the prompt so the process has what it needs.

Look for the issue located in the input-process-output chain and a fix that targets that stage.

- Targets: identifying issues across a wider output set and workflow steps by ranking with evidence (AIF·CE·G7·E - Identifies issues in outputs or workflow steps).
- Check: completed ranking strip with an evidence note per placement, plus a proposed improvement for the weakest output.
- Success criteria: student ranks the set, justifies each placement with specific evidence, and locates one issue in the workflow with a fix.
- Evidence: the ranking strips and the improvement notes collected at the close.
- Feedback / adjustment: if rankings rest on preference, re-run the tie-breaker with usefulness modelled before the next evaluation task.

Differentiation

Criterion Emerging (expected) Approaching Not yet

Ranking a wider set Orders the full set best to worst Orders most of the set with some gaps

Cannot yet order beyond a pair

Justifying with evidence Cites evidence for each placement Cites evidence for some

placements

Ranks by preference,

with no evidence

Weighing and improving Uses accuracy and usefulness and

proposes a targeted fix

Uses one lens or a vague fix No fix, or a fix unrelated
to the issue

Representation: provide a physical ranking strip with numbered slots and a criteria reminder card. Model one full ranking before pairs begin.

Action & expression: allow ranking by sliding cards, numbering a list, or explaining the order aloud; evidence may be spoken to the teacher.

People of Determination: fewer, high-contrast cards, sentence starters (I placed this above that because ...), extra time, and a scribe for the improvement note.

Week 5 - Core | Risks, Ethics & Responsible Use | Interactive lesson: 'Spotting the Risks'

Confidence-score checkpoint (2 min): ask 'What is the confidence score here? Is a high confidence the same as accuracy?' Reinforce that confidence is per-input sureness and a model can be confidently wrong (link back to the Foundation lesson).

Strand: Ethics & Governance | Framework ref(s): AIF·GE·G7·E | 1 x 45 min

Objectives & success criteria

- Identify risks in AI-generated outputs, such as factual errors, bias and misleading information.
- Analyse a short scenario for its risks and ethical or governance concerns, then propose a responsible-use measure and explain the reasoning behind it.
- Use the key vocabulary of Ethics and Governance accurately, and connect a named risk to how it would affect building or evaluating an AI workflow.

Student-facing success criteria:

- I can spot risks in an AI output, such as a mistake, an unfair bias, or a made-up fact.
- I can read a scenario, name a risk, and suggest a responsible-use rule with a clear reason.
- I can use words like bias, risk and responsible use correctly, and explain how a risk affects an AI workflow.

Live bias demonstration (teacher-led, 5-8 min). Make bias observable, not abstract. Before the lesson, train a

Teachable Machine image classifier on a deliberately skewed set - for example 15 clear photos of Class A but

only 3 of Class B, all in one lighting. In class: (1) See the skew - test the model live on fresh items and record the

count correct for each class; the rare class scores far lower (e.g. Class B: 1 of 4 correct = 25%).

(2) Name the

cause - students explain that the training data under-represented Class B, so the model learned it poorly; the

bias is in the data, not 'the computer'. (3) Fix it - add balanced examples of Class B, retrain and re-test; the rare-

class score rises (e.g. 3 of 4 = 75%). Students state the before/after accuracy and the one change that improved

fairness. Unplugged fallback: bias-scenario cards showing the same skew as counts, with a 'what to add' fix

card. No personal data - objects, drawings or gestures only.

Prior knowledge & Prerequisite Primer (10 min)

Assumes: In earlier Grade 7 lessons you built simple AI workflows with an input, a process and an output, and you

practised checking an output against the task before trusting it.

Activate: Show two versions of an AI-written fun fact about the UAE on the board - one correct, one with a wrong date. Ask students

which one they would trust and how they could tell the difference.

Teach: Remind students that an AI output is a prediction based on patterns, not a guaranteed truth. The output stage of any

workflow is where mistakes and unfairness are most likely to reach a real person.

Apply: In pairs, students underline the part of an output they are unsure about and write one reason for their doubt.

Check: Cold-call two pairs. The prior rung is secure if they can point to the output stage and restate the rule verify before trust.

Critical & analytical thinking focus

Evaluate & justify - reason about risk, ethics and governance (Evaluate | DOK 2-3 | SOLO: Relational)

The technical concept (teacher background)

AI systems generate outputs by predicting the most likely response from patterns in their training data. This creates three

recurring risks. First, factual error, sometimes called hallucination: a model can state something false with complete

confidence. Second, bias: if the training data over-represents some groups or views, the outputs can be unfair or

stereotyped. Third, misuse and over-reliance: people may enter personal data, or accept an output without checking it. In a

workflow these risks concentrate at the output stage, because that is what people act on.

Responsible use means treating

every output as a draft to be verified against a trusted source, never entering personal data, and being honest about the

tool's limitations. Governance is the set of rules and roles that keeps all of this safe.

Lesson plan

Phase | time Teacher (I do / we do) Students Key questions (targeted thinking)

Hook / 6 min Display three short pre-generated AI outputs about a school topic; one has a

factual error, one shows a stereotype,
one leaks a made-up personal detail.
Read them aloud without labelling the
problems.

Students vote with thumbs
on which outputs they would
act on and note a first hunch
about what feels wrong.

- Which of these would be risky to
just copy and use, and why?

Teach / 10 min Introduce three risk types - error, bias,
and misuse or over-reliance - and the
verify-before-trust rule. Model naming
the risk in the first output and stating
who it could affect.

Students record the three
risk types and the definition
of responsible use in their
own words.

- What makes a confident-
sounding output still risky?

Guided practice /
12 min

Hand out curated scenario cards. Work
one card together: name the risk, say
who is affected, and propose one
responsible-use measure with a
reason.

In pairs, students annotate
the shared card and
rehearse the sentence stem

A risk here is... so a
responsible measure is...
because...

- Does your measure actually
reduce the risk you named?

Independent / 12
min

Circulate as pairs analyse two more
scenario cards, prompting them to
justify each measure and to link the risk
back to a workflow stage.

Pairs complete the analysis
grid for two cards, naming

risk, impact, and a justified responsible-use measure for each.

- Which workflow stage - input, process or output - does your risk sit at?

Check & close / 5 min

Collect one analysis grid per pair as the exit ticket. Restate that responsibility sits with the human user, not the tool.

Students write one sentence: the responsible-use rule they will always apply and why.

- Name one risk and the one measure you would never skip.

Anticipated misconceptions

Misconception How to address it

If the AI sounds confident, it must be right. Show a confident but wrong output beside a trusted source.

Explain that fluency is a style of writing, not evidence of truth; confidence and accuracy are separate. Reinforce verify before trust.

Bias means the AI was being rude or unfair on purpose. Explain that bias usually comes from patterns in the training data, not from intent. Use a simple example where over-represented data leads to an unfair output, and separate the idea of a mistake from the idea of blame.

Staying safe is the tool's job, so responsible use just means being polite to the AI.

Reframe responsibility as belonging to the human user and the school's rules. Responsible use means checking outputs, protecting personal data, and following governance, not the tone you use in a prompt.

Hands-on activity guide

Overview: Students analyse curated scenario cards to identify risks and ethical or governance concerns, then propose a responsible-use measure with a reason for each, building toward justified evaluation.

Grouping / timing: Pairs for the scenario analysis, whole class for the shared worked card and the debrief.

Tool: Curated scenario cards plus a small set of pre-generated AI output samples shown on the board by the teacher only.

This is account-free and teacher-mediated: students never sign in or enter any data, they reason about outputs that have already been safely produced.

Facilitation steps:

1. Read your scenario card carefully and underline the AI output it describes.
2. Decide which risk type it shows - error, bias, or misuse and over-reliance - and write it down.
3. Name who could be affected and what the impact on them might be.
4. Identify which workflow stage the risk sits at: input, process or output.
5. Propose one responsible-use measure and finish the stem ...because... to justify it.
6. Swap cards with another pair and check whether their measure really reduces the risk they named.

Unplugged fallback: No live AI is used at any point. If the pre-generated output samples cannot be displayed, run the

whole activity from the printed scenario cards alone; the reasoning steps are identical.

Formative assessment (aligned to the objectives)

Task / item Model response Marking note

Read a short AI output that confidently states a wrong fact about a UAE landmark. Name the risk type and one responsible-use measure.

Risk type: factual error (hallucination).

Measure: verify the fact against a trusted source such as the official site before using it.

Award the mark for correctly naming error and a measure that actually checks accuracy. A measure that only reformats the text does not gain the mark.

A class chatbot answers questions about famous scientists but almost always describes them as men. Name the risk and say where it comes from.

Risk: bias. It comes from patterns in the training data over-representing men, not from intent. Measure: broaden examples and flag the limitation to users.

Full credit requires naming bias and locating its source in the data.

Answers blaming the tool's intent are only partially correct.

Explain how one identified risk would affect building or evaluating an AI workflow, using one key term correctly.

Example: over-reliance at the output stage
means we must add a human-check step
when evaluating the workflow, or errors reach
users.

Look for a correct key term (risk,
bias, responsible use,
governance) linked to a specific
workflow stage or design decision.

- Targets: Identifies risks in AI-generated outputs and proposes a justified responsible-use measure (AIF·GE·G7·E, Emerging).
- Check: Pair analysis grid plus a short cold-call justification using the ...because... stem.
- Success criteria: Student names a correct risk type, who it affects, and a responsible-use measure with a valid reason linked to a workflow stage.
- Evidence: One completed scenario analysis grid per pair and the closing exit sentence.
- Feedback / adjustment: If pairs name risks but cannot justify measures, model one more think-aloud with the stem; if justification is secure, extend to explaining bias and limitations.

Differentiation

Criterion Emerging (expected) Approaching Not yet

Identify risk Names the correct risk type in an
output and who it affects

Names a risk but not who it
affects

Cannot yet name a risk
in the output

Propose measure with
reason

Proposes a responsible-use
measure and justifies it with a clear
because

Proposes a measure but the
reason is vague or missing

Proposes no measure or
one unrelated to the risk

Use of vocabulary Uses risk, bias or responsible use
accurately and links to a workflow
stage

Uses a key term but with a
minor slip

Key terms are absent or
used incorrectly

Representation: provide the risk types on a reference strip with an icon and a one-line definition
each, and offer the scenario cards

as both text and a read-aloud so meaning does not depend on reading speed.

Action & expression: allow students to record their analysis as annotations, a spoken justification, or the written grid, and provide the ...because... sentence stem as scaffolding.

People of Determination: pre-teach the three risk words with matching symbols, pair a fluent reader with the student for the card, and accept a verbal or symbol-supported justification as full evidence.

Week 5 - Lab | Risks, Ethics & Responsible Use (Lab) | Interactive lesson: 'The Risk Audit'

Strand: Ethics & Governance | Framework ref(s): AIF·GE·G7·E | 1 x 45 min

Objectives & success criteria

- Carry out a risk-and-impact audit of a real AI workflow scenario, identifying the risks in its outputs.
- Rate each risk for likelihood and impact using a simple matrix, then propose a specific mitigation and explain the reasoning behind it.
- Use Ethics and Governance vocabulary to justify which mitigation the workflow should build in first, connecting the risk to the stage it affects.

Student-facing success criteria:

- I can audit a real scenario and list the risks in its AI outputs.
- I can rate each risk for how likely it is and how much it would hurt, then propose a mitigation with a reason.
- I can argue which fix to build in first, using words like bias, mitigation and governance correctly.

Prior knowledge & Prerequisite Primer (10 min)

Assumes: In the Core period you learned to name three risk types in AI outputs - error, bias, and misuse or over-reliance -

and to propose a responsible-use measure with a reason.

Activate: Recap the Core exit tickets on the board: show two or three responsible-use measures students wrote and ask which risk each one was fixing.

Teach: Introduce the audit idea: an audit does not just spot risks, it rates each one for likelihood and impact so a team knows which to fix first.

Apply: In pairs, students place one Core risk onto a two-by-two likelihood-and-impact grid and defend where they put it.

Check: Check that each pair can explain why a high-impact, high-likelihood risk is the priority to mitigate.

Critical & analytical thinking focus

Evaluate & justify - reason about risk, ethics and governance (Evaluate | DOK 2-3 | SOLO: Relational)

The technical concept (teacher background)

A risk-and-impact audit is how organisations turn a list of AI risks into an action plan. Each risk in an output is rated on two

axes: likelihood, how often it is expected to occur, and impact, how much harm it would cause when it does. Plotting risks on a likelihood-by-impact matrix reveals priorities: high-likelihood and high-impact risks are mitigated first. A mitigation is a specific change to the workflow that lowers the likelihood or the impact of a risk, for example adding a human-review step at the output stage, restricting inputs so no personal data is entered, or documenting a tool's limitations. Good governance records who is responsible for each mitigation. Auditing teaches students that responsible use is a design decision built into a workflow, not an afterthought.

Lesson plan

Phase | time Teacher (I do / we do) Students Key questions (targeted thinking)

Hook / 5 min Present one real-style scenario: a school uses an AI helper to draft parent notices. Show two pre-generated draft notices, one with a factual slip and one with an unfair assumption. Students quickly list every risk they can see in the two drafts.

- Which of these risks would matter most if the notice went home unchecked?

Model the audit /

10 min

Model rating one risk on a likelihood-and-impact matrix and writing a matched mitigation, thinking aloud about which stage of the workflow the fix belongs to.

Students copy the matrix template and follow along with the modelled risk.

- Why does rating likelihood and impact change which risk we fix first?

Team audit / 15

min

Give each team a scenario pack with pre-generated outputs. Circulate, pushing teams to justify each rating and to make mitigations specific to a Teams audit their scenario:

list risks, rate each on the matrix, and write a mitigation with a reason for each.

- Is your mitigation lowering the likelihood, the impact, or both? workflow stage.

Prioritise &
justify / 10 min

Facilitate teams choosing their top-priority mitigation and preparing a 30-second justification using governance vocabulary.

Each team ranks its mitigations and drafts the sentence We would build in... first because...

- Who in the workflow would be responsible for your top mitigation?

Share & close / 5
min

Two teams present their top mitigation; class critiques whether it truly reduces the rated risk. Collect the audit sheets.

Students write one sentence on how their audit would change the way the workflow is built.

- Name the one mitigation you would build in first and the risk it removes.

Anticipated misconceptions

Misconception How to address it

Every risk is equally important, so we should try to fix them all at once.

Use the matrix to show that risks differ in likelihood and impact.

Explain that auditing means prioritising: a high-impact, high-likelihood risk is fixed before a rare, minor one, because time and attention are limited.

A mitigation is just a warning label telling users to be careful.

Distinguish a real mitigation, which changes the workflow to lower likelihood or impact, from a mere notice. Model turning a vague warning into a concrete step such as adding a human-review

checkpoint at the output stage.

Once we pick a mitigation, the risk is gone for good. Explain that mitigations reduce risk, they rarely remove it entirely,

and governance means someone keeps monitoring. Residual risk

is normal and must still be tracked.

Hands-on activity guide

Overview: Teams run a full risk-and-impact audit of a real-style AI workflow scenario: listing risks, rating each for likelihood

and impact, proposing specific mitigations, and prioritising which to build in first.

Grouping / timing: Small teams of three or four for the audit, whole class for the prioritisation share-out.

Tool: Printed scenario packs with pre-generated AI output samples plus a paper likelihood-and-impact matrix. Everything is

teacher-mediated and account-free: no student logs in and no live prompt is sent; teams reason about outputs the teacher

has already produced safely.

Facilitation steps:

1. Read your scenario pack and list every risk you can find in the AI outputs.
2. For each risk, rate its likelihood as low, medium or high, and do the same for its impact.
3. Plot each risk on the likelihood-and-impact matrix.
4. For each risk, write a specific mitigation and name the workflow stage it changes.
5. Rank your mitigations and choose the one to build in first, with a because reason.
6. Note who in the workflow would be responsible for that top mitigation.

Unplugged fallback: The audit is already fully paper-based; no live AI is used. If the printed output samples are missing,

teams audit from the written scenario description alone using the same matrix and steps.

Lab tool integration (practical, hands-on)

Tool: Card decks (risk-scenario cards) + pre-generated output packs

Teacher setup: Print a risk-scenario deck (each card a short AI-use scenario) and include two flawed outputs from the output pack.

What students do:

1. Draw a scenario card and read it aloud in your group.
2. Name one risk in the scenario and say who it could affect.
3. Match the risk to a responsible-use rule from the class charter.
4. Check the two flawed outputs and label the risk each shows.
5. Write one rule that would reduce the biggest risk you found.

Students produce: A risk-spotting record with risks, who-is-affected and a proposed rule.

Data-safety: Unplugged; the outputs are pre-curated.

Unplugged fallback: This is the unplugged route; extend with a privacy-focused scenario card.

A risk-scenario card linked to who it affects and a matching responsible-use rule.

Formative assessment (aligned to the objectives)

Task / item Model response Marking note

Given three risks in a scenario, place

them on a likelihood-and-impact matrix and state which is the priority.
Correct plotting with the high-likelihood, high-impact risk placed top-right and named as the priority to mitigate first.
Award the mark for defensible placements and for choosing the top-right risk as priority, with a reason referring to likelihood and impact.

Turn this weak mitigation - tell users to be careful - into a real mitigation for an output-stage error risk.

Add a human-review checkpoint that verifies the output against a trusted source before it is sent, lowering impact.

Full credit for a concrete change to the workflow at the correct stage. A restated warning with no workflow change does not gain the mark.

Explain, using one key term, how bias, misuse or a limitation in this workflow would affect the solution you would build.

Example: bias in the training data at the process stage means the build needs a diverse-data check and a documented limitation, or outputs stay unfair.

Look for a correct term (bias, misuse, limitation, governance, mitigation) tied to a specific stage and a design consequence.

- Targets: Identifies risks in AI-generated outputs and audits them to a justified, prioritised mitigation (AIF·GE·G7·E, Emerging, stretching to Proficient).
- Check: Completed likelihood-and-impact audit sheet plus a 30-second team justification of the top mitigation.
- Success criteria: Team lists real risks, rates each defensibly, proposes a stage-specific mitigation, and justifies a first priority using key vocabulary.
- Evidence: One audit matrix and mitigation sheet per team, and the closing sentence on how the audit changes the build.
- Feedback / adjustment: If teams rate risks but write vague mitigations, model one concrete stage-specific fix; if audits are

strong, push them to explain bias, misuse and limitations across the workflow.

Differentiation

Criterion Emerging (expected) Approaching Not yet

Audit risks Lists real risks in the outputs and

rates each for likelihood and impact

Lists risks but rates only

some or rates them loosely

Cannot yet identify the

risks in the scenario

Propose mitigation Proposes a specific, stage-linked

mitigation with a clear reason

Proposes a mitigation but it

is vague or not linked to a

stage

Proposes no workable

mitigation

Prioritise & justify Chooses a top mitigation and

justifies it with likelihood, impact

and a key term

Chooses a top mitigation

but the justification is thin

Does not prioritise or

justify a choice

Representation: provide the likelihood-and-impact matrix pre-drawn with worked example placements, and offer scenario packs

as text and read-aloud so the audit does not depend on reading speed.

Action & expression: teams may present the audit as a completed matrix, a labelled poster, or a spoken justification, and the We

would build in... first because... stem is provided.

People of Determination: give a colour-free symbol key for low, medium and high, assign a clear team role such as matrix-plotter,

and accept a symbol-supported or spoken justification as full evidence.

Week 6 - Core | Bringing the Strands Together | Interactive lesson: 'Four Strands,

One Task'

Strand: Integration | Framework ref(s): AIF·CU·G7·E AIF·SD·G7·E AIF·CE·G7·E AIF·GE·G7·E |
1 x 45 min

Objectives & success criteria

- Complete a guided challenge that combines all four strands on one unit task.
- Explain how the input, process and output parts connect to produce and check a result.
- Identify one responsible-use choice made while completing the task.

Student-facing success criteria:

- I can complete a guided task that uses all four strands together.
- I can explain how the parts of my workflow connect to make and check a result.
- I can name one responsible choice I made while doing the task.

Prior knowledge & Prerequisite Primer (10 min)

Assumes: In earlier lessons this term you learned the three parts of an AI workflow - input, process and output - trained a simple classifier, checked an AI output against a trusted source, and named a responsible-use rule.

Activate: Show the four strand icons from earlier lessons - understand, build, check, use responsibly - and ask which lesson each one came from.

Teach: Explain that today is not a new idea; it is joining the four skills you already have into one short task, like clicking four puzzle pieces together.

Apply: In pairs, match four earlier lesson moments to the four strands on a quick sorting card.

Check: Each pair reads back one strand and the lesson it came from; the teacher listens for correct matches before moving on.

Critical & analytical thinking focus

Synthesise - connect all four strands (Analyse/Create | DOK 3 | SOLO: Relational)

The technical concept (teacher background)

An AI solution is a pipeline, not a single clever step. A task is broken into an input stage (the data you feed in), a process

stage (a model or rule that transforms it - here a simple classifier), and an output stage (the result). Two more layers wrap

around this: critical evaluation, where each output is checked against the task and a trusted source because a confident

model can still be wrong, and governance, where you decide what data is allowed in and how the tool is used responsibly.

Integration means these four concerns act on one task together. Students are not learning a new algorithm today; they are

seeing how understanding, building, checking and responsible use combine into a working, trustworthy workflow.

Lesson plan

Phase | time Teacher (I do / we do) Students Key questions (targeted thinking)

Hook / 6 min We-do: show a messy pile of six item

cards and ask how a machine could help sort them; draw the empty input-process-output frame on the board.

Watch and call out where sorting would help; copy the three-box frame.

- Which part of the workflow does the actual sorting - input, process or output?

Connect the

strands / 8 min

I-do: label the board frame with the four

strands - understand the parts, build the steps, check the result, keep it responsible - giving one example of each on today's task.

Annotate their own frame with the four strand labels.

- Which strand makes sure a confident answer is actually correct?

Guided build-
and-check / 18

min

We-do then you-do: run the simulated classifier once as a demo with no personal data, then release pairs to build and run their own six-item sort and check it against the trusted guide card.

In pairs: prepare the input list, run the simulated sort, write the output, then check each label against the trusted guide and mark agree or disagree.

- You found a label you disagree with - what do you do before trusting it?

Debrief / 8 min Facilitate a whole-class share: collect examples of outputs that had to be Report one correction they made and one responsible

- Why can a classifier be confident and still be wrong? corrected and one responsible-use choice each pair made. choice, such as using only object words and no names or photos.

Check and close

/ 5 min

Close with the exit ticket; restate that four strands ran on one task.

Complete the exit ticket individually.

- In one sentence, how did the parts connect to make and check your result?

Anticipated misconceptions

Misconception How to address it

The AI just knows the right bin, so I do not need to check. Show a deliberately wrong label - a greasy pizza box marked

Recycle. A confident output can still be wrong, so every result is checked against a trusted guide before it is trusted.

Integration means learning a brand-new tool today. Reframe: no new tool or concept - you are joining four skills you

already have (understand, build, check, use responsibly) into one short task.

Responsible use only matters at the end, as a rule to add on.

Point out the data choice made at the input stage - using only object words and no names or photos - so responsible use shapes the task from the start, not just the finish.

Hands-on activity guide

Overview: Pairs build and run a small three-step waste-sorting workflow, then check every output against a trusted guide

and note one responsible-use choice - a preview of the Term-1 signature activity.

Grouping / timing: Pairs for the build-and-check; whole class for the debrief.

Tool: A simulated AI classifier run as a teacher-mediated card routine - sorting cards plus a board demo - account-free, no

logins, and no personal data leaves the room. A school-approved, child-safe tool may be teacher-operated on the board instead.

Facilitation steps:

1. Prepare your input: choose six item cards, each showing only an object word (for example 'plastic bottle'), never a name, face or photo of a person.

2. Run the process: pass each card through the simulated classifier, which labels it Recycle, Compost or Landfill.

3. Record the output: write each item and its label in your workflow table.

4. Check the output: compare every label with the trusted recycling guide card and tick agree or flag disagree.

5. Correct and note: fix any flagged label using the trusted guide, and write down the one you changed.

6. Responsible-use note: record one responsible choice you made, such as using only object words and no personal data.

Unplugged fallback: This activity is already unplugged - the classifier is simulated with cards. If a board tool was used

instead and access fails, pairs sort the cards by hand using the trusted guide, which is the same input-process-output-check routine.

Formative assessment (aligned to the objectives)

Task / item Model response Marking note

Draw the three-box workflow for the sorting task and label each box with what goes there.

Input box: six item cards showing object words only. Process box: the classifier that labels each item. Output box: the sorted list with a bin for each item.

Award full credit when all three boxes are correctly labelled with the right content, not just named.

Explain in one or two sentences how checking the output connects to the rest of the workflow.

After the process gives an output, I compare each label to the trusted guide; if they disagree I correct it, so the checked output is more trustworthy than the raw one.

Look for the connection - the output is produced then checked against a trusted source - not just 'I checked it'.

Name one responsible-use choice you made and say why it mattered.

I used only object words, no names or photos, so no personal data went into the classifier.

Accept any valid choice (no personal data, verify before trust, teacher-operated tool) with a reason tied to safety or trust.

- Targets: objectives 1-3 - combining all four strands on one task, explaining how the parts connect, and naming a responsible-use choice (Analyse/Create | DOK 3 | SOLO: Relational).
- Check: observe pairs during the build-and-check and mark the exit ticket and workflow table.
- Success criteria: a completed three-step workflow, at least one output checked and corrected against the trusted guide, and one responsible-use choice named with a reason.
- Evidence: the pair's workflow table with agree/disagree flags and each student's exit ticket.

- Feedback / adjustment: if pairs skip the check step, re-model one comparison and ask them to re-run the check on two items before the close.

Differentiation

Criterion Emerging (expected) Approaching Not yet

Combining the strands on the task

Completes the three-step workflow and applies a check with guidance

Completes the workflow but the check is partial or only when prompted

Cannot yet complete the workflow steps without step-by-step help

Explaining how parts connect

Explains that the output is produced then checked against a trusted source

Names the parts but the connection is unclear

Lists parts with no connection between them

Responsible-use choice Names one responsible choice with a reason

Names a choice but gives no reason

Cannot yet name a responsible choice

Representation: provide the workflow frame pre-drawn, the four strand icons, and a worked example of one checked item so pairs can see the whole routine before starting.

Action & expression: allow the workflow to be shown by arranging cards, filling the table, or explaining aloud to the teacher instead of writing.

People of Determination: offer larger high-contrast item cards, a reduced set of four items, and a sentence frame for the responsible-use note ('I used ... so no ...'). Give extra processing time for the check step.

Week 6 - Lab | Bringing the Strands Together (Lab) | Interactive lesson: 'The Full Run'

Strand: Integration | Framework ref(s): AIF·CU·G7·E AIF·SD·G7·E AIF·CE·G7·E AIF·GE·G7·E | 1 x 45 min

Objectives & success criteria

- Run the integrated sorting workflow end-to-end on a larger item set, recording every check and correction made.
- Analyse the check log to judge how reliable the workflow was and where it failed.
- Adapt the workflow by adding a constraint - a second criterion or a risk rule - while keeping the responsible-use rules.

Student-facing success criteria:

- I can run my whole workflow from start to finish and write down every check I make.
- I can use my check log to say how reliable my workflow was and where it went wrong.
- I can add a new rule or risk to my workflow and run it again responsibly.

Prior knowledge & Prerequisite Primer (10 min)

Assumes: In the Core period you built and ran a small three-step sorting workflow, checked six outputs against a trusted

guide, and named a responsible-use choice.

Activate: Show one filled-in check log from the Core period and ask which labels the class had to correct, and why.

Teach: Explain that today you run the same workflow properly end-to-end on a bigger set, keep a full record of your checks, and then change one rule to see what happens.

Apply: In pairs, predict which kinds of items the classifier is most likely to get wrong before you start.

Check: Each pair states one prediction and one check they will record; the teacher confirms both are clear before the run begins.

Critical & analytical thinking focus

Synthesise - connect all four strands (Analyse/Create | DOK 3 | SOLO: Relational)

The technical concept (teacher background)

Running a workflow end-to-end means executing every stage in order on a full batch, not just demonstrating one example.

The value for students is the record they keep: a check log turns 'I checked it' into evidence - how many outputs matched

the trusted guide, which were corrected, and where errors clustered. Patterns in that log, such as greasy or mixed-material

items being misclassified most, are the beginning of critical evaluation and reliability thinking.

Adding a constraint - a

second criterion such as 'reusable' or a risk rule such as 'flag possible contamination' - forces students to adapt the

process and re-check, previewing the iteration in the signature activity. Throughout, governance holds: only non-personal

object data enters the tool, and every output is verified before it is trusted.

Lesson plan

Phase | time Teacher (I do / we do) Students Key questions (targeted thinking)

Recap and

predict / 6 min

We-do: restate the workflow and show

the Core check log; ask pairs to predict the trickiest items.

Recall the stages and write two prediction guesses.

- Which items do you expect the classifier to get wrong, and why?

Set up the full

run / 7 min

I-do: model logging one item cleanly - item, label, guide agrees, correction - and set the batch size at ten to twelve items.

Set up their check-log table and select their larger item set using object words only.

- What exactly will you write in each column of your log?

End-to-end run /

16 min

You-do: pairs run all items through the simulated classifier, record every output, check each against the trusted guide, and correct as needed.

Run the full batch, filling the check log for every item and flagging disagreements.

- You are halfway - how many have you had to correct so far?

Add a constraint

/ 10 min

Facilitate: introduce one constraint - a second criterion (reusable / not) or a risk rule (flag possible contamination) - and ask pairs to adapt and re-run the affected items.

Choose a constraint, adapt their process, and re-check the items it affects.

- How did adding the constraint change which items needed a second look?

Analyse and Close: pairs report their reliability count Complete the exit ticket, • Out of all items, how many did

close / 6 min (corrected out of total) and one thing

the constraint changed; run the exit ticket.

including their reliability count.

the workflow get right before you corrected it?

Anticipated misconceptions

Misconception How to address it

If the workflow ran without errors, the outputs must be right.

Distinguish 'ran' from 'correct' - the process always produces a label; only the check against the guide tells you if it is right. Point to the corrections in the log as proof.

A few wrong labels mean the whole workflow is useless. Introduce reliability as a count - most labels may be right; the log shows exactly how many needed fixing, so you can judge and improve rather than reject.

Adding a new rule means starting a completely new workflow.

Show that a constraint is added to the existing process; only the affected items are re-checked, so you adapt the workflow rather than rebuild it.

Hands-on activity guide

Overview: Pairs run the full sorting workflow end-to-end on ten to twelve items, keep a complete check log, then add one

constraint and re-check the items it affects - recording reliability as they go.

Grouping / timing: Pairs for the run and the constraint; whole class for the reliability share.

Tool: The same simulated AI classifier as a teacher-mediated card routine - account-free, no logins, no personal data. A

school-approved, child-safe tool may be teacher-operated on the board; the card version is the fallback.

Facilitation steps:

1. Select ten to twelve item cards, each showing only an object word - no names, faces or photos.
2. Run every item through the simulated classifier and record its label in the check log.
3. Check each label against the trusted guide; tick agree or flag disagree, and write the corrected label where needed.
4. Count your accuracy: how many the workflow got right out of the total before correcting (write it as a fraction and a percentage).
5. Add one constraint - a second criterion (reusable / not) or a risk rule (flag possible contamination) - and re-check only the items it affects.
6. Write one sentence on how the constraint changed your workflow and which items needed a second look.

Unplugged fallback: The routine is already card-based and unplugged. If a board tool is used and access fails, pairs run

the whole batch by hand with the trusted guide - the same end-to-end run and check log.

Lab tool integration (practical, hands-on)

Tool: Teachable Machine + Scratch + pre-generated packs (integrated run)

Teacher setup: Have the Week-2 Teachable Machine model and a Scratch starter ready; print one evaluation pack per group.

What students do:

1. Recall the trained model (input -> process -> output).
2. In Scratch, connect or simulate the model so a typed/selected input produces an output.
3. Run the mini solution on three inputs.
4. Use the evaluation pack criteria to judge each output.
5. Record one responsible-use choice you made while running it.

Students produce: A run-and-check record across all four strands.

Data-safety: Teachable Machine trains in the browser and the samples stay on the teacher's device. Use objects, hand gestures or drawings only - never faces, voices or names. Clear the project at the end of the lesson. Reuse only the non-personal model.

Unplugged fallback: Run the whole loop on paper: card-deck classifier + block-cards + printed outputs.

The four strands as one loop: build, run, evaluate, check the risk.

Formative assessment (aligned to the objectives)

Task / item Model response Marking note

Show your completed check log for at least ten items with the agree/disagree and correction columns filled.

A table with ten or more rows: each item, its classifier label, guide agrees Y/N, and a corrected label where N.

Full credit when every item has a check result and corrections are recorded wherever the guide disagreed.

State your workflow's reliability and one place it was weakest.

It got 8 of 12 right before correction; it was weakest on mixed-material items like a foil-lined carton.

Look for a count (right out of total) plus a specific weak spot, not a vague 'some were wrong'.

Describe the constraint you added and

how you adapted the workflow responsibly.

I added a 'flag possible contamination' risk rule, re-checked the food-touching items, and still used only object words - no personal data.

Accept any valid constraint with a re-check and an intact responsible-use rule.

- Targets: the same integrated outcome, deepened - running the workflow end-to-end, recording and analysing checks, and adapting it under a constraint (Analyse/Create | DOK 3 | SOLO: Relational).
- Check: review each pair's check log and reliability count, and mark the exit ticket.
- Success criteria: a complete log for ten or more items, a stated reliability count with a named weak spot, and one constraint added with the responsible-use rule kept.
- Evidence: the pair's full check log with corrections and the exit ticket showing the reliability count.
- Feedback / adjustment: if logs are incomplete, pause the run and re-model one clean log row; if reliability is stated without a weak spot, prompt pairs to point to a specific row.

Differentiation

Criterion Emerging (expected) Approaching Not yet
Running end-to-end and logging

Runs the full batch and records a check for every item with guidance

Runs the batch but the log has gaps

Cannot yet complete the run without step-by-step help

Analysing reliability States how many were right out of the total and names a weak spot

Gives a count but no weak spot, or a weak spot but no count

Cannot yet judge reliability from the log

Adapting under a constraint Adds a constraint and re-checks affected items responsibly

Adds a constraint but does not re-check, or drops a responsible-use rule

Cannot yet adapt the workflow

Representation: provide a pre-formatted check-log table, one completed sample row, and a small menu of constraint options so pairs choose rather than invent.

Action & expression: reliability can be reported as a count, a fraction, or a quick tally chart; the constraint can be shown by re-sorting cards rather than writing.

People of Determination: allow a reduced batch of eight items, pre-printed item labels, and a sentence frame for the reliability

statement ('We got ... out of ... right; the weakest was ...'). Give extra time for the re-check.

Week 7 - Core | Signature Solution - Planning | Interactive lesson: 'Architect First, Build Second'

Required Advanced-tier activity - Design two mitigations (10 min, all students). For a bias or misuse risk in the waste-sorting scenario (or their own signature solution), each student proposes TWO concrete strategies to reduce the risk (e.g. balance the training examples; add a confidence threshold that routes low-confidence items to a human) and explains, for each, how it would change the system's behaviour. Capture on the Worksheet; this feeds the Post-Assessment mitigation item.

Strand: Integration | Framework ref(s): AIF·CU·G7·E AIF·SD·G7·E AIF·CE·G7·E AIF·GE·G7·E | 1 x 45 min

Objectives & success criteria

- Plan the signature activity by stating the goal, the structure or steps, and the check steps.
- Identify the inputs, tools and data the solution will use, using only non-personal data.
- Describe one risk in the plan and explain how a check step reduces it.

Student-facing success criteria:

- I can write a clear goal for my workflow and list its input, process and output steps in order.
- I can name the inputs, the teacher-operated tool and the non-personal data my solution needs.
- I can point to one risk and show the check step that catches it before the output is trusted.

Prior knowledge & Prerequisite Primer (10 min)

Assumes: In Grade 6 you followed step-by-step instructions and, earlier this term, you traced simple input-process-output

workflows and spotted where an AI step sits inside them.

Activate: Show a three-box strip on the board labelled Input, Process, Output with the middle box empty. Ask the class to call out

what could go in each box for a familiar task such as sorting recycling photos.

Teach: Remind students that a workflow is a planned chain of steps, not a single magic answer, and that a good plan says what

goes in, what happens, and how we know the result is any good.

Apply: In pairs, students label a printed strip for one everyday task, writing one word in each of the three boxes.

Check: Cold-call three pairs to read their strip aloud; the rung is secure if the middle box describes an action, not just a tool name.

Critical & analytical thinking focus

Design & plan - architect a structured solution (Create (plan) | DOK 3 | SOLO: Relational)

The technical concept (teacher background)

A signature solution is a small multi-step AI workflow that completes one task. Planning it means designing an architecture

before building: a goal statement, an ordered structure of input, process and output steps, and check steps that test the

output against the task. The process step may use a teacher-operated, school-approved model, but the plan treats that

model as one component whose output must be verified, never trusted on sight. Good planning also names the resources

each step needs - inputs, the tool, and non-personal data - and anticipates at least one risk, such as a confident but wrong

output, then places a check step to catch it. This is design thinking applied to AI: architect first, build second, so the

solution is deliberate rather than accidental.

Lesson plan

Phase | time Teacher (I do / we do) Students Key questions (targeted thinking)

Hook / 6 min Show two plans for the same task on

the board: one is a single arrow labelled

ask the AI, the other is a labelled input-

process-output chain with a check.

Think-aloud about which you would

trust and why.

Students vote with thumbs

and give one reason for the

plan they trust more.

- Which plan tells you how we will

know the answer is good?

Model the plan / We-do: fill a planning frame on the Students follow along on • Where in this chain does the

10 min board for a shared example task - goal,

structure or steps, inputs, tool, data,

one risk, one check, success criteria -

narrating each choice.

their own blank planning

frame and copy the

headings.

check step belong, and what is it

checking?

Guided plan / 14

min

Circulate as pairs draft their own plan

for their signature activity, prompting

them to keep the process step teacher-operated and to use only non-personal data.

In pairs, students write a goal, order their input-process-output steps, and add at least one check step.

- What non-personal data does your process step actually need?

Risk and check /

10 min

Lead a short round: each pair names one risk in their plan; the class suggests where a check step could catch it.

Model adding a fallback idea as a stretch.

Students add one named risk and the matching check step to their plan, and note the success criteria.

- If that check fails, what is your plan B?

Check and close

/ 5 min

Collect the planning frames and run the exit ticket; flag any plan missing a check step for next lesson.

Students complete the exit ticket linking their goal to their check step.

- Read me the one sentence that proves your plan can tell good output from bad.

Anticipated misconceptions

Misconception How to address it

The AI does the whole task, so my plan is just ask it and copy the answer.

Show the same prompt giving a confident but wrong answer; make the point that the AI is one step inside the workflow and the plan must include a check that tests its output against the task.

A check step means checking that the tool ran, not whether the answer is right.

Model two checks side by side - did it produce something versus is what it produced correct and useful - and mark only the second

as a real check step.

More data always makes the plan better, so I should add student names or photos.

Reinforce the no personal data rule: the plan names only non-personal data, and adding personal data is never an improvement; it is a safeguarding failure.

Hands-on activity guide

Overview: Pairs architect the plan for their Term-1 signature solution, moving from a goal statement to an ordered input-

process-output structure with named resources, one risk and one check step.

Grouping / timing: Pairs for the planning build, whole class for the risk-and-check debrief.

Tool: A paper planning frame plus a teacher-operated, school-approved demo model shown on the board only; students

never log in, so the routine stays account-free and child-safe.

Facilitation steps:

1. Write your goal in one sentence: what task will your solution complete?
2. Draw three boxes - Input, Process, Output - and write the ordered steps inside them.
3. Under the boxes, list the inputs, the teacher-operated tool, and the non-personal data each step needs.
4. Add a check step after the output: what will you compare the result against to know it is good?
5. Name one risk in your plan and draw an arrow from it to the check step that catches it.
6. Write your success criteria: one sentence describing what a good output looks like.

Unplugged fallback: If no demo tool is available, run the process step as a simulated AI: a sealed envelope of pre-written

response cards that a student draws from, so the plan can still be built and pressure-tested with no tool access.

Formative assessment (aligned to the objectives)

Task / item Model response Marking note

Write a one-sentence goal for a signature solution that sorts photos of school lunches into healthy or less healthy, then list its three ordered steps.

Goal: decide whether each lunch photo is healthy or less healthy. Steps: Input - collect the lunch photos; Process - the teacher-operated classifier labels each photo; Output - a sorted list of the two groups.

Award the mark if the goal names one task and the three steps are correctly ordered as input, process, output.

Name one input, one tool and one type of non-personal data this solution needs.

Input: a lunch photo. Tool: the teacher-operated, school-approved image classifier.

Data: non-personal example photos with no faces or names used to guide the labels.

Full marks require all three named and the data described as non-personal.

Describe one risk in this plan and the check step that reduces it.

Risk: the classifier confidently mislabels a healthy lunch as less healthy. Check: compare a sample of outputs against a trusted food guide before trusting the result.

Award the mark if the check tests correctness of the output, not merely that the tool ran.

- Targets: the Integration outcome for planning a signature solution - a stated goal, an ordered input-process-output structure, named resources, and a check step that reduces a named risk.

- Check: a marked planning frame plus a one-sentence exit ticket linking the goal to the check step.

- Success criteria: the plan states a clear goal, orders the three steps correctly, names non-personal inputs and data, and includes at least one check step tied to a named risk.

- Evidence: the collected planning frames and exit tickets, annotated for a present or missing check step.

- Feedback / adjustment: return plans with a single next-step prompt; any plan missing a real check step is re-opened at the start of the Lab period.

Differentiation

Criterion Emerging (expected) Approaching Not yet

Goal and structure States a clear one-task goal and orders input, process and output steps correctly.

Goal present but broad, or one step out of order.

No clear goal or steps not in order.

Resources and data Names inputs, the teacher-operated tool, and non-personal data for the process step.

Names some resources but omits the tool or the data type.

Resources missing or
names personal data.

Risk and check Names one risk and a check step
that tests whether the output is
correct.

Names a risk or a check but
does not link them.

No risk or check step
identified.

Representation: provide the planning frame with the three boxes pre-drawn and labelled, plus a
small word bank of step verbs and
a worked example strip for reference.

Action & expression: let students plan by sketching, by sticky-note cards moved into order, or by
dictating steps to a partner
scribe; all routes produce the same frame.

People of Determination: offer a partially completed frame with the goal supplied so the learner
focuses on ordering steps and
placing the check; allow extra time and a quiet space for the risk-and-check round.

Week 7 - Lab | Signature Solution - Planning (Lab) | Interactive lesson: 'Run It,
Break It, Make It Safer'

Strand: Integration | Framework ref(s): AIF·CU·G7·E AIF·SD·G7·E AIF·CE·G7·E AIF·GE·G7·E |
1 x 45 min

Objectives & success criteria

- Prototype the structure of your signature solution by running each planned step by hand and checking the flow works end to end.
- Pressure-test the plan by feeding it a tricky input and confirming the check step catches a wrong output.
- Strengthen the plan by adding a fallback path for when a check step fails.

Student-facing success criteria:

- I can walk my whole workflow through once, step by step, and confirm the output reaches the check step.
- I can test my plan with a hard input and show that my check step catches the bad output.
- I can add a fallback path so my solution does something safer when a check fails.

Prior knowledge & Prerequisite Primer (10 min)

Assumes: In the Core period you planned your signature solution: a goal, an ordered
input-process-output structure,
named resources, one risk and a check step.

Activate: Ask each pair to hold up their Core planning frame and read their check step aloud in
ten seconds.

Teach: Explain that a plan on paper is a hypothesis; a prototype is a first walk-through that
shows whether the chain actually holds
together when a real input flows through it.

Apply: In pairs, students trace one input along their own chain with a finger and say each step out loud.

Check: Listen for pairs who reach the output but skip the check step; prompt them to finish the walk-through at the check.

Critical & analytical thinking focus

Design & plan - architect a structured solution (Create (plan) | DOK 3 | SOLO: Relational)

The technical concept (teacher background)

Prototyping is the bridge between a plan and a working solution: you run the planned steps by hand, with a real input, to

reveal gaps the paper plan hid. Pressure-testing goes further by deliberately feeding a hard or misleading input to see

whether the check step actually catches a wrong output rather than waving it through. This is how engineers gain

confidence in a workflow before scaling it. A key move is adding a fallback path - a defined action for when a check fails,

such as rerouting to a safer step, flagging for a human, or returning to the input - so the solution degrades gracefully instead

of trusting bad output. Throughout, the process step stays teacher-operated and account-free, and no personal data enters

the workflow; the prototype is tested on non-personal example inputs only.

Lesson plan

Phase | time Teacher (I do / we do) Students Key questions (targeted thinking)

Hook / 5 min Show a plan that looks perfect on paper

but breaks when a tricky input flows

through it - the check step passes a

wrong output. Think-aloud: paper plans

can hide gaps.

Students predict where this

plan will break when a hard

input arrives.

- What input would break your plan

the same way?

Model the

prototype walk /

8 min

We-do: take the shared example plan

and walk one input through every step

on the board, pausing at the check step

to test the output against the task.

Students set up their own

plan and a chosen test input,

ready to walk it through.

- At the check step, what exactly

are you comparing the output

against?

Prototype run /

12 min

Circulate as pairs run their plan by hand with a normal input, using the teacher-operated demo or the simulated-AI cards for the process step, and note any step that breaks.

In pairs, students walk a normal input end to end and mark every step that did not work as planned.

- Did the output actually reach your check step, or did it get stuck?

Pressure-test /

12 min

Hand each pair a tricky test input designed to produce a confident but Students feed the tricky input through, record

- Did your check catch the bad output, and if not, how will you wrong output; support them in confirming the check step catches it and logging the result.
- whether the check step caught the wrong output, and fix the check if it did not.

tighten it?

Fallback and

close / 8 min

Lead the fallback move: each pair adds a defined action for when the check fails, then run the exit ticket collecting the revised plan.

Students write a fallback path onto their plan and complete the exit ticket describing one fix they made.

- Read me your fallback: what does your solution do the moment a check fails?

Anticipated misconceptions

Misconception How to address it

If my plan looks complete on paper, the prototype will just work.

Show a tidy plan breaking on a real input; make the point that walking the steps by hand is what reveals the hidden gap a written plan cannot show.

Pressure-testing means picking an easy input so my plan passes.

Reframe testing as trying to break your own plan on purpose with a hard input; a plan that only survives easy inputs has not really been tested.

A fallback is giving up, so a good plan does not need one. Model a fallback as engineering for failure - rerouting to a safer

step or flagging for a human - so the solution stays safe when, not if, a check fails.

Hands-on activity guide

Overview: Pairs turn their Core plan into a working prototype: they walk a normal input through every step, then pressure-

test with a tricky input and add a fallback path so the solution handles failure safely.

Grouping / timing: Pairs for the prototype and pressure-test, whole class for the fallback share-out.

Tool: The teacher-operated, school-approved demo model on the board for the process step, or the simulated-AI response

cards; students stay account-free and enter no personal data, testing on non-personal example inputs only.

Facilitation steps:

1. Set up your Core plan and choose one normal, non-personal example input to test.
2. Walk that input through every step in order, saying each step aloud, until it reaches the output.
3. Run the output through your check step and record whether it passed the check.
4. Now take the tricky test input and send it through the same chain.
5. Confirm your check step catches the wrong output; if it does not, tighten the check and note the fix.
6. Add a fallback path: write exactly what your solution does the moment a check step fails.

Unplugged fallback: With no demo tool, run the whole prototype on paper: the process step is a simulated AI where a

partner draws a pre-written response card, including one deliberately wrong card so the pressure-test and fallback still work with no tool access.

Lab tool integration (practical, hands-on)

Tool: Card decks (plan cards) + Scratch storyboard

Teacher setup: Provide plan cards (Goal, Inputs, Steps, Checks, Risk) and an empty Scratch storyboard frame.

What students do:

1. Lay out the plan cards and fill each one for your signature solution.
2. Sequence the Steps cards into the order the workflow will run.
3. Sketch the same sequence as Scratch blocks on the storyboard.
4. Pressure-test: pick one step and ask what happens if it fails.
5. Add a fallback card for that step.

Students produce: A planned card layout plus a Scratch storyboard with a fallback path.

Data-safety: Planning only; no live tool or data.

Unplugged fallback: Cards alone are sufficient; the storyboard is optional.

Plan cards (goal, inputs, steps, checks, risk) sequenced into a workflow with a fallback branch.

Formative assessment (aligned to the objectives)

Task / item Model response Marking note

Walk one normal input through your workflow and record the output at each step. Did the output reach the check step?

Input photo entered; process labelled it healthy; output listed it in the healthy group; check step compared it to the food guide - yes, it reached the check step and passed.

Award the mark if every step is recorded in order and the walk-through ends at the check step.

Feed the tricky input through and state whether your check step caught the wrong output.

The tricky blurred photo was confidently labelled healthy in error; the check step compared it to the food guide, spotted the mismatch, and caught the wrong output.

Full marks require a clear yes or no plus what the check compared the output against; a caught error and a fixed check both earn credit.

Write the fallback path your solution follows when the check step fails.

If the check fails, do not use the label: send the photo back to the input step for a clearer photo and flag it for the teacher to review.

Award the mark if the fallback names a specific safer action rather than simply trusting or discarding the output.

- Targets: the Integration outcome applied practically - prototyping the planned structure and pressure-testing it so the check step reliably catches a wrong output, extended by a fallback path.
- Check: observed prototype walk-through plus the pressure-test log and the revised plan with a fallback path.
- Success criteria: the pair walks a normal input end to end to the check step, confirms the check catches a tricky wrong output, and adds a specific fallback action.
- Evidence: the annotated prototype log, the corrected check step, and the written fallback path on the revised plan.
- Feedback / adjustment: give one targeted prompt on the weakest step; pairs whose check step let the wrong output through re-run the pressure-test before moving on.

Differentiation

Criterion Emerging (expected) Approaching Not yet

Prototype walk-through Walks a normal input through every step in order and reaches the check step.

Walks most steps but skips one or does not reach the check.

Cannot trace an input through the steps.

Pressure-test Feeds a tricky input and shows whether the check step catches the wrong output.

Runs the tricky input but does not judge if the check caught it.

Only tests an easy input or does not test.

Fallback path Adds a specific safer action for Mentions a fallback but it is No fallback path added.

when a check fails. vague or unsafe.

Representation: provide a prototype log sheet with a row per step and a marked column for pass or fail at the check step, plus a ready-made tricky test input for pairs who need one.

Action & expression: pairs may prototype by moving physical cards along a printed chain, by narrating each step to a scribe, or by annotating their frame; the recorded log is the same.

People of Determination: supply a pre-set tricky input and a fallback sentence stem - 'If the check fails, my solution will ...' - and

allow extra time; a partner can operate the cards while the learner judges the check.

Week 8 - Core | Signature Solution - Building | Interactive lesson: 'Build,

Evaluate, Improve'

Required Advanced-tier activity - Compare two workflow designs (10 min, all students). Give the same task (e.g. sorting recyclables) as two designs: (A) a single step that decides everything at once, and (B) a multi-step workflow (pre-check -> classify -> confidence check -> human fallback). Each student runs both on 3-4 test items, records which produced better outcomes, and writes 2-3 sentences explaining WHY the design difference caused the difference. Capture on the Worksheet; this is required, not Stretch.

Strand: Integration | Framework ref(s): AIF·CU·G7·E AIF·SD·G7·E AIF·CE·G7·E AIF·GE·G7·E | 1 x 45 min

Objectives & success criteria

- Build the signature solution by following the planned input, process and output structure.
- Test the solution and evaluate its outputs against the agreed success criteria.
- Make at least one improvement to the inputs or structure based on an evaluation.

Student-facing success criteria:

- I can build my solution by following my input, process and output plan.
- I can test my solution and judge its outputs against our success criteria.
- I can make one clear improvement after looking at what my evaluation shows.

Prior knowledge & Prerequisite Primer (10 min)

Assumes: In earlier lessons this term you planned a multi-step AI workflow and wrote down its three parts: the input you give it, the process step that does the work, and the output you expect. You also wrote success criteria - the short list that tells you when an output is good enough.

Activate: Show one anonymous workflow plan on the board (input arrow, process box, output arrow) next to its success criteria.

Ask: if we ran this exactly as written, what is the very first thing we would type or place in the input box?

Teach: Model reading a plan like a recipe: point to each part in order and say aloud what you would do at that step. Stress that building means following the plan step by step, not inventing a new idea now.

Apply: In pairs, students point to input, process and output on their own plan and read their success criteria to each other. They tick that all three parts and the criteria are written and clear.

Check: Each pair states one sentence: our input is ____, our process step is ____, our output should be ____. If any part is missing, they patch the plan before the build begins.

Critical & analytical thinking focus

Create & evaluate - build, evaluate, iterate (Create/Evaluate | DOK 3 | SOLO: Relational)

The technical concept (teacher background)

A workflow solution is built by following a fixed order: an input is prepared, a process step transforms it, and an output is produced. Building is not designing a new idea - it is faithfully running the plan the students already made. Once an output

appears, it must be evaluated, not trusted on sight. Evaluation means comparing the output against the success criteria:

does it do the task, is it correct, is it complete and appropriate. A confident-looking AI output can still be wrong, so it is

checked against the task and a trusted source. Improvement is then targeted: if the output misses a criterion, the students

change the input wording or adjust one step of the structure, then run it again. This

build-evaluate-refine loop is the core

habit of solution building.

The build-evaluate-refine loop students follow during the build session.

Lesson plan

Phase | time Teacher (I do / we do) Students Key questions (targeted thinking)

Hook / 6 min Show two outputs from the same plan

on the board - one that meets the

criteria and one that does not. Do not

say which is which.

Students vote which output

passes the success criteria

and give one reason.

- Which criterion does the weaker output fail, and how can you tell?

Set up the build /

8 min

Model the build-evaluate-refine loop

once end to end using a teacher-run

demo of the class solution. Point to

each plan part as you go.

Students lay out their own

plan and success criteria,

ready to build.

- What is the first input you will give, exactly as your plan says?

Build session /

15 min

Circulate. For the AI step, run each pair's input on the approved child-safe tool at the board, or hand back a pre-generated output card. Remind: no names or personal details in any input.

Pairs build their solution

step by step, produce at

least one real output, and

keep their plan open beside

them.

- Did the output come from following your plan, or did the plan need a quick fix first?

Evaluate and refine / 12 min

Guide students to score their output against each success criterion and to verify one claim against a trusted source before trusting it.

Pairs mark the output against the criteria, identify one weakness, make one improvement to the input or structure, and re-run.

- Which single change did you make, and did the new output meet the criterion?

Check and close / 4 min

Collect the build log. Restate that every output is verified before trust and that improvement is targeted, not random.

Each pair writes the one improvement they made and why on their exit ticket.

- Name the criterion your improvement was aimed at.

Anticipated misconceptions

Misconception How to address it

If the AI gives me an answer, the solution must be working. Show a confident output that fails a success criterion. Establish

verify-before-trust: an output only counts when it is checked against the task and a trusted source, not because it sounds sure.

Building means I can keep changing my whole idea until I like it.

Reframe building as following the existing plan step by step.

Redesigning is a separate stage; today you run the plan, then improve one part at a time.

To make it better I should change lots of things at once. Model changing only one thing - the input wording or a single step

- then re-running, so you can see which change caused the improvement. Many changes at once hide the cause.

Hands-on activity guide

Overview: Pairs run their planned solution to produce a real output, evaluate it against their success criteria, and make one targeted improvement, logging each step.

Grouping / timing: Pairs for the build and evaluation; whole class for the loop demo and the closing debrief.

Tool: A teacher-operated, school-approved, child-safe AI tool shown on the board, or a simulated AI card set of pre-generated outputs. Students never log in and enter no personal data; the teacher runs every AI step so the activity stays account-free.

Facilitation steps:

1. Open your plan and success criteria side by side. Read your input, process and output aloud to check the plan is build-ready.
2. Prepare your input exactly as the plan says. Check it contains no names, faces or personal details before it goes anywhere near the tool.
3. Run the process step: give the input to the teacher for the board demo, or draw your output card. Record the exact output you get.
4. Evaluate: tick or cross the output against each success criterion, and verify one claim against a trusted source before you trust it.
5. Improve one thing: change the input wording or one step of the structure, then run it again and record the new output.
6. Log it: write what you changed, why, and whether the new output met the criterion it was aimed at.

Unplugged fallback: If no tool is available, use the simulated AI card set: the input is a question card, the process is a lookup table the pair applies by hand, and the output is the matching answer card. Pairs evaluate and refine the same way by swapping input cards, so the lesson never depends on tool access.

Formative assessment (aligned to the objectives)

Task / item Model response Marking note
Given a plan and an output, state whether the output meets the success criteria and give one reason.

The output meets criteria 1 and 2 but fails criterion 3 because it leaves out the required summary line.

Look for a criterion named and matched to evidence in the output, not just a like or dislike.

Describe one improvement you made and which criterion it was aimed at.

I shortened my input to ask only for three points, aimed at the criterion be concise; the new output listed three points instead of a paragraph.

Award if the change is specific and clearly linked to one criterion and one re-run.

Explain why a confident AI output still needs checking before you trust it.

An output can sound sure and still be wrong, so I check it against the task and a trusted source before using it.

Look for the verify-before-trust idea in the student's own words, not just repeating the output is fine.

- Targets: building the planned solution, evaluating its output against success criteria, and making one criterion-based improvement.
- Check: circulate during the build-evaluate-refine loop and review each pair's build log at the close.
- Success criteria: the pair produces an output by following their plan, scores it against every criterion, and makes one targeted improvement with a re-run.
- Evidence: the build log showing the first output, the criteria scores, the one change made, and the improved output.
- Feedback / adjustment: if a pair changed many things at once, prompt them to isolate one change and re-run; if they trusted an unchecked output, prompt a verify-against-source step.

Differentiation

Criterion Emerging (expected) Approaching Not yet

Follows the plan to build Builds the solution by following the input, process and output plan in order

Builds most of the solution but skips or reorders a step

Cannot yet run the plan without heavy support

Evaluates against criteria Scores the output against each Judges the output overall but States a like or dislike

success criterion with a reason names only one criterion with no criterion

Improves from evaluation Makes one targeted change and re-runs to check it

Suggests a change but does

not test it

Makes random changes

or none

Representation: provide the plan as a printed input-process-output strip with the success criteria as a tick-list, and show the loop

as a simple three-step diagram on the board.

Action & expression: pairs may record the build log by writing, by ticking a printed table, or by speaking their improvement to the teacher who scribes the key line.

People of Determination: offer a partly filled build log with sentence stems (I changed ____, aimed at ____, the new output ____), extra time for the re-run, and a quiet station for the evaluation step.

Week 8 - Lab | Signature Solution - Building (Lab) | Interactive lesson: 'Finish It, Then Prove It'

Strand: Integration | Framework ref(s): AIF·CU·G7·E AIF·SD·G7·E AIF·CE·G7·E AIF·GE·G7·E | 1 x 45 min

Objectives & success criteria

- Complete a working version of the signature solution by finishing the build across all steps.
- Run repeated tests and log each evaluation of the outputs against the success criteria.
- Refine the solution across more than one cycle, documenting each improvement and its effect.

Student-facing success criteria:

- I can finish a working version of my solution that runs from input to output.
- I can test it more than once and log how each output scores against the criteria.
- I can make and record several improvements and say what each one changed.

Prior knowledge & Prerequisite Primer (10 min)

Assumes: In the Core period you built your solution, produced a first output, evaluated it against your success criteria, and

made one improvement. You kept a short build log of what you changed.

Activate: Show a half-finished build log with only the first output and one change filled in. Ask: what would the next two rows of this log look like if we kept testing and improving?

Teach: Model extending the log: run again, score against the criteria, write the change, note the effect. Emphasise that the Lab is about repeating the loop with care, not starting a new idea.

Apply: In pairs, students find where their solution stopped in the Core period and name the very next step they need to complete it.

Check: Each pair states the one part of their solution that is not yet finished and what completing it will require. The teacher notes who needs the most support first.

Critical & analytical thinking focus

Create & evaluate - build, evaluate, iterate (Create/Evaluate | DOK 3 | SOLO: Relational)

The technical concept (teacher background)

The Lab is extended build and evaluation time: the goal is a complete, working solution rather than a first draft. Completion

means every step runs in order and produces the expected output without gaps. Beyond completion, repeated testing matters - running the same solution on several inputs, including awkward ones, reveals whether it is reliable or only worked once by chance. Each run is evaluated against the same success criteria and recorded in an evaluation log, so improvement can be traced over several cycles rather than a single change. A logged history also supports verify-before-trust: patterns of failure become visible when outputs are compared side by side. The habit being built is disciplined iteration - test, evaluate, refine, log, repeat - which is how real solutions are made dependable.

Lesson plan

Phase | time Teacher (I do / we do) Students Key questions (targeted thinking)

Hook / 5 min Show the same solution run on two

different inputs, one giving a good output and one a poor one. Ask why one build can pass and fail.

Students suggest what repeated testing would reveal that a single test hides.

- What would you learn by running your solution five times instead of once?

Plan the finish / 7 min

Help each pair name the exact steps left to complete their working solution and set an evaluation log target for the period.

Pairs mark on their plan where the build stopped and list the steps to completion.

- What is the single next step that gets your solution working end to end?

Extended build / 16 min

Circulate and run each AI step on the approved child-safe tool at the board, or issue pre-generated output cards. Reinforce no personal data in any input. Pairs finish the build so it runs input to output, then

test it on at least two different inputs.

- Does your solution still work when the input is different from the one you first tried?

Log the

evaluations / 12

min

Guide pairs to record each test in an evaluation log: input used, output, criteria met, change made, effect.

Pairs run repeated tests, score each output against the criteria, and log every

- Which change made the biggest difference across your logged runs?

Prompt verify-before-trust on at least one output.

cycle and its improvement.

Check and close

/ 5 min

Collect the completed solution and evaluation log. Restate disciplined iteration: test, evaluate, refine, log, repeat.

Each pair marks their log as complete and writes the improvement they are most confident in.

- Point to the log row that shows your solution became more reliable.

Anticipated misconceptions

Misconception How to address it

If it worked once, my solution is finished and reliable. Show a solution that passed one input but failed another.

Establish that reliability is shown by repeated tests on different inputs, not a single success.

The evaluation log is just extra paperwork. Compare two logged runs side by side to show a failure pattern

the log made visible. The log is how you trace which change actually helped.

An edge-case input that breaks my solution means I did it

wrong.

Reframe edge cases as useful: they show where the solution needs a graceful response. Finding and fixing one is a sign of strong building, not failure.

Hands-on activity guide

Overview: Pairs complete a working solution, run repeated tests on different inputs, and keep an evaluation log tracing several improvement cycles.

Grouping / timing: Pairs for the extended build and logging; whole class for the hook and the closing reliability debrief.

Tool: A teacher-operated, school-approved, child-safe AI tool shown on the board, or a simulated AI card set of pre-generated outputs. Students do not log in and enter no personal data; the teacher runs every AI step, keeping the Lab account-free.

Facilitation steps:

1. Find where your build stopped and finish every step so the solution runs from input to output.
2. Prepare a first test input with no names, faces or personal details, run the process step, and record the output.
3. Test again with a different input, including one awkward or edge-case input, and record that output too.
4. Score each output against your success criteria and verify at least one claim against a trusted source.
5. Refine one part, re-run, and add a new row to your evaluation log: input, output, criteria met, change, effect.
6. Repeat the test-evaluate-refine-log cycle until your solution meets the criteria across more than one input.

Unplugged fallback: If no tool is available, run the simulated AI card set: use a lookup table as the process and swap several input cards, including a tricky one, to generate different outputs. Pairs log and refine exactly as they would with a tool, so the Lab never depends on tool access.

Lab tool integration (practical, hands-on)

Tool: Scratch (build) with optional Teachable Machine model

Teacher setup: Open each group's Scratch project from L7; have the unit model available to connect if used.

What students do:

1. Build the workflow following your plan and block storyboard.
2. Connect or simulate the model at the process stage if your solution uses one.
3. Run it on a normal input and confirm the output.
4. Run a tricky input; note the failure and handle it (add a check or fallback).
5. Log one improvement you made from an evaluation.

Students produce: A working solution plus a build-and-test log with one logged fix.

Data-safety: Teachable Machine trains in the browser and the samples stay on the teacher's device. Use objects, hand gestures or drawings only - never faces, voices or names. Clear the project at the end of the lesson. Only the non-personal model is used.

Unplugged fallback: Build with block-cards and run by hand on printed inputs.

A build-and-test log: normal run, tricky input, the fix that was added.

Formative assessment (aligned to the objectives)

Task / item Model response Marking note

Show your evaluation log with at least three logged runs on different inputs.

Run 1 met 2 of 3 criteria; run 2 on a longer input failed to be concise; run 3 after shortening the input met all 3 criteria.

Award if the log shows different inputs, criteria scores, and a change traced to an effect across runs.

Identify one input that broke your solution and explain how you made it respond gracefully.

A blank input gave a confusing output, so I added a step that returns please enter a question, which is a sensible response.

Look for a real edge case, the failure named, and a documented fix - not just a claim it works.

Explain how repeated testing shows your solution is reliable, not just lucky once.

Because it met the criteria on three different inputs, I can trust it works in general, not only on the first one I tried.

Look for the reliability-through-repetition idea and reference to more than one logged run.

- Targets: completing a working solution, logging repeated evaluations against the criteria, and refining across more than one cycle.

- Check: circulate during the extended build and review each pair's completed solution and evaluation log at the close.

- Success criteria: the solution runs end to end, the log shows several tests on different inputs with criteria scores, and improvements are traced to their effects.

- Evidence: the working solution plus an evaluation log with at least three rows showing inputs, outputs, criteria met and changes made.

- Feedback / adjustment: if a pair tested only one input, prompt an edge-case test; if their log lacks effects, prompt them to compare two runs and note what changed.

Differentiation

Criterion Emerging (expected) Approaching Not yet

Completes a working solution

Finishes a solution that runs from input to output on its own

Completes most steps but one gap remains

Solution does not yet run end to end without heavy support

Logs repeated evaluations Logs several tests on different inputs, each scored against the criteria

Logs one or two tests with partial scoring

Keeps no usable log of testing

Refines across cycles Makes and records more than one Makes one improvement or Makes changes that are

improvement, noting each effect records changes without effects

not tested or logged

Representation: provide a printed evaluation-log template with headed columns and a worked example row, and show the test-

evaluate-refine-log cycle as a labelled strip on the board.

Action & expression: pairs may complete the log by writing, by filling a printed table, or by dictating each row to the teacher; the

working solution may be shown by running it or by walking through the steps aloud.

People of Determination: offer a log template with sentence stems and pre-headed columns, reduce the required number of

logged runs to two strong cycles, allow extra build time, and provide a quiet station for testing.

Week 9 - Core | Signature Solution - Refining & Finishing | Interactive lesson:

'Refining & Finishing'

Real-World Connection (UAE): UAE environmental monitoring (for example air-quality and marine-water networks) is refined over time as new data arrives - the same iterate-and-improve habit you apply to your solution this week.

Confidence-score checkpoint (2 min): ask 'What is the confidence score here? Is a

high confidence the same as accuracy?' Reinforce that confidence is per-input sureness and a model can be confidently wrong (link back to the Foundation lesson).

Strand: Integration | Framework ref(s): AIF·CU·G7·E AIF·SD·G7·E AIF·CE·G7·E AIF·GE·G7·E | 1 x 45 min

Objectives & success criteria

- Finalise the signature workflow solution so it meets every agreed success criterion.
- Evaluate the finished solution for accuracy, usefulness and risk against the shared criteria.
- Explain and justify one refinement that improved the solution.

Student-facing success criteria:

- I can finish my input-process-output workflow so it meets our success criteria.
- I can check my solution for accuracy, usefulness and risk.
- I can explain one change I made that improved it, and why it helped.

Applied numeracy (refinement). Quantify each improvement: record accuracy as $\text{correct} \div \text{total}$ before and after

a change (e.g. before $7 \div 10 = 70\%$; after adding two examples $9 \div 10 = 90\%$), so 'better' is evidenced by a number, not a feeling.

Prior knowledge & Prerequisite Primer (10 min)

Assumes: In earlier Term-1 lessons you built a simple three-step AI workflow (input, process, output) and drafted the

success criteria for it. In Grade 6 you learned to give a machine clear instructions and to notice when an output looks wrong.

Activate: Show two versions of the same workflow output on the board - an early rough one and a later improved one - and ask students to point to exactly what got better.

Teach: Remind students that refining means changing one thing at a time and re-checking it against the success criteria, not rebuilding the whole thing.

Apply: In pairs, students list their own three success criteria and tick which are already met.

Check: Each pair names one criterion not yet met and one small, testable change that could meet it; the teacher listens for a specific change, not 'make it better'.

Critical & analytical thinking focus

Evaluate & justify - quality and risk against criteria (Evaluate | DOK 3 | SOLO: Relational to Extended abstract)

The technical concept (teacher background)

Refining an AI solution is a controlled iteration loop: change one variable - usually the prompt, the input, or one process

step - re-run the workflow, then evaluate the output against fixed success criteria. Good evaluation separates three

qualities. Accuracy: is the output correct and verifiable against a trusted source. Usefulness: does it actually complete the

intended task. Risk: could it mislead, exclude or expose anyone. A confident-looking output can still be wrong, so verify-before-trust is the habit that matters most. A final risk check asks what happens if the output is wrong and who could be affected. Because students never enter personal data and never hold accounts, refinement here is about prompt and process quality, and every AI step stays teacher-operated on a school-approved tool.

Lesson plan

Phase | time Teacher (I do / we do) Students Key questions (targeted thinking)

Hook / 6 min Show the rough vs improved output pair; model naming one concrete improvement using the success criteria.

Compare the two outputs and name what changed and why it is better.

- Which success criterion does this change help meet?

Set the criteria /

8 min

We-do: rebuild the shared success-criteria checklist on the board and define accuracy, usefulness and risk.

Copy the checklist and mark which criteria their own solution already meets.

- How would we know, with evidence, that this criterion is met?

Refine / 14 min Circulate; coach each pair to change one thing at a time and re-check.

Operate any approved AI step at the board on request.

Make one targeted refinement, re-run their workflow (or the unplugged version) and record the before and after.

- What single change are you testing, and what result do you expect?

Evaluate and risk check / 12 min

Lead the accuracy-usefulness-risk check; model verify-before-trust

against a trusted source.

Score their output against each criterion and complete a short risk check.

- If this output were wrong, who could be affected and how would you catch it?

Justify and close

/ 5 min

Cold-call pairs to justify one improvement citing a check; preview the Lab showcase.

State one change that improved the solution and the evidence it worked.

- Which check proves your change actually helped?

Anticipated misconceptions

Misconception How to address it

Making more changes at once means I improve it faster. Show that changing several things at once hides which change

helped or hurt; require one change, then a re-check against the criteria before the next.

If the AI output sounds confident and neat, it must be correct.

Demonstrate a fluent but wrong output and check it against a trusted source; reinforce verify-before-trust as the rule, not the exception.

Low risk just means the tool did not break. Reframe risk as impact on people - could the output mislead,

exclude or expose someone? A working tool can still produce a risky output.

Hands-on activity guide

Overview: Pairs run one final refinement cycle on their signature workflow, then evaluate the finished output against the shared success criteria and complete a risk check, finishing with a one-line justification of what improved.

Grouping / timing: Pairs for the refine-and-evaluate cycle; whole class for the criteria set-up and the closing justifications.

Tool: A teacher-operated, school-approved AI step run at the board on request (or a pre-generated output set), plus a paper

success-criteria checklist - account-free, so no student logs in and no personal data is entered.

Facilitation steps:

1. Read your three success criteria aloud and tick the ones your solution already meets.

2. Choose ONE thing to change - the prompt wording, the input, or one process step.
 3. Re-run the workflow (your teacher operates any AI step) or run the unplugged version, and keep both the before and after output.
 4. Check the new output for accuracy against a trusted source - do not trust it just because it sounds confident.
 5. Complete the risk check: could this output mislead, exclude or expose anyone, and how would you catch it if it were wrong?
 6. Write one sentence - the change I made, and the check that proves it improved the solution.
- Unplugged fallback: Use the simulated-AI card routine - a pre-written set of rough and refined output cards - so pairs can run the full refine, evaluate and risk-check cycle on paper with no tool access.

Formative assessment (aligned to the objectives)

Task / item Model response Marking note

Tick which of your three success criteria are met and give one piece of evidence for each met criterion.

Criterion 2 (output is easy to read) met - evidence: a classmate read it aloud with no stumbles. Criterion 3 not yet met.

Look for evidence tied to the criterion, not just a tick; a 'met' claim needs observable proof.

Name one change you made and the check that shows it helped.

I shortened the prompt to one clear instruction; the re-run output answered the actual task, which the first one missed - checked against the task sheet.

Award if the change is specific AND linked to a check or evidence, not just 'it looks better'.

Complete the risk check: if your output were wrong, who could be affected and how would you catch it?

If the summary were wrong, a reader could act on false information; I catch it by checking the key facts against the source text before sharing.

Look for a named who plus a concrete catch method (verify against a trusted source).

• Targets: the Integration outcome - finalise against the success criteria and evaluate for accuracy, usefulness and risk (AIF·CU·G7·E, AIF·SD·G7·E, AIF·CE·G7·E, AIF·GE·G7·E).

- Check: pairs score their finished output against the shared criteria and justify one refinement in the closing cold-call.
- Success criteria: the student finishes the solution to meet the agreed criteria, evaluates it for accuracy, usefulness and risk, and explains one improving change.
- Evidence: the completed criteria checklist, the before/after output pair, and the one-sentence justification.
- Feedback / adjustment: if a pair cannot cite a check, re-model verify-before-trust on one criterion and have them re-test that single change before the Lab.

Differentiation

Criterion Emerging (expected) Approaching Not yet
Meets success criteria Finishes the solution so all agreed criteria are met, with evidence.

Meets most criteria;
evidence is partial.

Several criteria unmet or
unchecked.

Evaluates accuracy,
usefulness and risk

Judges the output on all three and
completes the risk check.

Judges one or two of the
three.

Judgement is 'looks fine'
with no check.

Explains an improving
change

Explains one change and cites the
check that proves it helped.

Names a change but the
justification is vague.

Cannot identify a change
or its effect.

Representation: provide the success criteria as a picture checklist and keep the before/after outputs side by side; pre-teach accuracy, usefulness and risk with one icon each.

Action & expression: allow the justification to be spoken, recorded or written; offer a sentence frame - 'I changed ... and I know it helped because ...'.

People of Determination: give a reduced two-criterion checklist, extra time and a partner reader; permit a verbal risk check scribed by the teacher.

Week 9 - Lab | Signature Solution - Refining & Finishing (Lab) | Interactive lesson:

'Lock It & Rehearse'

Strand: Integration | Framework ref(s): AIF·CU·G7·E AIF·SD·G7·E AIF·CE·G7·E AIF·GE·G7·E |
1 x 45 min

Objectives & success criteria

- Run a final iteration of the signature workflow and version-lock the solution to be showcased.
- Rehearse the showcase demonstration, narrating the input-process-output steps and the checks run.
- Justify why the finished solution is trustworthy and low-risk, citing specific checks.

Student-facing success criteria:

- I can run one final iteration and choose the version I will show.
- I can rehearse my demo and explain each step of my workflow clearly.
- I can justify that my solution is trustworthy and low-risk using the checks I ran.

Prior knowledge & Prerequisite Primer (10 min)

Assumes: In the Core period you finished your workflow against the success criteria and ran an accuracy, usefulness and risk check. You have a before/after output pair and one recorded improvement.

Activate: Play a 30-second model demo (teacher-led or recorded) of a workflow being shown well, then ask students to name the parts that made it clear.

Teach: Teach the demo spine - state the task, show the input, show the process, show the output, name one check you ran - all in under a minute.

Apply: In pairs, students draft their five demo sentences using that spine.

Check: Each pair does a 20-second dry run of just the opening two sentences; the teacher checks the task and input are stated clearly.

Critical & analytical thinking focus

Evaluate & justify - quality and risk against criteria (Evaluate | DOK 3 | SOLO: Relational to Extended abstract)

The technical concept (teacher background)

A showcase is where a solution is judged in public, so the final iteration focuses on reproducibility and clear

communication rather than new features. Version-locking means choosing one output you can reproduce on demand and

not changing it again - late changes reintroduce untested risk. A good demonstration narrates the workflow as input,

process and output, then shows the evidence: which checks were run and what they proved.

Justifying trustworthiness

means pointing to specific checks - accuracy verified against a trusted source, and a risk check naming who could be

affected - not simply asserting that the tool works. All AI steps stay teacher-operated and account-free, and no personal

data appears in the demo, so the showcase itself models safe practice.

Lesson plan

Phase | time Teacher (I do / we do) Students Key questions (targeted thinking)

Model demo / 6

min

Play a short model showcase; name the demo spine - task, input, process, output, one check.

Watch and list the parts that made the demo clear.

- Which single check made the demo feel trustworthy?

Final iteration /

12 min

Coach pairs through ONE last iteration, then version-lock; operate any approved AI step at the board.

Run one final change, confirm it meets the criteria, then lock the version they will show.

- Is this change worth the risk of re-testing, or should you lock now?

Build the demo /

12 min

We-do a sentence frame for the five-line demo; circulate to tighten wording.

Write and stage their five demo sentences with the before/after evidence to hand.

- Where in your demo do you show the check, not just the result?

Rehearse / 10

min

Run timed pair rehearsals; give one glow and one grow per pair.

Rehearse the full demo twice, swapping the speaker and checker roles.

- Can a listener restate your workflow and your check after one viewing?

Justify and lock / Cold-call the trustworthiness State why the solution is - Which two checks prove your

5 min justification; confirm each pair has a

locked version.

trustworthy and low-risk,

citing two checks.

solution is low-risk?

Anticipated misconceptions

Misconception How to address it

A better demo means adding more features right before we show it.

Show how a last-minute feature reintroduces untested risk;

require a version-lock and rehearsal instead of new building.

Rehearsing is just reading the output aloud. Model narrating the workflow and the checks; a demo explains

input-process-output and shows evidence, not only the final answer.

Saying 'it works' proves the solution is trustworthy. Require named checks - accuracy against a source and a risk

check - as the evidence; an assertion is not a justification.

Hands-on activity guide

Overview: Pairs run one final iteration and version-lock their solution, build a five-sentence demo that narrates the

workflow and its checks, then rehearse it twice and deliver a trustworthiness justification.

Grouping / timing: Pairs for iteration and rehearsal; brief whole-class rounds for the model demo and the closing justifications.

Tool: A teacher-operated, school-approved AI step at the board (or the reproducible pre-generated output for the locked

version), plus paper demo cards - account-free, no student login, no personal data in the demo.

Facilitation steps:

1. Run ONE final iteration and check it still meets every success criterion.
2. Version-lock: choose the output you can reproduce and agree not to change it again.
3. Write five demo sentences - the task, the input, the process, the output, and one check you ran.
4. Lay out your before/after evidence so you can point to what improved.
5. Rehearse the full demo twice, swapping who speaks and who runs the check.
6. Finish with your justification: why the solution is trustworthy and low-risk, citing two checks.

Unplugged fallback: Run the whole showcase on paper using the reproducible output cards and demo cards - pairs

rehearse the same five-sentence demo and justification with no live tool needed.

Lab tool integration (practical, hands-on)

Tool: Scratch + pre-generated comparison packs (refine)

Teacher setup: Have each solution ready and a small printed test-set of inputs plus a comparison pack of before/after outputs.

What students do:

1. Run your solution across the printed test-set and count how many outputs are correct.
2. Compare a before and after output from the pack and name what improved.
3. Make one final refinement to raise accuracy or usefulness.
4. Run a final risk check: confirm no personal data is used.
5. Rehearse a 60-second showcase of one design choice.

Students produce: A test-set accuracy count, a refinement note and a completed risk check.

Data-safety: No personal data; risk check is part of the task.

Unplugged fallback: Use printed inputs/outputs and tally by hand.

Before and after outputs compared, with an accuracy tally and a risk-check tick.

Formative assessment (aligned to the objectives)

Task / item Model response Marking note

Write your five demo sentences (task, input, process, output, one check).

Task: summarise a news paragraph. Input: the paragraph. Process: an AI summarise step run by our teacher. Output: a two-line summary. Check: we compared the key facts to the source.

Award if all five parts are present and the check is a real, named check.

Version-lock statement: which output are you showing and why that one?

We are showing the second version because it met all three criteria and we can reproduce it; the later draft was untested.

Look for a reproducible choice plus a reason tied to the criteria, not just 'the newest'.

Give your trustworthiness justification citing two checks.

It is trustworthy because we verified the facts against the source (accuracy) and our risk check found only low impact with a way to catch errors before sharing.

Award if trustworthiness is argued from two specific checks, not asserted.

- Targets: the Integration outcome at showcase - finalise, evaluate for accuracy, usefulness and risk, and justify trustworthiness

(AIF·CU·G7·E, AIF·SD·G7·E, AIF·CE·G7·E, AIF·GE·G7·E).

- Check: a timed pair rehearsal plus a cold-called trustworthiness justification citing two checks.

- Success criteria: the student runs a final iteration, version-locks, demos the workflow clearly, and justifies it is trustworthy

and low-risk from named checks.

- Evidence: the five-sentence demo script, the version-lock statement, and the recorded or observed justification.
- Feedback / adjustment: if a justification only asserts 'it works', prompt for the specific accuracy and risk checks and have the pair re-run the closing line before showcase day.

Differentiation

Criterion Emerging (expected) Approaching Not yet

Final iteration and version-lock

Runs one final iteration, confirms the criteria, and locks a reproducible version.

Locks a version but the criteria check is partial.

Keeps changing; no locked, reproducible version.

Clear demonstration Narrates task, input, process, output and one check in a tight demo.

Covers most parts; the check is missing or unclear.

Reads the output only; the workflow is not explained.

Justifies trustworthiness Justifies low-risk and trustworthy citing two specific checks.

Cites one check or the link is vague.

Asserts 'it works' with no check.

Representation: provide the five-sentence demo spine as a printed frame with one icon per line; show a captioned model demo before pairs draft.

Action & expression: let pairs deliver the demo live, as a recording, or read from cards; offer a justification frame - 'It is trustworthy because ... and low-risk because ...'.

People of Determination: allow a three-sentence demo, a partner co-presenter, extra rehearsal time and a teacher-scribed justification.

Week 10 - Core | Showcase, Assessment & Reflection | Interactive lesson:

'Showcase & Reflection'

Strand: Integration | Framework ref(s): AIF·CU·G7·E AIF·SD·G7·E AIF·CE·G7·E AIF·GE·G7·E |
1 x 45 min

Objectives & success criteria

- Present the finished AI workflow and justify at least two design or evaluation choices.
- Demonstrate the term's key outcomes across all four strands in a short sampling task.
- State one clear thing learned this term and set one specific goal for next term.

Student-facing success criteria:

- I can show my input-process-output workflow and explain two choices I made and why.
- I can answer a short task that samples what I learned about building, checking and using AI safely.
- I can name one thing I learned this term and write one goal for next term.

Prior knowledge & Prerequisite Primer (10 min)

Assumes: Across this term you built a simple multi-step AI workflow with an input, an AI process step, and an output, and

you learned to check the output before trusting it. In Grade 6 you described how a machine follows steps to turn an input

into an output.

Activate: Show the class the input-process-output diagram from Week 1 and one anonymised sample workflow on the board. Ask:

which part is the AI step, and how did we know the output was good enough?

Teach: Remind students that a showcase is not just showing the result - it is explaining the choices behind it. Model one sentence

starter on the board: I chose this input because... and I checked the output by...

Apply: In pairs, students each say one design choice they made in their own project and one way they checked the output, using the sentence starters.

Check: Cold-call two pairs. A rung is secure if a student can name a specific choice and a specific check, not just I made a robot that works.

Critical & analytical thinking focus

Evaluate & communicate - justify and reflect (Evaluate | DOK 3 | SOLO: Extended abstract (stretch))

The technical concept (teacher background)

This lesson integrates the term rather than adding a new idea. The real skill being assessed is end-to-end evaluation:

judging a whole input-process-output workflow, not one part in isolation. A strong showcase makes reasoning visible - why

an input was chosen, how the AI step behaves, how the output was verified against the task and a trusted source, and what

the known limits are. Reflection here is metacognition: students name what they learned and where they are still unsure,

which is what turns a one-off project into transferable skill. Assessment samples all four strands - conceptual

understanding, solution design, critical evaluation, and ethics and governance - so no single strong part hides a weak one.

Keep all AI teacher-mediated and account-free; the evidence is the student's explanation, not the tool.

Lesson plan

Phase | time Teacher (I do / we do) Students Key questions (targeted thinking)

Hook / 5 min Show one anonymised workflow and

think aloud: here is the input, the AI step, the output, and how I checked it.

Watch and identify the three parts of the workflow on their own sheet.

- Which part is the AI process step?

Showcase setup

/ 8 min

Explain the showcase format and share the two-choice rule: every presenter names two design or evaluation choices. Model a 30-second example.

Prepare their board demo or paper artefact and jot the two choices they will justify.

- What is one choice you will explain and why?

Showcase round

/ 14 min

Run short showcases; keep time; prompt each presenter to state their two choices and one verify step. No personal data shown.

Present for about one minute each; audience notes one strength and one question per presenter.

- How did you check the output was trustworthy?

Sampling task /

12 min

Give the four-item task that samples the strands. Circulate; do not coach

Complete the short task

individually, applying

- Which strand does this item test?

answers. vocabulary from the term.

Reflection and

close / 6 min

Model one learned-goal pair; collect reflection slips as exit evidence.

Write one thing learned and one specific goal for next term on the slip.

- Is your goal specific enough to act on next week?

Anticipated misconceptions

Misconception How to address it

A good showcase just means the output looks impressive. Reframe the success criteria: marks come from explaining

choices and verification, not from a flashy result. Ask each presenter to name a limitation of their workflow too.

If my project worked once, the AI is always right. Point back to verify-before-trust: one good run is not proof. Ask for

a case where the same workflow could give a wrong or unfair output, and how a check would catch it.

Reflection means saying it was fun or it was hard. Model the difference between a feeling and a learning: I learned

that a clearer input gives a more useful output is a learning; I found it fun is not. Require a specific, testable goal.

Hands-on activity guide

Overview: Students run a short showcase of their finished workflows, then complete a four-item task that samples all four

strands, then write a structured reflection and next-term goal.

Grouping / timing: Individuals or original project pairs for the showcase; whole class for the sampling task and reflection debrief.

Tool: Teacher-operated board demo or each student's own paper or slide artefact from the term.

Any live AI is school-

approved and teacher-run only, on non-personal example data; the assessed evidence is the student's explanation, so no student accounts are needed.

Facilitation steps:

1. Set out your workflow as input, AI process step, and output, using your project artefact.
2. Choose the two design or evaluation choices you will justify and write them on your card.
3. Present for about one minute: show the three parts and explain your two choices and one verify step.
4. As an audience member, note one strength and one question for each presenter.
5. Complete the four-item sampling task on your own, one item per strand.
6. Write your reflection: one thing you learned and one specific goal for next term.

Unplugged fallback: If no tools are available, students present entirely from a paper storyboard of their workflow and use a simulated-AI card set (pre-written inputs and outputs) to demonstrate the AI step and a verify check, with no device needed.

Formative assessment (aligned to the objectives)

Task / item Model response Marking note

Point to the AI process step in your workflow and say what it takes in and gives out.

The AI step takes my chosen input, for example a short description, and produces the output, for example a sorted list; the input and output are not the AI step itself.

Correct if the student separates the process step from input and output; a common slip is calling the whole project the AI.

Name one way you checked your output and one output you would not trust yet.

I compared the output to the task and to a trusted source; I would not trust an output that used personal data or that I could not verify.

Look for a real verification method plus a limit; reward verify-before-trust language over I just trusted it.

Write one learning from this term and one specific goal for next term.

Learning: a clearer input gives a more useful output. Goal: next term I will test each workflow with two different inputs before I trust it.

Goal must be specific and actionable, not I will do better; learning must be a statement about how the workflow behaves.

- Targets: the four strand outcomes for the term (CU, SD, CE, GE) evidenced through showcase, sampling task and reflection.
- Check: one-minute showcase with the two-choice rule, a four-item strand-sampling task, and a structured learned-goal reflection slip.
- Success criteria: student justifies two design or evaluation choices, names one verify step and one limit, and writes a specific

next-term goal.

- Evidence: presenter cards, completed sampling tasks, and reflection slips collected at the door.

- Feedback / adjustment: return one strand focus to each student for next term; re-teach verify-before-trust to any student who trusted an unchecked output.

Differentiation

Criterion Emerging (expected) Approaching Not yet

Justify design choices Explains two clear choices with reasons

Explains one choice with a reason

Describes the result but gives no reason

Evaluate the output Names a verify step and one real limit

Names a verify step but no limit

Assumes the output is correct

Reflect and set a goal One clear learning and one specific goal

A learning or a goal, but vague

A feeling only, no goal

Representation: provide the input-process-output diagram, a sentence-starter card (I chose... because...; I checked... by...), and

the strand labels so students can see which item tests which strand.

Action & expression: let students showcase by speaking, by a paper storyboard, or by a labelled slide; accept the sampling task as

spoken answers to the teacher where writing is a barrier.

People of Determination: allow extra presentation time, a printed script with blanks to complete, and a partner to hold the artefact;

reduce the sampling task to the two most secure strands while keeping the reflection.

Week 10 - Lab | Showcase, Assessment & Reflection (Lab) | Interactive lesson:

'Peer Review & Stress-Test'

Strand: Integration | Framework ref(s): AIF·CU·G7·E AIF·SD·G7·E AIF·CE·G7·E AIF·GE·G7·E | 1 x 45 min

Objectives & success criteria

- Give and receive structured peer feedback on a finished AI workflow using shared success criteria.

- Test the workflow against a tougher or higher-stakes case and record what changes are needed.

- Consolidate the term's reflections into one focused, measurable goal for next term.

Student-facing success criteria:

- I can review a peer's workflow against the criteria and give one strength and one specific improvement.
- I can run my workflow on a harder case and record where it holds up and where it fails.
- I can turn my term's reflections into one clear, measurable goal for next term.

Prior knowledge & Prerequisite Primer (10 min)

Assumes: In the Core period you showcased your workflow and justified two choices, and across the term you learned to verify outputs before trusting them. You know the input-process-output structure of your own project.

Activate: Post the shared success criteria from the Core rubric on the board and show one anonymised peer-review comment - a weak one (good job) and a strong one (your input was clear, but I would test it with a second example). Ask which is more useful and why.

Teach: Model the two-star-and-a-step feedback frame: two specific strengths tied to a criterion, one concrete step to improve.

Stress that feedback is about the workflow, never the person.

Apply: In pairs, students practise the frame on the anonymised sample workflow before the carousel starts.

Check: Listen for feedback that names a criterion and gives an actionable step. Redirect any comment that is only praise or is about the person, not the work.

Critical & analytical thinking focus

Evaluate & communicate - justify and reflect (Evaluate | DOK 3 | SOLO: Extended abstract (stretch))

The technical concept (teacher background)

The Lab deepens practice through structured peer review and stress-testing, not a new outcome. Peer review works when it is criterion-referenced: reviewers judge against the same success criteria the class used, which makes feedback specific and fair rather than a popularity vote. Stress-testing is the practical heart of critical evaluation - running a workflow on a harder or higher-stakes case (more ambiguous input, an edge case, a setting where a wrong output causes real harm) exposes limits that a single happy-path run hides. Students learn that robustness is designed, not assumed: a workflow that passes one easy case may need an extra verify step, a clearer input, or a human check before it is trusted more widely.

Consolidation then narrows many term reflections into one measurable goal. All AI remains teacher-mediated and account-free, tested only on non-personal example data.

Lesson plan

Phase | time Teacher (I do / we do) Students Key questions (targeted thinking)

Hook / 5 min Show a weak and a strong peer comment side by side; think aloud about why the specific one helps.

Decide which comment they would rather receive and note one reason.

- What makes feedback usable rather than just kind?

Model the frame

/ 7 min

Model two-star-and-a-step against the shared criteria on an anonymised workflow.

Practise the frame in pairs on the same sample before the carousel.

- Which criterion does your star point to?

Peer-review

carousel / 15 min

Run stations; each workflow gets timed review; enforce work-not-person and no personal data.

Rotate through stations, leaving one written star-star-step note per workflow reviewed.

- Is your improvement step concrete enough to act on?

Stress-test / 12

min

Give a harder or higher-stakes case; ask students to run their workflow and log where it holds or fails.

Run their workflow on the tougher case and record what breaks and what extra

- What safeguard would matter more in a high-stakes setting? check is needed.

Consolidate goal

/ 6 min

Help students merge peer feedback and stress-test findings into one

measurable goal; collect goals.

Write one measurable Term-

2 goal drawn from the

feedback and the stress-

test.

- How will you know next term if you have met this goal?

Anticipated misconceptions

Misconception How to address it

Peer feedback means saying nice things so nobody feels bad.

Reframe feedback as help for the work: model that a specific improvement step is the kindest thing to give. Require every note to name a criterion and one concrete step.

If my workflow passed the easy case, it will pass any case. Use the stress-test result: show that a harder or edge case often breaks it. Robustness is designed by adding checks, not assumed from one good run.

A good goal is just to get better at AI. Contrast vague and measurable goals: I will test each workflow

with two different inputs and log the result is measurable; get better is not. Require a way to know it is met.

Hands-on activity guide

Overview: Students run a criterion-referenced peer-review carousel, then stress-test their own workflow on a harder or

higher-stakes case, then consolidate all their feedback into one measurable Term-2 goal.

Grouping / timing: Rotating stations of three to four for the carousel; individuals for the stress-test and goal-setting.

Tool: Printed shared success criteria plus each student's own workflow artefact. Any live AI step is teacher-operated, school-approved and run on non-personal example data only; reviewers assess the artefact and its explanation, so no student accounts or personal data are used.

Facilitation steps:

1. Read the shared success criteria before you review anything.
2. At each station, use the two-star-and-a-step frame: two strengths tied to a criterion, one concrete improvement.
3. Leave your written note face up for the owner; keep it about the work, not the person.
4. Back at your own workflow, run it on the harder or higher-stakes case you are given.
5. Log where the workflow holds up, where it fails, and one extra check or safeguard it needs.
6. Merge your peer notes and your stress-test log into one measurable goal for next term.

Unplugged fallback: If no tools are available, run the whole carousel and stress-test on paper: reviewers annotate printed

workflow storyboards, and the harder case is a pre-written simulated-AI card whose output students must judge against the task by hand - no device needed.

Lab tool integration (practical, hands-on)

Tool: Showcase the Scratch / Teachable Machine solution + peer-review cards

Teacher setup: Set up a showcase carousel; print peer-review cards (two stars and a wish, strand-tagged).

What students do:

1. Demonstrate your solution's input -> process -> output to a visiting group.
2. Explain two design or evaluation choices you made.
3. Visit two other solutions and complete a peer-review card for each.
4. Collect your peer cards and read the feedback.
5. Write one Term-2 goal based on the feedback.

Students produce: A demonstrated solution, completed peer-review cards and a Term-2 goal.

Data-safety: Showcase uses only the non-personal solution; no data shared.

Unplugged fallback: Present the paper/block-card version of the solution.

A showcase carousel with a peer-review card: two stars and a wish, strand-tagged.

Formative assessment (aligned to the objectives)

Task / item Model response Marking note

Write one star-star-step note on a sample workflow using the shared criteria.

Star: your input is clear and matches the task. Star: you named a verify step. Step: test it with a second, messier input and log the result.

Both stars must tie to a criterion and the step must be concrete and actionable; reject notes that are only praise or aimed at the person.

Run your workflow on the harder case and record one thing that failed and one fix.

On the messier input the output was wrong; fix: add a check comparing the output to a trusted source before trusting it, and flag it for a human review.

Look for an honest failure plus a designed safeguard; reward it holds on easy cases but needs a check here over it still works.

Turn your feedback into one measurable Term-2 goal and say how you will know it is met.

Goal: test every workflow with two different inputs and log both outputs; met when I have a log for each project next term.

Goal must be measurable with a clear check of success; reject vague aims like get better at AI.

- Targets: the four strand outcomes for the term, deepened through criterion-referenced peer review, stress-testing and goal consolidation.
- Check: written star-star-step peer notes, a stress-test log on a harder or higher-stakes case, and one measurable consolidated goal.
- Success criteria: student gives criterion-linked feedback with a concrete step, records an honest failure plus a designed safeguard, and sets a measurable goal.
- Evidence: peer-review notes, completed stress-test logs, and consolidated goal slips collected at the end.
- Feedback / adjustment: pair students whose stress-tests exposed the same weakness for a targeted Term-2 start; re-teach the feedback frame to any student whose notes stayed vague.

Differentiation

Criterion Emerging (expected) Approaching Not yet

Give peer feedback Two criterion-linked strengths and one concrete step

One strength and a step, loosely linked

Only praise or comments about the person

Stress-test the workflow Records an honest failure and a designed safeguard

Records a failure but no fix Claims it works with no harder test

Set a measurable goal Goal is measurable with a clear success check

Goal is specific but hard to measure

Goal is vague, like get better

Representation: provide the printed success criteria, a star-star-step note template, and a two-column stress-test log (holds up / fails and fix) so the deeper task is scaffolded.

Action & expression: accept spoken peer feedback captured by a scribe, and let the stress-test log be recorded as annotations on

a printed storyboard rather than prose.

People of Determination: give a partially completed note template with sentence stems, allow a reader-writer partner in the

carousel, and let the student stress-test one case rather than several while still setting a goal.

Standards & references

This pack is designed against the following international standards for curriculum and assessment quality.

Reference Used for

ADEK (2026). K-12 AI Fluency Framework (Draft V4). Source of all learning outcomes, strands, tiers and codes.

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Biggs & Collis (1982). SOLO Taxonomy. Structure of observed learning; anchors success criteria.

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Grade 7 (Year 8) | Phase 3 / Cycle 2 | AI Solution Builder

Main Course, Term 1 - AI as Workflows - Input, Process, Output

Emerging tier | 10 weeks (Core + Lab)

Staying safe with AI in this course

- We never enter personal data - no names, faces or voices. We use objects, words, drawings or

neutral text only.

- The teacher runs any live AI tool. You direct it, but you do not log in or make accounts.

- We always check the output. AI can be confidently wrong, so we verify every result against the task

and a trusted source.

- We clear the tool at the end - trained models and projects are deleted after the lesson.

- If something looks wrong, unexpected or unfair, tell your teacher. That is exactly what a responsible

AI user does.

Foundation - How Models Learn (read before Week 1)

- I can say whether a system follows fixed rules or has learned from examples, and why.
- I can explain the difference between training a model and using it (inference).
- I can explain that a model stores learned patterns, not saved examples, and that a classifier repeats

the same answer for the same input.

- I can say what a train/test split is and read a confidence score.

Words to know: rule-based system - does exactly what a person wrote; learned model - adjusts to

many labelled examples; training - the one-off teaching phase; inference - using the trained model;

parameters - the patterns a model stores (not copies of the examples); confidence score - how sure

the model is about THIS answer; train/test split - holding back some labelled items to test on unseen

data.

Rules or learning? A rule-based system follows fixed rules a person wrote (if metal -> Recycle).

A

learned model is shown many labelled examples and adjusts inside until it predicts well - nobody

writes the rule by hand. Many real systems mix both.

Training vs using. Training teaches the model once from labelled examples. After that, every time you

use it (inference) it labels a new input without changing. It stores patterns, not your examples - that is

why it can answer about new inputs but cannot show you the originals.

Same input, same answer. A classifier gives the same output and the same confidence for the same

input. A confidence score like 0.90 tells you how sure it is about this one answer - it is NOT the same

as accuracy, and a confident answer can still be wrong. To check a model really learned, we test it on

2

unseen items we held back earlier - this is called held-out (or test) data - rather than the examples it

trained on (a train/test split). Held-out data matters because testing on items the model already trained on would make it look better than it really is.

Let's try it: sort the Deck A cards into 'follows fixed rules' and 'learns from examples', and say why.

Then take 12 labelled cards, hide 3, 'train' on 9 and test on the hidden 3 - how many did you get right?

My reflection: one new thing I understand about how models learn is ...

STUDENT LESSON CONTENT

3

Week 1 - Core | Unit Launch - AI as Workflows - Input, Process, Output

I can...

- I can ask at least two good questions about how AI turns an input into an output.
- I can sort examples into input, process and output and say why.
- I can help write our unit success criteria and our safe, responsible-use rules for AI.

Words to know

- input - what you give the system to start, like a question or a photo
- process - the steps inside that change the input into something new
- output - what the system gives back to you at the end
- model - the part of an AI that has learned patterns from lots of examples
- responsible use - using AI safely and fairly, with no personal data and always checking the result

Today we launch a new unit. AI tools follow a shape you already know from everyday machines: something goes

in, something happens, and something comes out.

AI is a workflow

Think about a vending machine. You choose a drink (input), the machine works inside (process), and your drink

drops down (output). AI works the same way. You give it something to start, it runs steps using a model that

learned from many examples, and it gives you a result. When you can spot the input, the process and the

output, you can understand almost any AI tool - and later you will build your own simple workflow.

Where this analogy breaks: a vending machine follows fixed rules a person wrote, so it always does exactly the

same thing. An AI model is different - it has learned its patterns from many examples rather than from rules

typed in by hand. Use the vending machine only to see the input-process-output shape, not to explain how the AI

'decides'.

A confident answer can still be wrong

AI outputs can look neat and sure of themselves, but that does not make them true. The model is guessing

patterns, so it can get things wrong. That is why we verify before we trust: we check the output against the task

and a source we trust before we use it. We also keep ourselves safe - no real names, faces or private details go

into any tool, and an adult always runs the tool for us.

4

Every AI tool follows the same shape: input, then process, then output.

Put a number on accuracy

“Accuracy” is not just a word – it is a number you can work out. Accuracy = number correct ÷ total tested,
written as a percentage.

Worked example: a classifier is tested on 12 items and gets 9 right, so accuracy = $9 \div 12 = 0.75$ = 75%.

Your turn: work out the accuracy of two sample output sets, then rank them by the number.

Then read the confidence score the model reports with each answer (a value from 0 to 1 – 0.90 means very sure, 0.55 means unsure). A high confidence score is not a guarantee that the answer is correct, so you still verify against a trusted source.

Why it matters

Why it matters - across the UAE, AI helps with things like translating signs for visitors, sorting recycling, and answering questions at government service desks; spotting the input, process and output helps you understand and question the tools around you.

Let's try it - hands-on

1. Watch the class stimulus and note one thing you noticed and one thing you wonder.
2. Turn your wondering into two clear questions about AI as a workflow.
3. Sort each example card into input, process or output with your partner.
4. Write one reason for each card you sorted.
5. Add one word to the vocabulary wall and suggest one safe-use rule.

Think about it

Think about it - a classifier gives the SAME output every time you give it the same input; that is what makes it a classifier. Only generative AI that samples is designed to vary. So if you re-run a build and the answer changes, what actually changed - the input, or the tool you used?

My reflection

My reflection: One question I still have about AI workflows is ...

5

Week 1 - Lab | Unit Launch - AI as Workflows - Input, Process, Output (Lab)

6

Human in the loop: the AI tool does the sorting or drafting, but a person decides what to give it, checks each output against the task, and makes the final call - especially on any low-confidence result. Notice where the human decides in today's workflow.

I can...

- I can compare two AI examples and write down how they differ.
- I can trace each example through input, process and output.
- I can use what I observed to say what made one output better.

Words to know

- compare - look closely at two things to find how they are alike and different
- difference - something that is not the same between the two examples
- log - a record where you write down what you observe
- evidence - what you actually saw, used to back up what you say
- conclusion - the idea you decide on after looking at the evidence

In the Core period you learned the shape of an AI workflow. Today you become an investigator - comparing real

examples and logging what makes one output better than another.

Same shape, different results

Two AI examples can share the same input-process-output shape but give different results.

Maybe one input

was clearer, or the process handled it differently. When you compare them side by side and trace each through

the three stages, you can spot exactly where the difference starts. That is detective work: you are not guessing

which is better, you are finding evidence for it. Logging what you see turns a feeling into a fact you can explain to someone else.

Log it, then trust it

A good investigator writes things down. In your log you note each difference and the stage it came from - input,

process or output. Then, before you trust a conclusion, you check it against the task: would it still be true for a

new example? Remember the charter - the examples are made by the teacher, with no personal data, and a

confident-looking output still has to be checked before you rely on it.

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Same input, small change, different output - your job is to find where the difference starts.

Why it matters

Why it matters - engineers in the UAE constantly compare AI outputs before they trust them, whether tuning a

smart-traffic system in Abu Dhabi or checking an Arabic-English translation; learning to log differences is the

first step of that job.

Let's try it - hands-on

1. Read the first example pair and predict which output is more useful.
2. Trace each example through input, process and output.
3. Log at least two differences and mark the stage each came from.
4. Rotate to a new pack and log again.
5. Write one conclusion about what made an output better.

Today's tool - hands-on

Tool: Card decks (workflow-example cards)

1. Deal the deck face-up between partners.

2. For each card, agree the input, the process and the output and write them on the card's three blanks.

3. Sort the cards into 'follows fixed rules' and 'learns from examples'.

4. Pick the two hardest-to-place cards and say why.

5. Log at least three differences you noticed between the two piles.

You will make: A sorted card layout plus a difference log.

Stay safe: Fully unplugged; no devices or data.

8

Card sort: each workflow card split into input, process and output, sorted into two piles.

Think about it

Think about it - if you changed only the input and kept everything else the same, could you make the weaker

output better? How would you test it?

My reflection

My reflection: One difference I logged today, and where it came from, was ...

9

Week 2 - Core | Understanding the System - Input, Process, Output

I can...

- I can name the three stages of an AI workflow - input, process and output - in order.
- I can trace a worked example and explain to a partner how the input became the output.
- I can use the words input, process and output correctly when I talk about a solution.

Words to know

- input - the data that goes into the system, such as text, a photo or a number
- process - what happens inside: a trained model or set of rules changes the input
- output - what comes out: a label, a prediction or a new piece of content
- workflow - the stages joined together in order, from input to output
- trace - to follow one piece of data through each stage

Every AI tool you use follows the same simple shape. Today you will find the three stages hiding inside it, so you

can explain how it works.

The three-stage shape

Look inside almost any AI tool and you find the same three stages. First an input goes in - the text you type, the

photo you choose or a number. Then a process changes it - this is the part you cannot see.

Last, an output

comes out - a label, a guess or something new. The arrows only point one way, so the order matters. For a

classifier the same input always gives the same output - it is deterministic; the answer only changes if the input

changes. (Generative tools that sample can vary, but that is a different kind of AI.) Once you can spot the three

stages, you can explain how any tool works.

The process is not magic

The process stage can feel like magic, but it is not. Inside, a model has learned patterns from many past examples, and it uses those patterns to turn your input into a likely output. It does not think or understand - it matches. That is why an input that is unusual or unclear can lead to a strange output. Knowing this helps you check the output instead of trusting it straight away.

One workflow, three stages: input flows into process, process produces output.

Why it matters

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AI apps used across the UAE - from Arabic-English translation to smart parking sensors - all follow input,

process, output. Knowing the shape lets you understand and question the tools around you.

Let's try it - hands-on

1. Look at the worked example your teacher shows on the board.
2. Find what goes in and write it next to Input.
3. Describe what happens inside and write it next to Process.
4. Find what comes out and write it next to Output.
5. Tell your partner how the input became the output, in order.

Think about it

If the input were wrong, would you still trust the output? Why or why not?

My reflection

My reflection: the stage I found easiest to spot was ... because ...

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Week 2 - Lab | Understanding the System - Input, Process, Output (Lab)

12

Human in the loop: the AI tool does the sorting or drafting, but a person decides what to give it, checks each output against the task, and makes the final call - especially on any low-confidence result. Notice where the human decides in today's workflow.

I can...

- I can build a workflow with an input, a process and an output, and trace one item through it.
- I can change the input, compare the two outputs, and explain why the output changed.
- I can point to where an error could enter the workflow and say how I would check the output.

Words to know

- trace - to follow one item of data through input, process and output
- compare - to run different inputs through the same process and see how the output changes
- test - to try an input on purpose to learn how the workflow behaves
- verify - to check an output against a trusted answer before trusting it
- black box - a process whose inside we cannot see, only its inputs and outputs

confidence score - a number the tool gives with its answer showing how sure it is (for example 0.90 = 90%); a

high confidence score is NOT proof the answer is correct

You already know the three stages. Today you become the tester - build a small workflow, then change the input and watch what happens.

Trace one item

Tracing means picking one item and following it all the way through. You watch it go in as the input, see what the process turns it into, and read what comes out as the output. Writing down each stage for the same item keeps

you honest - it stops you from mixing up two different items. A clear trace is how testers prove they know what a

workflow really does, not just what they hope it does.

Black box means you can see the input and the output but not the reason inside. Teachable Machine shows a

confidence score next to each label, but it does not tell you WHY it chose that label - that gap between seeing

the output and seeing the reason is what 'black box' means. It is also why we always verify the output instead of trusting the score.

Change the input, compare the output

Here is the tester's trick: keep the process exactly the same and change only the input. Run it again and

compare the two outputs. If the output changes, that tells you the process is responding to its input - which is

what it should do. A small change in input can cause a big change in output. Before you trust any output, check

it against something you know is correct.

Why it matters

Testers who change inputs on purpose find problems before real users do. This is how teams in the UAE check

an AI service behaves fairly before it is switched on for the public.

Let's try it - hands-on

1. Set up your three stages: input, process and output.
2. Send one item through and write what happens at each stage.
3. Change just the input and send it through again.
4. Compare the two outputs - what changed?
5. Check one output against a trusted answer before you trust it.

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Today's tool - hands-on

Tool: Google Teachable Machine (teacher-operated)

1. As a class, name the three classes (for example: open hand, fist, thumbs-up).
2. Students direct the teacher to add ~15 example images per class using the webcam pointed at objects or hands only.
3. Click Train. Narrate: these examples are the input, training is the process.

4. Test with a new, unseen example and read the confidence output aloud.

5. Trace the full input -> process -> output path on the board and label each stage.

You will make: A trained 3-class model and a completed input-process-output trace.

Stay safe: Teachable Machine trains in the browser and the samples stay on the teacher's

device. Use objects, hand

gestures or drawings only - never faces, voices or names. Clear the project at the end of the lesson.

Teachable Machine: three example classes feed training, which produces a confidence output.

Think about it

The process did not change - so why did the output change?

My reflection

My reflection: changing the input taught me that ...

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Week 3 - Core | Designing & Building a Workflow

I can...

- I can build a simple AI workflow by following guided steps for input, process and output.
- I can complete a hands-on task and explain why my workflow gave the result it did.
- I can use words like input, process, output and workflow correctly when I talk about my solution.

Words to know

- input — what you give the workflow to start it off
- process — the middle step where the AI does its job
- output — the result the workflow gives back to you
- workflow — an ordered set of steps that turns your input into an output
- verify — to check an output against the task and a trusted source before you trust it

Today you become a builder. You will take a task and build a small AI workflow - input, process, output - that

gets it done, then test whether it really works.

Building means putting steps in order

A workflow is just steps in the right order. To build one you decide three things: what goes in (input), what

happens in the middle (the process, done by the AI), and what should come out (output). The order matters -

each box feeds the next one. If you swap two boxes, the workflow breaks. When you build, you are not making

the AI from scratch; you are arranging the steps so the AI step has good input and gives a useful output.

If Scratch misbehaves - quick fixes

- Sprite not responding? Check it has a 'when green flag clicked' (or 'when this sprite clicked') hat block, and

that the script is on the right sprite.

- Variable not updating? Use 'set' or 'change [variable]' inside the running script, and tick the variable's box to

watch its value on the stage.

- Blocks won't snap together? Drag from the top-left of a block; a grey shadow shows where it will connect.

- Nothing happens? Make sure the green flag starts your script and you are editing the sprite whose name is highlighted.

- Still stuck? Save, tell your teacher, and use the unplugged card version so you keep learning. Our simulated model vs a real trained model

In these labs the 'AI step' is often a paper lookup table or a pre-made card. That is a simulated model: it can only

answer items already written on the table. A real trained model is different - it learns patterns from many

examples and can then generalise, giving a sensible answer to an input it has never seen before. Generalisation

to unseen inputs is the difference. When you design your build, know which part is real AI and which part is a stand-in.

A good builder tests before trusting

The first output is a draft, not a fact. Even when the AI sounds sure, it can be wrong. So a good builder runs the

workflow, then verifies: does the output match the task, and does a trusted source agree? If not, you go back

and fix a box - maybe the input was unclear. You also try a slightly different input to see if the workflow still

works. Testing twice is how you know your build is reliable, not just lucky.

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A three-box workflow: your input feeds the AI process, which gives the output.

Why it matters

Across the UAE, people use AI workflows to sort emails, plan deliveries and check documents.

Learning to build

and test one now means you can shape these tools instead of just using them.

Let's try it - hands-on

1. Pick one task from the menu.

2. Write your input in the first box - exactly what you will give it.

3. Put the matching process card in the middle box - this is the AI step.

4. Draw the output the cards give you in the last box.

5. Run it again with a new input, then check both outputs against the answer key and fix any box that is wrong.

Think about it

If two people gave your classifier the exact same input, it should give the same output every time - a classifier is

deterministic. What could make their outputs differ - a different input, a different rule, or using a generative tool

that samples instead?

My reflection

My reflection: The trickiest part of building my workflow was ... because ...

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Week 3 - Lab | Designing & Building a Workflow (Lab)

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Human in the loop: the AI tool does the sorting or drafting, but a person decides what to give it, checks each output against the task, and makes the final call - especially on any low-confidence result. Notice where the human decides in today's workflow.

I can...

- I can build a workflow and test it until it completes my task.
- I can find where my workflow goes wrong and change it to make the output better.
- I can use design words to explain why my steps are in that order and why the workflow is reliable.

Words to know

- iterate — to test, change one thing, and test again until it works
- reliable — the workflow completes the task for more than one input, not just once
- multi-step workflow — a workflow with more than one process step, where the output of one feeds the next
- hand-off — the point where the output of one step becomes the input of the next
- verify — to check an output against the task and a trusted source before trusting it

In the lab you get real build time. You will test the workflow you made, fix its weakest step, and then grow it into

a multi-step workflow that does more than one job.

Improve by changing one thing at a time

The fastest way to improve a workflow is not to redo it all. You test it, find the one box that weakens the output -

often the input - and change only that. Then you run it again and compare. If you change two boxes at once and

it works, you will not know which change helped. Keeping notes of before and after turns guessing into

evidence. This is iteration: a small, controlled loop that makes your build better and lets you explain exactly

why.

Chain steps to do more

A multi-step workflow does more than one job by chaining steps. The output of the first process becomes the

input of the second - that join is the hand-off. It only works if the first step gives the second step something it

can use. So when you test a chain you check two things: does each step complete, and does the hand-off hold.

A workflow is reliable when it completes the task for more than one input, so always test with at least two.

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A multi-step workflow: the output of the first AI step becomes the input of the second.

Why it matters

Real UAE workflows - checking passport photos, routing customer messages, planning deliveries - are almost

always multi-step and are tested before anyone trusts them. Building and testing a chain now is exactly this skill in miniature.

Let's try it - hands-on

1. Run your Core workflow and read the output against the task.
2. Mark the one box that weakens it, then change only that box.
3. Run it again and note the output before and after.
4. Add a second process step so your output becomes the next input.
5. Run the whole chain on two different inputs and verify each output against the answer key.

Today's tool - hands-on

Tool: Scratch (input -> process -> output build)

1. Decide a simple task, e.g. sort a typed word into 'question' or 'statement'.
2. Drag an input block (ask and wait) to capture the input.
3. Build the process: two or three 'if' blocks that apply your rule.
4. Add an output block (say) that reports the result.
5. Run it three times, including one tricky input, and note what happened.

You will make: A working three-stage Scratch project and a short run log.

Stay safe: No personal data in the project; type only neutral example words.

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A Scratch block stack arranged as input, then process (if-blocks), then output.

Think about it

Why is a workflow that works on two different inputs more trustworthy than one that worked only once?

My reflection

My reflection: The change that improved my workflow the most was ... and I know because ...

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Week 4 - Core | Judging & Improving Outputs

I can...

- I can spot where an AI output or a workflow step has gone wrong.
- I can judge an output against agreed criteria and back my decision with evidence.
- I can use words like criteria, evidence and accuracy correctly when I explain my judgement.

Words to know

- criteria - the agreed things you check an output against, like accuracy and clarity
- evidence - the exact word, fact or step you point to that proves your judgement
- accuracy - whether the output is actually correct when you check it against a trusted source
- verdict - your decision about an output: it meets, partly meets or fails a criterion
- verify - to check an answer against a trusted source before you trust it

Today you become an evaluator. Instead of trusting an AI answer because it sounds sure, you will judge it

against agreed criteria and prove your decision with evidence.

Judge with criteria, not gut feeling

When two outputs look different, it is tempting to just pick the one you like. That is not evaluation. First you agree criteria - the things everyone will check for, such as accuracy, relevance and clarity. Then you judge each output against every criterion and give a verdict. Because everyone uses the same criteria, your judgements are fair and someone else can check them. Liking an answer is an opinion; judging it against criteria is evaluation.

Confident is not the same as correct

AI outputs often sound smart and sure. But a model works by predicting likely words, not by knowing the truth, so a neat, confident answer can still be wrong. That is why you verify before you trust: check the key fact against a source you already trust. And when something is wrong, ask where the issue started - a weak input, a faulty process step, or the output itself. Naming where the problem sits helps you improve the whole workflow.

An issue can start at any stage - check the input and the process step, not only the output.

Why it matters

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Across the UAE, people use AI to draft messages, translate and answer questions in seconds - but a wrong figure in a school report or a mistranslation matters. Being able to judge an output before you use it keeps you in charge of the result.

Let's try it - hands-on

1. Agree the three criteria with your class: accuracy, relevance and clarity.
2. Read the task the outputs were meant to do, so you know what good looks like.
3. Judge each output against every criterion and write meets, partly or fails.
4. For every verdict, write one piece of evidence - the exact word, fact or step.
5. Verify one output against a trusted source, then say where any issue began: input, process or output.

Think about it

Think about it: could an output meet the accuracy criterion but still fail the relevance one? What would that look like?

My reflection

My reflection: the hardest output to judge was ... because ...

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Week 4 - Lab | Judging & Improving Outputs (Lab)

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Human in the loop: the AI tool does the sorting or drafting, but a person decides what to give it, checks each output against the task, and makes the final call - especially on any low-confidence result. Notice where the human decides in

today's workflow.

I can...

- I can apply agreed criteria to a larger, messier set of outputs.
- I can rank outputs from best to worst and justify each placement with evidence.
- I can weigh accuracy against usefulness and suggest one way to improve a weak output.

Words to know

- rank - to put outputs in order from best to worst against your criteria
- usefulness - how well an output actually helps someone complete the task
- tie-breaker - the extra reason you use to choose between two outputs that seem equal
- evidence - the exact fact, word or step you point to that justifies a placement
- improvement - a specific change that fixes an issue in the input, process or output

Welcome to the evaluation clinic. Today you do not just judge one output - you compare a whole set, rank them

best to worst, and prove every placement with evidence.

From verdicts to a ranking

Judging one output is a start; a clinic compares many. To rank them, line up your verdicts and ask which output

is stronger against the criteria. Often two are close - one is accurate but thin, another fuller but slightly wrong.

That is when you use a tie-breaker: which one is more useful for the actual task? Because every placement rests

on evidence, your ranking is not just an opinion - someone else could follow your evidence and reach the same order.

Accuracy and usefulness can pull apart

The best output is not always the longest or the most detailed. A short answer that gives the one correct fact can

be more useful than a long one that buries a mistake. So you weigh two things: accuracy (is it correct?) and

usefulness (does it help with this task?). When an output ranks low, do not just blame the output - trace back

through the workflow. Fixing a vague input or a weak process step can lift the whole result.

Why it matters

In UAE workplaces and schools, people often get several AI drafts and must pick one fast. The skill of ranking

outputs and improving the weakest is exactly what keeps the final choice accurate and genuinely useful.

Let's try it - hands-on

1. Lay out the wider set of outputs and re-read the task they were meant to do.
2. Judge each output against accuracy, relevance and clarity.
3. Rank them from best to worst on your ranking strip.
4. Write the evidence beside each placement, using usefulness to break any ties.
5. Take the weakest output, say where the issue began, and propose one improvement.

Today's tool - hands-on

Tool: Pre-generated output packs (evaluation clinic)

1. Agree the evaluation criteria as a class (accuracy, usefulness, clarity).
2. In groups, read each output and mark it correct / unclear / incorrect with a reason.

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3. Rank the pack from strongest to weakest.
4. Choose one weak output and write the one change that would most improve it.
5. Defend your ranking to another group.

You will make: A scored, ranked output pack with reasons and one improvement.

Stay safe: All outputs are teacher-curated in advance; students touch no live tool.

Two sample outputs side by side with a criteria checklist for judging each.

Think about it

Think about it: could improving the input change which output wins - or would you need to fix the process step

instead?

My reflection

My reflection: my ranking changed when I noticed ... , which taught me ...

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Week 5 - Core | Risks, Ethics & Responsible Use

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Confidence checkpoint: a confidence score (like 0.90) is how sure the model is about THIS answer - not how often it is right overall. A high score can still be wrong, so keep checking against the task.

I can...

- I can spot risks in an AI output, such as a mistake, an unfair bias, or a made-up fact.
- I can read a scenario, name a risk, and suggest a responsible-use rule with a clear reason.
- I can use words like bias, risk and responsible use correctly, and explain how a risk affects an AI workflow.

Words to know

- risk — the chance that an AI output causes harm, such as a mistake or something unfair
- bias — when an output is unfair because the data it learned from was one-sided
- hallucination — when an AI states something false as if it were true
- responsible use — checking outputs, protecting personal data, and following the rules
- governance — the rules and roles that keep AI use safe in a school or workplace

AI tools are fast and confident, but confident does not always mean correct. Today you learn to spot the risks

hidden in an AI output before anyone acts on it.

Where risks hide

An AI makes its output by predicting what usually comes next, based on patterns it has seen.

That is powerful,

but it means the output can be wrong, unfair, or misleading and still sound sure of itself. The riskiest moment in

a workflow is the output stage, because that is the part a real person reads and acts on. If nobody checks it, an

error or an unfair result travels straight to the user.

See bias in the data – and fix it

Bias is easiest to understand when you can see it. Watch a Teachable Machine image classifier trained on a

skewed set – say 15 clear photos of Class A but only 3 of Class B.

See the skew: test it on fresh items and count how many each class gets right. The rare class scores far lower –

for example Class B gets 1 of 4 correct = 25%.

Name the cause: the training data under-represented Class B, so the model learned it poorly – the bias is in the

data, not in “the computer”.

Fix it: add balanced examples of Class B, retrain, and re-test. The rare-class score rises – for example 3 of 4 =

75%. Say the before-and-after accuracy and name the one change that made it fairer. (No personal data – use

objects, drawings or gestures only.)

Responsible use is your job

Staying safe is not the tool's job, it is yours and the school's. Responsible use means three habits. Verify before

trust: check every output against a source you already trust. Protect people: never put someone's name, face or

private details into a tool. Be honest about limits: say what the tool cannot do well. Governance is simply the set

of shared rules that makes these habits normal for everyone.

Protecting people also has a technical side, not just a rule. De-identification means removing anything that

could name a person before data is used. Data minimisation means collecting only what the task truly needs

and nothing more. Retention and deletion mean keeping data only as long as it is needed and then deleting it. In

the UAE these ideas are backed by law and policy - the Personal Data Protection Law (PDPL) and ADEK's data-

protection rules.

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Risks cluster at the output stage of an AI workflow.

Why it matters

Schools, hospitals and companies across the UAE are starting to use AI in real workflows, so being the person

who can spot a risk and suggest a fix makes you genuinely useful and trusted.

Let's try it - hands-on

1. Read your scenario card and underline the AI output.
2. Decide the risk type: error, bias, or misuse and over-reliance.
3. Write down who could be affected and how.
4. Mark which workflow stage the risk sits at - input, process or output.
5. Finish this stem: A responsible measure is... because...

Think about it

If an AI output is confident and well written but you cannot verify it anywhere, should you use it?

Why or why

not?

My reflection

My reflection: the risk I will always check for is... because...

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Week 5 - Lab | Risks, Ethics & Responsible Use (Lab)

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Human in the loop: the AI tool does the sorting or drafting, but a person decides what to give it, checks each output against the task, and makes the final call - especially on any low-confidence result. Notice where the human decides in today's workflow.

I can...

- I can audit a real scenario and list the risks in its AI outputs.
- I can rate each risk for how likely it is and how much it would hurt, then propose a mitigation with a reason.
- I can argue which fix to build in first, using words like bias, mitigation and governance correctly.

Words to know

- audit — a careful check that lists risks and rates how serious each one is
- likelihood — how often a risk is expected to happen
- impact — how much harm a risk would cause when it happens
- mitigation — a specific change to a workflow that lowers a risk's likelihood or impact
- residual risk — the risk that is left over even after a mitigation is added

Spotting a risk is step one. Today you become an auditor: you rate how likely and how harmful each risk is, then

decide which fix a real AI workflow should build in first.

Rate before you fix

A long list of risks is not much use until you know which ones matter most. Auditors rate every risk two ways:

likelihood, how often it is expected to happen, and impact, how much harm it does when it does happen.

Plotting them on a simple grid makes the priority obvious. A risk that is both likely and harmful sits in the top

corner and gets fixed first. A rare, minor risk can wait. This is how real teams spend their limited time where it

protects people most.

A mitigation changes the workflow

A mitigation is not a warning label, it is a real change to how the workflow runs. It works by lowering either the

likelihood of a risk or its impact. Good examples are adding a human-review checkpoint at the output stage,

blocking any personal data at the input stage, or writing down what the tool cannot do well.

Even after a strong

mitigation, some residual risk usually remains, so governance means naming who keeps watching it.

Why it matters

When a UAE school, clinic or company adds an AI step to a real workflow, someone has to decide which risks to

fix first. That auditor role is exactly the skill you are practising today.

Let's try it - hands-on

1. List every risk you can find in your scenario's AI outputs.
2. Rate each risk: likelihood low, medium or high, and impact low, medium or high.
3. Plot each risk on the likelihood-and-impact matrix.
4. Write a specific mitigation for each risk and name the stage it changes.
5. Choose the one mitigation to build in first and finish: ...because...

Today's tool - hands-on

Tool: Card decks (risk-scenario cards) + pre-generated output packs

1. Draw a scenario card and read it aloud in your group.
2. Name one risk in the scenario and say who it could affect.

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3. Match the risk to a responsible-use rule from the class charter.
4. Check the two flawed outputs and label the risk each shows.
5. Write one rule that would reduce the biggest risk you found.

You will make: A risk-spotting record with risks, who-is-affected and a proposed rule.

Stay safe: Unplugged; the outputs are pre-curated.

A risk-scenario card linked to who it affects and a matching responsible-use rule.

Think about it

Two risks have the same impact, but one is far more likely. Which should the team mitigate first, and how would

you explain your choice to the person building the workflow?

My reflection

My reflection: the mitigation I would build in first is... because...

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Week 6 - Core | Bringing the Strands Together

I can...

- I can complete a guided task that uses all four strands together.
- I can explain how the parts of my workflow connect to make and check a result.
- I can name one responsible choice I made while doing the task.

Words to know

- workflow — the steps a task goes through, from input to process to output
- classifier — a process that sorts each item into a group, such as Recycle or Compost
- verify — to check an answer against a trusted source before you trust it
- personal data — information about a real person, such as a name, face or voice, which never goes into the tool
- integration — joining several skills into one working task

Today you bring together everything from this term - understanding, building, checking and using AI responsibly -

to run one small workflow that sorts items and checks its own answers.

Four skills, one task

Until now you have practised four skills in separate lessons: understanding the parts of a workflow, building the steps, checking the output, and using AI responsibly. Today they click together on a single task.

You prepare an

input, let a simple classifier process it, read the output, and then check that output against a trusted guide.

Nothing here is brand new - you are just using all four skills at once, which is exactly how real AI solutions are built.

Good data, good classifier: before you trust a classifier, look at the examples it learned from.

Three dataset rules

keep it fair - (1) balanced classes: roughly the same number of examples in each group; (2) no repeats: do not

use the same example twice; (3) a separate test set: always check the classifier on NEW examples it did not

learn from (a train/test split). Messy, one-sided or repeated data makes a classifier that looks fine but fails on

real items.

A confident answer can still be wrong

A classifier gives every item a label, and it can sound very sure. Sure is not the same as correct.

That is why the

check step matters: you compare each label with a trusted recycling guide, and if they disagree you fix the label.

You also keep the task safe by feeding in only object words - never a person's name, face or voice. Checking and

keeping data safe are part of the workflow, not extras added at the end.

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One task, four strands: prepare the input, process it, read the output, then check it.

Why it matters

Across the UAE, sorting and recycling systems use classifiers like this to reduce waste - and they only help if

their outputs are checked and their data is handled responsibly, just like in your task.

Cross-curricular links: Maths - when you count how many labels are correct 'out of the total', write it as a

fraction and convert it to a percentage. Science - checking a classifier by changing one thing at a time is a fair

test. Language - many UAE classifiers do Arabic-English translation, which links to your language study.

Let's try it - hands-on

1. Choose six item cards that show only object words - no names, faces or photos.
2. Pass each card through the simulated classifier to get a label: Recycle, Compost or Landfill.
3. Write each item and its label in your workflow table.

4. Check every label against the trusted guide and tick agree or flag disagree.

5. Fix any flagged label, then write down one responsible choice you made.

Think about it

If the classifier labels a greasy pizza box as Recycle but the guide says Landfill, which one should you trust, and

how do you decide?

My reflection

My reflection: the part of the workflow I trust most, and the part I trust least, is ... because ...

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Week 6 - Lab | Bringing the Strands Together (Lab)

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Human in the loop: the AI tool does the sorting or drafting, but a person decides what to give it, checks each output against the task, and makes the final call - especially on any low-confidence result. Notice where the human decides in today's workflow.

I can...

- I can run my whole workflow from start to finish and write down every check I make.
- I can use my check log to say how reliable my workflow was and where it went wrong.
- I can add a new rule or risk to my workflow and run it again responsibly.

Words to know

- end-to-end — running every stage of the workflow in order on a full set of items
- check log — a table where you record each output and whether the trusted guide agreed
- accuracy — how often the workflow got the answer right, counted out of the total; write it as a fraction or a percentage (keep 'reliable' for a workflow that stays consistent across different inputs)
- constraint — a new rule or limit you add, such as a second criterion or a risk to watch for
- iteration — improving a workflow by running it, checking it, and adjusting it

Today you run your whole sorting workflow properly from start to finish, keep a full record of every check, and

then add one new rule to see how your workflow copes.

From one example to a full run

In the Core lesson you tried the workflow on a few items. Today you run it end-to-end on ten or more, and you

write down every check in a log. That log is powerful: it turns 'I checked it' into real evidence.

You can count how

many labels the classifier got right, see which were corrected, and spot patterns - maybe messy or mixed items

trip it up most. That count is your workflow's accuracy, and it tells you how much to trust the output.

Adding a constraint

A real workflow rarely stays the same - you add rules as you learn. Today you add one constraint: either a

second criterion, like also labelling each item 'reusable' or 'not', or a risk rule, like flagging any item that might

be contaminated. Then you re-check only the items the new rule touches. Notice that you do not start over - you adapt the workflow you already have. All the while you keep the safe rules: only object words go in, and every output is verified before you trust it.

Why it matters

Recycling and sorting services in the UAE keep improving their systems by logging errors and adding new rules -

exactly the run, check and adapt loop you are practising today.

Let's try it - hands-on

1. Choose ten to twelve item cards that show only object words.
2. Run every item through the classifier and record its label in your check log.
3. Check each label against the trusted guide and mark agree or disagree, correcting where needed.
4. Count how many the workflow got right out of the total - that is your accuracy (say it as a fraction and a percentage).
5. Add one constraint, re-check the items it affects, and note what changed.

Today's tool - hands-on

Tool: Teachable Machine + Scratch + pre-generated packs (integrated run)

1. Recall the trained model (input -> process -> output).

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2. In Scratch, connect or simulate the model so a typed/selected input produces an output.
3. Run the mini solution on three inputs.
4. Use the evaluation pack criteria to judge each output.
5. Record one responsible-use choice you made while running it.

You will make: A run-and-check record across all four strands.

Stay safe: Teachable Machine trains in the browser and the samples stay on the teacher's device. Use objects, hand gestures or drawings only - never faces, voices or names. Clear the project at the end of the lesson. Reuse only the non-personal model.

The four strands as one loop: build, run, evaluate, check the risk.

Think about it

If your workflow gets 9 out of 12 right, is that reliable enough to use for real recycling? What would make you trust it more?

My reflection

My reflection: after running it end-to-end, the part of my workflow I would improve first is ... because ...

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Week 7 - Core | Signature Solution - Planning

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Design two mitigations (everyone does this): pick one risk in your system - bias or

misuse - and write TWO ways to reduce it (for example: balance your training examples; send low-confidence results to a person to check). For each one, say how it would change what the system does.

I can...

- I can write a clear goal for my workflow and list its input, process and output steps in order.
- I can name the inputs, the teacher-operated tool and the non-personal data my solution needs.
- I can point to one risk and show the check step that catches it before the output is trusted.

Words to know

- goal — the one task your solution is built to complete
- structure — the ordered steps of your workflow: input, process, output
- check step — a step that tests whether the output is correct, not just that the tool ran
- risk — something that could go wrong, such as a confident but wrong answer
- success criteria — one sentence describing what a good output looks like

Today you become the architect of your signature solution: before you build anything, you plan how it works and

how you will know it is any good.

Architect first, build second

A signature solution is a small workflow that finishes one task using input, process and output steps. Planning it

means designing the whole chain before you build. You write a clear goal, put your steps in order, and name

what each step needs - the inputs, the teacher-operated tool, and non-personal data only. A good plan is

deliberate: every step is there on purpose, and you could hand it to a classmate and they would know exactly

what to do.

The check step is the point

The most important part of your plan is the check step. An AI output can look confident and still be wrong, so

you never trust it on sight. Your check step compares the output against the task and a trusted source before

you accept it. Think about one risk in your plan - what could go wrong? - then place a check step to catch it. If

you want to go further, add a fallback: what your solution does when the check fails.

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A signature-solution plan: goal on top, an input-process-output chain, with a check step guarding the output.

Why it matters

Across the UAE, planners design smart systems - from traffic sensors in Abu Dhabi to farm-monitoring tools -

the same way you will today: goal first, checks built in, no personal data. Good planning is what makes an AI

solution trustworthy.

Let's try it - hands-on

1. Write your goal in one sentence: what task will your solution complete?
2. Draw three boxes - Input, Process, Output - and write the ordered steps inside them.
3. List the inputs, the teacher-operated tool, and the non-personal data each step needs.
4. Add a check step after the output and say what you will compare the result against.
5. Name one risk and draw an arrow from it to the check step that catches it.

Think about it

If your check step failed and the output was wrong, what should your solution do next instead of trusting it?

My reflection

My reflection: the part of my plan I am most confident about is ... and the check step I still need to strengthen is

...

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Week 7 - Lab | Signature Solution - Planning (Lab)

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Human in the loop: the AI tool does the sorting or drafting, but a person decides what to give it, checks each output against the task, and makes the final call - especially on any low-confidence result. Notice where the human decides in today's workflow.

I can...

- I can walk my whole workflow through once, step by step, and confirm the output reaches the check step.
- I can test my plan with a hard input and show that my check step catches the bad output.
- I can add a fallback path so my solution does something safer when a check fails.

Words to know

- prototype — a first working version you walk through by hand to see if it holds together
- walk-through — tracing one input through every step, in order, to the output
- pressure-test — feeding a hard or tricky input on purpose to see if your check catches a wrong output
- fallback path — what your solution does when a check step fails, instead of trusting the output
- test input — the example, non-personal input you use to try your workflow

Your plan meets reality today: you will run it as a prototype, try to break it on purpose, and make it safer when it fails.

A plan is only a hypothesis

Your Core plan looks neat on paper, but paper hides gaps. A prototype is the first time you actually run it: you

take one real, non-personal input and walk it through every step in order until it reaches the output and the

check step. Say each step out loud. Where the input gets stuck or a step does not do what you expected, you

have found a gap - and finding it now, by hand, is exactly the point. Prototyping turns a guess into something you

have seen work.

Try to break your own plan

Anyone can pass an easy test. To really trust your solution, feed it a tricky input designed to fool it - a blurry

photo, a strange example - and watch your check step. Did the check catch the confident but wrong output? If

not, tighten the check. Then build a fallback path: decide exactly what your solution does the moment a check

fails, such as rerouting to a safer step or flagging it for a human. That is how real engineers make solutions safe.

Why it matters

Engineers testing self-driving features or medical tools in the UAE deliberately try to break their systems before

people rely on them. Pressure-testing and fallbacks are how a smart solution earns trust.

Let's try it - hands-on

1. Choose one normal, non-personal test input and set up your Core plan.
2. Walk that input through every step aloud until it reaches the check step.
3. Now send a tricky input through the same chain.
4. Check whether your check step catches the wrong output; if not, tighten it.
5. Add a fallback path: write what your solution does the moment a check fails.

Today's tool - hands-on

Tool: Card decks (plan cards) + Scratch storyboard

1. Lay out the plan cards and fill each one for your signature solution.
2. Sequence the Steps cards into the order the workflow will run.

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3. Sketch the same sequence as Scratch blocks on the storyboard.
4. Pressure-test: pick one step and ask what happens if it fails.
5. Add a fallback card for that step.

You will make: A planned card layout plus a Scratch storyboard with a fallback path.

Stay safe: Planning only; no live tool or data.

Plan cards (goal, inputs, steps, checks, risk) sequenced into a workflow with a fallback branch.

Think about it

Which single input would be most likely to fool your solution, and does your check step catch it?

My reflection

My reflection: the gap my prototype revealed was ... and the fallback I added to handle it is ...

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Week 8 - Core | Signature Solution - Building

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Compare two designs (everyone does this): take the same task two ways - (A) one step that decides everything at once, and (B) a multi-step workflow (check -> classify -> confidence check -> ask a human if unsure). Run both on 3-4 items and write 2-3 sentences on WHY the design made the difference.

I can...

- I can build my solution by following my input, process and output plan.
- I can test my solution and judge its outputs against our success criteria.

- I can make one clear improvement after looking at what my evaluation shows.

Words to know

- build - follow your plan step by step to make the solution actually run
- evaluate - judge an output by checking it against your success criteria
- success criteria - the short list that says when an output is good enough
- verify - check an output against the task and a trusted source before you trust it
- refine - change one part of your input or structure to improve the output

Today you stop planning and start building. You will run your solution for real, look hard at what it produces, and

make it a little better.

Building means following your plan

Building is not the time to invent a brand new idea. It is the time to follow the plan you already made, like

following a recipe. You prepare your input, run the process step, and get your output. Keep your plan open

beside you and point to each part as you do it. If a part of the plan is unclear, fix that small part first, then keep

going. Remember: no names, faces or personal details ever go into your input.

Evaluate, then improve one thing

When your output appears, do not just trust it because it looks confident - a sure-sounding answer can still be

wrong. Check it against each success criterion and against a trusted source. Find one criterion it does not meet.

Then change one thing, either the words of your input or a single step, and run it again.

Changing only one thing

lets you see whether that change is what made the output better.

Why it matters

AI tools in the UAE, from smart government apps to school helpers, are only useful when someone builds them

carefully and checks their outputs - the same habit you are practising today.

Let's try it - hands-on

1. Put your plan and success criteria side by side and read them aloud.
2. Prepare your input exactly as planned, with no names or personal details in it.
3. Run the process step and write down the exact output you get.
4. Tick or cross the output against each success criterion and verify one claim against a trusted source.
5. Change one thing, run it again, and note whether the new output met the criterion.

Think about it

If your first output looked good but failed one criterion, why is it still not finished?

My reflection

My reflection: the one improvement I made today was ____ because ____.

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Week 8 - Lab | Signature Solution - Building (Lab)

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Human in the loop: the AI tool does the sorting or drafting, but a person decides what to give it, checks each output against the task, and makes the final call - especially on any low-confidence result. Notice where the human decides in today's workflow.

I can...

- I can finish a working version of my solution that runs from input to output.
- I can test it more than once and log how each output scores against the criteria.
- I can make and record several improvements and say what each one changed.

Words to know

- complete - the solution runs all the way from input to output with no gaps
- reliable - it keeps meeting the criteria on different inputs, not just one
- edge case - an unusual or awkward input that can break a solution
- evaluation log - a record of each test: input, output, criteria met and the change made
- iterate - repeat the test, evaluate, refine and log cycle to make it better

Today you turn your first build into a finished, working solution - and you test it enough times to know you can

trust it.

Finish it, then test it more than once

A solution that worked one time is not finished - it might have been lucky. In the Lab you complete every step so

it runs from input to output, then you test it again with different inputs. Try an awkward one on purpose, like a

very long or a blank input. Each time, check the output against your success criteria. Remember to keep names

and personal details out of every input, and let the teacher run the AI step.

Keep an evaluation log

Every test goes in your log: the input you used, the output you got, which criteria it met, the change you made,

and what that change did. A log lets you compare runs side by side and see which improvement really helped. It

also helps you stay honest - a confident output still gets checked against a trusted source before you trust it.

Aim for several logged cycles, not one.

Why it matters

Reliable apps that people across the UAE depend on are tested again and again before release; keeping an

evaluation log is how builders prove a solution really works.

Let's try it - hands-on

1. Finish every step so your solution runs from input to output.
2. Test it with one normal input and record the output.
3. Test it with an awkward or edge-case input and record that output too.
4. Score each output against your criteria and verify one claim against a trusted source.
5. Refine one part, run again, and add a new row to your evaluation log each time.

Today's tool - hands-on

Tool: Scratch (build) with optional Teachable Machine model

1. Build the workflow following your plan and block storyboard.
2. Connect or simulate the model at the process stage if your solution uses one.
3. Run it on a normal input and confirm the output.

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4. Run a tricky input; note the failure and handle it (add a check or fallback).
5. Log one improvement you made from an evaluation.

You will make: A working solution plus a build-and-test log with one logged fix.

Stay safe: Teachable Machine trains in the browser and the samples stay on the teacher's device. Use objects, hand gestures or drawings only - never faces, voices or names. Clear the project at the end of the lesson. Only the non-personal model is used.

A build-and-test log: normal run, tricky input, the fix that was added.

Think about it

If your solution passes one input but fails another, is it reliable yet - and how many tests would convince you?

My reflection

My reflection: across my logged tests, the improvement I trust most is ____ because ____.

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Week 9 - Core | Signature Solution - Refining & Finishing

48

Confidence checkpoint: a confidence score (like 0.90) is how sure the model is about THIS answer - not how often it is right overall. A high score can still be wrong, so keep checking against the task.

I can...

- I can finish my input-process-output workflow so it meets our success criteria.
- I can check my solution for accuracy, usefulness and risk.
- I can explain one change I made that improved it, and why it helped.

Words to know

- success criteria - the agreed list that says when your solution is good enough
- refine - to improve something by changing one part and re-checking it
- accuracy - whether the output is correct when you check it against a trusted source
- risk - the chance the output could mislead, exclude or expose someone
- verify-before-trust - checking an output before you believe it, even when it sounds confident

Today you finish your AI workflow and prove it is good enough to share. Finishing is not adding more - it is making

one smart change and checking that it worked.

Change one thing, then check

When you refine a workflow, change only one thing at a time - the prompt, the input, or a single process step.

Then run it again and compare the new output with the old one against your success criteria. If you change three

things at once, you will not know which one helped. Keep both the before and after so you can prove what

improved. Small, tested changes are how real solutions get finished.

Prove the improvement with a number

Do not just say a change made things “better” – show it. Record accuracy as correct ÷ total before and after

each change.

Example: before, $7 \div 10 = 70\%$; after adding two examples, $9 \div 10 = 90\%$. Now “better” is backed by evidence – a

number that went up – not just a feeling.

Good, useful and safe

A finished solution is judged three ways. Accuracy: is it correct when you check it against a trusted source? A

confident output can still be wrong. Usefulness: does it actually complete the task you set? Risk: could a wrong

output mislead, exclude or expose someone? Before you share, do a quick risk check - name who could be

affected and how you would catch a mistake. That is what makes a solution trustworthy.

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The refine loop: change one thing, re-run, then check against your criteria.

Why it matters

Across the UAE, teams finishing an AI tool - from a hospital scheduler to a customer chatbot - run the same

accuracy and risk checks before launch; learning to finish and check well is a real workplace skill.

Let's try it - hands-on

1. Read your three success criteria and tick the ones you already meet.
2. Pick ONE thing to change - the prompt, the input, or one step.
3. Ask your teacher to re-run the AI step (or run your paper version) and keep the before and after.
4. Check the new output against a trusted source - do not trust it just because it sounds sure.
5. Do the risk check and write one sentence about the change that improved your solution.

Think about it

Think about it: your output sounds confident and reads well, but you have not checked it - should you trust it yet, and why not?

My reflection

My reflection: the one change that most improved my solution was ... and I know it helped because ...

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Week 9 - Lab | Signature Solution - Refining & Finishing (Lab)

51

Human in the loop: the AI tool does the sorting or drafting, but a person decides what to give it, checks each output against the task, and makes the final call -

especially on any low-confidence result. Notice where the human decides in today's workflow.

I can...

- I can run one final iteration and choose the version I will show.
- I can rehearse my demo and explain each step of my workflow clearly.
- I can justify that my solution is trustworthy and low-risk using the checks I ran.

Words to know

- iteration - one round of changing and re-testing your solution
 - version-lock - choosing the output you will show and not changing it again
 - demo - a short, clear showing of your workflow from input to output
 - justify - to back up a claim with specific evidence, not just say it
 - trustworthy - deserving belief because it has been checked, not because it sounds confident
- Today you lock your final version and rehearse showing it well. A great demo does not just show the answer - it

shows the checks that make it trustworthy.

Lock it, do not add to it

Right before a showcase, the temptation is to add one more feature. Do not. A last-minute change is untested,

and untested means risky. Instead, run one final iteration, confirm it meets every success criterion, then

version-lock: pick the output you can reproduce and agree not to touch it again. A locked, reproducible version

is one you can show with confidence, because you already know exactly how it behaves and what checks it passed.

Show the checks, not just the answer

A strong demo narrates your workflow: here is the task, the input, the process step, and the output. Then it

shows evidence - the checks you ran. To justify that your solution is trustworthy and low-risk, point to two

specific checks: you verified the output against a trusted source, and your risk check named who could be

affected and how you would catch a mistake. 'It works' is not a justification; named checks are.

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The demo spine: five lines that show your workflow and its checks.

Why it matters

When a UAE team demos a new AI service to leaders, they are judged on the checks they can show, not on how

polished it looks; rehearsing an evidence-based demo builds exactly that skill.

Let's try it - hands-on

1. Run one final iteration and confirm it still meets all your criteria.
2. Version-lock the output you can reproduce and agree not to change it.
3. Write your five demo sentences - task, input, process, output, one check.
4. Rehearse twice with your partner, swapping speaker and checker.

5. Finish with your justification: why it is trustworthy and low-risk, citing two checks.

Today's tool - hands-on

Tool: Scratch + pre-generated comparison packs (refine)

1. Run your solution across the printed test-set and count how many outputs are correct.
2. Compare a before and after output from the pack and name what improved.
3. Make one final refinement to raise accuracy or usefulness.
4. Run a final risk check: confirm no personal data is used.
5. Rehearse a 60-second showcase of one design choice.

You will make: A test-set accuracy count, a refinement note and a completed risk check.

Stay safe: No personal data; risk check is part of the task.

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Before and after outputs compared, with an accuracy tally and a risk-check tick.

Think about it

Think about it: two teams show the same output, but only one names the checks they ran - which solution

would you trust, and why?

My reflection

My reflection: I can show my solution is trustworthy because ... and low-risk because ...

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Week 10 - Core | Showcase, Assessment & Reflection

I can...

- I can show my input-process-output workflow and explain two choices I made and why.
- I can answer a short task that samples what I learned about building, checking and using AI safely.
- I can name one thing I learned this term and write one goal for next term.

Words to know

- showcase — a short presentation where you show your work and explain your choices, not just the result
- justify — give a clear reason for a choice you made
- verify — check an output against the task and a trusted source before you trust it
- reflection — thinking about what you learned and where you are still unsure
- goal — a specific, doable target you set for yourself

Today you show what you built this term and explain the smart choices behind it, then set a goal for what comes

next. Your reasons matter more than a shiny result.

A showcase shows your thinking

A good showcase is not the flashiest output. It is you making your thinking visible. You show the three parts of

your workflow - the input, the AI process step, and the output - and then you justify two choices.

Maybe you

chose a certain input because it gave clearer results, or you checked your output against a trusted source.

When you explain why, other people can learn from you and you prove the workflow was not just luck.

Reflection turns a project into a skill

One project is a start; a skill is something you can use again. That is why you reflect. You name one thing you

learned about how the workflow behaves - for example, a clearer input gives a more useful output. Then you set

one specific goal for next term, like testing every workflow with two different inputs before trusting it. A goal you

can act on next week is far more useful than saying you will just do better.

A showcase makes your thinking visible: show the workflow, then justify your choices.

Why it matters

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Across the UAE, people who build with AI - in hospitals, banks and government services - have to explain and

check their work, not just show a result. Learning to justify and verify now is how you build trust in what you

make.

Let's try it - hands-on

1. Lay out your workflow as input, AI process step, and output.

2. Pick two choices you made and write a reason for each.

3. Present for about a minute: show the three parts, give your two reasons, and name one way you checked the output.

4. Complete the four short items, one for each strand you studied.

5. Write one thing you learned and one specific goal for next term.

Think about it

Think about it: if someone gave your workflow a harder or higher-stakes task, which part would you check first,

and why?

My reflection

My reflection: the most important thing I learned this term is ____, so next term my goal is ____.

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Week 10 - Lab | Showcase, Assessment & Reflection (Lab)

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Human in the loop: the AI tool does the sorting or drafting, but a person decides what to give it, checks each output against the task, and makes the final call - especially on any low-confidence result. Notice where the human decides in today's workflow.

I can...

- I can review a peer's workflow against the criteria and give one strength and one specific improvement.
- I can run my workflow on a harder case and record where it holds up and where it fails.
- I can turn my term's reflections into one clear, measurable goal for next term.

Words to know

- peer review — checking a classmate's work against shared criteria and giving useful feedback

- success criteria — the agreed list of what good work must show
- stress-test — running your workflow on a harder or riskier case to find its limits
- safeguard — an extra check you add so a wrong output causes less harm
- measurable goal — a goal with a clear way to know when you have met it

Today you go deeper: you review classmates' workflows against shared criteria, push your own workflow with a

harder case to see where it breaks, and turn everything you learned into one clear goal.

Good feedback is specific, not just kind

Saying good job feels nice but does not help. Useful feedback points to the shared criteria and gives a concrete

step. Use two-star-and-a-step: name two strengths tied to a criterion, then one clear thing to improve, like test it

with a messier input. Keep it about the work, never the person. When you review this way you help your

classmate and you also train your own eye to spot what makes a workflow trustworthy.

Stress-testing finds the limits

A workflow that passes one easy case is not proven. Stress-testing means running it on a harder or higher-

stakes case - a messier input, an edge case, or a setting where a wrong output could really matter, like a

hospital. Often it breaks, and that is useful: now you know where to add a check or a human review. Robustness

is something you design in, not something you assume. Finding a limit today is better than being surprised by it later.

Why it matters

In real UAE workplaces, AI tools are reviewed by other people and tested on hard cases before they are trusted

with anything important. Practising peer review and stress-testing now is exactly how professionals keep their AI safe and fair.

Let's try it - hands-on

1. Read the shared success criteria first.
2. At each station, write two stars tied to a criterion and one concrete step.
3. Keep every note about the work, not the person.
4. Run your own workflow on the harder case you are given and log where it holds and where it fails.
5. Turn your feedback and your log into one measurable goal for next term.

Today's tool - hands-on

Tool: Showcase the Scratch / Teachable Machine solution + peer-review cards

1. Demonstrate your solution's input -> process -> output to a visiting group.

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2. Explain two design or evaluation choices you made.
3. Visit two other solutions and complete a peer-review card for each.

4. Collect your peer cards and read the feedback.

5. Write one Term-2 goal based on the feedback.

You will make: A demonstrated solution, completed peer-review cards and a Term-2 goal.

Stay safe: Showcase uses only the non-personal solution; no data shared.

A showcase carousel with a peer-review card: two stars and a wish, strand-tagged.

Think about it

Think about it: what is one setting where a wrong output from your workflow would be serious, and what

safeguard would you add there?

My reflection

My reflection: my stress-test showed that ____, so my measurable goal for next term is ____.

Glossary - key words for this term

- workflow - a task done as input -> process -> output.
- input / process / output - what you give the system / the steps inside / what it gives back.
- rule-based system - follows fixed rules a person wrote; same input always gives the same result.
- learned model (model) - has learned patterns from many labelled examples and can handle new inputs.
- training - teaching a model from labelled examples (done once); inference - using the trained model on a new input.
- parameters - the patterns a trained model stores (not copies of the training examples).
- classifier - the step that sorts each input into one of a fixed set of classes.

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- class / label - one of the named groups a classifier chooses; the label is the answer it outputs.
- training data vs testing/input data - the labelled examples it learned from vs the new, unseen items you test or use it on.
- train/test split - holding back some labelled items to test the model on data it has not seen.
- accuracy - how many it got right out of the total, as a fraction or percentage.
- confidence score - how sure the model is about one answer (e.g. 0.90); not the same as accuracy.
- reliability - the workflow keeps working across more than one input.
- black box - you can see the input and output but not the reason inside for the decision.
- bias - a repeating, systematic error that makes a system work worse for one group than another.
- fairness - treating comparable cases or groups consistently; measured by comparing accuracy across groups.
- human in the loop - the person who decides, checks and can overrule the AI's output.
- responsible use - using AI safely and fairly: no personal data, always check the result.

Output

Emerging tier | 10 weeks (Core + Lab)

STUDENT WORKSHEETS

Week 1 - Core - Worksheet | Unit Launch - AI as Workflows - Input, Process,

Output

Name: _____ Class: _____ Date: _____

Sort each example card into the correct column and give a reason.

My two questions

Write at least two questions about AI as a workflow. Aim at least one at the input, the process or the output.

Sort it: input, process or output

Write each example in the right column and add one reason.

Example Input / Process / Output Reason

A typed question to a chatbot

Grouping photos by what is in them

A finished travel summary

Our safe-use charter

Write one responsible-use rule for the class and why it keeps us safe.

Exit ticket

Exit ticket - Name one example, say whether it is an input, a process or an output, and explain why. Then write one question you still have.

Challenge (stretch)

Challenge - Pick a real UAE situation (for example a smart parking app or an Arabic-English translator). Describe its input, process and output, then predict two things a strong AI solution would need to get right. This stretches toward the Proficient idea of explaining how workflows structure AI systems.

Week 1 - Lab - Worksheet | Unit Launch - AI as Workflows - Input, Process, Output
(Lab)

Name: _____ Class: _____ Date: _____

Trace each example through the three stages, then log the difference and where it began.

Comparison log

For each example pair, write one difference and tick the stage where it began.

Example pair Difference I observed Stage (input / process / output)

Pair 1

Pair 2

Pair 3

My conclusion

Use your log as evidence. What one change most improved the output?

Verify it

How could you check your conclusion is right for a new example?

Tool log - Card decks (workflow-example cards)

Record as you work. Goal: A sorted card layout plus a difference log.

Step What I did What I saw / result

1 Deal the deck face-up between partners.

2 For each card, agree the input, the process and the output

a

3 Sort the cards into 'follows fixed rules' and 'learns from e

4 Pick the two hardest-to-place cards and say why.

5 Log at least three differences you noticed between the two

p

Safety check (tick): I used no personal data (no faces, voices or names). ☐

Responsible-use & human-in-the-loop reflection

• Where did a person decide or check in today's workflow (not the AI)? Name the moment.

• One responsible-use point: could this workflow treat someone unfairly or be misused, and what would you check?

• Creator's reflection: what did you build or design today, and how is it different from what an AI would produce on its own?

Exit ticket

Exit ticket - Name one difference you logged and the workflow stage it came from, and say whether it made the output better or worse.

Challenge (stretch)

Challenge - Design your own fair test: choose one thing to change in the input, predict how the output should change, and describe how you would compare the before and after to prove it. Then explain what a strong UAE solution built on this idea would need to get right. This stretches toward the Proficient idea of evaluating outputs using criteria such as accuracy and usefulness.

Week 2 - Core - Worksheet | Understanding the System - Input, Process, Output

Name: _____ Class: _____ Date: _____

Trace the workflow

Fill in the three stages for the worked example your teacher showed.

Stage What happens in the worked example

Input

Process

Output

Explain the reasoning

Write one sentence explaining how the input became the output. Use the words because or so.

Spot the stage

Read each line and write which stage it describes: input, process or output.

Example Which stage?

A photo you choose in a filter app

The app finds the face and smooths it

The new smoothed photo appears

Exit ticket

Name the three stages of an AI workflow in order, and give one example of an input.

Challenge (stretch)

Challenge (stretch) - In two or three sentences, explain how splitting an AI system into input, process and output stages

helps a designer build it and fix it. Use the word workflow.

Week 2 - Lab - Worksheet | Understanding the System - Input, Process, Output

(Lab)

Name: _____ Class: _____ Date: _____

Same process, two inputs - watch the outputs change.

Build and trace

Set up your workflow and follow one item through it. Record each stage.

Stage Your workflow

Input

Process

Output

Change the input, compare the output

Keep the process the same. Change only the input, run it again, and compare.

First run Second run

Input

Output

What changed and why

Where could it go wrong?

Write one place an error could enter this workflow and how you would check the output.

Tool log - Google Teachable Machine (teacher-operated)

Record as you work. Goal: A trained 3-class model and a completed input-process-output trace.

Step What I did What I saw / result

1 As a class, name the three classes (for example: open

hand,

2 Students direct the teacher to add ~15 example images

per cl

3 Click Train. Narrate: these examples are the input, training

4 Test with a new, unseen example and read the confidence

outp

5 Trace the full input -> process -> output path on the board

Safety check (tick): I used no personal data (no faces, voices or names). ☐

Responsible-use & human-in-the-loop reflection

- Where did a person decide or check in today's workflow (not the AI)? Name the moment.

- One responsible-use point: could this workflow treat someone unfairly or be misused, and what would you check?

- Creator's reflection: what did you build or design today, and how is it different from what an AI would produce on its own?

Exit ticket

You changed the input and the output changed, but the process stayed the same. Explain in one sentence why.

Challenge (stretch)

Challenge (stretch) - A workflow is a set of stages joined in order. Explain how thinking in workflows lets you build a bigger AI system out of small, testable steps. Use input, process and output in your answer.

Week 3 - Core - Worksheet | Designing & Building a Workflow

Name: _____ Class: _____ Date: _____

Build your workflow

Fill in each box for the task you chose.

Box What goes here

Input

Process (AI step)

Output

Test and verify

Run your workflow twice and record what happened.

Run Input used Output Matches the answer key?

(tick or fix)

1

2

Explain it

Write one or two sentences saying why your workflow gave that output.

Exit ticket

Name your three workflow steps in order, and write one reason your output could be wrong even if it looks confident.

Challenge (stretch)

Challenge (stretch): Add a fourth step so your workflow does two things in a row - a multi-step workflow - for example sort the books and then count how many are fiction. Show how the output of your first process becomes the input of the second.

Week 3 - Lab - Worksheet | Designing & Building a Workflow (Lab)

Name: _____ Class: _____ Date: _____

The iterate loop: run, find the weak box, change one thing, run again.

Iteration log

Change one box, then record what happened.

Run Box you changed Output before Output after Better? (Y/N)

1

2

Your multi-step workflow

Draw your two process steps and label the hand-off arrow between them.

Prove it is reliable

Record the output for two different inputs and tick if each matches the answer key.

Input Output Matches key? (tick)

Input A

Input B

Tool log - Scratch (input -> process -> output build)

Record as you work. Goal: A working three-stage Scratch project and a short run log.

Step What I did What I saw / result

1 Decide a simple task, e.g. sort a typed word into 'question'

2 Drag an input block (ask and wait) to capture the input.

3 Build the process: two or three 'if' blocks that apply your

4 Add an output block (say) that reports the result.

5 Run it three times, including one tricky input, and note what

Safety check (tick): I used no personal data (no faces, voices or names). ☐

Responsible-use & human-in-the-loop reflection

• Where did a person decide or check in today's workflow (not the AI)? Name the moment.

• One responsible-use point: could this workflow treat someone unfairly or be misused, and what would you check?

• Creator's reflection: what did you build or design today, and how is it different from what an AI would produce on its own?

Exit ticket

Explain one change that improved your workflow and how you knew it helped, then say what makes a workflow reliable.

Challenge (stretch)

Challenge (stretch): Design a three-step workflow that completes a task on its own - designs multi-step AI workflows to complete tasks. Show every hand-off, add one verify step inside the chain, then test it on two different inputs.

Week 4 - Core - Worksheet | Judging & Improving Outputs

Name: _____ Class: _____ Date: _____

An evaluation grid: judge each output against every criterion, then record your evidence.
Evaluation grid

For each output, write a verdict (meets / partly / fails) for every criterion, then write your evidence.

Output Accuracy Relevance Clarity Evidence

A

B

C

Where did the issue start?

Pick one output with a problem. Tick where the issue began - input, process or output - and explain in one line.

Verify one fact

Choose one fact from an output. Write the trusted source you checked it against and whether it held up.

Exit ticket

Exit ticket: choose one output, give your verdict on accuracy, and write the single piece of evidence that decided it.

Challenge (stretch)

Challenge (stretch): rank two outputs from best to worst using accuracy and usefulness together, and explain why a less accurate output might still be more useful for a particular task - or why not.

Week 4 - Lab - Worksheet | Judging & Improving Outputs (Lab)

Name: _____ Class: _____ Date: _____

A ranking ladder: order the outputs top to bottom and clip the deciding evidence to each rung.

Ranking strip

Write each output in order from best (1) to worst, then give the evidence that decided each place.

Rank Output Evidence for this place

- 1
- 2
- 3

4

Break the tie

Pick two outputs that were close. Explain which is more useful for the task and why.

Improve the weakest

For your lowest-ranked output, tick where the issue began - input, process or output - and write one specific fix.

Tool log - Pre-generated output packs (evaluation clinic)

Record as you work. Goal: A scored, ranked output pack with reasons and one improvement.

Step What I did What I saw / result

1 Agree the evaluation criteria as a class (accuracy, usefulness)

2 In groups, read each output and mark it correct / unclear /

3 Rank the pack from strongest to weakest.

4 Choose one weak output and write the one change that would make it better

5 Defend your ranking to another group.

Safety check (tick): I used no personal data (no faces, voices or names). ☐

Responsible-use & human-in-the-loop reflection

• Where did a person decide or check in today's workflow (not the AI)? Name the moment.

• One responsible-use point: could this workflow treat someone unfairly or be misused, and what would you check?

• Creator's reflection: what did you build or design today, and how is it different from what an AI would produce on its own?

Exit ticket

Exit ticket: name your best and worst output and give one piece of evidence for each placement.

Challenge (stretch)

Challenge (stretch): write a short evaluation using criteria such as accuracy and usefulness that argues why your top output

would still win even if one criterion were weighted more heavily - and name the workflow fix that would most improve your bottom output.

Week 5 - Core - Worksheet | Risks, Ethics & Responsible Use

Name: _____ Class: _____ Date: _____

Three risk types to check in every AI output.

Scenario risk analysis

For each scenario, name the risk type, say who is affected, and mark the workflow stage where the risk sits.

Scenario Risk type Who is affected Workflow stage

1

2

Propose a responsible-use measure

Choose one scenario above. Write a responsible-use measure and justify it using the word because.

Verify-before-trust check

Tick or write how you would verify the AI output in scenario 1 against a trusted source.

De-identify a record

Here is a short student profile. Cross out or rewrite anything that could identify the real person - names, school and photo

details - replacing each with a safe, general description. This is de-identification and data minimisation in action.

Sample profile: Name: Salma Al Mheiri. School: Al Falah Middle School, Al Ain. Favourite subject: Robotics Club, after

school on Tuesdays. Photo note: a student in a blue school jumper, smiling, standing next to the 3D printer, with a name badge visible.

Rewrite it below with the identifying details removed or generalised (for example: 'a student', 'a middle school in Al Ain',

'wearing a school jumper, no name badge visible').

What did you remove or generalise, and why?

Exit ticket

Name one risk you found today and the one responsible-use measure you would never skip, and explain why in one sentence.

Challenge (stretch)

Challenge (stretch): pick one scenario and explain how bias, misuse and a tool limitation could each appear at a different stage of its workflow, and which one you would fix first and why.

Week 5 - Lab - Worksheet | Risks, Ethics & Responsible Use (Lab)

Name: _____ Class: _____ Date: _____

Plot each risk by how likely and how harmful it is; fix the top-right first.

Risk-and-impact audit

List each risk you find, then rate its likelihood and impact as low, medium or high and note the workflow stage it sits at.

Risk Likelihood Impact Workflow stage

Mitigation plan

For your two highest-priority risks, write a specific mitigation and justify it with because.

Priority and owner

State which mitigation you would build in first and who in the workflow would be responsible for it.

Tool log - Card decks (risk-scenario cards) + pre-generated output packs

Record as you work. Goal: A risk-spotting record with risks, who-is-affected and a proposed rule.

Step What I did What I saw / result

1 Draw a scenario card and read it aloud in your group.

2 Name one risk in the scenario and say who it could affect.

3 Match the risk to a responsible-use rule from the class
char

4 Check the two flawed outputs and label the risk each
shows.

5 Write one rule that would reduce the biggest risk you
found.

Safety check (tick): I used no personal data (no faces, voices or names). ☐

Responsible-use & human-in-the-loop reflection

- Where did a person decide or check in today's workflow (not the AI)? Name the moment.

- One responsible-use point: could this workflow treat someone unfairly or be misused, and what would you check?

- Creator's reflection: what did you build or design today, and how is it different from what an AI would produce on its own?

Exit ticket

Name the one mitigation you would build into this workflow first, the risk it reduces, and whether it lowers likelihood, impact or both.

Challenge (stretch)

Challenge (stretch): explain how bias, misuse and a tool limitation each appear at a different stage of your scenario's

workflow, then write one governance rule naming who monitors the residual risk after your top mitigation is in place.

Week 6 - Core - Worksheet | Bringing the Strands Together

Name: _____ Class: _____ Date: _____

Check each label before you trust it: agree, keep it; disagree, correct it.

My workflow map

Fill in what goes in each stage of your sorting task.

Stage What goes here

Input

Process

Output

My check log

Record each item, the label the classifier gave, and whether the trusted guide agrees.

Dataset rules (check before you trust your classifier)

☐ Balanced classes - roughly the same number of example items in each group (e.g. Recycle / Compost / Landfill).

☐ No repeats - no example item is used twice in the training set.

☐ Separate test set - the items you TEST with are new, not the same ones the classifier learned from (a train/test split).

Write one thing your dataset does NOT contain that could make it unfair:

Item Classifier label Guide agrees? (Y/N) Corrected label

My responsible-use choice

Write one responsible choice you made and why it mattered.

Exit ticket

In one sentence, explain how your workflow made a result and then checked it, and name one responsible choice you made.

Challenge (stretch)

Challenge (stretch): add a second criterion to your sort - for example, also label each item 'reusable' or 'not reusable' - or

add a risk rule, such as flagging any item that could be contaminated. Re-run the check and explain how adding the

constraint changed your workflow. This is a step toward the Proficient target of adapting an approach when a new constraint is added.

Week 6 - Lab - Worksheet | Bringing the Strands Together (Lab)

Name: _____ Class: _____ Date: _____

Run every item, log each check, count what was right, then add a rule and re-check.

Full check log

Record every item you run: its label, whether the guide agrees, and any correction.

Item Classifier label Guide agrees? (Y/N) Corrected label

Accuracy count (reliability = consistency across inputs)

Write how many items the workflow got right out of the total (as a fraction and a percentage) - that is your accuracy - and name the weakest kind of item.

Fairness check - accuracy by category

Split your test items by a characteristic (for example: clean items vs greasy/messy items, OR metal vs paper vs plastic). Count correct in each group, then compare the rates.

Group A: _____ correct ____ out of ____ = ____ %

Group B: _____ correct ____ out of ____ = ____ %

Compare the two rates. If one group is clearly worse, that is the start of a fairness problem - name the likely

cause in the data. If the numbers are too close or the test set is too small, write 'cannot determine'.

Maths link: turn each 'correct out of total' into a fraction and then a percentage before you compare.

My added constraint

Name the constraint you added and describe how you re-checked the affected items responsibly.

Tool log - Teachable Machine + Scratch + pre-generated packs (integrated run)

Record as you work. Goal: A run-and-check record across all four strands.

Step What I did What I saw / result

1 Recall the trained model (input -> process -> output).

2 In Scratch, connect or simulate the model so a

typed/selecte

3 Run the mini solution on three inputs.

4 Use the evaluation pack criteria to judge each output.

5 Record one responsible-use choice you made while running it.

Safety check (tick): I used no personal data (no faces, voices or names). ☐

Responsible-use & human-in-the-loop reflection

- Where did a person decide or check in today's workflow (not the AI)? Name the moment.
-

- One responsible-use point: could this workflow treat someone unfairly or be misused, and what would you check?
-

- Creator's reflection: what did you build or design today, and how is it different from what an AI would produce on its own?
-

Exit ticket

State your workflow's accuracy (right out of total), name one weak spot, and describe the constraint you added and how you kept the task responsible.

Challenge (stretch)

Challenge (stretch): add a second constraint on top of the first - for example, combine a 'reusable' criterion with a 'flag contamination' risk rule - then predict, and test, which change affects accuracy more. Explain how you would adapt the workflow further. This pushes toward the Proficient target of adapting an approach as constraints grow.

Week 7 - Core - Worksheet | Signature Solution - Planning

Design two mitigations (required)

- Risk you are reducing (bias or misuse):
-

- Mitigation 1 and how it changes the system's behaviour:

• Mitigation 2 and how it changes the system's behaviour:

Name: _____ Class: _____ Date: _____

My signature solution: goal

Write the goal of your solution in one sentence. It should name one task.

My workflow structure

Complete the table: write the ordered steps and what each step needs. Use non-personal data only.

Step What happens What it needs (input / tool / data)

Input

Process

Output

Check step

Risk and check

Name one risk in your plan, then explain how your check step reduces it.

Exit ticket

In one sentence, link your goal to your check step: what is your solution for, and how will you know its output is good?

Challenge (stretch)

Challenge (stretch): design a fallback path - write what your solution should do when the check step fails, so it reroutes to a safer step instead of trusting a wrong output.

Week 7 - Lab - Worksheet | Signature Solution - Planning (Lab)

Name: _____ Class: _____ Date: _____

Pressure-testing a workflow: a tricky input flows to the check step, which either passes it or triggers a fallback path.

Prototype log

Walk one normal input through your workflow. Tick pass or fail for each step and note what happened at the check step.

Step What happened Pass / Fail

Input

Process

Output

Check step

Pressure-test

Describe the tricky input you used and say whether your check step caught the wrong output. If it did not, write how you tightened the check.

Fallback path

Write exactly what your solution does when a check step fails.

Tool log - Card decks (plan cards) + Scratch storyboard

Record as you work. Goal: A planned card layout plus a Scratch storyboard with a fallback path.

Step What I did What I saw / result

1 Lay out the plan cards and fill each one for your signature

2 Sequence the Steps cards into the order the workflow will

run

3 Sketch the same sequence as Scratch blocks on the storyboard

4 Pressure-test: pick one step and ask what happens if it fail

5 Add a fallback card for that step.

Safety check (tick): I used no personal data (no faces, voices or names). ☐

Responsible-use & human-in-the-loop reflection

• Where did a person decide or check in today's workflow (not the AI)? Name the moment.

• One responsible-use point: could this workflow treat someone unfairly or be misused, and what would you check?

• Creator's reflection: what did you build or design today, and how is it different from what an AI would produce on its own?

Exit ticket

In one sentence, describe one gap your prototype revealed and the fallback path you added to handle it.

Challenge (stretch)

Challenge (stretch): design two fallback paths for different failures - one for a wrong output and one for a missing or unusable input - and state which check step triggers each.

Week 8 - Core - Worksheet | Signature Solution - Building

Compare two workflow designs (required)

• Design A (single step) result on your test items:

• Design B (multi-step: check -> classify -> confidence -> human) result:

• Which was better, and WHY did the design difference cause it? (2-3 sentences)

Name: _____ Class: _____ Date: _____

Score each output against your success criteria before you trust it.

Build log

Record your first output, then your improved output after one change.

Step What you did / got

Input used

First output

One change made

Improved output

Evaluate against your criteria

Tick or cross each success criterion for your best output and give the reason.

Success criterion Met? (tick/cross) and why

Verify before trust

Name one claim in your output and the trusted source you checked it against.

Exit ticket

Write the one improvement you made today and which success criterion it was aimed at.

Challenge (stretch)

Challenge (stretch): make your solution handle a tricky input gracefully. Feed it an incorrect or edge-case input, note how the output goes wrong, then adjust your structure so it responds sensibly - and document the fix.

Week 8 - Lab - Worksheet | Signature Solution - Building (Lab)

Name: _____ Class: _____ Date: _____

An evaluation log traces each test and improvement across several cycles.

Evaluation log

Fill one row for each test you run. Aim for at least three rows on different inputs.

Input used Output Criteria met Change made Effect

Edge-case test

Describe one awkward input that broke your solution and the fix you made so it responds gracefully.

Reliability check

Tick the box that matches your solution and explain in one line.

Statement Tick and why

Worked on one input only

Worked on two or more different inputs

Tool log - Scratch (build) with optional Teachable Machine model

Record as you work. Goal: A working solution plus a build-and-test log with one logged fix.

Step What I did What I saw / result

1 Build the workflow following your plan and block storyboard.

2 Connect or simulate the model at the process stage if your s

3 Run it on a normal input and confirm the output.

4 Run a tricky input; note the failure and handle it (add a ch

5 Log one improvement you made from an evaluation.

Safety check (tick): I used no personal data (no faces, voices or names). ☐

Responsible-use & human-in-the-loop reflection

• Where did a person decide or check in today's workflow (not the AI)? Name the moment.

• One responsible-use point: could this workflow treat someone unfairly or be misused, and what would you check?

• Creator's reflection: what did you build or design today, and how is it different from what an AI would produce on its own?

Exit ticket

Point to the log row that shows your solution became more reliable, and say what change made it so.

Challenge (stretch)

Challenge (stretch): design a second edge-case input that you think could still break your solution, test it, and either show it handles the case gracefully or document exactly what you would change next to fix it.

Week 9 - Core - Worksheet | Signature Solution - Refining & Finishing

Name: _____ Class: _____ Date: _____

Three checks before you share: accuracy, usefulness, risk.

Success-criteria check

Write your three criteria, tick if met, and give one piece of evidence for each.

Success criterion Met? (Y/N) Evidence

Refine and re-check

Record the one change you made and what happened when you re-ran it.

Iteration Impact Log (complete for at least one change)

- 1) Change made - which tool or step did you change?

- 2) Before -> After - what did the output look like before, and after?

- 3) Why it worked - why did that change cause the difference you saw?

The one change I made Before output (short) After output (short)

Risk check

Answer both questions in full sentences: who could be affected if the output were wrong, and how would you catch it?

Exit ticket

Exit ticket: name one change that improved your solution and the check that proves it helped.

Challenge (stretch)

Challenge: write a short justification arguing that your solution is trustworthy and low-risk, citing at least two specific

checks you ran - accuracy against a source, and your risk check.

Week 9 - Lab - Worksheet | Signature Solution - Refining & Finishing (Lab)

Name: _____ Class: _____ Date: _____

My five-sentence demo

Write one sentence for each part of your demo.

Part My sentence

Task

Input

Process

Output

One check I ran

Version-lock

Record the version you will show and why you chose it.

Rehearsal feedback

After each rehearsal, note one glow and one grow from your partner.

Rehearsal Glow (worked) Grow (fix)

1

2

Tool log - Scratch + pre-generated comparison packs (refine)

Record as you work. Goal: A test-set accuracy count, a refinement note and a completed risk check.

Step What I did What I saw / result

1 Run your solution across the printed test-set and count how

2 Compare a before and after output from the pack and name wha

3 Make one final refinement to raise accuracy or usefulness.

4 Run a final risk check: confirm no personal data is used.

5 Rehearse a 60-second showcase of one design choice.

Safety check (tick): I used no personal data (no faces, voices or names). □

Responsible-use & human-in-the-loop reflection

• Where did a person decide or check in today's workflow (not the AI)? Name the moment.

• One responsible-use point: could this workflow treat someone unfairly or be misused, and what would you check?

• Creator's reflection: what did you build or design today, and how is it different from what an AI would produce on its own?

Exit ticket

Exit ticket: in one sentence, justify why your solution is trustworthy and low-risk, naming the two checks you ran.

Challenge (stretch)

Challenge: prepare a 20-second answer to a tough question - 'How do you know your AI output is not misleading anyone?' - citing your accuracy check and your risk check by name.

Week 10 - Core - Worksheet | Showcase, Assessment & Reflection

Name: _____ Class: _____ Date: _____

My workflow and my choices

Fill in the three parts of your workflow, then write the two choices you will justify and a reason for each.

Part What it is in my project

Input

AI process step

Output

Choice 1 and my reason

Choice 2 and my reason

Verify and limits

Write one way you checked your output was trustworthy, and one output you would not trust yet and why.

Learned and goal

Complete the reflection: one thing you learned about your workflow, and one specific goal for next term.

Exit ticket

Name the AI process step in your workflow, one way you verified its output, and one specific goal you set for next term.

Challenge (stretch)

Challenge (stretch): explain how your solution would change for a harder task or a higher-stakes context - for example, a hospital or exam setting. Name one extra check or safeguard you would add and why it would matter more there.

Week 10 - Lab - Worksheet | Showcase, Assessment & Reflection (Lab)

Name: _____ Class: _____ Date: _____

Stress-testing: run the same workflow on an easy case and a hard case, then log what breaks.

Peer-review note

For one workflow you reviewed, write two stars linked to a criterion and one concrete improvement step.

Part of the note What I wrote

Star 1 (criterion + strength)

Star 2 (criterion + strength)

Step (one concrete improvement)

Stress-test log

Run your workflow on the harder or higher-stakes case. Record where it holds up, where it fails, and the safeguard you would add.

Question My answer

Where did it hold up?

Where did it fail?

Safeguard I would add

My measurable goal

Write one measurable Term-2 goal from your feedback and log, and say how you will know it is met.

Tool log - Showcase the Scratch / Teachable Machine solution + peer-review cards

Record as you work. Goal: A demonstrated solution, completed peer-review cards and a Term-2 goal.

Step What I did What I saw / result

1 Demonstrate your solution's input -> process -> output to a

2 Explain two design or evaluation choices you made.

3 Visit two other solutions and complete a peer-review card
fo

4 Collect your peer cards and read the feedback.

5 Write one Term-2 goal based on the feedback.

Safety check (tick): I used no personal data (no faces, voices or names). ☐

Responsible-use & human-in-the-loop reflection

- Where did a person decide or check in today's workflow (not the AI)? Name the moment.

- One responsible-use point: could this workflow treat someone unfairly or be misused, and what would you check?

- Creator's reflection: what did you build or design today, and how is it different from what an AI would produce on its own?

Exit ticket

Give one star-star-step note in short, name one thing your stress-test exposed, and write your one measurable goal for next term.

Challenge (stretch)

Challenge (stretch): explain in full how your solution would change for a higher-stakes context such as a hospital, exam or

bank. Name two safeguards you would add, say which strand each protects, and explain why a single verify step is not enough there.

New Core Lesson - How Models Learn

Grade 7 (Year 8) | Phase 3 / Cycle 2

Emerging tier · Core period · 1 x 45 min (+10-min Prerequisite Primer as pre-lesson time)

Strand: AI Conceptual Understanding | Framework ref(s): AIF·CU·G7·E | Place: insert as a W1/W2 Core lesson so the rule-based vs learned distinction is taught BEFORE the first Lab sort depends on it.

Objectives

- Distinguish a rule-based system from a learned (trained) model, with an example of each.
- Explain the difference between training (teaching the model) and inference (using it).
- State what a trained model stores - learned patterns/parameters - and that it does NOT keep copies of the training examples.
- Explain that a classifier gives the same output for the same input (deterministic), and that variation belongs to generative models that sample.
- Describe a train/test split and why a model is tested on unseen data.
- Read a confidence score as a distinct idea from accuracy.

I can... (student-facing)

- I can say whether a system follows fixed rules or has learned from examples, and why.
- I can explain the difference between training a model and using (inference) a model.
- I can explain that a model stores learned patterns, not saved examples.
- I can explain why a classifier gives the same answer for the same input.
- I can say what a train/test split is and read a confidence score.

Teacher background (the technical concept)

Rule-based vs learned. A rule-based system does exactly what a person wrote (if metal → Recycle). A learned model is shown many labelled examples and adjusts internal values (parameters) until it can map inputs to the right class. Nobody writes the rule by hand; the model finds patterns in the data.

Training vs inference. Training is the one-off teaching phase where the model sees labelled examples and updates its parameters. Inference is every later use: you give a new input and the finished model produces an output. During inference the model does NOT change - this is why a

Teachable Machine model does not 'keep learning' from students as they test it.

What a model stores. A trained model does not save the training photos or sentences. It stores learned patterns as parameters (numbers). That is why it can answer inputs it never saw (generalisation) - and also why you cannot simply 'look up' why it gave an answer (the black-box gap).

Determinism. A classifier is deterministic: the same input gives the same output and the same confidence score every time. Variation is a deliberate property of generative models that sample from a range of likely outputs (which is why a chatbot can word an answer differently twice). Students who re-run their own classifier build should get an identical result - so if the answer changes, the input or the tool changed.

Train/test split. To know if a model really learned (rather than memorised), you hold back some labelled examples as a test set and check accuracy only on those unseen items. Testing on the training data would flatter the model.

Confidence score. Many classifiers output a confidence score (e.g. 0.90) showing how sure the model is for THIS input. It is not the same as accuracy (a count over many items), and a high confidence score can still be wrong.

Anticipated misconceptions

Misconception How to address it

A lookup table (or matching cards) is machine learning.

Remove all matching rows for one item and run it - the lookup fails

or guesses. A trained model can still give a sensible answer because it generalises. That gap is the difference.

The model keeps learning while we use it.

Show a trained Teachable Machine model: test several new inputs and confirm it does not update. Learning happened only at Train; inference does not change the model.

The same input can give a different answer each time.

Re-run the class classifier on the identical input - identical output and confidence. Only generative models that sample are designed to vary.

The model saves our example photos. Explain it stores learned patterns (numbers), not the images; that is

how it handles new photos it never saw.

Lesson plan (45 minutes)

Phase / time Teacher (I do / we do) Students Key question

Hook / 6 min Show two machines: a coin-sorter

(rule) and a photo-tagger that

improved after training. Ask what is different.

Predict which one

'learned'.

What makes one of these learn and the other not?

Rule vs learned / 10

min

Define rule-based vs learned; sort

Deck A cards on the board.

Sort 4 cards into fixed-

rules / learns-from-

examples with a

reason.

How can you tell a

learned system

from a rule?

Train vs inference /

10 min

Train a 2-class Teachable Machine

model live; narrate 'this is training'.

Then test new inputs; narrate 'this is

inference - the model is not

changing'.

Note the confidence score; predict if a re-run changes it.

Does the model keep learning while we test it?

Stores patterns + determinism / 9 min

Explain the model stores patterns not photos; re-run the same input to show identical output.

Explain to a partner why the same input gives the same output.

Why does the same input give the same answer?

Train/test split + close / 10 min

Model a train/test split with cards; hold back 3 as the test set; check accuracy on them. Run exit check.

Compute accuracy on the held-out set as a fraction and %.

Why test on examples the model never saw?

Timing note: the 10-minute Prerequisite Primer runs as separate pre-lesson time (see Grade7_Revised_Lesson_Timings). The five phases above total 45 minutes.

Formative check (exit ticket)

1. Name one rule-based system and one learned system, and say how you can tell them apart.
2. In one sentence, what is the difference between training and inference?
3. A classifier gives 'cat, 0.95' for a photo. If you run the same photo again, what will it give - and why?
4. We had 12 labelled cards. Why hold back 3 as a test set instead of testing on all 12?

Look for: rule = written by a person / learned = from examples; training teaches once, inference uses without changing; identical output because deterministic; test set checks generalisation on unseen data.

Cross-curricular links

- Maths - accuracy on the test set as a fraction and a percentage (x out of total).
- Science - the train/test split as a fair test: change one thing, hold the rest constant.
- Language - classifiers behind Arabic-English translation.

Fairness & Bias - Technical Note

Grade 7 (Year 8) | Phase 3 / Cycle 2

Emerging tier · supports Main L5/L6 and Bridging W4/W6

This note gives teachers (1) a concrete procedure to MEASURE fairness, and (2) the technical sources of bias. It turns fairness from 'read a scenario and name who is affected' into something students can count and compare.

Part 1 - A fairness-evaluation procedure students can run

Fairness can be measured by comparing how well a classifier works for one group of cases against another. Use the existing W6 check log - just add the group column.

1. Pick a characteristic that splits your TEST set into groups (e.g. clean items vs greasy items; metal vs paper vs plastic; short words vs long words).
2. Run every test item through the classifier and mark each correct or incorrect (as usual).
3. Count correct per group and write it as 'correct out of total' for each group.
4. Turn each count into a fraction, then a percentage - this is the group's accuracy.
5. Compare the rates. A clear, repeated gap is the start of a fairness problem; trace the likely cause in the data.
6. If the groups are too small or the rates are too close to tell, record 'cannot determine' - do not over-claim.

Worked example

Group Correct / total Fraction Percentage Read-out

Clean items 5 / 6 5/6 83% Works well

Greasy items 2 / 6 2/6 33% Much worse - possible bias

Gap - - 50 points Investigate the training data

Interpretation: the classifier is far less accurate on greasy items. Likely cause: the training data had few greasy examples (training-data bias). Fix: add balanced greasy examples and re-test. If instead we had 5/6 vs 4/6 on tiny groups, the honest answer is 'cannot determine'.

Teacher cue: model this calculation aloud once - count correct per group, write each as a fraction then a percentage, then compare - before students run their own evaluation. The 'cannot determine' line shows the honest result when groups are too small or too close.

Part 2 - Technical sources of bias

Bias is a repeating, systematic error that makes a system perform worse for one group of cases than another. It can enter at four points:

2

Source What it is Classroom-scale example Fix

Training-data bias The examples over-represent some groups and under-represent others.

40 apple photos, 3 mango photos → misses mangoes.

Balance the classes; add representative examples.

Selection bias The data collected does

not match who/what the tool will actually meet.

Trained only on adult voices, used by children.

Collect data that represents the real users/cases.

Label / measurement bias

The labels or measurements are wrong or inconsistent.

Cards labelled in a rush; many mislabelled.

Re-label carefully; check a sample for errors.

Design bias The way the task or input is designed disadvantages a group.

A form that only fits some users' answers.

Redesign the input/task to serve all cases.

Key terms (consistent with the packs)

- bias - a repeating, systematic error that makes a system perform worse for one group of cases than another.
- fairness - treating comparable cases or groups consistently; measurable by comparing accuracy across groups.
- accuracy - correct out of total (fraction/percentage) - the measure used above.
- 'cannot determine' - an allowed result when the evidence is too thin to judge.

Cross-curricular links

- Maths - fractions and percentages to compare group rates.
- Science - splitting the test set by one characteristic is a fair test.

Revised Lesson Timings - Fitting Every Lesson into 45 Minutes

Grade 7 (Year 8) | Phase 3 / Cycle 2

Resolves the '55 minutes in a 45-minute period' defect (reviewer 4.1 / 6.3)

The problem

Each lesson as written listed a 10-minute 'Prior knowledge & Prerequisite Primer' PLUS five in-period phases that already summed to 45 minutes (Hook 6 + Frame 8 + Explore 14 + Co-build 12

+ Check 5 = 45). Total = 55 minutes in a 45-minute period. The primer must not be counted inside the teaching period.

The rule

Adopt ONE of these two options for every lesson; Option 1 is the default.

Option How it works When to use

1. Primer as separate

pre-lesson time

(default)

The 10-minute Prerequisite Primer is delivered as a starter/Do-Now BEFORE the timed period (e.g. as students arrive, or the last 10 min of the previous session). The 45-minute period then runs the five phases exactly as written.

Most schools - keeps the rich phase plan intact.

2. Primer folded into

the 45 minutes

Compress the primer to a 5-minute recall and trim the five phases to 40 minutes using the compressed template below.

When no pre-lesson time exists.

Corrected phase-timing template - Option 1 (45-min period, primer separate)

Apply this to every Core and Lab lesson. The named phases differ by lesson, but the minutes stay fixed.

Phase Minutes Purpose

Pre-lesson: Prerequisite Primer 10 (separate) Activate prior knowledge; not part of the 45-min period

Hook 6 Curated stimulus; surface noticing/wondering

Frame / Teach 8 Introduce or model the lesson's core move

Explore / Build (main activity) 14 Pairs do the hands-on task (ONE activity - see note)

Apply / Consolidate 12 Extend, log evidence, or co-build the artefact

Check & close 5 Exit ticket + formative check

Period total 45

Corrected phase-timing template - Option 2 (primer inside 45 min)

Phase Minutes

Compressed recall (primer) 5

Hook 5

Frame / Teach 7

Explore / Build 13

Apply / Consolidate 10

Check & close 5

Period total 45

One activity per Lab, not two (reviewer 4.1 / 6.3)

Each Lab lists a 'Hands-on activity guide' (unplugged card routine) and a 'Lab tool integration' (Teachable Machine / Scratch). These are the SAME task at two levels, not two tasks. Run ONE within the 14-minute Explore/Build slot: use the tool integration when the tool is available; otherwise run the unplugged activity guide (the fallback). Never schedule both in one 45-minute period.

Foundation lesson placement - 'How Models Learn' (reviewer 1.2)

The new Core lesson Foundation - How Models Learn

(Grade7_New_Concept_Lesson_How_Models_Learn.docx; interactive lesson 'How Models Learn' HTML) is scheduled as a stand-alone Foundation session immediately before Week 1 Core

- not as a numbered Week 1 or Week 2 lesson, so it does not renumber or conflict with any existing Week 1-10 content. It runs as its own 45-minute period using the Option 1 template above, with the same 10-minute Prerequisite Primer pattern (delivered pre-lesson, not counted in the 45 minutes). It appears as row 'F' in the Program-at-a-glance table in the Teacher Pack, directly before L1 (Week 1 Core - Unit Launch), and Week 1 Core's own Prerequisite Primer assumes this Foundation session has already been taught.

Worked example - Week 1 Core, Option 1

Phase Min What happens

Pre-lesson primer 10 Recall Grade 6: data trains models; activate with a vending-machine input/output question

Hook 6 Curated AI stimulus; note one noticing + one wondering

Frame 8 Introduce input-process-output; model on the vending machine (and where it breaks)

Explore & sort 14 Pairs sort example cards; ONE activity (cards OR board demo)

Co-build wall & charter 12 Shared vocabulary wall + safe-use charter

Check & close 5 Exit ticket: classify one example + one question

Teachers: replace the '(10 min)' label on each lesson's 'Prior knowledge & Prerequisite Primer' with '(10 min, pre-lesson - not counted in the 45-minute period)'.

Printable Card Decks - Term 1 Labs

Grade 7 (Year 8) | Phase 3 / Cycle 2 |

Rule-vs-Learning · Output/Verdict · Bias & Fairness · Workflow-Step · Plan · Risk-Scenario · Block

Print one set per pair unless noted. Card size: print 2 columns x 4 rows per A4 page and cut. All content is non-personal. These decks supply the physical cards named in the Tooling Guide and the Lab lessons (Main L1, L5, L7 and Bridging W4, W6, plus the unplugged fallback everywhere).

Deck A - Rule-based vs Learned System cards (Main L1)

Students read each card, agree the input/process/output, then sort into two piles: 'follows fixed rules' or 'learns from examples'. Answer key in the right column - do not print the key onto the student cards.

Borderline cards (A9-A11) can be argued either way - use them to push the discussion toward

WHY the categorisation matters and how systems often mix fixed rules with learned parts (Advanced-tier reasoning).

System card (front) Sort → pile Why (teacher key)

A1 Vending machine: coin + choice in → snack out Fixed rules A person wrote the rule; same input, same output

A2 Spam filter that improved after seeing many emails

Learns from

examples

Learned patterns from labelled data

A3 Calculator: $7 \times 8 \rightarrow 56$ Fixed rules Exact arithmetic rule, no learning

A4 Photo app that tags cats after training on cat photos

Learns from

examples

Trained on examples;

generalises to new photos

A5 Traffic light on a fixed timer Fixed rules Pre-set schedule, no data

A6 Voice assistant that recognises spoken Arabic Learns from examples

Trained on many voice samples

A7 Automatic door with a motion sensor Fixed rules If motion → open; a written rule

A8 Translation app: Arabic $\rightleftharpoons$ English Learns from examples

Learned from many translated sentence pairs

A9 Music app playlist: hand-picked 'rules' (skip live tracks) PLUS a learned 'you may like' list

Borderline -

argue both

Hybrid: part is fixed rules a

person wrote, part is

learned from listening

history. Advanced point:

most real systems mix

both - what matters is

which part decides.

A10 Spam filter that starts with a fixed block-list AND keeps learning from what users mark as spam

Borderline -

argue both

A fixed rule-list is

combined with a model

that updates from

2

examples. Ask: which part

changed the verdict for

THIS message?

A11 Thermostat that follows a set schedule

but 'learns' your typical times over a

week

Borderline -

argue both

Schedule = fixed rule;

'learns your times' =

learned pattern. Push

students to justify their

pile with evidence, not the

label alone.

Deck B - Output / Verdict cards (Main L4, L6, L9)

Each card is one AI output for a stated task. Students give a verdict per criterion (accuracy / relevance / clarity) and cite one piece of evidence. Print the output side only.

Task Output on the card Hidden fault (key)

B1 Translate 'exit' to Arabic 'خروج' (correct) None - correct, high accuracy

B2 Sort 'greasy pizza box' Recycle (confident) Wrong - should be Landfill;
confident but wrong

B3 Summarise a 3-line notice Adds a fact not in the notice Hallucination - invented detail

B4 Classify 'battery' Landfill Debatable - many councils say
special disposal

B5 Answer 'capital of the UAE' Abu Dhabi Correct

B6 Translate 'bank' (river) 'مرصف' (money bank) Wrong sense - context error

Deck C - Bias & Fairness Scenario cards (Main L5; Bridging W4, W6)

Each card describes a workflow, its training data and an output. Students name who is treated unfairly, trace the cause to a bias type, and propose one fix. Bias types: training-data, selection, label/measurement, design.

Scenario Bias type (key) Sample fix

C1 Fruit classifier trained on 40 apples, 3
mangoes; misses mangoes

Training-data

(imbalance)

Add more mango examples;

balance classes

C2 Voice tool trained only on adult voices;

fails for children

Selection Include child-voice samples
representative of users

C3 Handwriting labels typed in a rush, many
wrong

Label / measurement Re-label carefully; check a
sample for errors

C4 Form only offers 'male/female/other' as
tiny checkbox last

Design Redesign the input so all users
are served

C5 Recycling model tested only on clean
items; fails on messy

Selection (test set) Test on a representative,
messier set too

C6 Translation trained mostly on formal
news text; poor on slang

Training-data Broaden data to match real use

C7 Email sorter marks Arabic-language
messages as spam more often than
English ones

Training-data / label
bias

Train on balanced Arabic +
English examples; measure
accuracy per language and

re-test. (Groups =
languages, not people.)

C8 Recycling-photo classifier trained Selection bias Collect representative
3

mostly on large-supermarket
packaging misreads traditional-
market items

photos of local/traditional
items across the UAE
context; re-balance the
classes.

C9 Tourist-info helper trained mostly
on questions about one emirate
gives weaker answers about the
others

Selection bias Include questions from
across all seven emirates so
the data matches the real

users; re-test per emirate.

Deck D - Workflow-Step cards (Main L1, L3, L7)

Cards are arranged left-to-right into an input → process → output sequence, with a check step and a fallback. Mix and match for the sorting task.

Category Cards to print (one per card)

INPUT 'item card (object word)' · 'typed word' · 'photo of an object' · 'guided form: material/size/clean?'

PROCESS 'classifier: pick a class' · 'lookup table (simulated)' · 'rule: if metal → Recycle' · 'trained model'

OUTPUT 'label: Recycle' · 'label: Compost' · 'label: Landfill' · 'confidence score 0.____'

CHECK 'compare with trusted guide' · 'verify against a source' · 'agree / disagree?'

FALLBACK 'not sure → ask a human' · 'flag for review' · 'return to input'

Deck E - Plan cards (Main L7) & Risk-scenario cards (Main L5, Bridging W4)

Plan cards (fill in for a signature solution) Risk-scenario cards (draw and respond)

GOAL: one task my solution completes R1: A confident output is wrong and nobody checks it

INPUTS: what goes in (non-personal) R2: Someone types a real name into the tool

STEPS: ordered process steps R3: The training data left out one group of items

CHECK: how I test the output R4: The tool is used for something it was not built for

RISK: what could go wrong R5: The model is trusted because it 'sounds' sure

FALLBACK: what happens if the check fails R6: Data is kept forever with no deletion plan

Deck F - Peer-review cards (Main L10)

Two stars and a step, tied to the shared criteria.

Field Prompt on card

Star 1 (criterion) One strength tied to a criterion: _____

Star 2 (criterion) A second strength tied to a criterion: _____

A step One concrete improvement (about the work, not the person): _____

Strand tag Understand ☐ Build ☐ Evaluate ☐ Responsible use ☐

Deck G - Block cards (Main L3, L4, L7, L8, L10)

Cards printed as Scratch-style instruction blocks. Students arrange them left-to-right into one working input → process → output sequence on the desk, then read the stack aloud as a program. This is the unplugged fallback for every Scratch build in the unit (see the Tooling Guide).

Category Cards to print (one per card)

4

INPUT 'when green flag clicked' · 'ask ... and wait' · 'sensing: [item] input card'

PROCESS / IF 'if metal → Recycle' · 'if paper → Recycle' · 'else → Landfill' · 'repeat until done'

OUTPUT 'say [label]' · 'say the result' · 'show confidence: 0.____'

CHECK 'compare with trusted guide' · 'agree / disagree?'

FALLBACK 'not sure → ask a human' · 'return to input'

Data-safety: every card uses object words, gestures or neutral text only - no names, faces or voices. Print, laminate and reuse across the term.

Sample Core Lesson:

Grade 7 · Main Course · Term 1

Week 1 · Core

AI as a Workflow

Something goes in, something happens, and something comes out - every AI tool follows a shape you already know. This unit you become a Systems Detective who can spot the input, the process and the output inside any tool. | About 45 minutes

What you will learn

✓

I can ask at least two good questions about how AI turns an input into an output.

✓

I can sort examples into input, process and output and say why.

✓

I can help write our unit success criteria and our safe, responsible-use rules for AI.

Lesson route

Work through each part in order.

1

The three-stage shape
Input, process, output inside any tool

2

Ask sharp questions
What a real workflow question sounds like

3

Our responsible-use rules
Agree the class safe-use charter

AI as a Workflow

Something goes in, something happens, and something comes out - every AI tool follows a shape you already know. This unit you become a Systems Detective who can spot the input, the process and the output inside any tool. | About 45 minutes

Lesson

Part 1

Part 2

Part 3

Summary

Worksheet

Cases cracked:



Full lesson · Week 1 · Core

AI as a Workflow

Read the whole lesson, then try the practice below and complete your worksheet.

◀ Previously

Welcome, Systems Detective - this term you learn to see the input, process and output inside any AI tool.

Next ▶

Next you compare two AI examples side by side and log exactly where their outputs start to differ.



I can...

I can ask at least two good questions about how AI turns an input into an output.

I can sort examples into input, process and output and say why.

I can help write our unit success criteria and our safe, responsible-use rules for AI.



Words to know

input - what you give the system to start, like a question or a photo

process - the steps inside that change the input into something new

output - what the system gives back to you at the end

model - the part of an AI that has learned patterns from lots of examples

responsible use - using AI safely and fairly, with no personal data and always checking the result

Today we launch a new unit. AI tools follow a shape you already know from everyday machines: something goes in, something happens, and something comes out.

AI is a workflow

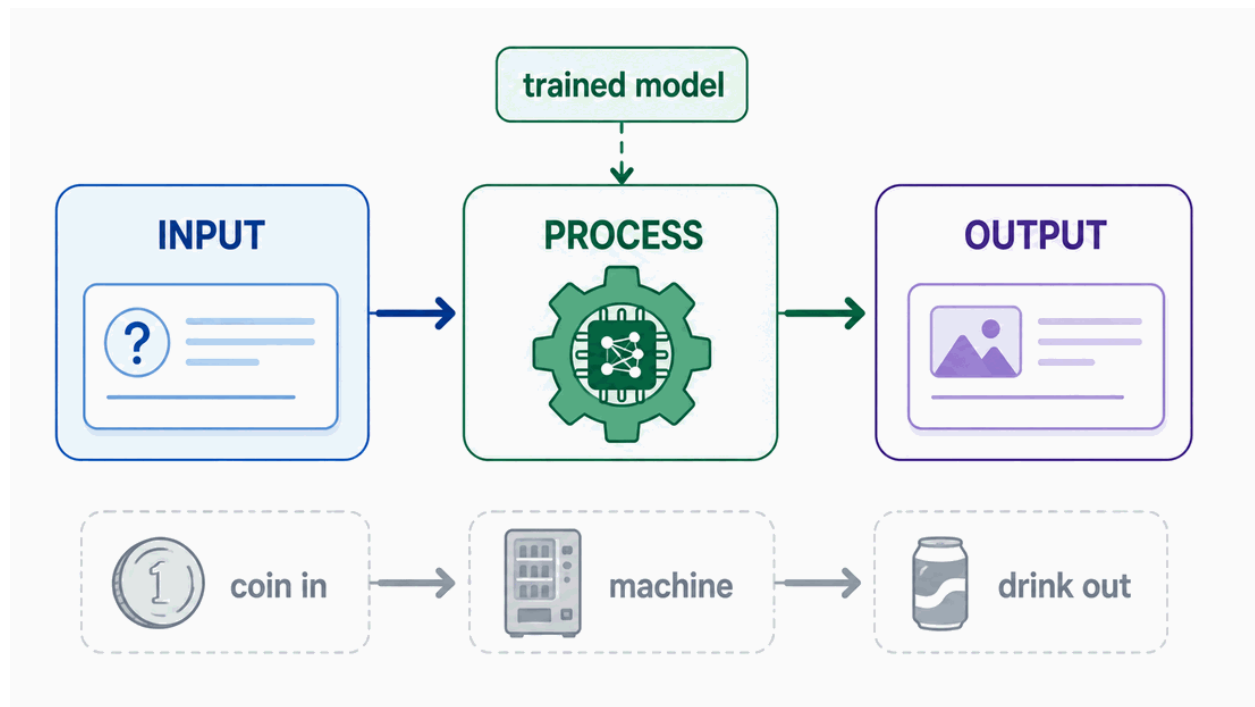
Think about a vending machine. You choose a drink (input), the machine works inside (process), and your drink drops down (output). AI works the same way. You give it something to start, it runs steps using a model that learned from many examples, and it gives you a result. When you can spot the input, the process and the output, you can understand almost any AI tool - and later you will build your own simple workflow.

Where this analogy breaks: a vending machine follows fixed rules a person wrote, so it always does exactly the same thing. An AI model is different - it has learned its patterns from many examples rather than from rules typed in by hand. Use the vending machine only to see the input-process-output shape, not to explain how the AI 'decides'.

A confident answer can still be wrong

AI outputs can look neat and sure of themselves, but that does not make them true. The model is guessing patterns, so it can get things wrong. That is why we verify before we trust: we check the output against the task and a source we trust before we use it. We also keep ourselves safe - no real names, faces or private details go into any tool, and an adult always runs the tool for us.

Lesson figure



Every AI tool follows the same shape: input, then process, then output.

🌐 Why it matters

Why it matters - across the UAE, AI helps with things like translating signs for visitors, sorting recycling, and answering questions at government service desks; spotting the input, process and output helps you understand and question the tools around you.

🛠️ Let's try it - hands-on

Watch the class stimulus and note one thing you noticed and one thing you wonder.

Turn your wondering into two clear questions about AI as a workflow.

Sort each example card into input, process or output with your partner.

Write one reason for each card you sorted.

Add one word to the vocabulary wall and suggest one safe-use rule.

💭 Think about it

Think about it - a classifier gives the SAME output every time you give it the same input; that is what makes it a classifier. Only generative AI that samples is designed to vary. So if you re-run a build and the answer changes, what actually changed - the input, or the tool you used?

📝 My reflection

My reflection: One question I still have about AI workflows is ...

 Cross-curricular link · Science

In Science, photosynthesis is an input-process-output system: sunlight, water and carbon dioxide go in, the leaf's reactions transform them, and glucose and oxygen come out. Have students label a plant diagram with the same three stages they use for an AI workflow.



Tool for this week

Unplugged - simulated-AI cards (teacher-led)

This week: a 'simulated AI' paper routine

This is an unplugged concept lesson: the teacher runs a simulated-AI card routine on the board (no live tool, no logins).

Students reason about pre-prepared example outputs on the cards - no personal data is used.

Use the worksheet to sort or judge the cards and explain the thinking with the key vocabulary.

The live tools (Teachable Machine and Scratch) are used in the matching Lab period.

Cards

input

process

output

Sort / match /

sequence

record on worksheet,

give the reason



Practice · interactive

Try it yourself

Three quick activities and case check-ins to lock in the lesson.

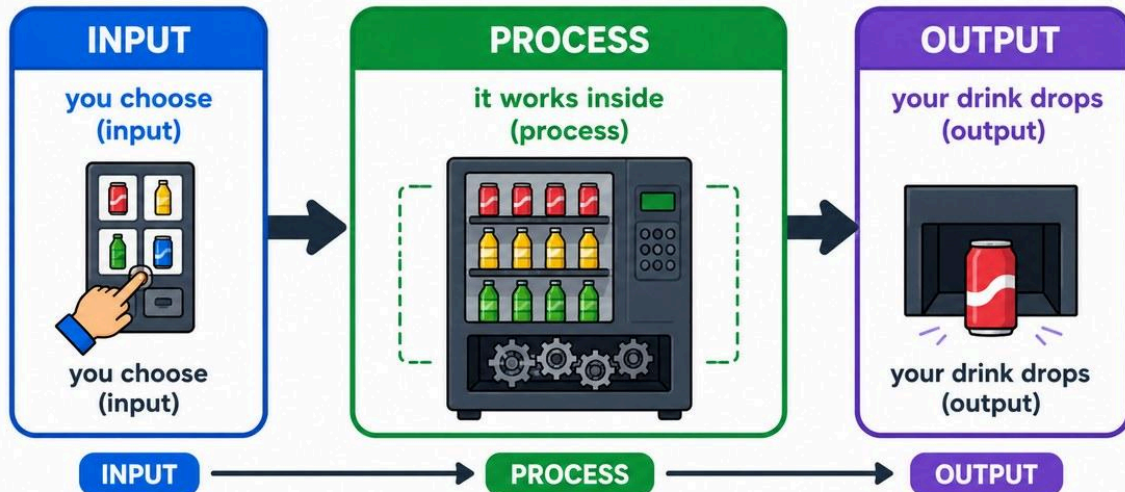
Part 1 of 3

The three-stage shape

What goes in, what happens, what comes out?

VENDING MACHINE WORKFLOW

A vending machine shows the shape: you choose (input), it works inside (process), your drink drops (output).



1Figure 1. A vending machine shows the shape: you choose (input), it works inside (process), your drink drops (output).

Figure 1. A vending machine shows the shape: you choose (input), it works inside (process), your drink drops (output).

AI is a workflow

Think about a vending machine. You choose a drink (input), the machine works inside (process), and your drink drops down (output). AI follows the same shape: you give it something to start, it runs steps using a model that learned from many examples, and it gives you a result.

Where the analogy breaks: a vending machine follows fixed rules a person wrote, so it always does the same thing. An AI model has learned its patterns from examples instead. Use the machine only to see the input-process-output shape, not to explain how the AI 'decides'.



Input, process or output?

Objective 2 · classify examples and justify

Do it

Tap a card, then drop it in the right stage. Ask: is this what goes IN, the steps INSIDE, or what comes OUT?



A question you ask a voice assistant



The English translation shown on screen



The hidden steps that change your input



A label the tool returns, like 'desert'



A learned model matching patterns inside



The words you type for the tool to answer



A photo you point at an Arabic sign



Input



Process



Output



Systems Detectives · Case 1

The Translated Sign

“

A visitor points a translate app at an Arabic sign, and English words appear on the screen. Detective, which part of this workflow is the input?”



goes in



changes



comes out

Which part is the input?

A

The photo of the Arabic sign the app was given.

B

The model matching the patterns that it learned.

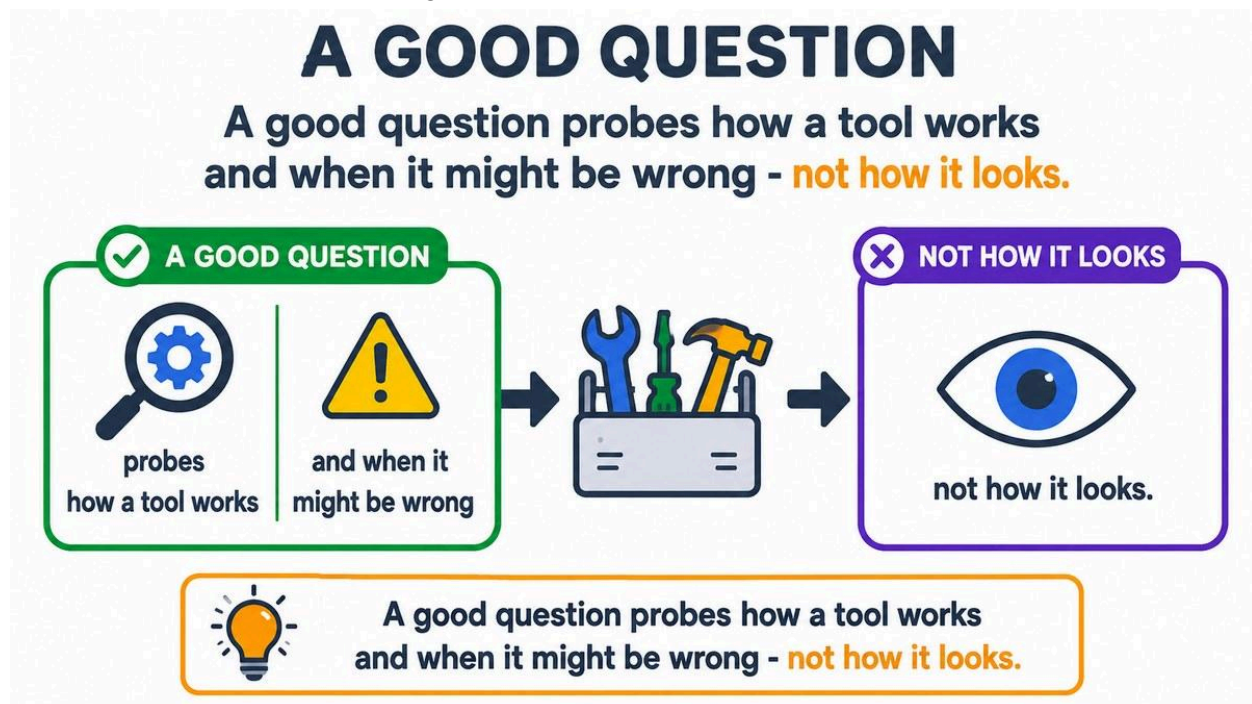
C

Nothing at all - the app made the words itself.

D

The English words the app showed on the screen.

Part 2 of 3
Ask sharp questions
When is an answer worth trusting?



2Figure 2. A good question probes how a tool works and when it might be wrong - not how it looks.

Figure 2. A good question probes how a tool works and when it might be wrong - not how it looks.

A confident answer can still be wrong

AI outputs can look neat and sure of themselves, but that does not make them true. The model is guessing patterns, so it can get things wrong. That is why we verify before we trust: we check the output against the task and a source we trust before we use it.

A Systems Detective asks sharp questions about how a tool decides - what goes in, what the process does, and how to check the result - rather than accepting an answer just because it sounds confident.



Which are genuine detective questions?

Objective 1 · ask focused questions

Do it

Tap the **THREE** questions that really probe how the workflow works.



How much did the app cost to make?



What colour is the app's logo?



What happens in the process before the output?



What input does this tool need to start?



How could I check whether the output is true?



How many stars did it get in the store?



Systems Detectives · Case 2

The Confident Answer

“

An AI tool gives a neat, confident-sounding answer, but a fact inside it looks off. Detective, what should you do before you use it?”



neat answer



looks off



decide

What is the smart next move?

A

Delete the tool for good and never open it again.

B

Trust it only if the font on screen looks nice.

C

Verify it against the task and a trusted source.

D

Trust it at once because it just sounds confident.

Part 3 of 3

Our safe-use charter

What makes a rule keep us safe?

Responsible use

Responsible use means using AI safely and fairly. We keep ourselves safe: no real names, faces or private details go into any tool, and an adult always runs the tool for us.

We also check results instead of trusting them blindly. Today the class agrees a safe-use charter - a short list of rules everyone can follow - and adds new words to the vocabulary wall.



Does it belong in our charter?

Objective 3 · agree responsible-use rules

Do it

Tap a rule, then choose a bin. Ask: does this keep us safe and can everyone follow it?



Never type real names, faces or private details



An adult runs the AI tool for us



Share your password so a partner can log in



Always check the output before we trust it



Upload a photo of a classmate to test it



Trust any answer that sounds confident



Belongs in our charter



Does not keep us safe



Systems Detectives · Case 3

The Test Photo

“

A pair wants to try a new AI tool by uploading a photo of their friend's face along with the friend's real name. Detective, what is the responsible choice?”



a face



personal data



decide

What should the pair do?

A

Upload the face and the real name to test the tool.

B

Type the friend's home address into the tool too.

C

Share the photo publicly to collect more likes.

D

Use no personal data and let an adult run it.



Away from the screen · Hands-on mission

Systems Detectives: sort, question, charter

Open the unit as a Systems Detective. Sort example cards into input, process and output, post the questions you most want answered, and help agree one class rule for safe AI use.

What you need:

Example workflow cards

A mini whiteboard

Sticky notes

A pencil

Watch the class stimulus and note one thing you noticed and one thing you wonder.

Turn your wondering into at least two clear questions about AI as a workflow.

With your partner, sort each example card as input, process or output.

Write one reason for each card - why does it belong there?

Add one word and its meaning to the class vocabulary wall.

Propose one responsible-use rule for the safe-use charter.



No stimulus clip? Describe one worked input-output example aloud and sort the printed cards by hand - the mission runs fully unplugged.

Lesson summary

You launched the unit and started thinking like a Systems Detective.

Key ideas to remember

Every AI tool follows the same shape: input, then process, then output.

The process uses a model that learned patterns from examples - it does not think like a person.

A confident-looking output can still be wrong, so we verify before we trust.

A strong responsible-use rule is clear, keeps data private, and can be followed.

What to do next

Next you compare two AI examples side by side and log exactly where their outputs start to differ.



Print / Save worksheet as PDF

Type your answers in the boxes below, then use this button to print the completed worksheet or choose "Save as PDF".



Student worksheet

AI as a Workflow

Print this page or save it as a PDF to complete by hand.

Name: Class: Date:

Lesson figure

	INPUT	PROCESS	OUTPUT
 typed question			
 sorting photos			
 finished summary	reason: _____	reason: _____	reason: _____

Sort each example card into the correct column and give a reason.

My two questions

Write at least two questions about AI as a workflow. Aim at least one at the input, the process or the output.

Sort it: input, process or output

Write each example in the right column and add one reason.

Example Input / Process / Output Reason

A typed question to a chatbot

Grouping photos by what is in them

A finished travel summary

Our safe-use charter

Write one responsible-use rule for the class and why it keeps us safe.

 Exit ticket

Exit ticket - Name one example, say whether it is an input, a process or an output, and explain why. Then write one question you still have.

★ Challenge (stretch)

Challenge - Pick a real UAE situation (for example a smart parking app or an Arabic-English translator). Describe its input, process and output, then predict two things a strong AI solution would need to get right. This stretches toward the Proficient idea of explaining how workflows structure AI systems.

Extension & reflection

Local Connection · UAE

AI in the UAE

The UAE set a National Strategy for Artificial Intelligence looking to 2031 and, in 2017, appointed a Minister of State for AI. Many everyday UAE services already run as input -> process -> output AI workflows.

Sample Core Lab:

Grade 7 · Main Course · Term 1

Week 1 · Lab

Compare and Log the Difference

Two AI examples can share the same input-process-output shape yet give different results. Today you become an investigator - comparing real examples and logging what makes one output better than another. | About 45 minutes

What you will learn

☑

I can compare two AI examples and write down how they differ.

☑

I can trace each example through input, process and output.

☑

I can use what I observed to say what made one output better.

Lesson route

Work through each part in order.

1

Evidence, not feelings
Log a difference you can point to

2

Find where it starts
Locate the difference by stage

3

Draw a conclusion
Say what made an output better

Grade 7 · Main Course · Term 1

Week 1 · Lab

Compare and Log the Difference

Two AI examples can share the same input-process-output shape yet give different results. Today you become an investigator - comparing real examples and logging what makes one output better than another. | About 45 minutes

M1L · Compare and Log the Difference

Lesson

Part 1

Part 2

Part 3

Summary

Worksheet

Cases cracked:



Full lesson · Week 1 · Lab

Compare and Log the Difference

Read the whole lesson, then try the practice below and complete your worksheet.

◀ Previously

Previously you met the input-process-output shape - now you use it to compare two real AI examples.

Next ▶

Next you name the three stages precisely and trace one worked example from input to output.



I can...

I can compare two AI examples and write down how they differ.

I can trace each example through input, process and output.

I can use what I observed to say what made one output better.



Words to know

compare - look closely at two things to find how they are alike and different

difference - something that is not the same between the two examples

log - a record where you write down what you observe

evidence - what you actually saw, used to back up what you say

conclusion - the idea you decide on after looking at the evidence

In the Core period you learned the shape of an AI workflow. Today you become an investigator - comparing real examples and logging what makes one output better than another.

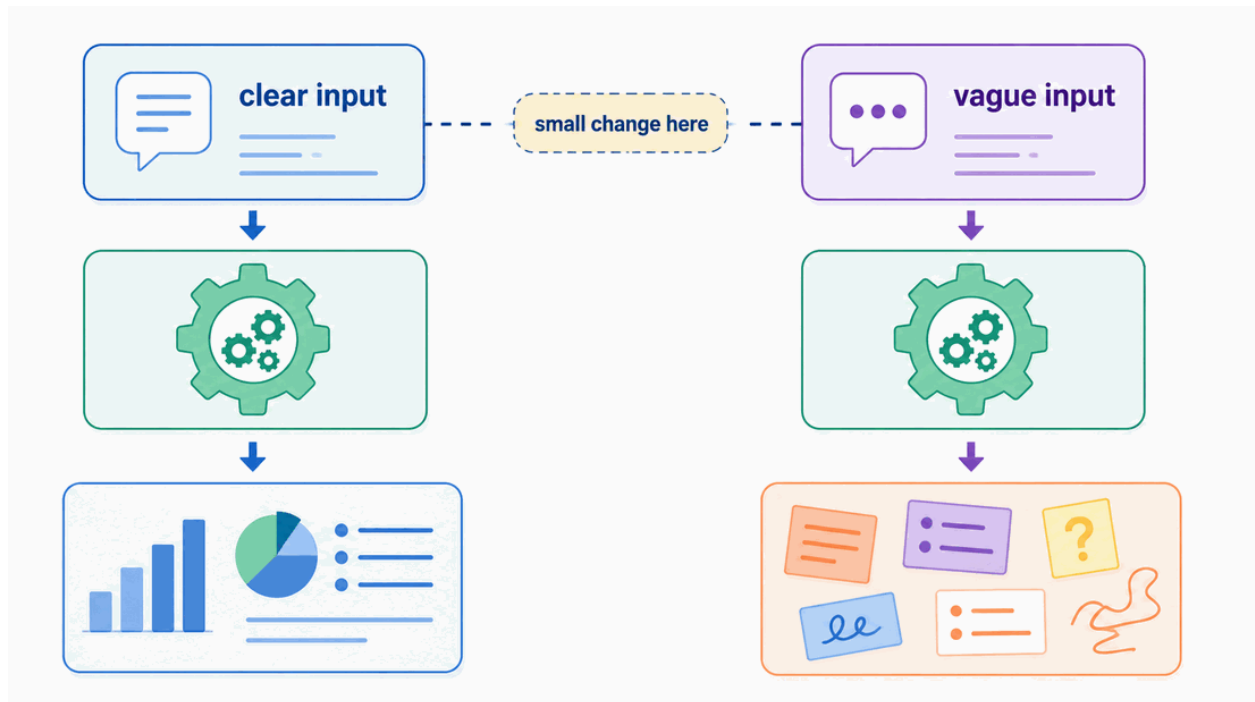
Same shape, different results

Two AI examples can share the same input-process-output shape but give different results.

Maybe one input was clearer, or the process handled it differently. When you compare them side by side and trace each through the three stages, you can spot exactly where the difference starts. That is detective work: you are not guessing which is better, you are finding evidence for it. Logging what you see turns a feeling into a fact you can explain to someone else.

Log it, then trust it

A good investigator writes things down. In your log you note each difference and the stage it came from - input, process or output. Then, before you trust a conclusion, you check it against the task: would it still be true for a new example? Remember the charter - the examples are made by the teacher, with no personal data, and a confident-looking output still has to be checked before you rely on it.



Lesson figure

Same input, small change, different output - your job is to find where the difference starts.

🌐 Why it matters

Why it matters - engineers in the UAE constantly compare AI outputs before they trust them, whether tuning a smart-traffic system in Abu Dhabi or checking an Arabic-English translation; learning to log differences is the first step of that job.

🛠️ Let's try it - hands-on

Read the first example pair and predict which output is more useful.

Trace each example through input, process and output.

Log at least two differences and mark the stage each came from.

Rotate to a new pack and log again.

Write one conclusion about what made an output better.

🛠️ Today's tool - hands-on

Tool: Card decks (workflow-example cards)

Deal the deck face-up between partners.

For each card, agree the input, the process and the output and write them on the card's three blanks.

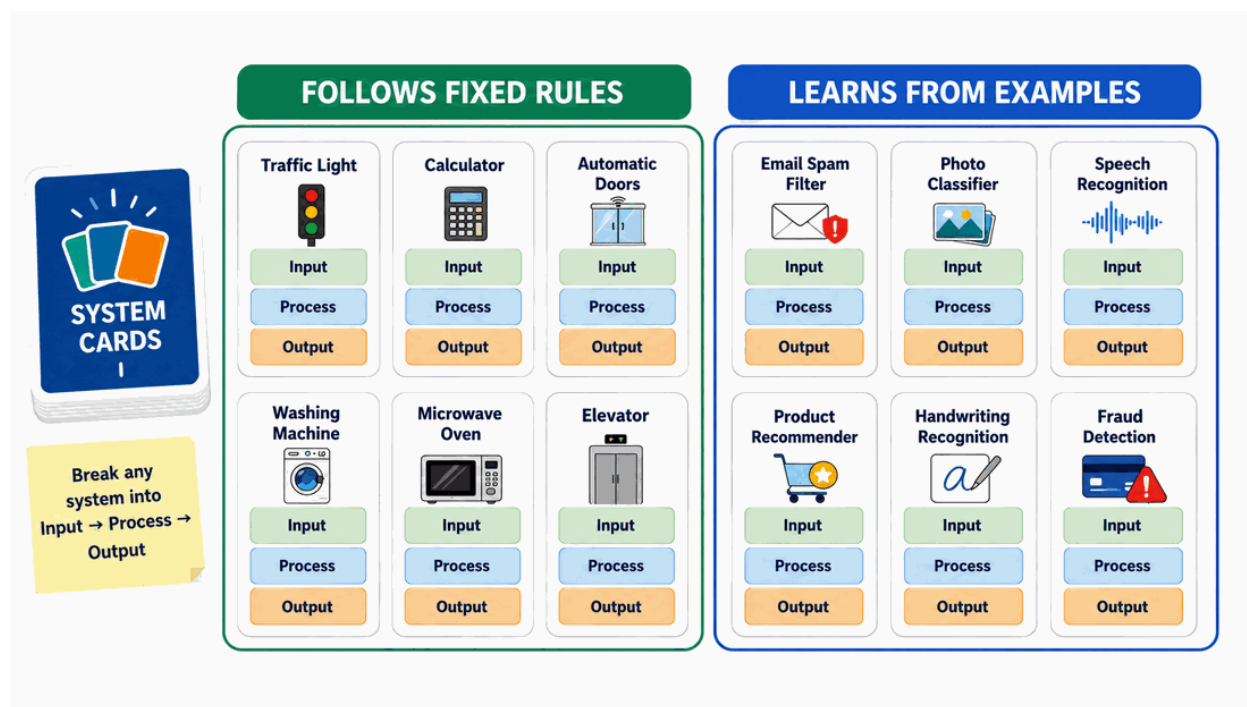
Sort the cards into 'follows fixed rules' and 'learns from examples'.

Pick the two hardest-to-place cards and say why.

Log at least three differences you noticed between the two piles.

You will make: A sorted card layout plus a difference log.

Stay safe: Fully unplugged; no devices or data.



Lesson figure

Card sort: each workflow card split into input, process and output, sorted into two piles.

💭 Think about it

Think about it - if you changed only the input and kept everything else the same, could you make the weaker output better? How would you test it?

👉 My reflection

My reflection: One difference I logged today, and where it came from, was ...

🔗 Cross-curricular link · English

In English, students compare two versions of a paragraph and log which communicates better, quoting the exact words as evidence. That is the same investigate-and-log habit used here to explain, with evidence, why one AI output beats another.



Tool for this week

Unplugged - card decks / output packs

This week: workflow-example cards

Teacher lays out the pre-printed card deck / output pack for this task (account-free, no devices).

In pairs, students sort, match or sequence the cards as the worksheet directs.

For each card read its input, process and output and record it on the worksheet.

Compare piles, then agree the reason using the lesson's key words.

Cards

input

process
output
Sort / match /
sequence
record on worksheet,
give the reason



Practice · interactive

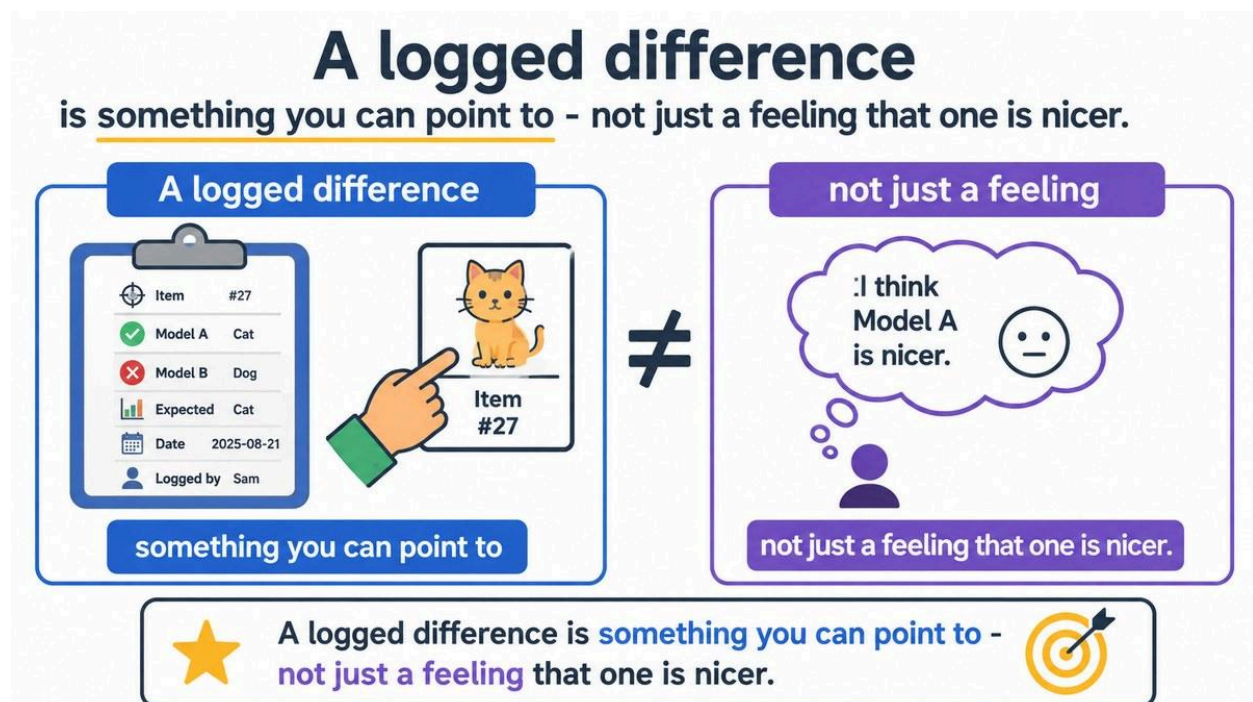
Try it yourself

Three quick activities and case check-ins to lock in the lesson.

Part 1 of 3

Evidence, not feelings

What counts as a real difference?



1Figure 1. A logged difference is something you can point to - not just a feeling that one is nicer.

Figure 1. A logged difference is something you can point to - not just a feeling that one is nicer.

Same shape, different results

Two AI examples can share the same input-process-output shape but give different results.

Maybe one input was clearer, or the process handled it differently. When you compare them side by side, you can spot exactly where the difference starts.

That is detective work: you are not guessing which is better, you are finding evidence for it.

Logging what you see turns a feeling into a fact you can explain to someone else.



Evidence or just a feeling?

Objective 1 · record differences as evidence

Do it

Tap a log note, then choose a bin. Ask: could I point to this in the examples, or is it only an opinion?



'Pack A named the city, Pack B did not.'



'This one seems nicer somehow.'



'Pack A gave three steps, Pack B gave one.'



'Pack B's answer was longer by two lines.'



'I like the blue card better.'



'Pack A just feels smarter to me.'



Evidence I can point to



Just a feeling



Systems Detectives · Case 1

The Two Packs

“

Two packs used almost the same photo input, but one answer named the city and the other did not. Detective, which log note is proper evidence?”



two packs



compare



log it

Which note belongs in the log?

A

'Pack A just feels smarter than Pack B to me.'

B

'Both packs are fine, there is nothing to note.'

C

'Pack A named the city, Pack B did not.'

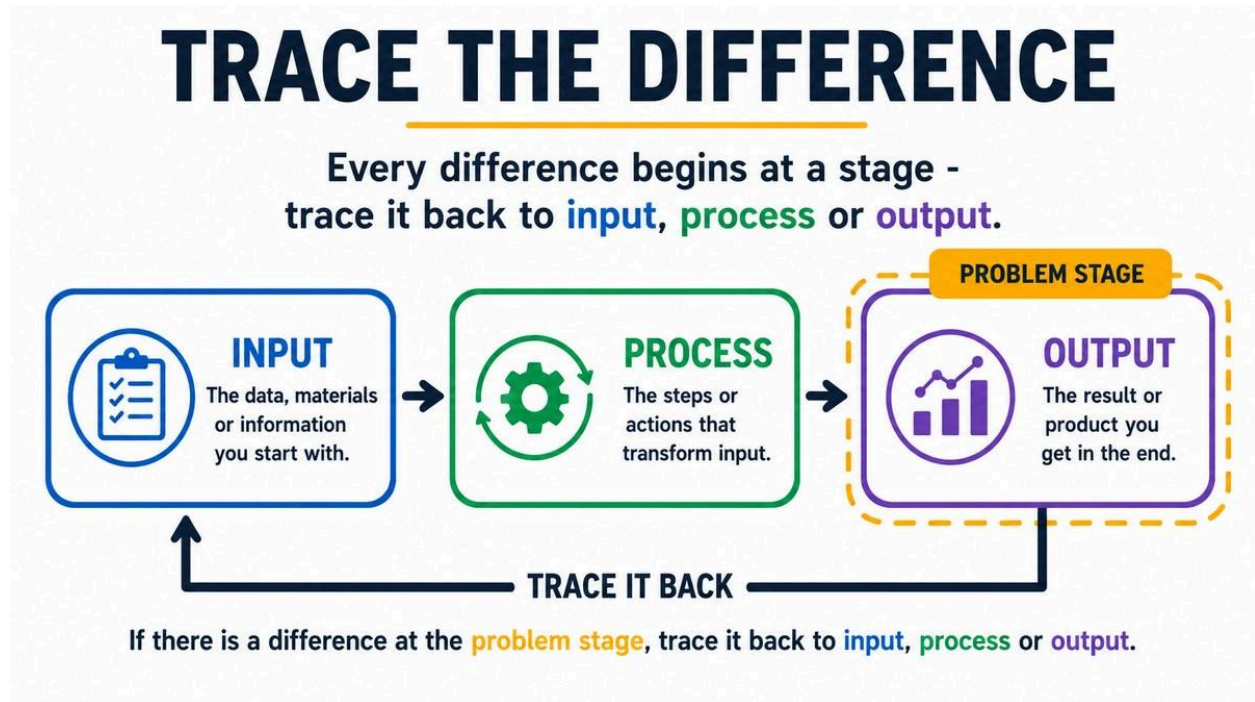
D

'I just prefer the look of the blue card here.'

Part 2 of 3

Find where it starts

Which stage caused the difference?



2Figure 2. Every difference begins at a stage - trace it back to input, process or output.

Figure 2. Every difference begins at a stage - trace it back to input, process or output.

Trace, then locate

A good investigator writes things down. In your log you note each difference and the stage it came from - input, process or output. Tracing means following one item all the way through the three stages so you can see where the two examples split apart.

Often the difference begins at the input (one prompt was clearer) or the process (it handled the input differently). Locating the stage is more useful than only noticing that the outputs look different.



Where did the difference begin?

Objective 2 · locate the difference by stage

Do it

Tap a logged difference, then drop it at the stage where it first appeared.



One pack used a model trained on more examples



One process kept extra steps the other skipped



One prompt was clear, the other was vague



One photo was sharp, the other was blurry



One answer was printed longer than the other



One label was correct, the other was wrong



Input



Process



Output



Systems Detectives · Case 2

The Split Point

“

One pair typed a clear prompt and the other typed a vague one; the two outputs came out different. Detective, where did the difference begin?”



two prompts



same process



different output

Where did the difference start?

A

At the output, because it is the very last stage.

B

Nowhere at all, since two outputs can never differ.

C

At the input, where one prompt was clearer.

D

In the paper the cards happen to be printed on.

Part 3 of 3

Draw a conclusion

What made one output better?

Log it, then trust it

Before you trust a conclusion, check it against the task: would it still be true for a new example?

A conclusion backed by what you actually logged is evidence-based; a conclusion based on a hunch is not.

Remember the charter - the examples are made by the teacher, with no personal data, and a confident-looking output still has to be checked before you rely on it.



Which conclusions are backed by evidence?

Objective 3 · draw an evidence-based conclusion

Do it

Tap the THREE conclusions that are supported by what you actually logged.



This tool is just better - no need for evidence.



The nicer-looking answer must be the true one.



The output changed by pure luck.



A vague prompt led to a vaguer answer in both packs.



The difference began at the input, not the output.



The clearer input gave the more useful output.



Systems Detectives · Case 3

The Final Report

“

After logging across several packs, a detective must write one conclusion in the report.
Detective, which conclusion is safe to write?"



many logs



check task



conclude

Which conclusion is evidence-based?

A

A clearer input led to a more useful output.

B

Better outputs just happen by luck, not by reason.

C

Never write anything down in the report at all.

D

The tool with the neatest handwriting is the best.



Away from the screen · Hands-on mission

Systems Detectives: compare, trace, log

Work through workflow-example card packs. Trace each example through input, process and output, log the differences as evidence, then decide what made one output better.

What you need:

Workflow-example card decks

A difference-log sheet

Two blank cards

A pencil

Read the first example pair and predict which output is more useful.

For each card, agree the input, the process and the output and write them on the blanks.

Sort the cards into 'follows fixed rules' and 'learns from examples'.

Log at least two differences and mark the stage each began at.

Rotate to a new pack and log differences again.

Write one conclusion about what improved the output and check it against the task.



Cards are already paper-based, so no device is needed. Extend by adding two blank cards and inventing new systems to sort.

Lesson summary

You investigated real examples and logged where their differences begin.

Key ideas to remember

Two examples can share the same shape yet give different outputs.

Evidence is something you can point to; a feeling is not.

Trace a difference back to the stage where it started - usually input or process.

A strong conclusion points back to a difference you actually logged.

What to do next

Next you name the three stages precisely and trace one worked example from input to output.



Print / Save worksheet as PDF

Type your answers in the boxes below, then use this button to print the completed worksheet or choose "Save as PDF".



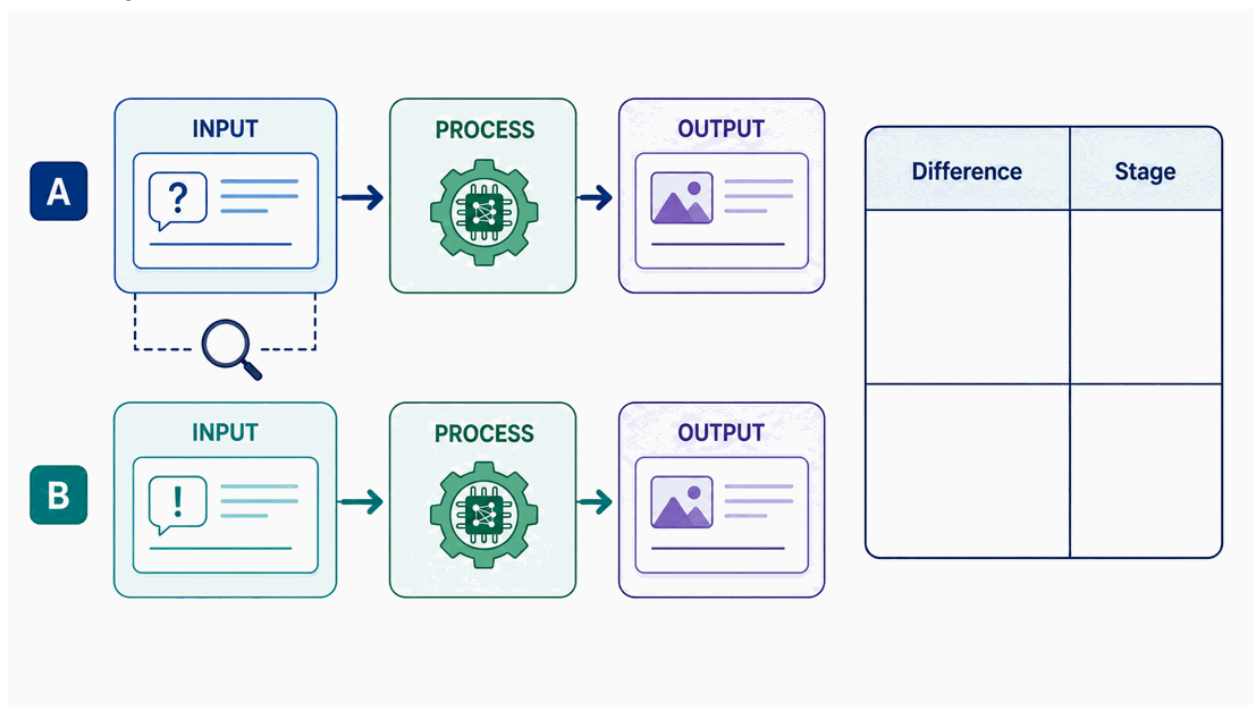
Student worksheet

Compare and Log the Difference

Print this page or save it as a PDF to complete by hand.

Name: Class: Date:

Lesson figure



Trace each example through the three stages, then log the difference and where it began.

Comparison log

For each example pair, write one difference and tick the stage where it began.

Example pair Difference I observed Stage (input / process / output)

Pair 1

Pair 2

Pair 3

My conclusion

Use your log as evidence. What one change most improved the output?

Verify it

How could you check your conclusion is right for a new example?

Tool log - Card decks (workflow-example cards)

Record as you work. Goal: A sorted card layout plus a difference log.

Step	What I did	What I saw / result
1	Deal the deck face-up between partners.	
2	For each card, agree the input, the process and the output a	
3	Sort the cards into 'follows fixed rules' and 'learns from e	
4	Pick the two hardest-to-place cards and say why.	
5	Log at least three differences you noticed between the two p	

Safety check (tick): I used no personal data (no faces, voices or names). ☐

 Exit ticket

Exit ticket - Name one difference you logged and the workflow stage it came from, and say whether it made the output better or worse.

★ Challenge (stretch)

Challenge - Design your own fair test: choose one thing to change in the input, predict how the output should change, and describe how you would compare the before and after to prove it.

Then explain what a strong UAE solution built on this idea would need to get right. This stretches toward the Proficient idea of evaluating outputs using criteria such as accuracy and usefulness.

Extension & reflection

Local Connection · UAE

AI in the UAE

The UAE set a National Strategy for Artificial Intelligence looking to 2031 and, in 2017, appointed a Minister of State for AI. Many everyday UAE services already run as input -> process -> output AI workflows.

Reflect

Responsible use & human in the loop

The AI tool does the sorting or drafting, but a person stays in charge. Answer in your worksheet or below:

Where did a person decide or check in today's workflow (not the AI)?

Could this workflow treat someone unfairly, or be misused - and what would you check?
Creator's reflection: what did you build or design today, and how is it different from what an AI would produce on its own?

Assessment documents

FINAL ASSESSMENT

Grade 7 (Year 8)

Unit: AI as Workflows - Input, Process, Output

ASSESSMENT IDENTIFICATION

Assessment End-of-module final assessment

Module 9 of 14

Phase / Cycle Phase 3 - AI Solution Builder | Cycle 2

Term / Tier Term 1 | Emerging tier

Competency strands AI Conceptual Understanding; AI Solution Design &

Development; Critical Evaluation & Informed

Interaction; Ethics & Governance

Framework reference ADEK K-12 AI Fluency Framework - AIF - Grade 7 -

Emerging

Format Performance-based (formative checkpoints +
summative task)

Administration 55-65 minutes (may be split into two sessions) |

individual work with structured pair discussion

Version / Date Version 1.0 | 2026

Prepared by AI Curriculum Unit

A curriculum-aligned, performance-based assessment of the four AI Fluency competency strands for Module 9, in which students map an AI workflow, design a multi-step input-process-output solution with a check step and fallback, evaluate its outputs against criteria, and audit its risks. It includes formative checkpoints, an authentic summative task, an analytic rubric, moderation guidance, annotated exemplars, and a full teacher guide.

For teacher use in ADEK schools. Aligned to the delivered Grade 7 Main Course (Term 1, 'AI as Workflows - Input,

Process, Output'). Not for distribution to students.

ADEK K-12 AI Fluency | Final Assessment | Grade 7 - Module 9: AI as Workflows

1. Assessment overview and design rationale

This final assessment measures how well Grade 7 students have met the four AI Fluency competency outcomes taught across the Term 1 Main Course, 'AI as Workflows - Input, Process, Output'. It sits in Phase 3, AI Solution Builder, where the demand moves from understanding AI systems to designing the workflows that underlie them: students describe and design a multi-step workflow (input to process to output), evaluate its outputs against criteria such as accuracy and usefulness, improve it through iteration, and reason about its risks and governance. The assessment is built to be valid (it measures the workflow design and evaluation that was taught), reliable (two teachers reach the same

judgement from the same evidence), and fair (every learner has an equitable route to show what they know).

The assessment is performance-based and account-free. Consistent with the delivered course, all live AI use is teacher-mediated and no student logs in or enters personal data; students reason with a workflow map, a planning frame, pre-generated output packs, an evaluation grid and a risk-and-impact matrix rather than typing into a live model. Students demonstrate learning by designing and judging a real workflow, not by recalling isolated facts.

What this assessment contains, at a glance

Component	Purpose	Evidence type	Time
-----------	---------	---------------	------

Formative checkpoints			
-----------------------	--	--	--

(2)

Assessment FOR learning - confirm

workflow reasoning and evaluation habits

mid-unit and adjust teaching before the

summative task.

Teacher observation

and short student

products

10-15 min

each, in-

lesson

Summative

performance task (4

parts)

Assessment OF learning - one authentic

workflow-building scenario covering all four

strands.

Workflow map,

planning frame,

evaluation grid, risk

audit; observation;

conversation

55-65 min

(may be

split)

Analytic rubric Judge each strand at Emerging / Proficient /

Advanced against observable descriptors

keyed to the scenario.

Teacher judgement

record

During or

right after

the task

Evidence and
moderation tools

Record evidence consistently and agree
judgements across teachers and classes.

Observation record,
triangulation table,
moderation protocol

As needed

Annotated exemplars Anchor judgements to worked examples at
each level.

Reference material Reference

Research-based design principles

Page 2 of 23

ADEK K-12 AI Fluency | Final Assessment | Grade 7 - Module 9: AI as Workflows

The design draws on well-established assessment research so that results can be trusted
and used to move learning forward:

Backward design (Wiggins and McTighe). The task and rubric were built from the
four framework outcomes first, then the evidence each requires, then the scenario, so
every part earns its place and nothing is assessed that was not taught.

Constructive alignment (Biggs). Each task part is aligned to one primary strand, its
success criteria, and a cognitive-demand level, shown in the alignment map in Section
2 and the blueprint in Section 4.

Formative assessment (Black and Wiliam). Two checkpoints make workflow
design and evaluation visible early and carry specific feedback moves, so teaching can
adapt before the summative task.

Triangulation of evidence. Judgements draw on three sources - what students
produce (workflow map, planning frame, evaluation grid, risk audit), what they do
while building and checking, and what they say in the justification conversation - not a
single worksheet.

Cognitive demand (Webb Depth of Knowledge and the SOLO taxonomy). Parts
progress from describing how a workflow turns an input into an output (DOK 1-2) to
designing, evaluating and improving a workflow and reasoning about its risks (DOK 2-3
/ SOLO Relational), matching the demand levels used in the lessons.

Universal Design for Learning (CAST). Multiple means of engagement,
representation and expression are built in, so access arrangements are part of the
design rather than an afterthought.

Academic integrity. Parts 1, 2 and 4 are recorded individually so each student's map, design
and audit are their own; structured pair talk is expected during exploration in Part 3 but recorded
judgements must be independent. All output packs and scenario materials are teacher-curated
and account-free; students enter no personal data and use no live AI tool without teacher
mediation. Teachers may scribe or accept alternative response modes without changing the
substance of a student's answer.

Page 3 of 23

ADEK K-12 AI Fluency | Final Assessment | Grade 7 - Module 9: AI as Workflows

2. Curriculum alignment map

This section makes the link between the delivered curriculum and every part of the assessment explicit and auditable. A teacher or reviewer can trace each strand from the framework outcome, to where it was taught, to where it is assessed.

Module identification

Field Detail

Programme Grade 7 Main Course, Term 1 (Emerging tier)

Module 9 of 14

Phase / Cycle Phase 3 - AI Solution Builder | Cycle 2

Unit theme AI as Workflows - Input, Process, Output

Signature activity

assessed

Plan, build, evaluate and govern a multi-step input-process-output AI workflow (a classifier solution) with a check step and a fallback path

Weeks covered Foundation (How Models Learn) and Weeks 1-9 (Core + Lab, 45-minute periods)

Framework reference AIF - Grade 7 - Emerging (CU, SD, CE, GE)

Framework outcomes assessed (the four strands)

Competency strand Emerging outcome
(target)

Proficient outcome Advanced outcome
(stretch)

AI Conceptual

Understanding

Describes how AI

systems follow a

sequence of steps (input
to process to output).

Explains how workflows
structure AI systems.

Explains how different
workflow designs affect
outcomes.

AI Solution Design &
Development

Builds simple AI solutions
using guided steps.

Designs multi-step AI
workflows to complete
tasks.

Refines workflows by
improving structure,
sequencing, and tool
selection.

Critical Evaluation & Informed Interaction

Identifies issues in outputs or workflow steps.

Evaluates outputs using criteria such as accuracy and usefulness.

Improves workflows through iteration and explains the impact of changes.

Ethics & Governance Identifies risks in AI-generated outputs.

Explains bias, misuse, and limitations in workflows.

Proposes ways to reduce risks and improve responsible use of AI systems.

Emerging is the target standard for this Term 1 tier. Proficient and Advanced descriptors are shown

so teachers can recognise and credit students who reason beyond the target, and to keep vertical

continuity with Grade 8.

Alignment matrix: taught, then assessed

Page 4 of 23

ADEK K-12 AI Fluency | Final Assessment | Grade 7 - Module 9: AI as Workflows

Each strand is mapped to the weeks that teach it and the assessment component that provides evidence, with the cognitive-demand level for the summative part.

Strand Where

taught

Success criteria assessed Assessed by Demand

AI Conceptual

Understanding

Foundation;

W1-W2

Name the input, process and output stages in order; explain how the workflow structures the system; show how a change to the design changes the outcome

Checkpoint 1;

Summative Part 1

DOK 1-2 /

SOLO

Relational

AI Solution Design &

Development

W3, W7-W8 Design an ordered multi-step

workflow with a check step and

a fallback path; name inputs,

tool and non-personal data

Checkpoint 1;

Summative Part 2

DOK 2-3 /

SOLO

Relational

Critical Evaluation &

Informed Interaction

W4 Judge outputs against agreed

criteria with evidence; compute

accuracy as correct over total;

propose one iteration and its

effect

Checkpoint 2;

Summative Part 3

DOK 2-3 /

SOLO

Relational

Ethics & Governance W5 Identify risks in the outputs;

name bias, misuse or a

limitation; rate on a likelihood-

and-impact matrix; propose a

prioritised mitigation

Checkpoint 2;

Summative Part 4

DOK 2-3 /

SOLO

Relational

Vertical coherence (progression)

From Grade 6: students distinguished rule-based from learning-based AI systems, designed structured prompts, evaluated outputs for bias and privacy, and reasoned about multiple stakeholders. Grade 7 shifts the focus to the workflow itself - how a system turns an input into an output through a process, how to design and iterate a multi-step workflow, how to judge its outputs against criteria such as accuracy and usefulness, and how to audit and reduce its risks.

Towards Grade 8: students will compare competing workflow designs and argue why one is better than another for a given task. A strong Advanced performance here - explaining how a different workflow design would change the outcome - is early evidence of readiness for that comparative reasoning.

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3. Learning outcomes and student-facing success criteria

Teachers share these student-facing 'I can' statements before and during the unit so learners know what success looks like. They are written in the same language used in the lessons.

Strand Student-facing success criteria ('I can...')

AI Conceptual

Understanding

I can name the three stages of an AI workflow - input, process and output - in the correct order.

I can trace a worked example and explain how the input became the output.

I can explain how the way a workflow is designed changes the result it gives.

AI Solution Design &

Development

I can design a multi-step workflow by putting input, process and output steps in the right order.

I can name the input, the teacher-operated tool and the non-personal data my solution needs.

I can add a check step and a fallback path so a wrong output is caught before it is trusted.

Critical Evaluation &

Informed Interaction

I can judge an output against agreed criteria - accuracy, usefulness and clarity - and back my verdict with evidence.

I can work out accuracy as the number correct out of the total.

I can name one improvement to a weak step and say what it would change.

Ethics & Governance I can spot risks in an AI output, such as a factual error, bias or over-reliance.

I can rate a risk for how likely it is and how much harm it would do.

I can propose a mitigation that changes the workflow, and say why it is the one to build in first.

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4. Assessment blueprint

The blueprint shows how the summative task distributes emphasis and cognitive demand across the four strands. Each strand carries roughly equal weight so no single strand dominates the overall judgement.

Summative part Primary strand Cognitive demand

Evidence Weight

Part 1 - Map the workflow

AI Conceptual Understanding

DOK 1-2 Completed workflow map +

talk

25%

Part 2 - Design the workflow

AI Solution Design & Development

DOK 2-3 Planning frame (individual)

25%

Part 3 - Judge and improve

Critical Evaluation & Informed Interaction

DOK 2-3 Evaluation grid + iteration note

25%

Part 4 - Audit the risks Ethics & Governance DOK 2-3 Risk-and-impact audit +

conversation

25%

Cognitive-demand spread. The task rises from naming and tracing the stages of a workflow (DOK 1-2) to designing a multi-step workflow, evaluating its outputs with evidence, and reasoning

about its risks (DOK 3). Advanced probes in each part give every student a route to demonstrate

higher-order reasoning without penalising those still consolidating the Emerging outcome.

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5. Formative checkpoints (Assessment FOR learning)

Run these two short checkpoints during the unit, not on the summative day. Each names when to use it, what to look for at each level, and a specific feedback move that closes the gap before the summative task.

Checkpoint 1 - 'Trace and build a workflow' (use after Week 3)

Strands: AI Conceptual Understanding and AI Solution Design & Development.

Activity. Show a three-box input-process-output strip on the board for a simple classifier (for example, sorting typed words into 'question' or 'statement'). Ask students to (a) label the three stages and say which step the AI does, and (b) on a mini-whiteboard, order a set of scrambled step cards into a working workflow and name what the input must look like for the output to be useful. Allow three minutes.

Look-for Emerging Proficient Advanced

Tracing the stages Names input and output
but muddles the process
step

Names input, process and
output in order and points
to the AI step

Explains how the input
shapes the output as it
moves through the
process

Ordering the build Places most step cards in
order with support

Independently orders the
steps into a working input-
process-output chain

Orders the chain and
names where a check step
should go

Feedback move. If a student cannot name the process step, re-model the three-box strip with the

everyday-machine analogy (input in, the machine works, output out) from Week 2. If the order is shaky, hand them the step cards and have them run one real input along the chain with a finger, saying each step aloud.

Checkpoint 2 - 'Judge it, then check the risk' (use after Week 5)

Strands: Critical Evaluation & Informed Interaction, and Ethics & Governance.

Activity. Give pairs two pre-generated outputs for the same classifier task, one confidently wrong. Ask them to (a) judge each against the agreed criteria (accuracy, relevance, clarity) and write one piece of evidence per verdict, and (b) name one risk in the wrong output, say who it could affect, and propose one responsible-use measure using the stem '...because...'.
Look-for Emerging Proficient Advanced

Judging with
evidence

Gives an opinion without
naming a criterion

Names the criterion and
points to the exact word
or fact as evidence

Weighs accuracy against

usefulness to decide which
output better serves the
task

Naming and Says 'it might be wrong' Names the risk type Locates the risk at a

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reducing risk without naming the risk (error, bias, over-reliance)

and who it affects, with a

measure

workflow stage and

proposes a measure that

changes that stage

Feedback move. If a verdict rests on 'I like it', insist on a named criterion and a pointed piece of evidence, modelling one row of the grid. If a risk stays at 'it might be wrong', prompt 'what type of

risk is it, who does it hurt, and which stage - input, process or output - does it sit at?' to move the reasoning onto the workflow.

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6. Summative performance task - 'The Sort-It

Station'

An authentic task built directly on the unit's signature activity: a multi-step AI workflow that classifies items to help a school Sustainability Week. AI Noor School is setting up a 'Sort-It Station' where an AI helper takes a photo of a dropped item and sorts it into Recycle, Compost or Landfill so the bins stay clean. The four parts move through all four strands. Read the scene-setting script, then run the parts in order.

Scene-setting script (read aloud). 'For Sustainability Week, our school is building a Sort-It Station. When someone drops an item at the station, an AI helper takes a photo, a trained classifier decides Recycle, Compost or Landfill, and the helper shows the answer so the item goes

in the right bin. The classifier was trained on example photos, so it is usually right but not always.

Your job is to map how this workflow works, design it step by step with a check and a safe fallback, judge how good its outputs are, and audit its risks so the school can use it responsibly. There are no wrong answers when you explain your thinking - show it by writing, drawing, arranging cards, or telling me.'

Materials to prepare (see Appendix)

Sort-It Station Workflow Map (Appendix A) - a partly completed input-process-output diagram with two boxes and one arrow-label left blank.

Workflow Planning Frame and plan cards (Appendix B) - a printed frame with Goal, Input, Process steps, Output, Check step and Fallback, plus a small word bank of step verbs.

Sort-It Output Pack and Evaluation Grid (Appendix C) - six pre-generated classifier outputs (some correct, some flawed) for the same set of items, with a criteria grid and

an accuracy box. Account-free and teacher-curated.

Risk-and-Impact Audit sheet (Appendix D) - a two-by-two likelihood-and-impact matrix and a risk table.

A pencil, and coloured pens or highlighters; the input-process-output diagram and the criteria strip (accuracy, relevance, clarity) on display.

Part 1 - Map the workflow

Strand: AI Conceptual Understanding | DOK 1-2

Give each student the Sort-It Station Workflow Map (Appendix A) and read the stages aloud. Ask:

'Fill the two blank boxes and the missing arrow-label so the workflow runs from input to output in order. Which single step does the AI do?'

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'Trace one item - a plastic bottle - through your map. How did the input become the output?'

Advanced probe: 'Suppose we redesigned the workflow so the photo went straight to the bin with no check step. How would that change the outcome, and for which items would it matter most?'

Part 2 - Design the workflow

Strand: AI Solution Design & Development | DOK 2-3 | Completed individually

Give each student the Workflow Planning Frame (Appendix B). Point to the word bank, then say: 'Design the Sort-It Station workflow. Write a one-sentence goal, then order your input, process and output steps. Name the input, the teacher-operated classifier tool and the non-personal data it uses. Add a check step after the output, and a fallback path for when the check fails.' When finished, ask each student: 'Why are your steps in that order, and what does your fallback do when the check fails?'

What a complete design shows (for the teacher). Goal names one task (sort a dropped item into Recycle, Compost or Landfill).

Steps are ordered so the output of one feeds the next; the AI classifier is the process step.

Resources name the input (a photo of an object, no faces or names), the teacher-operated classifier, and the non-personal training data.

A check step compares the label against a trusted recycling guide; a fallback path (for example, 'if the check fails, send the item to a human to sort') keeps the station safe.

Part 3 - Judge and improve the outputs

Strand: Critical Evaluation & Informed Interaction | DOK 2-3

Give each student the Sort-It Output Pack and Evaluation Grid (Appendix C). Say: 'Here are six outputs the classifier gave for the same items, with the correct answers. Judge each against our criteria - accuracy, usefulness and clarity - and write one piece of evidence per verdict. Work out the accuracy: how many did it get right out of six? Then pick the weakest step and write one improvement and what it would change.' Two to three minutes may be spent comparing in pairs, but the grid is recorded individually.

Planted results in the pack (for the teacher). The classifier gets four of six items right, so accuracy is 4/6, about 67 per cent.

One error is a confident wrong label - a greasy pizza box marked Recycle - to test the 'confident

does not mean correct' idea.

The classifier is weakest on items it saw few of in training (for example, food-soiled card), pointing

the iteration at the input or training data.

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Part 4 - Audit the risks

Strand: Ethics & Governance | DOK 2-3

Give each student the Risk-and-Impact Audit sheet (Appendix D) and say: 'The classifier was trained mostly on clean, common items, so it does worse on messy or unusual ones.

Audit the Sort-It Station.' Then ask:

'List two risks in the classifier's outputs, and say who each could affect.'

'Rate each risk on the matrix for how likely it is and how much harm it would do.'

'Propose one mitigation that changes the workflow, and say why it is the one to build in first.'

Advanced probe: 'The classifier is worse on food-soiled items because it saw few of them in training. What is really causing that - the items, or the training data - and how would you fix the cause, not just the symptom?'

How the task maps to the strands

Part Strand What strong evidence looks like Demand

1 AI Conceptual Understanding Completes the map in order, names the AI process step, and explains how a design change would change the outcome

DOK 1-2

2 AI Solution Design & Development

Designs an ordered multi-step workflow with named resources, a check step and a working fallback path

DOK 2-3

3 Critical Evaluation & Informed Interaction

Judges every output against the criteria with evidence, computes accuracy, and proposes a targeted iteration

DOK 2-3

4 Ethics & Governance Names and rates real risks, ties the weak spot to skewed training data, and proposes a prioritised mitigation

DOK 2-3

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7. Analytic rubric

Judge each strand independently against the descriptors below. Circle the observed level.

The rubric describes observable performance, so two teachers using it on the same

evidence should reach the same level.

Competency

strand

Emerging Proficient Advanced

AI Conceptual

Understanding

(Part 1)

Completes part of the workflow map but leaves a stage blank or out of order, or cannot say which step the AI does.

Completes the map in the correct order, names the AI process step, and explains how the workflow structures the system so the input becomes the output.

Also explains how a different workflow design - such as removing the check step - would change the outcome, and for which items it would matter most.

AI Solution Design

& Development

(Part 2)

Fills some boxes but the workflow is missing a step, out of order, or has no check step or fallback.

Designs an ordered multi-step workflow that completes the task, names the input, tool and non-personal data, and adds a working check step and fallback path.

Refines the design as they build - improving the sequencing, the check, or the tool choice - and

explains why the change makes the workflow more reliable.

Critical Evaluation

& Informed

Interaction

(Part 3)

Says an output is right or wrong but does not name a criterion or give evidence, and cannot compute accuracy.

Judges each output against accuracy, usefulness and clarity with a piece of evidence, computes accuracy as correct over total, and names one improvement.

Weighs accuracy against usefulness to decide which output best serves the task, and explains what the proposed iteration would change and why.

Ethics &

Governance

(Part 4)

Identifies at least one risk in the outputs but does not name who it affects or how to reduce it.

Names bias, misuse or a limitation, says who it affects, rates it on the matrix, and proposes a stage-linked mitigation.

Traces the weak spot to skewed training data, prioritises the mitigation using likelihood and impact, and proposes a way to improve

responsible use of the
whole workflow.

Overall level judgement

Overall level Decision rule

Emerging

(expected)

0-2 strands at Proficient, the rest at Emerging. The student describes the stages of the workflow, builds a solution with guided steps, spots issues in outputs, and identifies a risk, but does not yet independently design a multi-step workflow, evaluate with criteria and evidence, or reduce a risk.

Proficient At least 3 of 4 strands at Proficient or above, none below Emerging. The student explains how the workflow structures the system, designs an ordered multi-step workflow with a check and fallback, evaluates outputs against criteria with accuracy computed, and explains bias or a limitation

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with a mitigation.

Advanced At least 3 strands at Advanced, with no strand below Proficient. The student reasons about how a different design changes the outcome, refines the workflow as they build, explains the impact of an iteration, and traces a risk to the training data with a prioritised mitigation.

Working towards Emerging. If a student cannot yet meet the Emerging descriptor in a strand even with support, record 'working towards' and use the linked bridging content (Section 11) before re-checking. This is diagnostic information, not a failure. For this Term 1 tier, Emerging is the expected standard; Proficient and Advanced credit students who reason beyond it.

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8. Evidence recording and triangulation

Base each judgement on three sources of evidence, not one worksheet. This raises reliability and gives students more than one way to show what they know.

Evidence source What to collect Strands it best
evidences

Products (what students
make)

The completed workflow map (Part 1), the
planning frame (Part 2), the evaluation grid
(Part 3) and the risk-and-impact audit (Part 4)

All four strands

Observations (what
students do)

How students order the steps, place the check
and fallback, judge outputs against criteria, and
plot risks on the matrix

AI Solution Design &

Development; Critical
Evaluation & Informed
Interaction

Conversations (what
students say)

Answers to the Part 1 trace, the Part 2 'why this
order and fallback' question, and the Part 4
advanced probe

AI Conceptual

Understanding; Ethics &
Governance

Teacher observation record (one per student)

Field Record

Student name / Class _____ Date: _____

AI Conceptual Understanding E P A Note: _____

AI Solution Design &
Development

E P A Note: _____

Critical Evaluation &
Informed Interaction

E P A Note: _____

Ethics & Governance E P A Note: _____

Overall judgement EMERGING PROFICIENT ADVANCED

Access arrangements used _____

Next step / bridging _____

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9. Standardisation and moderation guide

Moderation makes judgements consistent across teachers and classes. Run a short moderation meeting after the first set of students has been assessed.

1. Each teacher independently levels the same 2-3 anonymised student sets (map, planning frame, evaluation grid and audit) using the rubric in Section 7.
2. Compare levels. Where they match, note the rubric language that decided it. Where they differ, return to the observable evidence and the annotated exemplars in Section 10 to agree a level.
3. Agree borderline decisions: a student Proficient on three strands but Emerging on one is Proficient overall; a student Advanced on two strands and Proficient on two is Proficient (Advanced needs three).
4. Record any task-specific decisions - for example, how much scaffolding on the planning frame still counts as Proficient, or whether a fallback that is safe but vague counts - so the whole team applies them the same way.

Common pitfalls to avoid

Rewarding fluency over reasoning. A confident talker who cannot order the workflow or link the output to a criterion is still Emerging on that strand.

Crediting a plan with no check or fallback as Proficient. Part 2 reaches Proficient only when the design includes a working check step and a fallback path, not just three ordered boxes.

Accepting a verdict without evidence. In Part 3, credit Proficient only when each verdict names a criterion and points to the exact word, fact or item that decided it, and accuracy is computed.

Halo effect. Level each strand on its own evidence; a strong Part 2 design does not lift a weak Part 4 audit.

Counting scaffolding as independence. If the workflow order or the risk rating needed heavy teacher prompting, it is Emerging, not Proficient.

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10. Annotated exemplars

Use these worked responses to anchor judgements. Each shows a plausible Grade 7 performance and the annotations that place it at a level. The scenarios are constructed, not real children.

Exemplar - EMERGING

Student response (observed) Annotation (why this level)

Part 1: fills the input and output boxes but leaves the process box blank; when asked which step the AI does, points to the whole map. Part 2: writes three boxes in order but adds no check step or fallback. Part 3: says 'output 3 is wrong' but names no criterion and cannot work out the accuracy. Part 4: 'It might sort things wrong,' but does not say who it affects or how to reduce it.

Describes the workflow partly and builds with guided steps, spots an issue and names a risk - the Emerging outcomes across the four strands.

No independent multi-step design, no criteria-based evaluation, and no mitigation. 0-2 strands at Proficient - Emerging overall (the expected Term 1 standard). Next step: re-run the Week 2 three-stage trace and the Week 3 check-and-fallback build.

Exemplar - PROFICIENT

Student response (observed) Annotation (why this level)

Part 1: completes the map in order and says 'the classifier is the process step - it decides the label.'

Part 2: designs an ordered workflow, names the photo input, the teacher-operated classifier and the non-personal training data, adds a check against the recycling guide and a fallback ('if the check fails, a person sorts it'). Part 3: judges all six outputs

against accuracy, usefulness and clarity with evidence, computes accuracy as 4/6, and proposes clearer wording on the output. Part 4: names bias, says it affects students with unusual items, rates it on the matrix, and proposes adding a human-review checkpoint.

Independent workflow design with a check and fallback, criteria-based evaluation with accuracy computed, and a stage-linked mitigation with who-is-affected. At least 3 strands at Proficient - Proficient overall, beyond the Term 1 target.

Exemplar - ADVANCED

Student response (observed) Annotation (why this level)

As Proficient, and: Part 1 explains that removing the check step would let confident wrong labels through and would matter most for messy items. Part 2 refines the design mid-build, moving the check earlier and swapping to a clearer output message, and explains why this makes it more reliable. Part 3 weighs accuracy against usefulness to choose the best output and predicts what the iteration would change. Part 4 traces the weak spot to too few food-soiled examples in the training data, prioritises adding balanced training images using likelihood Reasons about how a different design changes the outcome, refines as they build, explains the impact of an iteration, and fixes the cause (training data) not just the symptom. At least 3 strands at Advanced with none below Proficient - Advanced overall.

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11. Reporting, feedback and next steps

Report each student against the four strands and the overall level, and turn the result into a next step. Feedback is most useful when it is specific and forward-looking.

If the student is... Feed-forward for the student Teacher next step / bridging link

Working towards /

Emerging

'You can name the parts of a workflow and spot when something is wrong. Next, put

the steps in order yourself and add a check step that catches a bad output.'

Re-run the Week 2 three-stage trace and the Week 3 'test, fix and chain' build; use the plan cards (Goal, Inputs, Steps, Checks, Risk) with the goal supplied.

Proficient (beyond target)

'You design a working workflow and judge outputs with evidence. Next, weigh accuracy against usefulness and say what each improvement would change.'

Extend with the Week 4 evaluation clinic ranking and the Week 8 build-evaluate-improve loop; introduce comparing two designs to prepare for Grade 8.

Advanced 'You reason about design choices and trace risks to the data. Next, argue why one whole workflow design is better than another for a task.'

Offer the Week 5 risk-audit matrix stretch and the required 'compare two workflow designs' activity; bridge to Grade 8 comparative reasoning.

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12. Accessibility and Students of Determination

Access is designed in, following Universal Design for Learning and the school's Best Endeavours and Reasonable Adjustments (BERA). None of these changes the construct being assessed - they change how a student accesses the task or shows what they know.

Universal Design for Learning - built-in options

UDL principle Options offered in this assessment

Multiple means of engagement

A familiar UAE context (Sustainability Week, recycling); a low-stakes 'no wrong answers when you explain' framing; choice of which item to trace and which risks to audit.

Multiple means of representation

The input-process-output diagram and criteria strip on display; the workflow map and plan cards as both text and pictures; key terms

(workflow, input, process, output, classifier, accuracy, bias, mitigation, fallback) on a word card.

Multiple means of action
and expression

Students may write, draw, arrange plan cards, highlight, or explain aloud; a teacher may scribe; the workflow can be built by moving physical step cards instead of writing.

Reasonable adjustments by need (BERA)

Need Adjustment

Visual impairment Large-print, high-contrast workflow map, output pack and matrix; read all items and outputs aloud; allow a magnifier or screen reader; use tactile step cards.

Hearing impairment Provide the scene-setting script and all prompts in writing; face the student when speaking; check understanding visually.

Physical / fine-motor Allow pointing or verbal answers scribed by an adult; build the workflow with

cut-and-place step cards instead of writing in Part 2.

Attention / cognitive
load

Chunk into two sessions (Parts 1-2, then 3-4); one part at a time; extra processing time; a checklist of the four parts.

Language / EAL Sentence stems ('The input is ____; the process is ____; the output is ____', 'This fails accuracy because ____', 'If the check fails, my solution will ____'); a bilingual key-word card; accept answers in the student's stronger language and scribe.

Sensory needs A quiet space; a movement break between parts; reduced visual clutter on the printed sheets.

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13. Administration guide

Field Guidance

Time 55-65 minutes total, about 12-15 minutes per part. May be split into two sessions (Parts 1-2, then 3-4).

Grouping Part 1 talk and Part 3 exploration may be in pairs; Parts 2 and 4 are recorded individually. The Part 1 trace and Part 2 'why this order' question are short one-to-one conversations.

Conditions Calm and low-stakes. Reassure students that exploration and pair talk are expected; recorded answers are their own. All AI use is teacher-mediated; no student logs in or enters personal data.

When to run After Week 8, once students have planned and built the signature solution, so the workflow-design and evaluation habits are in place before Part 2 and Part 3.

Preparation Print the workflow map, planning frame and plan cards, output pack with evaluation grid, and risk-and-impact audit (Appendix); prepare coloured pens; display the input-process-output diagram and the criteria strip.

Integrity Keep the output pack and its planted results consistent across the cohort; do not

pre-mark the map blanks, the weak step, or the risks for students.

Before / during / after checklist

Before: share the success criteria; print materials; plan access arrangements; run both checkpoints earlier in the unit.

During: read the script; run parts in order; observe and note; ask the trace and 'why this order' probes; scribe where needed.

After: complete the rubric and observation record; moderate a sample; record next steps and bridging links.

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Appendix. Printable stimulus materials

Print one copy of each per student. The materials are teacher-curated and account-free; the classifier is simulated with a pre-generated output pack, so no student uses a live AI tool. Do not mark the blanks, the weak step, or the risks for students.

A. Sort-It Station Workflow Map

Complete the two blank boxes and the missing arrow-label so the workflow runs in order. Then circle

the one step the AI does.

Stage What happens here Complete this

Input A photo is taken of the dropped item. (given)

Process (step

1)

_____ Fill in: which step does the AI classifier do?

Arrow label The output of step 1 becomes the _____ of the next step.

Fill in the missing word.

Output _____ Fill in: what does the station show the user?

Check step The label is compared against the trusted recycling guide.

(given)

Teacher note: Process step 1 = the classifier decides Recycle, Compost or Landfill; the arrow label =

'input'; Output = the bin label shown to the user. The AI step is the classifier (Process step 1).

B. Workflow Planning Frame (plan cards)

Word bank of step verbs: take, prepare, classify, compare, check, show, flag, send.

Plan card Your design

Goal (one sentence) _____

Input (and non-personal data)

Process steps (in order) 1. _____ 2. _____ 3. _____

Tool (teacher-operated) _____

Output _____

Check step Compare the output against _____

Fallback path If the check fails, my solution will _____

One risk _____

C. Sort-It Output Pack and Evaluation Grid

The classifier gave these outputs for six dropped items. The correct answer is shown so you can judge accuracy. For each output, write a verdict (meets / partly / fails) against each criterion and one piece of evidence.

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Item Classifier output Correct answer Right?

1 Plastic water bottle (empty) Recycle Recycle yes

2 Banana peel Compost Compost yes

3 Greasy pizza box Recycle Landfill no

4 Aluminium can Recycle Recycle yes

5 Food-soiled paper napkin Recycle Compost no

6 Glass jar (clean) Recycle Recycle yes

Criterion Verdict (meets / partly / fails)

Evidence (the exact item or reason)

Accuracy (correct / total)

Accuracy = _____ / 6 = _____ %

Which items did it get wrong?

Usefulness (does it help sort the bin?)

Clarity (is the label clear?)

Then: pick the weakest step and write one improvement, and say what it would change. Teacher note:

accuracy is 4/6, about 67 per cent; items 3 and 5 (messy / food-soiled) are the failures, pointing the

iteration at the input photo or the training data.

D. Risk-and-Impact Audit sheet

List two risks in the classifier's outputs, say who each affects, then rate each on the matrix and propose a mitigation. Rate likelihood and impact as Low, Medium or High.

Risk (in the output) Who it affects Likelihood

Impact Mitigation (changes which stage?)

1. _____ L M H L M H _____
2. _____ L M H L M H _____

Build in first (why?) This mitigation,
because

Teacher note: a strong audit names a real risk (for example, a confident wrong label at the output stage, or bias against messy items from skewed training data), rates a high-likelihood, high-impact risk as the priority, and proposes a mitigation that changes a workflow stage - such as a human-review check step or adding balanced training images - not just a warning label.